The Little Book of Valuation

Aswath Damodaran
Stern School of Business
44 West Fourth Street, 9-96
New York, NY 10012
Email: adamodar@stern.nyu.edu
Phone: 212-998-0340

First draft: October 14, 2010

Preface

Knowing the value of an asset may not be a prerequisite for investing or a guarantee for success, but it does help us make more informed judgments. For most investors, though, valuing an asset seems to be a task that is far too complex and complicated for their skill sets. Consequently, they either depend upon those that they regard as professionals (equity research analysts, appraisers) for their valuations or ignore value entirely when investing.

In this book, I hope to show that valuation, at its core, is simple and that anyone who is willing to spend some time collecting information and analyzing it, can value an asset. I also hope to strip the mystique away from valuation practices and provide ways in which we can look at valuation judgments made by analysts and appraisers and decide whether they make sense or not.

Valuation models, in their full form, are filled with details. In fact, I have written other books for practitioners who do valuation for a living that are immersed in these details. However, here is an admission. Not all of the details in a valuation are equally important and some matter more than others. In fact, most valuations rest on one of two key drivers, though these drivers may vary from company to company. If you can identify these value drivers and estimate them with any success, you can value a company. That is what I hope to do in this book and I will leave it to you to judge whether I succeed.

Table of Contents

The Little Book of Valuation 1

First draft: October 14, 2010 1

Preface 2

Chapter 1: Introduction to Value 6

What is value? 6

Estimating Value: Intrinsic versus Relative Value 7

Why should we care? 7

The Accountant 10

Some truths about valuation 11

Conclusion 16

Section 1: Laying the groundwork 17

Chapter 2: The Tools of the trade 18

Valuation Tools: An overview 18

The Time Value of Money 19

Risk Measurement 23

Accounting data 27

Statistics 35

Looking for Relationships in the Data 37

Conclusion 40

Chapter 3: Intrinsic Value 41

Discounted Cash flow (DCF) Valuation 41

What do intrinsic valuation models tell us? 63

Conclusion 65

Chapter 4: Relative Valuation 66

What is relative valuation? 66

Standardized Values and Multiples 68

The Four Basic Keys to Using Multiples 69

Reconciling Relative and Intrinsic Valuations 85

Conclusion 86

Section 2: Valuing across the life cycle 87

Chapter 5: Valuing young growth companies and businesses 88

What is a young growth company? 88
Characteristics of young growth companies 89
Valuation Issues 90
Valuation Solutions 95
Value Plays 108
Conclusion 110

Chapter 6: Valuing growth companies 112

What is a growth company? 112
Characteristics of growth companies 114
Valuation Issues 115
Valuation Solutions 120
Value Plays 139
Conclusion 140

Chapter 7: Valuing Mature Companies 142

What is a mature company? 142
Characteristics of Mature Companies 143
Valuation Issues 145
Valuation Solutions 151
Value Plays 176
Conclusion 177

Chapter 8: Valuing Declining and Distressed Companies 179

What is a declining company? 179
Characteristics of Declining Companies 180
Valuation Issues 181
Relative Valuation 185
Valuation Solutions 186
Value Plays 209
Conclusion 211

Section 3: Special Situations in Valuation 213

Chapter 9: Valuing Financial Service Companies 214

What is a financial service firm? 214
Characteristics of financial service firms 215
Valuation Issues 217
Valuation Solutions 219
Value Plays 238
Conclusion 239

Chapter 10: Valuing Cyclical and Commodity Companies 241

What makes commodity and cyclical companies different? 241
Characteristics of commodity & cyclical firms 242
Valuation Issues 243
Valuation Solutions 244
Value Plays 262
Conclusion 264

Chapter 12: Valuing companies with intangible assets: Technology, Brand name and Patents 265

What are intangible assets? 265
Characteristics of firms with intangible assets 266
Valuation Issues 267
Valuation Solutions 269
Value Plays 296
Conclusion 298

Section 4: Mopping up 299

Chapter 12: Valuation Lessons 300

Enlightening Propositions 300
Conclusion 311

Chapter 1: Introduction to Value

It was Oscar Wilde who defined a cynic as one who "knows the price of everything and the value of nothing". That characterization would fit many investors who regard investing as a game and define winning as staying ahead of the pack. In this chapter, we will define value and follow up by drawing a distinction between the intrinsic and relative value of an asset. We will also establish a rationale for why estimating value is an integral part of success for all investors, no matter what their philosophy.

What is value?

A postulate of sound investing is that an investor does not pay more for an asset than it is worth. This statement may seem logical and obvious, but it is forgotten and rediscovered at some time in every generation and in every market. There are those who are disingenuous enough to argue that value is in the eyes of the beholder, and that any price can be justified if there are other investors willing to pay that price. That is patently absurd. Perceptions may be all that matter when the asset is a painting or a sculpture, but you do not and should not buy most assets for aesthetic or emotional reasons; you buy financial assets for the cash flows you expect to receive from them. Consequently, perceptions of value have to be backed up by reality, which implies that the price you pay for any asset should reflect the cashflows it is expected to generate. The models of valuation described in this book attempt to relate value to the level of, uncertainty about and expected growth in these cashflows.

There are many aspects of valuation where we can agree to disagree, including estimates of true value and how long it will take for prices to adjust to that true value. But there is one point on which there can be no disagreement. Asset prices cannot be justified by merely using the argument that there will be other investors around who will pay a higher price in the future. That is the equivalent of playing a very expensive game of musical chairs, where every investor has to answer the question, "Where will I be when the music stops?" before playing. The

problem with investing with the expectation that there will be a bigger fool around to sell an asset to, when the time comes, is that you might end up being the biggest fool of all.

Estimating Value: Intrinsic versus Relative Value

In intrinsic valuation, we begin with a simple proposition. The value of an asset is a function of the expected cash flows on that asset. Put simply, assets with high and predictable cash flows should be worth more than assets with low and volatile cash flows. In its most widely used version, the intrinsic value of an asset is estimated in a discounted cash flow (DCF) model, where the value is set equal to the expected cash flows on the asset.

While the focus in theory remains on discounted cash flow valuation, the reality is that most assets are valued on a relative basis. In relative valuation, we value an asset by looking at how the market prices similar assets. Thus, when determining what to pay for a house, we look at what similar houses in the neighborhood sold for rather than doing an intrinsic valuation. Extending this analogy to stocks, investors often decide whether a stock is cheap or expensive by comparing its pricing to that of similar stocks, usually in its peer group.

While there are purists in each camp who argue that the other approach is useless, we will take the middle ground. While intrinsic valuation often allows us to get a fuller sense of what drives the value of a business or stock, there are times when relative valuation will yield a more realistic estimate of value. In general, there is no reason to choose, since nothing stops us from using both approaches on the same investment. If investing is a game of odds, we can improve our odds significantly by investing in stocks that are under valued not only on an intrinsic basis but also on a relative one.

Why should we care?

Investors come to the market with a wide range of investment philosophies. Some are market timers, while others believe in picking stocks. Some pore over price charts and classify themselves as technicians, whereas others compute

financial ratios and swear by fundamental analysis. Some invest for short-term profits and other for long term gains. We believe that knowing how to value assets is useful to all of these investors, though its place in the process will vary.

The Market timer

Market timers note, with some legitimacy, that the payoff to calling turns in markets is much greater than the returns from stock picking. They argue that it is easier to predict market movements than to select stocks and that these predictions can be based upon factors that are observable. While valuation of individual stocks may not be of much direct use to a market timer, market-timing strategies can use valuation in one of at least two ways:

(a) The overall market itself can be valued, either on an intrinsic or relative basis, and compared to the current level.

(b) Valuation models can be used to value a large number of stocks, and the results from the cross-section can be used to determine whether the market is over or under valued. For example, as the number of stocks that are overvalued, using the valuation model, increases relative to the number that are undervalued, there may be reason to believe that the market is overvalued.

In effect, market timers can use valuation models to determine when to enter a market (stocks, bonds, real estate) and how much of their portfolio to allocate to an asset class.

The fundamentalist

Stock pickers believe that while timing the market is a fruitless exercise, picking the right stocks to invest in can generate profits over the long term, especially when the choices are based upon "fundamentals". The underlying theme in "fundamental" stock picking is that the true value of the firm can be related to its financial characteristics -- its growth prospects, its risk profile and its cashflows. Any deviation from this true value is a sign that a stock is under or overvalued.

Fundamental analysts can include both value and growth investors. The key difference between the two is in where the valuation focus lies. Value investors are

primarily interested in valuing a firm's existing asset base and acquiring it at less than its true value. Growth investors, on the other hand, are far more focused on valuing a firm's growth potential and buying these growth assets at a discount. While valuation is the central focus in fundamental analysis, some analysts use discounted cashflow models to value firms, while others use multiples and comparable firms.

The chartist

Chartists believe that prices are driven as much by investor psychology as by any underlying financial variables. The information available from trading measures -- price movements, trading volume, short sales and other indicators -- gives an indication of investor psychology and future price movements. The assumptions here are that prices move in predictable patterns and that the average investor in the market is driven more by emotion than by rational analysis. While valuation does not play as much of a role in charting as it does in fundamental analysis, there are ways in which an enterprising chartist can incorporate valuation views into analysis. For instance, valuation can be used to determine support and resistance lines on price charts .[1] It can be also useful in determining when bubbles will burst and how much risk there is in a momentum investing strategies.

The information trader

Prices move on information about the firm. Information traders attempt to trade in advance of new information or shortly after it is revealed to financial markets. The underlying assumption is that these traders can anticipate information announcements and gauge the market reaction to them better than the average investor in the market. For an information trader, the focus is on the relationship between information and changes in value, rather than on value, per se. Thus an information trader may buy an 'overvalued' firm if he believes that the next

---

[1] On a chart, the support line usually refers to a lower bound below which prices are unlikely to move and the resistance line refers to the upper bound above which prices are unlikely to venture. While these levels are usually estimated using past prices, the range of values obtained from a valuation model can be used to determine these levels, i.e., the maximum value will become the resistance level and the minimum value will become the support line.

information announcement is going to cause the price to go up, because it contains better than expected news. If there is a relationship between how undervalued or overvalued a company is, and how its stock price reacts to new information, then valuation could play a role in investing for an information trader.

The corporate manager

There is a role for valuation at every stage of a firm's life cycle. For small private businesses thinking about expanding, valuation plays a key role when they approach venture capital and private equity investors for more capital. The share of a firm that a venture capitalist will demand in exchange for a capital infusion will depend upon the value she estimates for the firm. As the companies get larger and decide to go public, valuations determine the prices at which they are offered to the market in the public offering. Once established, decisions on where to invest, how much to borrow and how much to return to the owners will be all decisions that are affected by valuation. If the objective in corporate finance is to maximize firm value[2], the relationship between financial decisions, corporate strategy and firm value has to be delineated.

The Accountant

The most significant trend in accounting standards globally is the shift towards fair value accounting, where assets are valued on balance sheets at their fair values rather than at their original cost. This shift, which we view with trepidation, has required accountants to hone their valuation skills, as they are charged with valuing brand name, customer lists and other intangible assets. In addition, both US and international accounting standards now require that acquisitions be followed immediately by a reassessment of the values of the assets of the target company and evaluation of goodwill "impairment" in future periods; the latter task requires that the target company be revalued on a continuing basis, with any decline in value below the original purchase price being recorded as impairment.

---

[2] Most corporate financial theory is constructed on this premise.

Some truths about valuation

Before you delve into some of the details of valuation, it is worth noting some truths about valuation in general that will not only provide you with perspective when looking at valuations done by others but also some comfort, when doing your own.

All valuations are biased

You almost never start valuing a company with a blank slate. All too often, uour views on a company are formed before you start inputting the numbers into the models and metrics that you use and not surprisingly, your conclusions tend to reflect your biases. The sources of the biases can be manifold:

a. The bias in valuation starts with the companies you choose to value. These choices are almost never random, and how you make them can start laying the foundation for bias. It may be that you have read something in the press (good or bad) about the company or heard from an expert that it was under or over valued. Thus, you already begin with a perception about the company that you are about to value. You add to the bias when you collect the information you need to value the firm. The annual report and other financial statements include not only the accounting numbers but also management discussions of performance, often putting the best possible spin on the numbers. With many larger companies, it is easy to access what other analysts following the stock think about these companies. Zacks, I/B/E/S and First Call, to name three services among many, provide summaries of how many analysts are bullish and bearish about the stock. Finally, you have the market's own estimate of the value of the company- the market price – adding to the mix. Valuations that stray too far from this number make investors uncomfortable.

b. There are institutional factors that add to this already substantial bias. For instance, it is an acknowledged fact that equity research analysts are more likely to issue buy rather than sell recommendations, i.e., that they are more

likely to find firms to be undervalued than overvalued.[3] This can be traced partly to the difficulties analysts face in obtaining access and collecting information on firms that they have issued sell recommendations on, and partly to pressure that they face from portfolio managers, some of whom might have large positions in the stock, and from their own firm's investment banking arms which have other profitable relationships with the firms in question.

c. The reward and punishment structure associated with finding companies to be under and over valued is also a contributor to bias. An analyst whose compensation is dependent upon whether she finds a firm is under or over valued will be biased in her conclusions. This should explain why acquisition valuations are so often biased upwards. The analysis of the deal, which is usually done by the acquiring firm's investment banker, who also happens to be responsible for carrying the deal to its successful conclusion, can come to one of two conclusions. One is to find that the deal is seriously over priced and recommend rejection, in which case the analyst receives the eternal gratitude of the stockholders of the acquiring firm but little else. The other is to find that the deal makes sense (no matter what the price) and to reap the ample financial windfall from getting the deal done.

There are three ways in which your views on a company (and the biases we have) can manifest themselves in value.

a. The first is in the inputs that you use in the valuation. When you value companies, you constantly come to forks in the road where you have to make assumptions to move on. These assumptions can be optimistic or pessimistic. For a company with high operating margins now, you can either assume that competition will drive the margins down to industry averages very quickly (pessimistic) or that the company will be able to maintain its margins for an extended period (optimistic). The path you

[3] There are approximately five times as many buy recommendations issued by analysts on Wall Street as there are sell recommendations.

choose will reflect our prior biases. It should come as no surprise then that the end value that you arrive at is reflective of the optimistic or pessimistic choices we made along the way.

b. The second is in what we will call post-valuation garnishing, where analysts revisit assumptions after a valuation in an attempt to get a value closer to what they had expected to obtain starting off. Thus, an analyst who values a company at $ 15 per share, when the market price is $ 25, may revise his growth rates upwards and his risk downwards to come up a higher value, if she believed that the company was under valued to begin with.

c. The third is to leave the value as is but attribute the difference between the value you estimate and the value you think is the right one to a qualitative factor such as synergy or strategic considerations. This is a common device in acquisition valuation where analysts are often called upon to justify the unjustifiable.

Bias cannot be regulated or legislated out of existence. Analysts are human and bring their biases to the table. As investors, what are the lessons you can drawn from this discussion? If you are valuing companies, try to be honest about your own biases; in fact, put them down on paper, if possible, before you start. In addition, confine your background research on the company to information sources rather than opinion sources; in other words, spend more time looking at a company's financial statements than reading equity research reports about the company. If you are looking at someone else's valuation of a company, always consider the source of the valuation and potential biases that may affect their judgments. As a general rule, the more bias there is in the process, the less weight you should attach to the valuation.

All valuations are wrong, but it is not always your fault

Starting early in life, you are taught that if you do the right things, you will get the right answers. In other words, the precision of the answer is used as a measure of the quality of the process that yielded the answer. While this may be appropriate

in mathematics or physics, it is a poor measure of quality in valuation. Barring a very small subset of assets, there will always be uncertainty associated with valuations, and even the best valuations come with a substantial margin for error.

When valuing an asset at any point in time, you make forecasts for the future. Since none of us possess crystal balls, you have to make your best estimates, given the information that you have at the time of the valuation. Our estimates of value can be wrong for a number of reasons, and you can categorize these reasons into three groups.

a. Estimation Uncertainty: Even if our information sources are impeccable, you have to convert raw information into inputs and use these inputs in models. Any mistakes that we make at either stage of this process will cause estimation error.

b. Firm-specific Uncertainty: The path that you envision for a firm can prove to be hopelessly wrong. The firm may do much better or much worse than you expected it to perform, and the resulting earnings and cash flows will be very different from your estimates.

c. Macroeconomic Uncertainty: Even if a firm evolves exactly the way you expected it to, the macro economic environment can change in unpredictable ways. Interest rates can go up or down and the economy can do much better or worse than expected. These macro economic changes will affect value.

The contribution of each type of uncertainty to the overall uncertainty associated with a valuation can vary across companies. When valuing a mature cyclical or commodity company, it may be macroeconomic uncertainty that is the biggest factor causing actual numbers to deviate from expectations. Valuing a young technology company can expose you to far more estimation and firm-specific uncertainty.

Even if you feel comfortable with your estimates of an asset's values at any point in time, that value itself will change over time, as a consequence of new information that comes out both about the firm and about the overall market.. Given the constant flow of information into financial markets, a valuation done on a firm ages quickly, and has to be updated to reflect current information. Thus, technology companies that were valued highly in late 1999, on the assumption that the high

growth from the nineties would continue into the future, would have been valued much less in early 2001, as the prospects of future growth dimmed. With the benefit of hindsight, the valuations of these companies (and the analyst recommendations) made in 1999 can be criticized, but they may well have been reasonable, given the information available at that time.

The advantage of breaking uncertainty down into estimation uncertainty, firm-specific and macroeconomic uncertainty is that it gives us a window on what you can manage, what you can control and what you should just let pass through into the valuation. Building better models and accessing superior information will reduce estimation uncertainty but will do little to reduce exposure to firm-specific or macro-economic risk. Even the best-constructed model will be susceptible to these uncertainties.

More detail and complexity does not always result in better valuations

Valuation models have become more and more complex over the last two decades, as a consequence of two developments. On the one side, computers and calculators have become far more powerful and accessible in the last few decades. With technology as our ally, tasks that would have taken us days in the pre-computer days can be accomplished in minutes. On the other side, information is both more plentiful, and easier to access and use. We can download detailed historical data on thousands of companies and use them as we see fit. The complexity, though, has come at a cost.

A fundamental question that we all face when doing valuations is how much detail we should break a valuation down into. There are some who believe that more detail is always better than less detail and that the resulting valuations are more precise. We disagree. The trade off on adding detail is a simple one. On the one hand, more detail gives you a chance to use specific information to make better forecasts on each individual item. On the other hand, more detail creates the need for more inputs, with the potential for error on each one, and generates more complicated models. In the physical sciences, the principle of parsimony dictates that we try the simplest possible explanation for a phenomenon before we move on

to more complicated ones. We would be well served adopting a similar principle in valuation. When valuing an asset, use the simplest model that you can get away with. In other words, if you can value an asset with three inputs, you should not be using five. If you can value a company with 3 years of cash flow forecasts, forecasting ten years of cash flows is asking for trouble.

Conclusion

When faced with the question of whether to invest in a stock, a bond or any asset, you can either choose to make your decisions based upon the actions or recommendations of others that you view as more informed or make your own assessment of value. Most investors choose not to do the latter and offer a variety of excuses: that valuation models are too complex, that there is insufficient information or that there is too much uncertainty about the future. While all of these reasons have a core of truth to them, there is no reason why they should stop you from valuing assets. Valuation models can be simplified, you can make do with the information that you have (rather than wish you had) and you can make your best estimates about an uncertain future. Will you be wrong? Of course, but so will everyone else. Success in investing comes not from being right but from being less wrong than everyone else playing the game.

Section 1: Laying the groundwork

Chapter 2: The Tools of the trade

There are a few basic tools that should be part of every valuation toolkit. While all of these tools have common sense underpinnings, they can take complex forms in some analyses. In this chapter, we will navigate our way through the basics of these tools, while steering away from the complexities that may or may not improve valuations at the margin.

Valuation Tools: An overview

Before we start looking at valuation models and metrics, there are four tools that we focus on as essential for our pursuit:

a. Estimating time value of money: An investment generates cash flows over many years and a dollar today is worth more than a dollar in the future. True, but to convert this common sense proposition to value, we have to first understand why time has value and then develop ways in which we can make this notion specific.
b. Measuring risk and estimating expected return: When investing, we face uncertainty about future cash flows. The basic principle that more risky or uncertainty cash flows should be worth less than less risky cash flows is intuitive. To put this principle to work in valuation, we need to be clear about what comprises risk, how to measure that risk and how we adjust value for that risk.
c. Making sense of accounting data: When valuing companies, much of the information that we use comes from financial statements. To the extent that this data is misread, our valuations will run off course. We look at the questions that we would like accounting statements to address and how we might be able to eke out answers from the numbers.
d. Understanding relationships between data: The biggest problem that we face in investing today is not that we have too little information but that we have too much. The data is often contradictory and pulls us in different directions on whether an asset is under or over valued. Statistical measures such as

standard deviation can help us consolidate data and understand relationships.

The Time Value of Money

The simplest tools in finance are often the most powerful. The notion that a dollar today is preferable to a dollar some time in the future is intuitive enough for most people to grasp without the use of models and mathematics. The principles of present value enable us to calculate exactly how much a dollar some time in the future is worth in today's dollars and to move cash flows across time.

Why money has time value

There are three reasons why a cash flow in the future is worth less than a similar cash flow today.

1. ***Individuals prefer present consumption to future consumption.*** People would have to be offered more in the future to give up present consumption. If the preference for current consumption increases, individuals will have to be offered much more in terms of future consumption to give up current consumption, a trade-off that is captured by a high "real" rate of return or discount rate.
2. ***When there is monetary inflation, the value of currency decreases over time.*** As inflation rises, the purchasing power of a currency decreases. To compensate, we will value reduce the value that we attribute to cash flows in the future.
3. ***A promised cash flow might not be delivered for a number of reasons:*** Any uncertainty (risk) associated with the cash flow in the future reduces the value of the cash flow.

The process by which future cash flows are adjusted to reflect these factors is called discounting, and the magnitude of these factors is reflected in the discount rate. The discount rate can be viewed as a composite of the expected real return (reflecting consumption preferences in the aggregate over the investing population), the expected inflation rate (to capture the deterioration in the purchasing power of the cash flow), and a premium for the uncertainty associated with the cash flow.

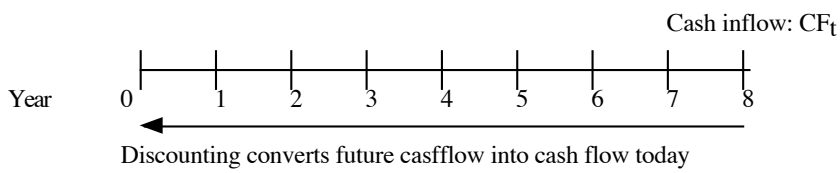
How we compute time value?

The process of discounting converts future cash flows into cash flows in today's terms. There are five types of cash flows—simple cash flows, annuities, growing annuities, perpetuities, and growing perpetuities—which we discuss next.

Simple Cash Flows

A simple cash flow is a single cash flow in a specified future time period; it can be depicted on a time line as in Figure 2.1.

Figure 2.1: Present Value of a Cash Flow



Discounting a cash flow converts it into present value dollars and enables the user to do several things. First, once cash flows are converted into present value dollars, they can be aggregated and compared. Second, if present values are estimated correctly, the user should be indifferent between the future cash flow and the present value of that cash flow. The present value of a cash flow can be written as follows

$$\text{Present Value of Simple Cash Flow} = \frac{CF_t}{(1+\text{Discount Rate})^t}$$

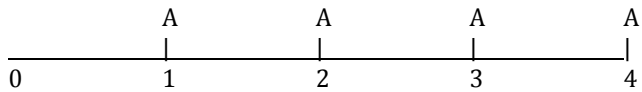
Thus, the present value of $ 1000 in ten years, with a discount rate of 8% can be written as:

$$\text{Present value of \$ 1000 in ten years @ 8\%} = \frac{1000}{(1.08)^{10}} = \$463.19$$

Other things remaining equal, the present value of a cash flow will decrease as the discount rate increases and the further into the future the cash flow occurs.

Annuities

An *annuity* is a constant cash flow that occurs at regular intervals for a fixed period of time. Defining A to be the constant cash flow, the time line for an annuity may be drawn as follows:



The value of an annuity can be calculated by discounting each cash flow back to today and then adding up the present values. Alternatively, an equation can be used in the calculation.

$$PV \text{ of an Annuity} = \text{Annual Cash flow} \left[\frac{1 - \dfrac{1}{(1+\text{Discount Rate})^{\text{No of periods}}}}{\text{Discount Rate}} \right]$$

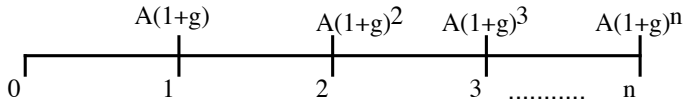
To illustrate, assume again that you are have a choice of buying a car for $10,000 cash down or paying $3,000 a year, at the end of each year, for five years for the same car. If the discount rate is 12 percent, which would you rather do?

$$PV \text{ of \$3000 each year for next 5 years} = \$3000 \left[\frac{1 - \dfrac{1}{(1.12)^5}}{.12} \right] = \$10{,}814$$

The present value of the installment payments exceeds the cash-down price; therefore, you would want to pay the $10,000 in cash now.

Growing Annuities

A *growing annuity* is a cash flow that grows at a constant rate for a specified period of time. If A is the current cash flow, and *g* is the expected growth rate, the time line for a growing annuity appears as follows:



Note that to qualify as a growing annuity, the growth rate in each period has to be the same as the growth rate in the prior period. In most cases, the present value of a growing annuity can be estimated by using the following formula:[4]

---

[4] The present value of a growing annuity can be estimated in all cases, but one—where the growth rate is equal to the discount rate. In that case, the present value is equal to the nominal sums of the annuities over the period, since the growth effect will exactly offset the discounting effect.

$$PV \text{ of a Growing Annuity} = \text{Cashflow}(1+\text{Growth rate})\left[\frac{1 - \frac{(1+\text{Growth rate})^n}{(1+\text{Discount rate})^n}}{\text{Discount rate} - \text{Growth rate}}\right]$$

To illustrate a growing annuity, suppose you have the rights to a gold mine for the next twenty years, over which time you plan to extract 5,000 ounces of gold every year. The current price per ounce is $300, but it is expected to increase 3 percent a year. The appropriate discount rate is 10 percent. The present value of the gold that will be extracted from this mine can be estimated as follows:

$$PV \text{ of extracted gold} = \$300 * 5000 * (1.03)\left[\frac{1 - \frac{(1.03)^{20}}{(1.10)^{20}}}{.10 - .03}\right] = \$16{,}145{,}980$$

The present value of the gold expected to be extracted from this mine is $16.146 million; it is an increasing function of the expected growth rate in gold prices.

Perpetuities and Growing Perpetuities

A *perpetuity* is a constant cash flow at regular intervals forever. The present value of a perpetuity can be written as

$$PV \text{ of Perpetuity} = \frac{\text{Annual Cash flow}}{\text{Discount Rate}}$$

The most common example offered for a perpetuity is a console bond. A console bond is a bond that has no maturity and pays a fixed coupon. Assume that you have a 6 percent coupon console bond. The value of this bond, if the interest rate is 9 percent, is as follows:

Value of Console Bond = $60/0.09 = $667

The value of a console bond will be equal to its face value (which is usually $1,000) only if the coupon rate is equal to the interest rate.

A *growing perpetuity* is a cash flow that is expected to grow at a constant rate forever. The present value of a growing perpetuity can be written as:

$$PV \text{ of Growing Perpetuity} = \frac{\text{Expected Cash flow next year}}{(\text{Discount Rate} - \text{Expected growth rate in perpetuity})}$$

Although a growing perpetuity and a growing annuity share several features, the fact that a growing perpetuity lasts forever puts constraints on the growth rate. It has to be less than the discount rate for this formula to work but an even tighter constraint is that it has to be lower than the nominal growth rate of the economy. Consider a simple example. In 1992, Southwestern Bell paid dividends per share of $2.73. Its earnings and dividends had grown at 6 percent a year between 1988 and 1992 and were expected to grow at the same rate in the long run. The rate of return required by investors on stocks of equivalent risk was 12.23 percent. With these inputs, we can value the stock using a perpetual growth model:

$$\text{Value of Stock} = \$2.73 * 1.06/(0.1223 - 0.06) = \$46.45$$

As an aside, the stock was actually trading at $70 per share. This price could be justified by using a higher growth rate (of about 8%). This growth rate is often referred to as an *implied growth rate*.

The Bottom Line

Almost every investment that we value is come combination of the five cash flows outlined above. Take, for instance, a corporate or government bond, where the cash flows take the form of fixed coupons every period for the life of the bond and the face value of the bond is paid out at maturity. This bond can be written as the sum of the present values of an annuity (coupon) and a single cash flow (face value of bond). Equity in a mature company, illustrated with the valuation of Southwestern Bell, will offer cash flows (dividends) growing at a constant rate forever and can be valued as a growing perpetuity. Equity in a high growth company, where cash flows are expected to grow at a high rate for a initial period, before subsiding to a stable rate forever is the sum of the present values of a growing annuity (dividends during high growth) and a growing perpetuity (dividends afterwards).

Risk Measurement

When stocks were first traded in the 16th and 17th century, there was little access to information and few ways of processing even that limited information. Only the

very wealthy invested in stocks, and even they were lead down the proverbial garden path. The resulting scams and scandals of that day are well described by Charles Mackey in his book on bubbles titled, "The Extraordinary Popular Delusions and the Madness of Crowds". As new investors entered into the financial markets at the start of the twentieth century, the demand for risk measures also surged. In response, services were already starting to collect return and price data on individual securities and computing basic statistics such as the expected return and standard deviation in returns. By 1915, services including the Standard Statistics Bureau (the precursor to Standard and Poor's), Fitch and Moody's were processing accounting information to provide bond ratings as measures of credit risk in companies. Concurrently, there were other services that provided risk measures for individual stocks that used a combination of accounting data, market data (stock price volatility) and qualitative information on the company (products, management etc.) to form assessments of risk.

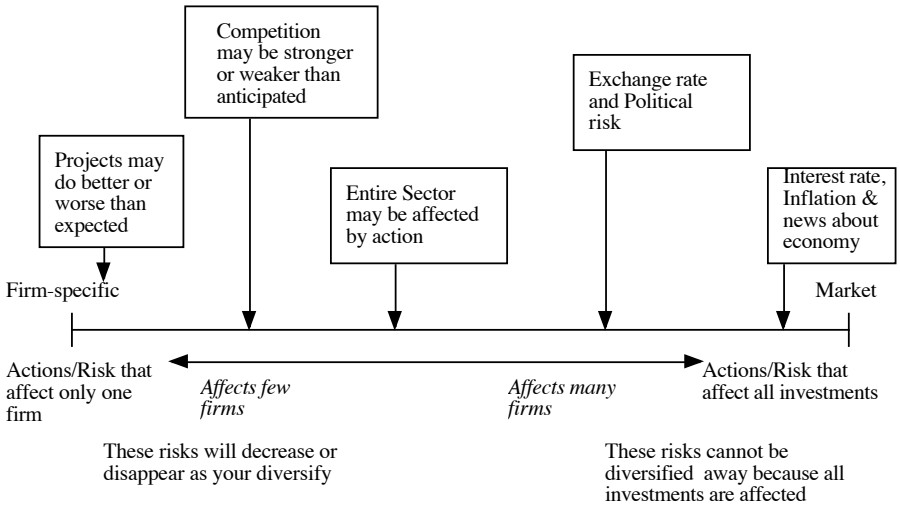
The Markowitz Revolution: Diversification and Risk

Harry Markowitz revolutionized both how we think about risk and how we measure that risk in the early 1950s by wrestling with a conundrum as a doctoral student in the University of Chicago. Markowitz noted that if the value of a stock is the present value of its expected dividends and an investor were intent on only maximizing returns, he or she would invest in the one stock that had the highest expected dividends, a practice that was clearly at odds with both practice and theory at that time, which recommended investing in diversified portfolios. So, what were we missing?

Markowitz reasoned that investors must diversify because they care about risk, and the risk of a diversified portfolio must therefore be lower than the risk of the individual securities that went into it. His key insight was that the variance of a portfolio could be written as a function not only of how much was invested in each security and the variances of the individual securities but also of the correlation between the securities. Markowitz then went one step further. If investors are diversified, the risk in any individual investment (stock, bond or real asset) has to be

the risk added to the diversified portfolio. Consequently, risks that are specific to a firm (and thus affect only that firm) will be averaged out in the portfolio. To illustrate this concept, consider the variety of risks that any modern business is exposed to in figure 2.2:

Figure 2.2: Breaking down risks in a company



At one extreme, there are some risks that affect only the firm whereas at the other, there are risks that affect all or most firms at the same time. When you invest all your money in one stock, you are exposed to all of these risks. When you spread your money out across many stocks, the risks that affect one or a few firms will get averaged out in your portfolio: for every company, where something worse than expected happens on a project, there will be another company, where something better than expected will happen. Risk that affects many or most firms cannot be diversified away. In the Markowitz world, this is the only risk that you should consider, as an investor in a publicly traded company.

Finance enters the mix: Risk and Return Models

If we accept the Markowtiz proposition that the only risk that we care about in an investment is the risk that you cannot diversify away, the question then becomes one of measurement. How do you measure this non-diversifiable risk? This is where different models part ways.

The most widely used and critiqued model in practice is the capital asset pricing model or the CAPM. In this model, we assume that investors face no transactions costs and have no way of separating good investments from bad ones. Consequently, they end up holding supremely diversified composed of all traded assets (called the market portfolio). The risk of any asset then becomes the risk added to this "market portfolio" and that is what a beta measures. The beta has the useful property of being standardized around one; a stock with a beta above one is above average risk and one with a beta below one is below average. The expected return on the investment can then be written as:

Expected return = Risk free rate + Beta (Risk premium for an average risk investment)

In the years since the introduction of the CAPM, two classes of models have developed as alternatives:

a. Multi beta models: For the most part, though, these models (the arbitrage pricing model and the multi factor model) expand on the basic theme of the CAPM and measure the risk added by an investment to a diversified portfolio, with multiple betas (rather than the single beta) measuring exposure to different types of market risk.
b. Proxy models: In proxy models, we look at the characteristics of companies that have historically generated high returns for stockholders and use them as risk measures. One of the earliest studies in this vein, for instance, found that small market capitalization companies and companies that traded at low market values, relative to accounting book values, generated higher returns than large market capitalization companies and companies that trade at high market values, relative to book value.

There are many practitioners and some academics who argue that all risk and return models in finance are flawed, either because of unrealistic assumptions made to arrive at the models or because the parameters for these models cannot be estimated precisely.

Bottom Line

We agree with critics of conventional risk and return models that these models are flawed, but we should also be able to agree on core propositions about risk:

1. Risk matters: We can take issue with whether beta is a good measure of risk but there should be no debate that "riskier" investments need to earn higher returns, to pass muster, than safer investments.
2. Not all investments are equally risky: Some investments are riskier than others. One of the advantages of beta is that it provides a measure of relative risk, but even in its absence, we need a measure of relative risk in an investment. That measure of relative risk can be obtained by looking at stock prices (like betas), earnings variability or even be subjective.
3. The price of risk matters: Investors set prices for taking risk, in both equity and bond markets and these prices vary across time. In good times, prices tend to drift down and in bad times, they tend to drift up. Holding the risk free rate and relative risk (beta) constant, we should expect to see required returns move concurrently.

Thus, whatever our views on the efficacy or otherwise of conventional risk and return models, we have to devise ways in which we capture all three factors in our required returns.

Accounting data

Financial statements provide the fundamental information that we use to analyze and answer valuation questions. Therefore, it is important that we understand the principles governing these statements by looking at three questions:

- How valuable are the assets of a firm?
- How did the firm raise the funds to finance these assets? In particular, how much of the funding came from owners' funds (equity) or borrowed money (debt)?
- How profitable are these assets?

As accounting standards have evolved over time, often in response to accounting scandals, financial statements have become more complicated. Ironically, it has

become more difficult to use these statement to answer these questions, rather than less difficult.

The Basic Accounting Statements

There are three basic accounting statements that summarize information about a firm. The first is the *balance sheet*, shown in Figure 2.3, which summarizes the assets owned by a firm, the value of these assets, and the mix of financing, debt, and equity used to finance these assets at a point in time.

Figure 2.3: The Balance Sheet

Assets		Liabilities	
Long Lived Real Assets	Fixed Assets	Current Liabilties	Short-term liabilities of the firm
Short-lived Assets	Current Assets	Debt	Debt obligations of firm
Investments in securities & assets of other firms	Financial Investments	Other Liabilities	Other long-term obligations
Assets which are not physical, like patents & trademarks	Intangible Assets	Equity	Equity investment in firm

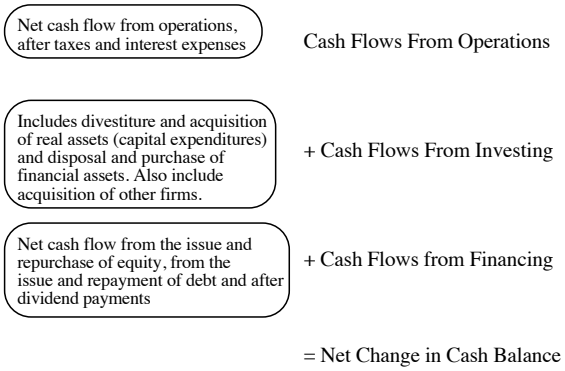
The *income statement*, shown in Figure 2.4, provides information on revenues and expenses of the firm and the resulting income made during a period.

Figure 2.4: Income Statement

Gross revenues from sale of products or services	Revenues
Expenses associates with generating revenues	- Operating Expenses
Operating income for the period	= Operating Income
Expenses associated with borrowing and other financing	- Financial Expenses
Taxes due on taxable income	- Taxes
Earnings to Common & Preferred Equity for Current Period	= Net Income before extraordinary items
Profits and Losses not associated with operations	- (+) Extraordinary Losses (Profits)
Profits or losses associated with changes in accounting rules	- Income Changes Associated with Accounting Changes
Dividends paid to preferred stockholders	- Preferred Dividends
	= Net Income to Common Stockholders

The *statement of cash flows*, shown in Figure 2.5, specifies the sources and uses of cash of the firm from operating, investing, and financing activities during a period.

Figure .2.5: Statement of Cash Flows



The statement of cash flows can be viewed as an attempt to list the cash flows during a period and explain why the cash balance changed during the period.

Assets Owned: Measures of asset value

The accounting view of asset value is to a great extent grounded in the notion of *historical cost*, which is the original cost of the asset, adjusted upward for improvements made to the asset since purchase and downward for loss in value associated with the aging of the asset. This historical cost is called the *book value*. To examine how asset value is measured on a balance sheet, let us begin with the way assets are categorized in the balance sheet.

- First, there are the *fixed assets*, which include the long-term assets of the firm, such as plant, equipment, land, and buildings. Generally accepted accounting principles (GAAP) in the United States require the valuation of fixed assets at historical cost, adjusted for any estimated gain and loss in value from improvements and the aging, respectively, of these assets. Although in theory the adjustments for aging should reflect the loss of earning power of the asset as it ages, in practice they are much more a product of accounting rules and convention, and these adjustments are called *depreciation*.
- Next, we have the short-term assets of the firm, including inventory (such as raw materials, works in progress, and finished goods), receivables (summarizing

moneys owed to the firm), and cash; these are categorized as *current assets.* It is in this category accountants are most amenable to the use of market value.

- In the category of *investments and marketable securities*, accountants consider investments made by firms in the securities or assets of other firms and other marketable securities. The way these assets are valued depends on the way the investment is categorized and the motive behind the investment. If the investment is viewed as a trading investment, as is the case with financial investments, the value is marked to market. If the investment is viewed as a strategic and long-term investment, the investment is usually recorded at what was originally paid for the asset, with adjustments. In the special case, where the holdings comprise more than 50% of the value of a subsidiary, the firm has to consolidate, i.e., record all of the subsidiary's assets and liabilities on its balance sheet, with a "minority interest" item capturing the percentage of the subsidiary that does not belong to it.
- Finally, we have what is loosely categorized as *intangible assets*. While we would normally consider assets such as brand names, customer loyalty and a well-trained work force as intangible assets, the most commonly encountered intangible asset in accounting is goodwill. When a firm acquires another firm, the purchase price is first allocated to the existing assets of the acquired firm. Any excess paid becomes goodwill and is recorded as an asset. This goodwill has to be written off, if the accountants deem it to be impaired, i.e., if they determine that the value of the target company has dropped since the acquisition.[5]

Funds Raised: Measures of Financing Mix

Just as with the measurement of asset value, the accounting categorization of liabilities and equity is governed by a set of fairly rigid principles. Accountants categorize liabilities into current liabilities, long-term debt, and long-term liabilities

[5] Toi make this judgment, accountants have to value the acquired company at regular intervals and compare the value that they get to the price paid. If the value is substnatilaly lower than the price, the company has to write off an equivalent portion of the goodwill.

that are neither debt nor equity; the last category includes leases, underfunded pension and health care obligations and deferred taxes.

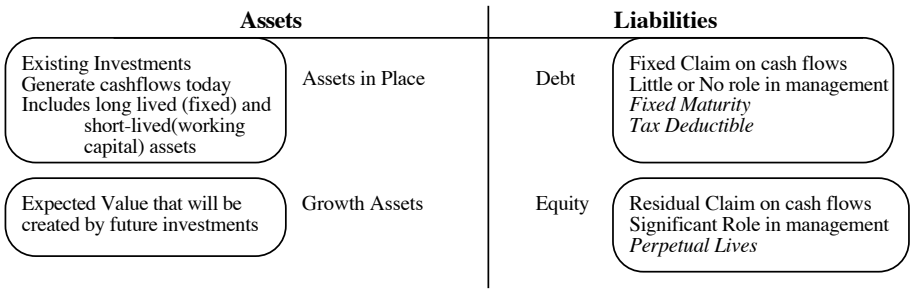
- *Current liabilities* include all obligations that the firm has coming due in the next accounting period. These generally include accounts payable (representing credit received from suppliers and other vendors to the firm), short term borrowing (representing short-term loans taken to finance the operations or current asset needs of the business) and the short-term portion of long-term borrowing (representing the portion of the long-term debt or bonds that is coming due in the next year). As with current assets, these items are usually recorded at close to their current market value.
- *Long-term debt* for firms can take one of two forms: long-term loans from banks or other financial institutions, or long-term bonds issued to financial markets. These are generally recorded at the face value at the time of issue and are generally not marked to market.
- In *a pension plan*, the firm agrees to provide certain benefits to its employees, either by specifying a "defined contribution" (wherein a fixed contribution is made to the plan each year by the employer, without any promises as to the benefits to be delivered in the plan) or a "defined benefit" (wherein the employer promises to pay a certain benefit to the employee). In the latter case, the employer has to put sufficient money into the plan each period to meet the defined benefits. A pension fund whose assets exceed its liabilities is an overfunded plan, whereas one whose assets are less than its liabilities is an underfunded plan, and disclosures to that effect have to be included in financial statements, generally in the footnotes.
- Firms often use different methods of accounting for tax and financial reporting purposes, leading to a question of how tax liabilities should be reported. Because accelerated depreciation and favorable inventory valuation methods for tax accounting purposes lead to a deferral of taxes, the same principles of matching expenses to income that underlie accrual accounting require that the *deferred income tax* be recognized in the financial statements, as a liability (if the firm underpaid taxes) or as an asset (if the firm overpaid taxes).

The accounting measure of equity is a historical cost measure. The value of equity shown on the balance sheet reflects the original proceeds received by the firm when it issued the equity, augmented by any earnings made since then (or reduced by losses, if any) and reduced by any dividends paid out during the period. A sustained period of negative earnings can make the book value of equity negative.

A Financial Balance Sheet

An accounting balance sheet is useful because it provides us with information about a firm's history of investing and raising capital. However, it is focused on the past. To provide a more forward-looking picture, consider an alternative, the financial balance sheet, illustrated in figure 2.6:

Figure 2.6: A Financial Balance Sheet



While a financial balance sheet bears a superficial resemblance to the accounting balance sheet, it differs on two important counts. The first is in the classification of assets. Rather than classify assets based on asset life or tangibility, it categorizes them into investments already made by the company (assets in place) and investments that you expect the company to make in the future (growth assets). The second is that the values reflect not what has already been invested in these assets, but their current values, based upon expectations for the future. In fact, the entire value of growth assets rests on perceptions and expectations. Since the assets are recorded at current value, the debt and equity values also reflect that choice. Thus, the value of equity in a financial balance sheet is an estimate of what the value of the equity in the firm is today and is directly comparable to the market value.

It should be noted here that both US and international accounting standards are pushing towards "fair value" accounting. Put simply, this would lead to all assets being revalued at each year and recorded at that value on the balance sheet, rather than at original cost. While this will push accounting balance sheets towards financial balance sheets, they will not converge, since it is unlikely that growth assets will ever find their rightful place in an accounting balance sheet, even in fair value terms.

Profitability: Measures of Earnings

How profitable is a firm? What did it earn on the assets in which it invested? Two primary principles underlie the measurement of accounting earnings and profitability. The first is the principle of accrual accounting. In accrual accounting, the revenue from selling a good or service is recognized in the period in which the good is sold or the service is performed (in whole or substantially). A corresponding effort is made on the expense side to match expenses to revenues.[6] This is in contrast to cash accounting, wherein revenues are recognized when payment is received and expenses are recorded when they are paid. Accounting principles require publicly traded companies to use accrual accounting to record earnings from continuing operations. The second principle is the categorization of expenses into operating, financing, and capital expenses. Operating expenses are expenses that at least in theory provide benefits only for the current period; the cost of labor and materials expended to create products that are sold in the current period is a good example. Financing expenses are expenses arising from the non-equity financing used to raise capital for the business; the most common example is interest expenses. Capital expenses are expected to generate benefits over multiple periods; for instance, the cost of buying land and buildings is treated as a capital expense.

Much of financial analysis is built around the expected future earnings of a firm, and many of these forecasts start with the current earnings. It is therefore important that we know how much of these earnings come from the ongoing operations of the firm and how much can be attributed to unusual or extraordinary

---

[6]If a cost (such as an administrative cost) cannot be easily linked with a particular revenues, it is usually recognized as an expense in the period in which it is consumed.

events that are unlikely to recur on a regular basis. Consequently, accounting principles require that income statements be classified into four sections: income from continuing operations, income from discontinued operations, extraordinary gains or losses, and adjustments for changes in accounting principles.

Although the income statement allows us to estimate how profitable a firm is in absolute terms, it is just as important that we gauge the profitability of the firm in comparison terms or percentage returns. The simplest and most useful gauge of profitability is relative to the capital employed to get a rate of return on investment. To measure the overall profitability of a firm, we can look at the magnitude of operating income relative to the capital invested in the firm, where capital is defined as the sum of the book value (BV) of debt and equity, net of cash and marketable securities. This is the *return on capital* (ROC) or return on invested capital (ROIC).

$$\text{After - Tax ROC} = \frac{\text{Operating income } (1 - \text{tax rate})}{\text{BV of Debt} + \text{BV of Equity - Cash}}$$

$$\text{Pre - Tax ROC} = \frac{\text{Operating Income}}{\text{BV of Debt} + \text{BV of Equity - Cash}}$$

We can also measure overall profitability relative to revenues generated and the efficiency of the firm in using existing capital invested, and write the return on capital as a function of the two:

$$\text{Pre - Tax Operating Margin} = \frac{\text{Operating Income}}{\text{Revenues or Sales}}$$

$$\text{Capital Turnover Ratio} = \frac{\text{Sales}}{\text{BV of Debt} + \text{BV of Equity - Cash}}$$

$$\text{Pre - Tax ROC} = \text{Pre – tax Operating Margin} * \text{Capital Turnover Ratio}$$

The return on capital varies widely across firms in different businesses, largely as a consequence of differences in profit margins and capital turnover ratios.

Although ROC measures the profitability of the overall firm, the *return on equity* (ROE) examines profitability from the perspective of the equity investor by relating profits to the equity investor (net profit after taxes and interest expenses) to the book value of the equity investment.

$$\text{ROE} = \frac{\text{Net Income}}{\text{Book Value of Common Equity}}$$

Because preferred stockholders have a different type of claim on the firm than common stockholders, the net income should be estimated after preferred dividends, and the book value of common equity should not include the book value of preferred stock.

The bottom line

While the value of a business is ultimately determined by expectations about the future, accounting statements provide the basis for these forecasts. Put another way, to estimate future revenues and earnings, we have to begin with current earnings and revenues, and to make sense of current earnings, we have to understand how accountants measure earnings. Furthermore, the measures of return that we obtain for a firm are central inputs in determining its value.

Statistics

The problem that we face in financial analysis today is not having too little information but too much. Making sense of large and often contradictory information is part of what we are called on to do when analyzing companies. Basic statistics can make this job easier. In this appendix, we consider the most fundamental tools available in data analysis.

Summarizing Data

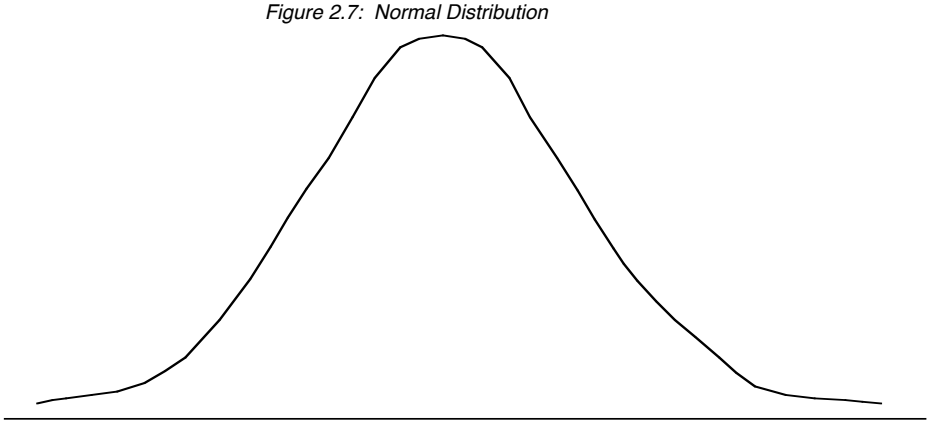
Large amounts of data are often compressed into more easily assimilated summaries, which provide the user with a sense of the content, without overwhelming him or her with too many numbers. There a number of ways data can be presented. We will consider two here—one is to present the data in a distribution, and the other is to provide summary statistics that capture key aspects of the data.

When presented with thousands of pieces of information, you can break the numbers down into individual values (or ranges of values) and indicate the number of individual data items that take on each value or range of values. This is called a

frequency distribution. If the data can only take on specific values, as is the case when we record the number of goals scored in a soccer game, you get a *discrete distribution*. When the data can take on any value within the range, as is the case with income or market capitalization, it is called a *continuous distribution*.

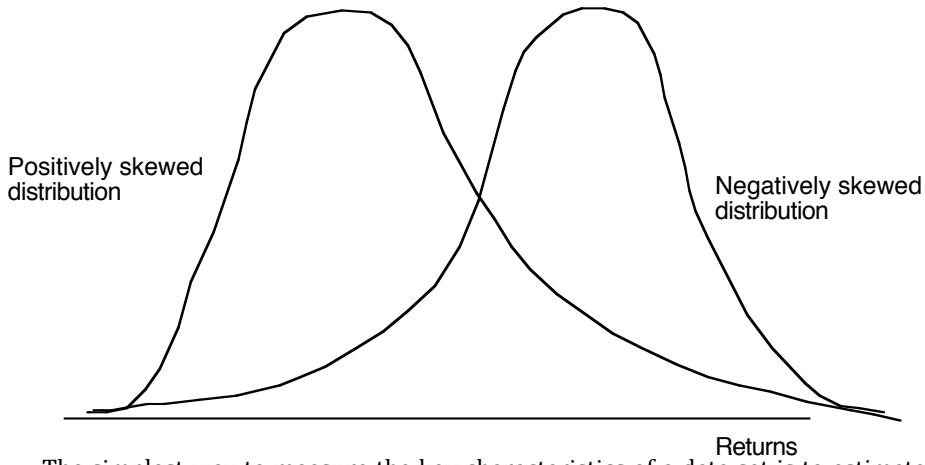
The advantages of presenting the data in a distribution are twofold. For one thing, you can summarize even the largest data sets into one distribution and get a measure of what values occur most frequently and the range of high and low values. The second is that the distribution can resemble one of the many common ones about which we know a great deal in statistics. Consider, for instance, the distribution that we tend to draw on the most in analysis: the normal distribution, illustrated in Figure 2.7.

Figure 2.7: Normal Distribution



A normal distribution is symmetric, has a peak centered around the middle of the distribution, and tails that are not fat and stretch to include infinite positive or negative values. Not all distributions are symmetric, though. Some are weighted towards extreme positive values and are positively skewed, and some towards extreme negative values and are negatively skewed. Figure 2.8 illustrates positively and negatively skewed distributions.

Figure 2.8: Skewed Distributions



The simplest way to measure the key characteristics of a data set is to estimate the summary statistics for the data. For any data series, the most widely used summary statistics are as follows:

- The mean is the simple average of the data points in a sample.
- The median is the midpoint of the series; half the data in the series is higher than the median and half is lower.
- The variance measures how far individual observations fall from the average and the standard deviation is the square root of the variance.

The mean and the standard deviation are the called the first two moments of any data distribution.

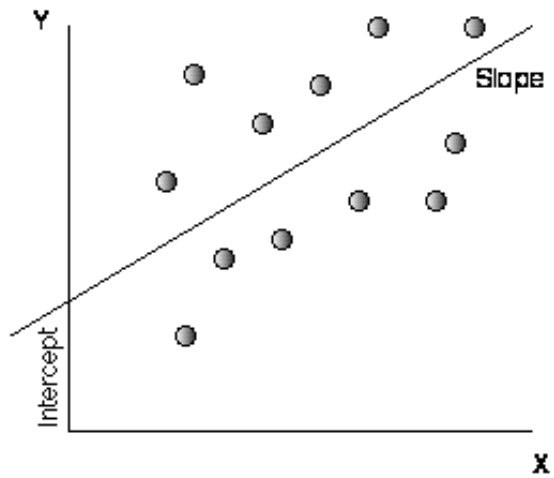
Looking for Relationships in the Data

When there are two series of data, there are a number of statistical measures that can be used to capture how the series move together over time. A positive *correlation* between two variables indicates that the variables move together and a negative sign that they move in opposite directions. The correlation can never be greater than one or less than negative one; the closer the value is to one, the stronger the correlation. A correlation close to zero indicates that the two variables are unrelated. Two variables that are perfectly positively correlated essentially

move in perfect proportion in the same direction, whereas two variables that are perfectly negatively correlated move in perfect proportion in opposite directions. The covariance is a closely related measure to correlation, measuring how much two variables move together but is not standardized (to fall between -1 and +1).

A *simple regression* is an extension of the correlation/covariance concept. It attempts to explain one variable, the dependent variable, using the other variable, the independent variable. Keeping with statistical tradition, let *Y* be the dependent variable and *X* be the independent variable. If the two variables are plotted against each other with each pair of observations representing a point on the graph, you have a *scatterplot*, with *Y* on the vertical axis and *X* on the horizontal axis. Figure 2.9 illustrates a scatter plot.

Figure 2.9: Scatter Plot of Y versus X



In a regression, we attempt to fit a straight line through the points that best fits the data. In its simplest form, this is accomplished by finding a line that minimizes the sum of the squared distance of the points from the line. Consequently, it is called an *ordinary least squares* (OLS) regression. When such a line is fit, two parameters emerge—one is the point at which the line cuts through the *Y*-axis, called the intercept of the regression, and the other is the slope of the regression line:

$$Y = a + bX$$

The slope (b) of the regression measures both the direction and the magnitude of the relationship between the dependent variable (Y) and the independent variable (X). When the two variables are positively correlated, the slope will also be positive, whereas when the two variables are negatively correlated, the slope will be negative. The magnitude of the slope of the regression can be read as follows: For every unit increase in the dependent variable (X), the independent variable will change by b (slope). Although regressions mirror correlation coefficients and covariances in showing the strength of the relationship between two variables, they also serve another useful purpose. The regression equation described in the last section can be used to estimate predicted values for the dependent variable, based on assumed or actual values for the independent variable. In other words, for any given Y, we can estimate what X should be:

$$X = a + b(\mathrm{Y})$$

How good are these predictions? That will depend entirely on the strength of the relationship measured in the regression. When the independent variable explains a high proportion of the variation in the dependent variable (R^2 is high), the predictions will be precise. When the R^2 is low, the predictions will have a much wider range.

The regression that measures the relationship between two variables becomes a *multiple regression* when it is extended to include more than one independent variable in trying to explain a dependent variable. Although the graphical presentation becomes more difficult, the multiple regression yields output that is an extension of the simple regression. The R^2 still measures the strength of the relationship, but an additional R^2 statistic called the adjusted R^2 is computed to counter the bias that will induce the R^2 to keep increasing as more independent variables are added to the regression. If there are k independent variables in the regression, the adjusted R^2 is computed as follows:

There are implicit statistical assumptions behind every multiple regression that we ignore at our own peril. For the coefficients on the individual independent variables to make sense, the independent variable needs to be uncorrelated with each other, a condition that is often difficult to meet. When independent variables

are correlated with each other, the statistical hazard that is created is called *multicollinearity*. In its presence, the coefficients on independent variables can take on unexpected signs (positive instead of negative, for instance) and unpredictable values. There are simple diagnostic statistics that allow us to measure how far the data may be deviating from our ideal.

Conclusion

The tools in this chapter are indispensable aids to the process of valuation. We can use time value concepts to move cash flows across time on investments, to arrive at a value. The measures that are part of "risk and return" models in finance can be used to derive costs of equity and capital, and by extension, to value companies in different markets. We derive much of the earnings and cash flow data from financial statement; it behooves us to keep in touch with FASB requirements on leases. Finally, given the quantity and quality of information that we have to access, statistical tools that compress the data and provide a sense of the relationships between data items can all make us better investors.

Chapter 3: Intrinsic Value

What is intrinsic value? Consider it the value that would be attached to an asset by an all-knowing analyst with access to all information available right now and a perfect valuation model. No such analyst exists, of course, but we all aspire to be as close as we can to this perfect analyst. Every asset that has an intrinsic value that reflects its cash flow potential and its risk. While estimating this value poses challenges, we believe that it is important that we look past market perceptions and gauge the intrinsic value of a business or asset.

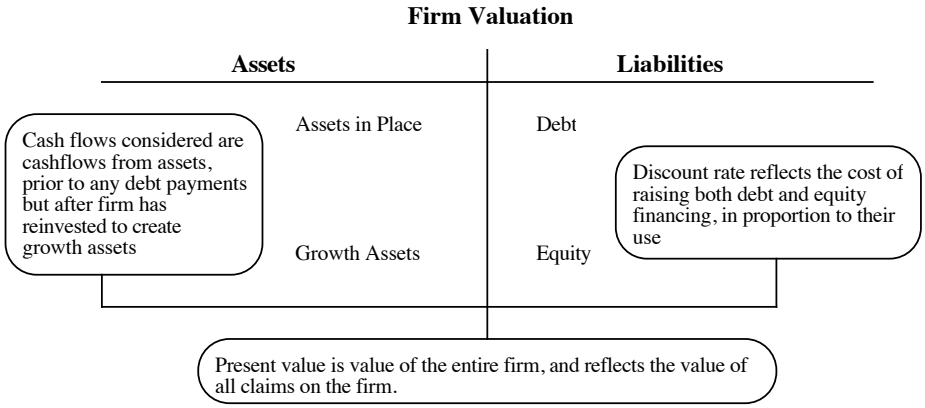
Discounted Cash flow (DCF) Valuation

In discounted cash flow valuation, we begin with a simple proposition. The value of an asset is not what someone perceives it to be worth but it is a function of the expected cash flows on that asset. Put simply, assets with high and predictable cash flows should have higher values than assets with low and volatile cash flows. Putting this principle into practice requires us to grapple with how to define cash flows and what we mean by risk.

Equity versus Firm Valuation

While there are many approaches for adjusting for risk in discounted cash flow valuation, the most common one is the risk adjusted discount rate approach, where we use higher discount rates to discount expected cash flows when valuing riskier assets, and lower discount rates when valuing safer assets. There are two ways in which we can approach discounted cash flow valuation. The first is to value the entire business, with both existing assets (assets-in-place) and growth assets; this is often termed firm or enterprise valuation. The best way to frame this choice is to think in terms of a financial balance sheet, which we introduced in chapter 2.

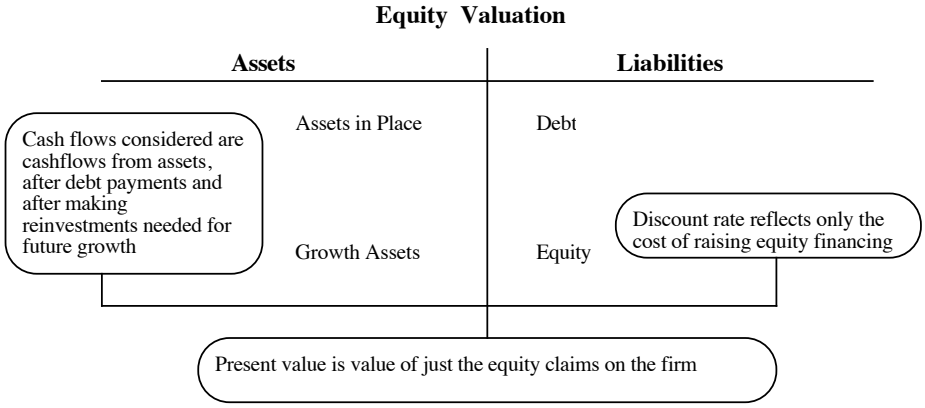
Figure 3.1: Valuing a Firm (Business)



The cash flows before debt payments and after reinvestment needs are termed free cash flows to the firm, and the discount rate that reflects the composite cost of financing from all sources of capital is the cost of capital.

The second way is to just value the equity stake in the business, and this is termed equity valuation.

Figure 3.2: Valuing Equity



The cash flows after debt payments and reinvestment needs are called free cash flows to equity, and the discount rate that reflects just the cost of equity financing is the cost of equity. With publicly traded firms, it can be argued that the only cash flow equity investors get from the firm is dividends and that discounting expected dividends back at the cost of equity should yield the value of equity in the firm. T

Note also that we can always get from the former (firm value) to the latter (equity value) by netting out the value of all non-equity claims from firm value. Done right, the value of equity should be the same whether it is valued directly (by discounting cash flows to equity a the cost of equity) or indirectly (by valuing the firm and subtracting out the value of all non-equity claims).

Inputs to a DCF Valuation

While we can choose to value just the equity or the entire business, we have four basic inputs that we need for a value estimate, though how we define the inputs will be different depending upon whether you do firm or equity valuation. The first input is the cashflow from existing assets, defined either as pre-debt (and to the firm) or as post-debt (and to equity) earnings, net of reinvestment to generate future growth. The second input is growth, with growth in operating income being the key input when valuing the entire business and growth in equity income (net income or earnings per share) becoming the focus when valuing equity. The third input is the discount rate, defined as the cost of the overall capital of the firm, when valuing the business, and as cost of equity, when valuing equity. The final input, allowing for closure, is the terminal value, defined as the estimated value of firm (equity) at the end of the forecast period in firm (equity) valuation.

Cash Flows

Leading up to this section, we noted that cash flows can be estimated to either just equity investors (cash flows to equity) or to all suppliers of capital (cash flows to the firm). In this section, we will begin with the strictest measure of cash flow to equity, i.e. the dividends received by investors, and then progressively move to more expansive measures of cash flows, which generally require more information.

Dividends

When an investor buys stock, he generally expects to get two types of cash flows - dividends during the holding period and an expected price at the end of the holding period. Since this expected price is itself determined by future dividends,

the value of a stock is the present value of just expected dividends. If we accept this premise, the only cash flow to equity that we should be considering in valuation is the dividend paid, and estimating that dividend for the last period should be a simple exercise. Since many firms do not pay dividends, this number can be zero, but it should never be negative.

Augmented Dividends

One of the limitations of focusing on dividends is that many companies, especially in the United States but increasingly around the world, have shifted from dividends to stock buybacks as their mechanism for returning cash to stockholders. While only those stockholders who sell their stock back receive cash, it still represents cash returned to equity investors. In 2007, for instance, firms in the United States returned twice as much cash in the form of stock buybacks than they did in dividends, and focusing only on dividends will result in the under valuation of equity. One simple way of adjusting for this is to *augment the dividend* with stock buybacks and look at the cumulative cash returned to stockholders.

Augmented Dividends = Dividends + Stock Buybacks

One problem, though, is that unlike dividends that are smoothed out over time, stock buybacks can spike in some years and be followed by years of inaction. We therefore will have to normalize buybacks by using average buybacks over a period of time (say, 5 years) to arrive at more reasonable annualized numbers.

Potential Dividends (Free Cash flow to Equity)

With both dividends and augmented dividends, we are trusting managers at publicly traded firms to return to pay out to stockholders any excess cash left over after meeting operating and reinvestment needs. However, we do know that managers do not always follow this practice, as evidenced by the large cash balances that you see at most publicly traded firms. To estimate what managers could have returned to equity investors, we develop a measure of potential dividends that we term the *free cash flow to equity*. Intuitively, the free cash flow to equity measures the cash left over after taxes, reinvestment needs and debt cash flows have been met. It is measured as follows:

FCFE = Net Income – Reinvestment Needs – Debt Cash flows

= Net Income + (Capital Expenditures - Depreciation + Change in non-cash working Capital)- (Principal Repayments + New Debt Issues)

Consider the equation in pieces. We begin with net income, since that is the earnings generated for equity investors; it is after interest expenses and taxes. We compute what the firm has to reinvest in two parts:

a. *Reinvestment in long-lived assets* is measured as the difference between capital expenditures (the amount invested in long lived assets during the period) and depreciation (the accounting expense generated by capital expenditures in prior periods). We net the latter because it is not a cash expense and hence can be added back to net income.
b. *Reinvestment in short-lived assets* is measured by the change in non-cash working capital. In effect, increases in inventory and accounts receivable represent cash tied up in assets that do not generate returns – wasting assets. The reason we done consider cash in the computation is because we assume that companies with large cash balances generally invest them in low-risk, marketable securities like commercial paper and treasury bills; these investments earn a low but a fair rate of return and are therefore not wasting assets.[7] To the extent that they are offset by the use of supplier credit and accounts payable, the effect on cash flows can be muted. The overall change in non-cash working capital therefore is investment in short-term assets.

Reinvestment reduces cash flow to equity investors, but it provides a payoff in terms of future growth. We will come back and consider whether the net effect is positive or negative after we consider how best to estimate growth. The final input into the process are the negative cash flows associated with the repayment of old debt and the positive cash flows to equity investors from raising new debt. If old debt is replaced with new debt of exactly the same magnitude, this term will be zero,

---

[7] Note that we do not make the distinction between operating and non-operating cash that some analysts do (they proceed to include operating cash in working capital). Our distinction is between wasting cash (which would include currency or cash earning below-market rate returns) and non-wasting cash. We are assuming that the former will be a small or negligible number at a publicly traded company.

but it will generate positive (negative) cash flows when debt issues exceed (are less than) debt repayments. A more conservative version of FCFE, which Warren Buffett calls "owners' earnings" ignores the net cash flow from debt. Unlike dividends, the FCFE can be negative for many reasons: the firm may have negative net income, it may be reinvesting far more than it generates in income or it has significant debt repayments scheduled. Since FCFE have to be funded with new equity, existing equity investors will see their stake diluted and have to factor that into value.

Cash Flow to the Firm

The cash flow to the firm is the cash left over after taxes and after all reinvestment needs have been met. Since a firm raises capital from debt and equity investors, the cash flow to the firm should be before interest and principal payments on debt. The cash flow to the firm can be measured in two ways. One is to add up the cash flows to all of the different claim holders in the firm. Thus, the cash flows to equity investors (estimated using one of the three measures described in this section) are added to the cash flows to debt holders (interest and net debt payments) to arrive at the cash flow. The other approach is to start with operating earnings and to estimate the cash flows to the firm prior to debt payments but after reinvestment needs have been met:

Free Cash flow to firm (FCFF) = After-tax Operating Income - Reinvestment
= After-tax Operating Income– (Capital Expenditures – Depreciation + Change in non-cash Working Capital)

It is easiest to understand FCFF by contrasting it with FCFE. First, we begin with after-tax operating income instead of net income; the former is before interest expenses whereas the latter is after interest expenses. Second, we adjust the operating income for taxes, computed as if you were taxed on the entire income, whereas net income is already an after-tax number.[8] Third, while we subtract out

[8] In effect, when computing taxes on operating income, we act like we have no interest expenses or tax benefits from those expenses while computing the cash flow. That is because we will be counting the tax benefits from debt in the cost of capital (through the use of an after-tax cost of debt). If we use actual taxes paid or reflect the tax benefits from interest expenses in the cash flows, we will be double counting its effect.

reinvestment, just as we did to arrive at free cash flows to equity, we do not net out the effect of debt cash flows, since we are now looking at cash flows to all capital and not just to equity.

Another way of presenting the same equation is to cumulate the net capital expenditures and working capital change into one number, and state it as a percentage of the after-tax operating income. This ratio of reinvestment to after-tax operating income is called the reinvestment rate, and the free cash flow to the firm can be written as:

$$\text{Reinvestment Rate} = \frac{(\text{Capital Expenditures} - \text{Depreciation} + \Delta\,\text{Working Capital})}{\text{After-tax Operating Income}}$$

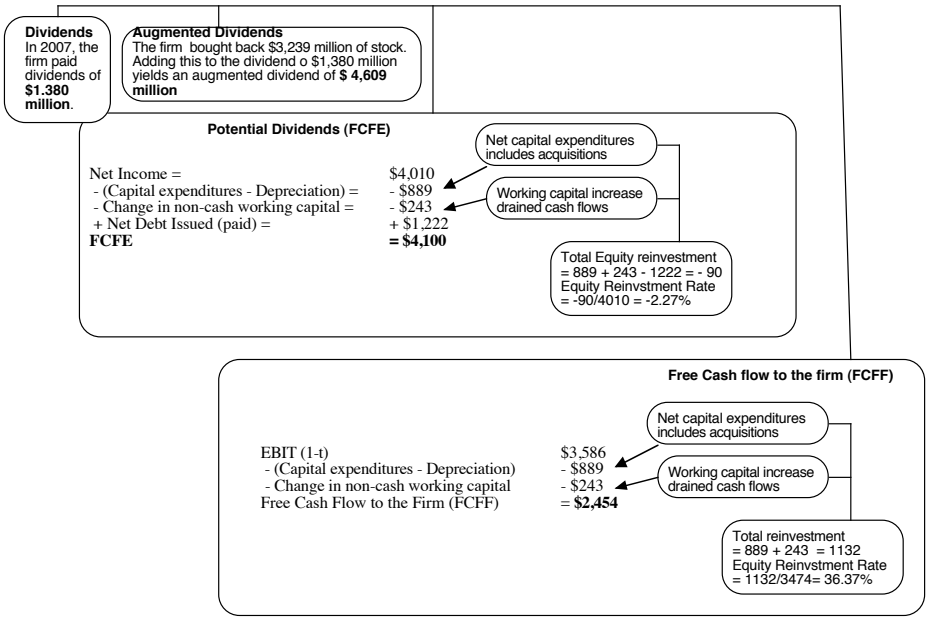
Free Cash Flow to the Firm = EBIT (1-t) (1 – Reinvestment Rate)

Note that the reinvestment rate can exceed 100%[9], if the firm has substantial reinvestment needs. The reinvestment rate can also be less than zero, for firms that are divesting assets and shrinking capital.

How different can these estimates of cash flows be? To illustrate, we looked at Minnesota Mining and Manufacturing (3M, a large market capitalization company, with operations in transportation, health care, office supplies and electronics, in 2007. Exhibit 3.1 provides the estimates of dividends, augmented dividends, free cash flows to equity and free cash flows to the firm.

[9] In practical terms, this firm will have to raise external financing, either from debt or equity or both, to cover the excess reinvestment.

Exhibit 3.1: Estimates of Cash Flows - 3M in 2007



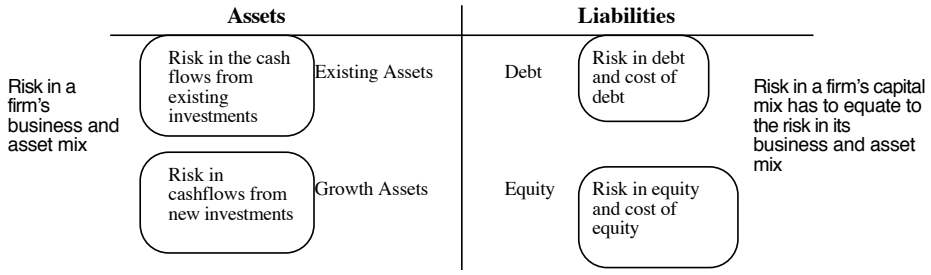
Risk

Cash flows that are riskier should be assessed a lower value than more stable cashflows, but how do we measure risk and reflect it in value? In conventional discounted cash flow valuation models, the discount rate becomes the vehicle for conveying our concerns about risk. We use higher discount rates on riskier cash flows and lower discount rates on safer cash flows. In this section, we will begin be contrasting how the risk in equity can vary from the risk in a business, and then consider the mechanics of estimating the cost of equity and capital.

Business Risk versus Equity Risk

Before we delve into the details of risk measurement and discount rates, we should draw a contrast between two different ways of thinking about risk that relate back to the financial balance sheet that we presented in chapter 1. In the first, we think about the risk in a firm's operations or assets, i.e., the risk in the business. In the second, we look at the risk in the equity investment in this business. Figure 3.3 captures the differences between the two measures:

Figure 3.3: Risk in Business versus Risk in Equity



As with any other aspect of the balance sheet, this one has to balance has well, with the weighted risk in the assets being equal to the weighted risk in the ingredients to capital – debt and equity. Note that the risk in the equity investment in a business is partly determined by the risk of the business the firm is in and partly by its choice on how much debt to use to fund that business. The equity in a safe business can be rendered risky, if the firm uses enough debt to fund that business.

In discount rate terms, the risk in the equity in a business is measured with the cost of equity, whereas the risk in the business is captured in the cost of capital. The latter will be a weighted average of the cost of equity and the cost of debt, with the weights reflecting the proportional use of each source of funding.

Measuring Equity Risk and the Cost of Equity

Measuring the risk in equity investments and converting that risk measure into a cost of equity is rendered difficult by two factors. The first is that equity has an implicit cost, which is unobservable, unlike debt, which comes with an explicit cost in the form of an interest rate. The second is that risk in the eyes of the beholder and different equity investors in the same business can have very different perceptions of risk in that business and demand different expected returns as a consequence.

In the risk and return models that we described in chapter 2, the inputs that we need for the expected return are straightforward. We need to come up with a risk free rate and a price for risk (equity risk premium) to use across all investments. Once we have these market-wide estimates, we then have to measure the relative risk in individual investments. In this section, we will lay out the broad

principles that will govern these estimates but we will return in future chapters to the details of how best to make these estimates for different types of businesses:

- The riskfree rate is the expected return on a long-term investment with guaranteed returns; in effect, you expected return is also your actual return. Since only entities that cannot default can issue risk free securities, we generally use ten or thirty year government bonds rates as risk free rates, implicitly assuming that government don't default, since they can print currency to pay off obligations.
- The equity risk premium is the premium that investors demand for investing in risky assets (or equities) as a class, relative to the riskfree rate. It will be a function not only of how much risk investors perceive in equities, as a class, but the risk aversion that they bring to the market. It also follows that the equity risk premium can change over time, as market risk and risk aversion both change. Many analysts look at the returns stocks have earned, relative to government bonds, over long periods to estimate a historical equity risk premium. An alternative is to back out a forward looking premium (called an implied equity risk premium) from current stock price levels and expected future cash flows.
- To estimate the relative risk (beta), the standard approach for estimating the CAPM beta is to run a regression of returns on a stock against returns on a broad equity market index, with the slope capturing how much the stock moves, for any given market move. As a consequence, the beta estimates that we obtain will always be backward looking (since they are derived from past data) and noisy (they are estimated with errors). In addition, these approaches clearly will not work for investments that do not have a trading history (young companies, divisions of publicly traded companies). One solution is to replace the regression beta with a sector-average beta, if the firm operates in only one sector or a weighted sum of multiple sector betas, if the firm operates in many businesses.

The Cost of Debt

While equity investors receive residual cash flows and bear the bulk of the operating risk in most firms, lenders to the firm also face the risk that they will not

receive their promised payments – interest expenses and principal repayments. It is to cover this default risk that lenders add a "default spread" to the riskless rate when they lend money to firms; the greater the perceived risk of default, the greater the default spread and the cost of debt. The other dimension on which debt and equity can vary is in their treatment for tax purposes, with cashflows to equity investors (dividends and stock buybacks) coming from after-tax cash flows, whereas interest payments are tax deductible.

To estimate the cost of debt for a firm, we need three components. The first is the riskfree rate, an input to the cost of equity as well. As a general rule, the riskfree rate used to estimate the cost of equity should be used to compute the cost of debt as well; if the cost of equity is based upon a long-term riskfree rate, as it often is, the cost of debt should be based upon the same rate. The second is the default spread and there are two approaches that are used, depending upon the firm being analyzed. If the firm has a bond rating from an established ratings agency such as S&P or Moody's, we can estimate a default spread based upon the rating. In September 2008, for instance, the default spread for BBB rated bonds was 2% and would have been used as the spread for any BBB rated company. If the firm is unrated and has debt outstanding (bank loans), we can estimate a "synthetic" rating for the firm, based upon its financial ratios. A simple, albeit effective approach for estimating the synthetic ratio is to base it entirely on the interest coverage ratio (EBIT/ Interest expense) of a firm; higher interest coverage ratios will yield higher ratings and lower interest coverage ratios.

The final input needed to estimate the cost of debt is the tax rate. Since interest expenses save you taxes at the margin, the tax rate that is relevant for this calculation is not the effective tax rate but the marginal tax rate. In the United States, where the federal corporate tax rate is 35% and state and local taxes add to this, the marginal tax rate for corporations in 2008 was close to 40%, much higher than the average effective tax rate, across companies, of 28%. The after-tax cost of debt for a firm is therefore:

After-tax cost of debt = (Riskfree Rate + Default Spread) (1- Marginal tax rate)

The after-tax cost of debt for most firms will be significantly lower than the cost of equity for two reasons. First, debt in a firm is generally less risky than its equity, leading to lower expected returns. Second, there is a tax saving associated with debt that does not exist with equity.

Debt Ratios and the Cost of Capital

Once we have estimated the costs of debt and equity, we still have to assign weights for the two ingredients. In making this estimate, the values that we should use are market values, rather than book values. For publicly traded firms, estimating the market value of equity is usually a trivial exercise, where we multiply the share price by the number of shares outstanding. Estimating the market value of debt is usually a more difficult exercise, since most firms have some debt that is not traded and many practitioners fall back on book value of debt.

Once we have the current market value weights for debt and equity for use in the cost of capital, we have a follow up judgment to make in terms of whether these weights will change or remain stable. If we assume that they will change, we have to specify both what the right or target mix for the firm will be and how soon the change will occur. In an acquisition, for instance, we can assume that the acquirer can replace the existing mix with the target mix instantaneously. As companies change over time, we should expect the cost of capital to change as well.

Using 3M again as our illustrative example, we estimate the cost of equity, debt and capital for the firm in exhibit 3.2. To estimate the cost of equity, we used the sector average betas of the businesses that 3M operates in and to estimate the cost of debt, we used a synthetic rating, based upon the interest coverage ratio:

Exhibit 3.2: Estimate of Cost of Capital - 3M iin Setpember 2008

	Market value	*Proportion of capital*	*Cost*
Equity	$57,041	91.50%	9.16%
Debt	$5,297	8.50%	2.91%
Capital	$62,338	100.00%	8.63%

Cost of Equity

3.72% + 1.36 X 4.00% = **9.16%**

Riskfree Rate
Treasury Bond rate of 3.72% in September 2008 used as riskfree rate.

Mature market premium
Estimated using S&P 500 and expected cash flows on equities.

Beta based upon business mix

Business	*Estimated Value*	*Weight in firm*	*Beta*
Industrial & Transportation	$8,265	27.42%	0.86
Health Care	$7,261	24.09%	1.48
Display & Graphics	$6,344	21.04%	2.08
Consumer & Office	$2,654	8.80%	1.04
Safety, Security & Protection	$3,346	11.10%	1.22
Electro & Communications	$2,276	7.55%	1.39
	$30,146	100.00%	1.36

Cost of debt

(3.72% + 0.75%) (1 - .35) = 2.91%

Riskfree Rate
Treasury Bond rate of 3.72% in September 2008 used as riskfree rate.

Marginal tax rate
Federal corporate tax rate for US

3M's interest coverage ratio (2007) - $\frac{\text{Operating Income}}{\text{Interest Expenses}} = \frac{\$5,361}{\$227} = 23.63$

Interest Coverage Ratio	*Rating*	*Typical default spread*
> 12.5	AAA	0.75%
9.50 - 12.50	AA	1.25%
7.50 – 9.50	A+	1.40%
6.00 – 7.50	A	1.50%
4.50 – 6.00	A-	1.70%
4.00 – 4.50	BBB	2.50%
3.50 - 4.00	BB+	3.20%
3.00 – 3.50	BB	3.65%
2.50 – 3.00	B+	4.50%
2.00 - 2.50	B	5.65%
1.50 – 2.00	B-	6.50%
1.25 – 1.50	CCC	7.50%
0.80 – 1.25	CC	10.00%
0.50 – 0.80	C	12.00%
< 0.65	D	20.00%

Synthetic Rating = AAA
Default spread = 0.75%

Historical and Forecasted Growth Rates

When confronted with the task of estimating growth, it is not surprising that analysts turn to the past. In effect, they use growth in revenues or earnings in the recent past as a predictor of growth in the future. Before we put this practice under the microscope, we should add that the historical growth rates for the same company can yield different estimates, depending upon computational choices: how far back to go, which measure of earnings (net income, earnings per share, operating income) to use and how to compute the average (arithmetic or geometric). A debate how best to estimate historical growth makes sense only if it is a good predictor of future growth. Unfortunately, studies that have looked at the relationship have generally concluded that (a) the relationship between past and future growth is a very weak one, (b) scaling matters, with growth dropping off significantly as companies grow and (c) firms and sectors grow through growth cycles, with high growth in one period followed by low growth in the next.

If historical growth is not a useful predictor of future growth, there is another source that we can use for future growth. We can draw on those who know the firm better than we do – equity research analysts who have tracked the firm for years or the managers in the firm – and use their estimates of growth. On the plus side, these forecasts should be based upon better information than we have available to us. After all, managers should have a clearer sense of how much they will reinvest in their own businesses and what the potential returns on investments are when they do, and equity research analysts have sector experience and informed sources that they can draw on for better information. On the minus side, neither managers nor equity research analysts are objective about the future; managers are likely to over estimate their capacity to generate growth and analysts have their own biases. In addition, both analysts and managers can get caught up in the mood of the moment, over estimating growth in buoyant times and under estimating growth in down times. As with historical growth, studies indicate that neither analyst estimates nor management forecasts are good predictors of future growth.

Fundamental Growth Rates

If we cannot draw on history or trust managers and analysts, how then do we estimate growth? The answer lies in the fundamentals within a firm that ultimately determine its growth rate. In this section, we will consider the two sources for growth – new investments that expand the business and improved efficiency on existing investments.

Growth from new investments

The growth rate from new investments can be written as a function of two variables - the percentage of earnings that is reinvested back in the business and the return earnings on that investment. While investment and return on investment are generic terms, the way in which we define them will depend upon whether we are looking at equity earnings or operating income. When looking at equity earnings, our focus is on the investment in equity and the return is the return on equity. When looking at operating earnings, the focus is on the investment in capital and the return is the return on capital. Central to any estimate of fundamental

growth is the estimate of return on capital or equity, measures that we introduced in chapter 2. Table 3.3 summarizes the inputs for each measure depending on the measure of cash flow that we are focused on:

Table 3.3: Measuring Investment and Return on Investment

Growth in earnings	=	*Proportion of earnings reinvested*	*X*	*Return on Investment*
Operating Income	=	Reinvestment Rate = $\frac{(\text{Cap Ex} - \text{Deprec'n} + \Delta\text{WC})}{\text{EBIT}(1-t)}$	X	Return on Invested Capital (ROC or ROIC)
Net Income (Non-cash)	=	Equity Reinvestment Rate = $\frac{(\text{Cap Ex} - \text{Deprec'n} + \Delta\text{WC} - \Delta\text{Debt})}{\text{Net Income}}$	X	Non-cash Return on Equity (NCROE)
Earnings per share	=	Retention Ratio = $1 - \frac{\text{Dividends}}{\text{Net Income}}$	X	Return on Equity (ROE)

It is conventional practice to use accounting measures of investment and return on investment. Thus, the book values of equity and invested capital and accounting earnings are used to compute returns on equity and capital:

$$\text{Return on Capital (ROIC)} = \frac{\text{Operating Income}_t\ (1 - \text{tax rate})}{\text{Book Value of Invested Capital}_{t-1}}$$

$$\text{Non-cash Return on Equity(NCROE)} = \frac{\text{Net Income}_t - \text{Interest Income from Cash}_t\ (1 - \text{tax rate})}{\text{Book Value of Equity}_{t-1} - \text{Cash}_{t-1}}$$

$$\text{Return on Equity (ROE)} = \frac{\text{Net Income}_t}{\text{Book Value of Equity}_{t-1}}$$

The problem with accounting measures on both dimensions is well documented, with accounting choices on restructuring charges, amortization and capitalization all making a difference in the final numbers.[10]

Efficiency Growth

For many mature firms with limited investment opportunities, the potential for growth from new investments is limited. These firms cannot maintain a high

[10] To get a sense of the problems with using accounting numbers, and how best to correct for them, see: Damodaran, A., 2007, Return on capital, Return on Invested Capital and Return on Equity: Measurement and Implications, Working Paper, SSRN.

reinvestment rate and deliver a high return on capital with that reinvestment. However, they can still grow at healthy rates if they can improve the returns that they earn on existing assets. Thus, a firm that generates a 5% return on capital this year can generate a growth rate of 40% next year, if it can improve that return on capital to 7%.[11] Conversely, declines in returns on existing assets can translate into drops in earnings growth rates.

When valuing companies, efficiency growth is pure gravy in terms of value created, since the growth comes with no concurrent cost. Unlike growth from new investments, where the positive effects of growth have to be offset against the negative effect of more investment, improving the return on capital on existing assets increases the growth rate without adversely affecting the cash flows. It should as come as no surprise, then, that analysts who want to increase the value of a company draw on the efficiency argument to justify much higher growth rates than those estimated using fundamentals.

While the potential for efficiency growth is always there, we should put some common sense constraints on how much we can draw on this growth.

1. There is more potential for efficiency growth at mature firms, with poor returns on capital (equity), than there is at firms that are performing well, for two reasons. First, improving the return on capital is a much more feasible option for a firm that generates a return on capital that is well below the sector average than at a firm that already outperforms the sector. Second, the effect of an improvement in returns on growth is much greater when the return on capital is low than when it is high. A firm that improves its return on capital from 5% to 6% will report a 20% growth rate from efficiency in that period, whereas a firm that improves its return on capital from 25% to 26% will generate a 4% growth rate from efficiency in that period.
2. You can draw on increased efficiency to justify growth only for finite periods. After all, a firm cannot be infinitely inefficient. Once the inefficiencies, no matter

[11] The easiest way to see this is to act like you have $ 100 invested in capital in this company. By improving its return on capital from 5% to 7%, the earnings will increase from $5 to $7, which is a 40% increase.

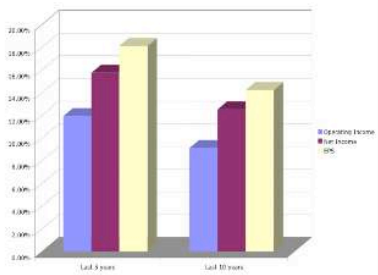
how significant, are fixed, the firm will have to revert back to its sustainable growth rate, based upon new investments.

In closing, growth in a specific firm can come from new investments or improved efficiency, but it has to be earned either way. None of us has the power to endow companies with higher growth rates, just because we like the managers or want to make it value increase.

Figure 3.3 provides estimates of historical, analyst and fundamental growth rates for 3M in September 2008. Note that we are assuming no efficiency growth for 3M, because its return on capital is already high and has been stable over time.

Exhibit 3.3: Estimate of Growth - 3M iin Setpember 2008

Historical Growth



Fundamental Growth

	In last financial year - 2007			Expected for next 5 years		
	Reinvestment	Return	Expected growth	Reinvestment	Return	Expected growth
Non-cash net income	-2.27%	47.65%	-1.08%	25.00%	30.00%	7.50%
After-tax Operating Income	36.37%	25.31%	9.21%	30.00%	25.00%	7.50%

Analyst Estimate

Forecasted growth in Earnings per share next 5 years = 8%-9%

Terminal Value

Publicly traded firms do not have finite lives. Given that we cannot estimate cash flows forever, we generally impose closure in valuation models by stopping our estimation of cash flows sometime in the future and then computing a terminal value that reflects all cash flows beyond that point. The two legitimate ways of estimating terminal value are to estimate a liquidation value for the assets of the firm, assuming that the assets are sold in the terminal year, or to estimate a going concern or a terminal value. Applying a multiple to estimate terminal value, a technique used by many analysts, is incompatible with intrinsic value and makes the valuation a relative one.

1. Liquidation Value

If we assume that the business will be ended in the terminal year and that its assets will be liquidated at that time, we can estimate the proceeds from the liquidation. This liquidation value still has to be estimated, using a combination of market-based numbers (for assets that have ready markets) and cashflow-based estimates. For firms that have finite lives and marketable assets (like real estate), this represents a fairly conservative way of estimating terminal value. For other firms, estimating liquidation value becomes more difficult to do, either because the assets are not separable (brand name value in a consumer product company) or because there is no market for the individual assets. One approach is to use the estimated book value of the assets as a starting point, and to estimate the liquidation value, based upon the book value.

2. Going Concern or Terminal value

If we treat the firm as a going concern at the end of the estimation period, we can estimate the value of that concern by assuming that cash flows will grow at a constant rate forever afterwards. This perpetual growth model draws on a simple present value equation to arrive at terminal value:

$$\text{Terminal Value}_n = \frac{\text{Cashflow in year n} + 1}{(\text{Discount rate} - \text{Perpetual growth rate})}$$

Our definitions of cash flow and growth rate have to be consistent with whether we are valuing dividends, cash flows to equity or cash flows to the firm; the discount rate will be the cost of equity for the first two and the cost of capital for the last. The perpetual growth model is a powerful one, but it can be easily misused. In fact, analysts often use it is as a piggy bank that they go to whenever they feel that the value that they have derived for an asset is too low or high. Small changes in the inputs can alter the terminal value dramatically. Consequently, there are three key constraints that should be imposed on its estimation:

a. <u>Cap the growth rate</u>: Small changes in the stable growth rate can change the terminal value significantly and the effect gets larger as the growth rate approaches the discount rate used in the estimation. The fact that a stable growth rate is constant forever, however,

puts strong constraints on how high it can be. Since no firm can grow forever at a rate higher than the growth rate of the economy in which it operates, the constant growth rate cannot be greater than the overall growth rate of the economy. Setting the stable growth rate to be less than or equal to the growth rate of the economy is not only the consistent thing to do but it also ensures that the growth rate will be less than the discount rate. In fact, a simple rule of thumb on the stable growth rate is that it should not exceed the riskless rate used in the valuation.[12]

b. Use mature company risk characteristics: As firms move from high growth to stable growth, we need to give them the characteristics of stable growth firms. A firm in stable growth is different from that same firm in high growth on a number of dimensions. In general, you would expect stable growth firms to be less risky and use more debt. In practice, we should move betas for even high risk firms towards one in stable growth and give them debt ratios, more consistent with larger, more stable cashflows.

c. Reinvestment and Excess Return Assumptions: Stable growth firms tend to reinvest less than high growth firms and it is critical that we both capture the effects of lower growth on reinvestment and that we ensure that the firm reinvests enough to sustain its stable growth rate in the terminal phase. Given the relationship between growth, reinvestment rate and returns that we established in the section on expected growth rates, we can estimate the reinvestment rate that is consistent with expected growth in table 3.5:

Table 3.5: Reinvestment in Stable Growth

Model	Reinvestment Rate in stable growth
Dividend	$\frac{\text{Stable Growth rate}}{\text{Return on Equity in stable growth}}$
FCFE	$\frac{\text{Stable Growth rate}}{\text{Non - cash Return on Equity in stable growth}}$

[12] This is because of the relationship between the riskless rate that goes into the discount rate and the growth rate of the economy. Note that the riskless rate can be written as:
Nominal riskless rate = Real riskless rate + Expected inflation rate
In the long term, the real riskless rate will converge on the real growth rate of the economy and the nominal riskless rate will approach the nominal growth rate of the economy.

FCFF	$\frac{\text{Stable growth rate}}{\text{Return on capital in stable phase}}$

Linking the reinvestment rate and retention ratio to the stable growth rate also makes the valuation less sensitive to assumptions about stable growth. While increasing the stable growth rate, holding all else constant, can dramatically increase value, changing the reinvestment rate as the growth rate changes will create an offsetting effect.

$$\text{Terminal Value} = \frac{\text{EBIT}_{n+1}(1-t)(1-\text{Reinvestment Rate})}{\text{Cost of Capital}_n - \text{Stable Growth Rate}}$$

The gains from increasing the growth rate will be partially or completely offset by the loss in cash flows because of the higher reinvestment rate. Whether value increases or decreases as the stable growth increases will entirely depend upon what you assume about excess returns. If the return on capital is higher than the cost of capital in the stable growth period, increasing the stable growth rate will increase value. <u>If the return on capital is equal to the stable period cost of capital, increasing the stable growth rate will have no effect on value</u>. You could establish the same propositions with equity income and cash flows and show that the terminal value of equity is a function of the difference between the return on equity and cost of equity.

In closing, the key assumption in the terminal value computation is not what growth rate you use in the valuation, but what excess returns accompany that growth rate. If you assume no excess returns, the growth rate becomes irrelevant. There are some valuation experts who believe that this is the only sustainable assumption, since no firm can maintain competitive advantages forever. In practice, though, there may be some wiggle room, insofar as the firm may become a stable growth firm before its excess returns go to zero. If that is the case and the competitive advantages of the firm are strong and sustainable (even if they do not last forever), we may be able to give the firm some excess returns in perpetuity. As a simple rule of thumb again, these excess returns forever should be modest (<4-5%) and will affect the terminal value.

Using 3M as the illustrative example again, we assumed that the firm would be in stable growth after the fifth year and grow 3% a year forever. Table 3.6 lists how we expect the other inputs to change in stable growth:

Table 3.6: Valuing 3M – High versus Stable Growth Phases

	High Growth	Stable Growth
Length of High Growth Period =	Next 5 years	After year 5
Growth Rate =	7.50%	3.00%
Debt Ratio used in Cost of Capital Calculation=	8.48%	20.00%
Beta used for stock =	1.36	1.00
Riskfree rate =	3.72%	3.72%
Risk Premium =	4.00%	4.00%
Cost of Debt =	4.47%	4.47%
Tax Rate =	35.00%	35.00%
Cost of capital	8.63%	6.76%
Return on Capital =	25.00%	6.76%
Reinvestment Rate =	30.00%	44.40%

Note that as the growth declines after year 5, the beta is adjusted towards one and the debt ratio is raised to the industry average of 20% to reflect the overall stability of the company. Since the cost of debt is relatively low, we leave it unchanged, resulting in a drop in the cost of capital to 6.76%. We do change the reinvestment rate in stable growth to reflect the assumption that there will be no excess returns in stable growth (return on capital = cost of capital). Using the predicted stable growth rate of 3% and the return on capital of 6.76% (equal to cost of capital), we derive a reinvestment rate of 44.4%:

$$\text{Reinvestment Rate in stable growth} = \frac{\text{Expected Growth}}{\text{Stable ROC}} = \frac{3.00\%}{6.76\%} = 44.40\%$$

Tying up loose ends

We have covered the four inputs that go into discounted cash flow valuation models – cash flows, discount rates, growth rates and the terminal value. The present value we arrive at, when we discount the cash flows at the risk-adjusted rates should yield an estimate of value, but getting from that number to what we

would be willing to pay per share for equity does require use to consider a few other factors.

a. Cash and Marketable Securities: Most companies have cash balances that are not insignificant in magnitude. Is this cash balance already incorporated into the present value? The answer depends upon how we estimated cash flows. If the cash flows are based on operating income (free cash flow to the firm) or non-cash net income, we have not valued cash yet and it should be added on to the present value. If, on the other hand, we estimate cash flows from the cumulative net income or use the dividend discount model, cash already has been implicitly valued; the income from cash is part of the final cash flow and the discount rate presumably has been adjusted to reflect the presence of cash

b. Cross Holdings in other companies: Companies sometimes invest in other firms, and these cross holdings can generally be categorized as either minority or majority holdings. With the former, the holdings are usually less than 50%, and the income from the holdings are reported in the income statement below the operating income line. If we use free cash flow to the firm to value the operating assets, we have not valued these minority holdings yet, and they have to be valued explicitly and added to present value. With majority holdings, which generally exceed 50%, firms usually consolidate the entire subsidiary in their financials, and report 100% of the operating income and assets of the subsidiary. To reflect the portion of the subsidiary that does not belong to them, they report the book value of that portion as minority interest in a balance sheet. If we compute cash flows from consolidated financial statements, we have to subtract out the estimated market value of the minority interest.

c. Potential liabilities (not treated as debt): Since we are interested in the value of equity in the firm, we have to consider any potential liabilities that we may face that reduce that value. Thus, items like under funded pension obligations and health care obligations may not meet the threshold to be categorized as debt for cost of capital purposes but should be considered when valuing equity. In other words, we would subtract out the values of these and other claims (such as

potential costs from lawsuits against the firm) on equity from firm value to arrive at equity value.

d. Employee Options: Having arrived at the value of equity in the firm, there is one final estimate that we have to make, especially if the firm has made it a practice to grant options to managers. Since many of these options will be still outstanding, we have to consider them as another (and different) claim on equity. While analysts often use short cuts (such as adjusting the number of shares for dilution) to deal with these options, the right approach is to value the options (using an option pricing model), reduce the value of equity by the option value and then divide by the actual number of shares outstanding.

Table 3.7 summarizes the loose ends and how to deal with them in the different models.

Table 3.7: Dealing with loose ends in valuation

Loose End	*Dividend Discount Model*	*FCFE Model*	*FCFF Model*
Cash and Marketable Securities	Ignore, since net income includes interest income from cash.	Ignore, if FCFE is computed using total net income. Add, if FCFE is computed using non-cash net income	Add. Operating income does not include income from cash.
Cross Holdings	Ignore, since net income includes income from cross holdings.	Ignore, since net income includes income from cross holdings.	Add market value of minority holdings and subtract market value of minority interests.
Other Liabilities	Ignore. The assumption is that the firm is considering costs when setting dividends.	Subtract out expected litigation costs.	Subtract out under funded pension obligations, health care obligations and expected litigation costs.
Employee options	Ignore.	Subtract out value of equity options outstanding	Subtract out value of equity options outstanding

Figure 3.4 summarizes the valuation of 3M, bringing together all of the key assumptions on cash flow, growth and risk.

Exhibit 3.4: A Valuation of 3M

Income from existing assets
The current after-tax operating income measures how much income is generated by existing investments.

How high is the growth rate?
Higher growth rates lead to higher operating income in the future but growth rate itself is a product of how much you reinvest and how well you reinvest.
Growth = Reinvestment rate * ROC

What is the quality of growth?
Not all growth is value adding. It is only if you earn a return on capital that exceeds your cost of capital.

How risky are the cash flows?
The cost of capital is the device we use to capture the risk in the cost of capital. As cash flows become less risky, the cost of capital should decrease.

Year	After-tax Operating Income	Expected Growth rate	ROC	Reinvestment Rate	Reinvestment	FCFF	Cost of capital	Present Value
Current	$3,586		25.31%	31.57%	1132	$2,454		
1	$3,854	7.50%	25.00%	30.00%	$1,156	$2,698	8.63%	$2,484
2	$4,144	7.50%	25.00%	30.00%	$1,243	$2,900	8.63%	$2,484
3	$4,454	7.50%	25.00%	30.00%	$1,336	$3,118	8.63%	$2,484
4	$4,788	7.50%	25.00%	30.00%	$1,437	$3,352	8.63%	$2,484
5	$5,147	7.50%	25.00%	30.00%	$1,544	$3,603	8.63%	$2,484
Beyond	$5,302	3.00%	6.76%	44.40%	$2,354	$78,464	6.76%	$51,872
Value of operating assets =								$64,291
(Add) Cash								$3,253
(Subtract) Debt								$5,297
(Subtract) Management Options								$1,216
Value of Equity in Common Stock								$60,776
Value per share								$86.95

When will it become a mature firm?
This is a function of both your growth rate and your competitive advantages. The stronger the barriers to entry, the longer you can put off becoming a mature company

What is the value as a mature firm?
When you make your firm a mature firm and give it a stable growth rate, you should give it the characteristics of a mature firm - a mature company cost of capital and excess returns.

Are there other claims on equity?
If you use convertible securities or give options to managers, you are creating claims on equity that take away from your value as a stockholder. You need to net the value of these options out, before dividing by the actual number of shares.

What do intrinsic valuation models tell us?

When we estimate intrinsic value, we are estimating what an asset or business is worth, given its cash flows and the risk in those cash flows. To the extent that the value is dependent upon the assumptions we make about cash flows, growth and risk, it represents what we think the intrinsic value is at any point in time. So, what if the intrinsic value that we derive is very different from the market price? There are several possible explanations. One is that we have made erroneous or unrealistic assumptions about a company's future growth potential or riskiness. A second and related explanation is that we have made incorrect assessments of risk premiums for the entire market. A third is that the market is, in fact, making a mistake in its assessment of value.

Even in the last scenario, where our assessment of value is right and the market price is wrong, there is no guarantee that we can make money of our valuations. For that to occur, markets have to correct their mistakes and that may not happen in the near future. In fact, we can buy stocks that we believe are under valued and find them become more under valued over time. That is why a long time horizon is almost a pre-requisite for using intrinsic valuation models. Giving the

market more time (say 3 to 5 years) to fix its mistakes provides better odds than hoping that it will happen in the next quarter or the next six months.

Conclusion

The intrinsic value of a company reflects its fundamentals. The primary tool for estimating intrinsic value is the discounted cash flow model. There are four ingredients that go into estimating this value. The first is the cash flows from existing assets, before debt payments (to the firm) and after debt payments (to equity). The second is the discount rate, with the cost of equity used to discount cash flows to equity and the cost of capital to discount cash flows to the firm. The third is expected growth, which is driven by how much and how well firms reinvest. Finally, we applied closure to the models by assuming that cash flows will settle into stable growth, sometime in the future, but imposed constraints on what this growth rate can be and the characteristics of stable growth companies.

Chapter 4: Relative Valuation

In discounted cash flow valuation, the objective is to find the value of an asset, given its cash flow, growth and risk characteristics. In relative valuation, the objective is to value an asset, based upon how similar assets are currently priced by the market. Consequently, there are two components to relative valuation. The first is that to value assets on a relative basis, prices have to be standardized, usually by converting prices into multiples of some common variable. While this common variable will vary across assets, it usually takes the form of earnings, book value or revenues for publicly traded stocks.. The second is to find similar assets, which is difficult to do since no two assets are exactly identical. With real assets like antiques and baseball cards, the differences may be small and easily controlled for when pricing the assets. In the context of valuing equity in firms, the problems are compounded since firms in the same business can still differ on risk, growth potential and cash flows.

What is relative valuation?

In relative valuation, we value an asset based upon how similar assets are priced in the market. A prospective house buyer decides how much to pay for a house by looking at the prices paid for similar houses in the neighborhood. A baseball card collector makes a judgment on how much to pay for a Mickey Mantle rookie card by checking transactions prices on other Mickey Mantle rookie cards. In the same vein, a potential investor in a stock tries to estimate its value by looking at the market pricing of "similar" stocks.

Embedded in this description are the three essential steps in relative valuation. The first step is <u>finding comparable assets that are priced by the market</u>, a task that is easier to accomplish with real assets like baseball cards and houses than it is with stocks. All too often, analysts use other companies in the same sector as comparable, comparing a software firm to other software firms or a utility to other utilities, but we will question whether this practice really yields similar companies later in this chapter. The second step is <u>scaling the market prices to a common</u>

variable to generate standardized prices that are comparable. While this may not be necessary when comparing identical assets (Mickey Mantle rookie cards), it is necessary when comparing assets that vary in size or units. Other things remaining equal, a smaller house or apartment should trade at a lower price than a larger residence. In the context of stocks, this equalization usually requires converting the market value of equity or the firm into multiples of earnings, book value or revenues. The third and last step in the process is adjusting for differences across assets when comparing their standardized values. Again, using the example of a house, a newer house with more updated amenities should be priced higher than a similar sized older house that needs renovation. With stocks, differences in pricing across stocks can be attributed to all of the fundamentals that we talked about in discounted cash flow valuation. Higher growth companies, for instance, should trade at higher multiples than lower growth companies in the same sector. Many analysts adjust for these differences qualitatively, making every relative valuation a story telling experience; analysts with better and more believable stories are given credit for better valuations.

There is a significant philosophical difference between discounted cash flow and relative valuation. In discounted cash flow valuation, we are attempting to estimate the intrinsic value of an asset based upon its capacity to generate cash flows in the future. In relative valuation, we are making a judgment on how much an asset is worth by looking at what the market is paying for similar assets. If the market is correct, on average, in the way it prices assets, discounted cash flow and relative valuations may converge. If, however, the market is systematically over pricing or under pricing a group of assets or an entire sector, discounted cash flow valuations can deviate from relative valuations.

Notwithstanding the focus on discounted cash flow valuation in classrooms and in theory, there is evidence that most assets are valued on a relative basis for many reasons. They usually require less information and are easier to do than intrinsic valuations and are more likely to reflect the market mood of the moment. The strengths of relative valuation are also its weaknesses. First, the ease with which a relative valuation can be put together, pulling together a multiple and a group of comparable firms, can also result

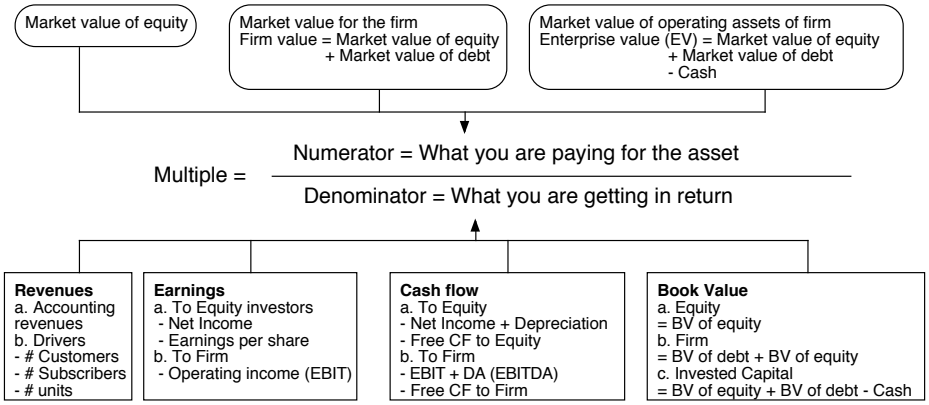
in inconsistent estimates of value where key variables such as risk, growth or cash flow potential are ignored. Second, the fact that multiples reflect the market mood also implies that using relative valuation to estimate the value of an asset can result in values that are too high, when the market is over valuing comparable firms, or too low, when it is under valuing these firms. Third, while there is scope for bias in any type of valuation, the lack of transparency regarding the underlying assumptions in relative valuations make them particularly vulnerable to manipulation.

Standardized Values and Multiples

When comparing identical assets, we can compare the prices of these assets. Thus, the price of a Tiffany lamp or a Mickey Mantle rookie card can be compared to the price at which an identical item was bought or sold in the market. However, comparing assets that are not exactly similar can be a challenge. If we have to compare the prices of two buildings of different sizes in the same location, the smaller building will look cheaper unless we control for the size difference by computing the price per square foot.

Things get even messier when comparing publicly traded stocks across companies. After all, the price per share of a stock is a function both of the value of the equity in a company and the number of shares outstanding in the firm. Thus, a stock split that doubles the number of units will approximately halve the stock price. To compare the values of "similar" firms in the market, we need to standardize the values in some way by scaling them to a common variable. In general, values can be standardized relative to the earnings firms generate, to the book value or replacement value of the firms themselves, to the revenues that firms generate or to measures that are specific to firms in a sector. Figure 4.1 lists the alternatives:

Figure 4.1: Choices in Standardization



To provide a couple of illustrations, we can divide the market value of equity by the net income to estimate how much a dollar in earnings today is costing us, as equity investors, or the enterprise value to EBITDA to get a sense of the market value of operating assets, relative to operating cash flows. We can even estimate what we are paying as a market value per subscriber in a cable company or what the market value is relative to the accounting estimate of value (book value). The central reason for standardizing, though, does not change. We want to compare these numbers across companies.

The Four Basic Keys to Using Multiples

Multiples are easy to use and easy to misuse. There are four basic steps to using multiples wisely and for detecting misuse in the hands of others. The first step is to ensure that the multiple is defined consistently and that it is measured uniformly across the firms being compared. The second step is to be aware of the cross sectional distribution of the multiple, not only across firms in the sector being analyzed but also across the entire market. The third step is to analyze the multiple and understand not only what fundamentals determine the multiple but also how changes in these fundamentals translate into changes in the multiple. The final step is finding the right firms to use for comparison and controlling for differences that may persist across these firms.

Definitional Tests

Even the simplest multiples are defined differently by different analysts. Consider, for instance, the price earnings ratio (PE), the most widely used valuation

multiple in valuation. Analysts define it to be the market price divided by the earnings per share but that is where the consensus ends. There are a number of variants on the PE ratio. While the current price is conventionally used in the numerator, there are some analysts who use the average price over the last six months or a year. The earnings per share in the denominator can be the earnings per share from the most recent financial year (yielding the current PE), the last four quarters of earnings (yielding the trailing PE) and expected earnings per share in the next financial year (resulting in a forward PE). In addition, earnings per share can be computed based upon primary shares outstanding or fully diluted shares and can include or exclude extraordinary items. Not only can these variants on earnings yield vastly different values for the price earnings ratio but the one that gets used by analysts depends upon their biases. For instance, in periods of rising earnings, the forward PE will yield consistently lower values than the trailing PE, which, in turn, will be lower than the current PE. A bullish analyst will tend to use the forward PE to make the case that the stock is trading at a low multiple of earnings, while a bearish analyst will focus on the current PE to make the case that the multiple is too high.

Consistency

Every multiple has a numerator and a denominator. The numerator can be either an equity value (such as market price or value of equity) or a firm value (such as enterprise value, which is the sum of the values of debt and equity, net of cash). The denominator can be an equity measure (such as earnings per share, net income or book value of equity) or a firm measure (such as operating income, EBITDA or book value of capital). One of the key tests to run on a multiple is to examine whether the numerator and denominator are defined consistently. *If the numerator for a multiple is an equity value, then the denominator should be an equity value as well. If the numerator is a firm value, then the denominator should be a firm value as well*. To illustrate, the price earnings ratio is a consistently defined multiple, since the numerator is the price per share (which is an equity value) and the denominator is earnings per share (which is also an equity value). So is the Enterprise value to EBITDA multiple, since the numerator and

denominator are both firm value measures; the enterprise value measures the market value of the operating assets of a company and the EBITDA is the cashflow generated by the operating assets, prior to taxes and reinvestment needs. In contrast, the price to sales ratio and price to EBITDA are not consistently defined, since they divide the market value of equity by a measure of operating revenue or earnings. Using these multiples will lead you to find any firm with a significant debt burden to be cheap.

Uniformity

In relative valuation, the multiple is computed for all of the firms in a group and then compared across these firms to make judgments on which firms are over priced and which are under priced. For this comparison to have any merit, the multiple has to be defined uniformly across all of the firms in the group. Thus, if the trailing PE is used for one firm, it has to be used for all of the others as well. In fact, one of the problems with using the current PE to compare firms in a group is that different firms can have different fiscal-year ends. This can lead to some firms having their prices divided by earnings from July 2007 to June 2008, with other firms having their prices divided by earnings from January 2008 to December 2008. While the differences can be minor in mature sectors, where earnings do not make quantum jumps over six months, they can be large in high-growth sectors.

With both earnings and book value measures, there is another component to be concerned about and that is the accounting standards used to estimate earnings and book values. Differences in accounting standards can result in very different earnings and book value numbers for similar firms. This makes comparisons of multiples across firms in different markets, with different accounting standards, very difficult. Even with the same accounting standards governing companies, there can be differences in firms that arise because of discretionary accounting choices. There is also the additional problem posed by the fact that some firms use different accounting rules (on depreciation and expensing) for reporting purposes and tax purposes and others do not.[13] In summary, companies that

---

[13] Firms that adopt different rules for reporting and tax purposes generally report higher earnings to their stockholders than they do to the tax authorities. When they are compared on a price earnings basis to firms that do not maintain different reporting and tax books, they will look cheaper (lower PE).

use aggressive assumptions in measuring earnings will look cheaper on earnings multiples than firms that adopt conservative accounting practices.

Descriptive Tests

When using a multiple, it is always useful to have a sense of what a high value, a low value or a typical value for that multiple is in the market. In other words, knowing the distributional characteristics of a multiple is a key part of using that multiple to identify under or over valued firms. In addition, we need to understand the effects of outliers on averages and unearth any biases in these estimates, introduced in the process of estimating multiples. In the final part of this section, we will look at how the distributions of multiples shift over time.

Distributional Characteristics

Many analysts who use multiples have a sector focus and have a good sense of how different firms in their sector rank on specific multiples. What is often lacking, however, is a sense of how the multiple is distributed across the entire market. Why should a software analyst care about price earnings ratios of utility stocks? Because both software and utility stocks are competing for the same investment dollar, they have to, in a sense, play by the same rules. Furthermore, an awareness of how multiples vary across sectors can be very useful in detecting when the sector we are analyzing is over or under valued.

What are the distributional characteristics that matter? The standard statistics – the average and standard deviation – are where we should start, but they represent the beginning of the exploration. In markets like the United States, characterized by diverse companies in very different businesses there will be significant variation across companies on any multiple at any point in time. Table 4.1 summarizes the average and standard deviation for three widely used multiples -price earnings ratios, price to book value ratios and enterprise value to EBITDA multiple – in January 2010 in the United States. In addition, the maximum and minimum values on

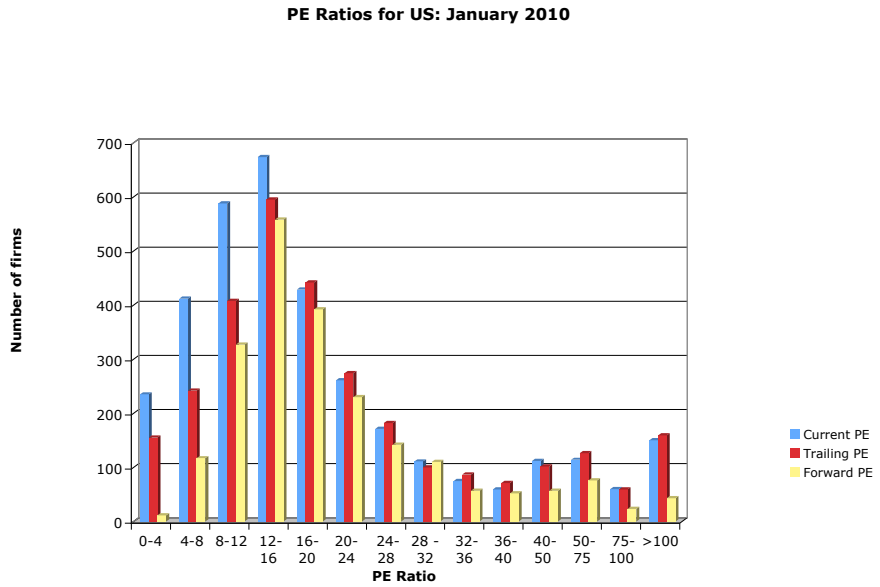
Table 4.1:Summary Statistics on Multiples

	Current PE	Price to Book Equity	EV/EBITDA	EV/Sales
Mean	29.57	3.81	36.27	13.35
Standard Error	1.34	0.30	17.04	5.70

Median	14.92	1.51	5.86	1.13
Skewness	12.12	41.64	64.64	68.99
Maximum	1570.00	1294.00	5116.05	28846.00
Number of firms	3457	5399.00	5429.00	5130.00
Sample Size	7036	7036.00	7036.00	7036.00

Note that the lowest value that any company can register on any of these multiples is zero whereas the highest values are unbounded. As a result, the distributions for these multiples are skewed towards the positive values as evidenced by the distribution of PE ratios of US companies in January 2010 in figure 4.2:

Figure 4.2: PE Ratio Distribution: US stocks in January 2010



The consequences of skewed distributions for investors and analysts are significant:

a. Average versus Median values: As a result of the positively skewed distributions, the average values for multiples will be higher than median values[14]. For instance, the median PE ratio in January 2010 was about 15, well below the average PE reported in table 4.1 and this is true for all multiples. The median value is much more representative of the typical firm in the group and any comparisons should be made to medians. The standard sales pitch of a stock being cheap because it trades at a

[14] With the median, half of all firms in the group fall below this value and half lie above.

multiple less than the average for the sector should be retired in favor of one which compares the stock's pricing to the median for the sector.

b. Probabilistic statements: As a result of the focus on normal distributions in most statistics classes, we begin attributing its properties to all distributions. For instance, it is true that the probability of values in a normal distribution falling more than two standard deviations away from the mean is very small. In the case of the PE ratio, this rule would suggest that few companies should have PE ratios that fall below 27.89 (which is the average of 29.57 minus two standard errors) or above 32.25 (the average plus two standard errors). The reality is that there are thousands of firms that fall outside this range because the distributions are not normal.

Outliers and Averages

As noted earlier, multiples are unconstrained on the upper end and firms can trade at multiples of 500 or 2000 or even 10000. This can occur not only because of high stock prices but also because earnings at firms can sometime drop to a few cents or even a fraction of a cent. These outliers will result in averages that are not representative of the sample. In many cases, data reporting services (such as Value Line and Standard and Poors) that compute and report average values for multiples either throw out these outliers when computing the averages or constrain the multiples to be less than or equal to a fixed number. For instance, any firm that has a price earnings ratio greater than 500 will be assumed to have a price earnings ratio of 500. The consequence is that the averages reported by two services for the same sector or market will almost never match up because they deal with outliers differently. In November 2008, for instance, the average PE reported for the S&P 500 varied widely across services from a low value of 11.5 on Yahoo! Finance to 14.2 on Morningstar. It is incumbent on those investors using these numbers to be clear about how they are computed and consistent in their comparisons.

Biases in Estimating Multiples

With every multiple, there are firms for which the multiple cannot be computed. Consider again the price-earnings ratio. When the earnings per share are negative, the price earnings ratio for a firm is not meaningful and is usually not reported. When

looking at the average price earnings ratio across a group of firms, the firms with negative earnings will all drop out of the sample because the price earnings ratio cannot be computed. Why should this matter when the sample is large? The fact that the firms that are taken out of the sample are the firms losing money creates a bias in the selection process. In fact, the average PE ratio for the group will be biased upwards because of the elimination of these firms.

There are three solutions to this problem. The first is to be aware of the bias and build it into the analysis. In practical terms, this will mean adjusting the average PE down to reflect the elimination of the money-losing firms. The second is to aggregate the market value of equity and net income (or loss) for all of the firms in the group, including the money-losing ones, and compute the price earnings ratio using the aggregated values. The third choice is to use a multiple that can be computed for all of the firms in the group. The inverse of the price-earning ratio, which is called the earnings yield, can be computed for all firms, including those losing money and is not exposed to the same biases as the price earnings ratio is.

Time Variation in Multiples

As any investor who has tracked the market for any length of time knows, multiples change over time for the entire market and for individual sectors. To provide a measure of how much multiples can change over time, we have computed the average and median PE ratios each year from 2000 to 2008 for the United States in table 4.2:

Table 4.2: PE Ratios across time: US Stocks

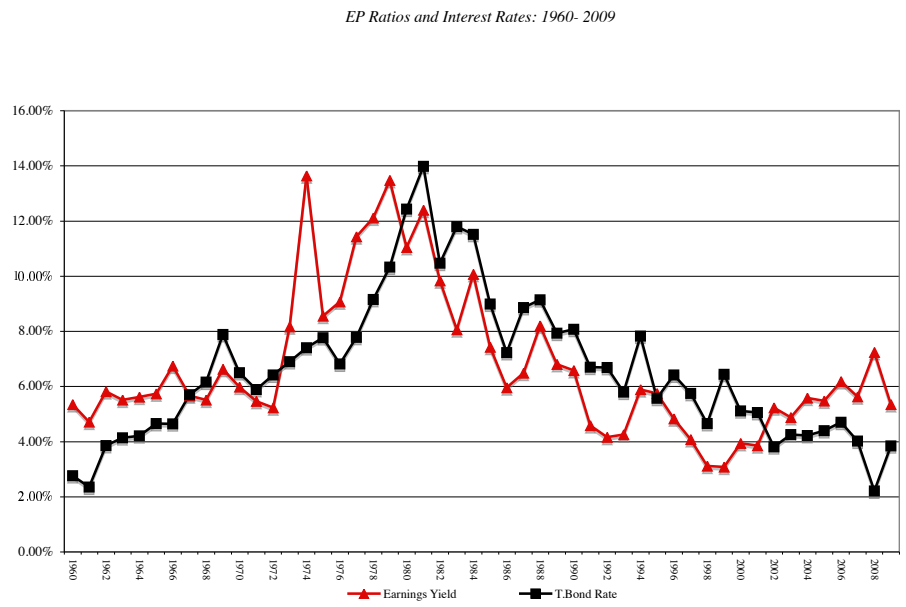
	Average	*Median*	*% of firms with PE ratios*
Jan-00	52.16	24.55	65.33%
Jan-01	44.99	14.74	63.00%
Jan-02	43.44	15.5	57.06%
Jan-03	33.36	16.68	49.99%
Jan-04	41.4	20.76	58.18%
Jan-05	48.12	23.21	56.43%
Jan-06	44.33	22.40	56.89%
Jan-07	40.77	21.21	57.50%

Jan-08	45.02	18.16	56.42%
Jan-09	18.91	9.80	58.36%
Jan-10	29.57	14.92	49.13%

In the last column, we note the percentage of firms in the overall sample for which we were able to compute PE ratios. Note that more than half of all US firms had negative earnings in January 2010, making the PE ratio not meaningful.

Why do multiples change over time? Some of the change can be attributed to fundamentals. As interest rates and economic growth shift over time, the pricing of stocks will change to reflect these shifts; lower interest rates, for instance, played a key role in the rise of earnings multiples through the 1990s. Some of the change, though, comes from changes in market perception of risk. As investors become more risk averse, which tends to happen during recessions, multiples paid for stocks will decrease. This is captured in figure 4.4, which shows earnings yield (Earnings/Price) ratio for the S&P 500 and the T.bond rate from 1960 to 2009.

Figure 4.4: Earnings to Price Ratios over time: 1960-2009



Note that the earnings yield rose with the treasury bond rate in the 1970s and declined as rates went down in the 1980s and 1990s. From a practical standpoint, what are the consequences? The first is that comparisons of multiples across time are fraught with danger. For instance, the common practice of branding a market to be under or over valued based upon comparing the PE ratio today to historical PE ratios will lead to misleading judgments when interest rates are higher or lower than historical norms. The second is that relative valuations have short shelf lives. A stock may look cheap relative to comparable companies today but that assessment can shift dramatically over the next few months. Intrinsic valuations are inherently more stable than relative valuations.

Analytical Tests

In discussing why analysts were so fond of using multiples, we argued that relative valuations require fewer assumptions than discounted cash flow valuations. While this is technically true, it is only so on the surface. In reality, we make just as many assumptions when we do a relative valuation as we do in a discounted cash flow valuation. The difference is that the assumptions in a relative valuation are implicit and unstated, whereas those in discounted cash flow valuation are explicit and stated. The two primary questions that we need to answer before using a multiple are: What are the fundamentals that determine at what multiple a firm should trade? How do changes in the fundamentals affect the multiple?

Determinants

In the introduction to discounted cash flow valuation, we observed that the value of a firm is a function of three variables – it capacity to generate cash flows, its expected growth in these cash flows and the uncertainty associated with these cash flows. Every multiple, whether it is of earnings, revenues or book value, is a function of the same three variables – risk, growth and cash flow generating potential. Intuitively, then, firms with higher growth rates, less risk and greater cash flow generating potential should trade at higher multiples than firms with lower growth, higher risk and less cash flow potential.

The specific measures of growth, risk and cash flow generating potential that are used will vary from multiple to multiple. To look under the hood, so to speak, of equity

and firm value multiples, we can go back to fairly simple discounted cash flow models for equity and firm value and use them to derive the multiples.

In the simplest discounted cash flow model for equity, which is a stable growth dividend discount model, the value of equity is:

$$\text{Value of Equity} = P_0 = \frac{DPS_1}{k_e - g_n}$$

where DPS_1 is the expected dividend in the next year, k_e is the cost of equity and g_n is the expected stable growth rate. Dividing both sides by the earnings, we obtain the discounted cash flow equation specifying the PE ratio for a stable growth firm.

$$\frac{P_0}{EPS_0} = PE = \frac{\text{Payout Ratio} * (1 + g_n)}{k_e - g_n}$$

The key determinants of the PE ratio are the expected growth rate in earnings per share, the cost of equity and the payout ratio. Other things remaining equal, we would expect higher growth, lower risk and higher payout ratio firms to trade at higher multiples of earnings than firms without these characteristics.

Dividing both sides by the book value of equity, we can estimate the price/book value ratio for a stable growth firm.

$$\frac{P_0}{BV_0} = PBV = \frac{\text{ROE} * \text{Payout Ratio} * (1 + g_n)}{k_e - g_n}$$

where ROE is the return on equity and is the only variable in addition to the three that determine PE ratios (growth rate, cost of equity and payout) that affects price to book equity.

Dividing by the Sales per share, the price/sales ratio for a stable growth firm can be estimated as a function of its profit margin, payout ratio, risk and expected growth.

$$\frac{P_0}{Sales_0} = PS = \frac{\text{Profit Margin} * \text{Payout Ratio} * (1 + g_n)}{k_e - g_n}$$

The net margin is the new variable that is added to the process. While all of these computations are based upon a stable growth dividend discount model, we will show that the conclusions hold even when we look at companies with high growth potential and with other equity valuation models.

We can do a similar analysis to derive the firm value multiples. The value of a firm in stable growth can be written as:

$$\text{Value of Firm} = V_0 = \frac{FCFF_1}{k_c - g_n}$$

Dividing both sides by the expected free cash flow to the firm yields the Value/FCFF multiple for a stable growth firm.

$$\frac{V_0}{FCFF_1} = \frac{1}{k_c - g_n}$$

The multiple of FCFF that a firm commands will depend upon two variables – its cost of capital and its expected stable growth rate. Since the free cash flow the firm is the after-tax operating income netted against the net capital expenditures and working capital needs of the firm, the multiples of revenues, EBIT and after-tax EBIT can also be estimated similarly.

$$\frac{V_0}{EBIT_1(1-t)} = \frac{(1 - \text{ Reinvestment Rate})}{k_c - g_n}$$

$$\frac{V_0}{EBIT_1} = \frac{(1 - \text{ Reinvestment Rate})(1 - t)}{k_c - g_n}$$

$$\frac{V_0}{Sales} = \frac{\text{After - tax Operating Margin } (1 - \text{ Reinvestment Rate})}{k_c - g_n}$$

Table 4.3 summarizes the multiples and the key variables that determine each multiple, with the sign of the relationship in brackets next to each variable: ⇧ indicates that an increase in this variable will increase the multiple, whereas 🡻 indicates that an increase in this variable will decrease the multiple, holding all else constant.

Table 4.3: Fundamentals Determining Multiples

Multiple	*Fundamental Determinants*
Price Earnings Ratio	Expected Growth(⇧), Payout(⇧), Risk(🡻)
Price to Book Equity Ratio	Expected Growth(⇧), Payout(⇧), Risk(🡻), ROE(⇧)
Price to Sales Ratio	Expected Growth(⇧), Payout(⇧), Risk(🡻), Net

	Margin(⇧)
EV to FCFF	Cost of capital(🡻), Growth Rate(⇧)
EV to EBITDA	Expected Growth(⇧), Reinvestment Rate(🡻), Risk(🡻), ROC(⇧), Tax rate(🡻)
EV to Capital Ratio	Expected Growth(⇧), Reinvestment Rate(🡻), Risk(🡻), ROC(⇧)
EV to Sales	Expected Growth(⇧), Reinvestment Rate(🡻), Risk(🡻), Operating Margin(⇧)

The point of this analysis is not to suggest that we go back to using discounted cash flow valuation, but to understand the variables that may cause these multiples to vary across firms in the same sector. If we ignore these variables, we might conclude that a stock with a PE of 8 is cheaper than one with a PE of 12 when the true reason may be that the latter has higher expected growth or we might decide that a stock with a P/BV ratio of 0.7 is cheaper than one with a P/BV ratio of 1.5 when the true reason may be that the latter has a much higher return on equity.

Companion Variable

While the variables that determine a multiple can be extracted from a discounted cash flow model and the relationship between each variable and the multiple can be developed by holding all else constant and asking what-if questions, there is a single variable that dominates when it comes to explaining each multiple (and it is not the same variable for every multiple). This variable, which is called the companion variable, is critical to using multiples wisely in making valuation judgments and can be identified by looking for the variable that best explain differences across firms using a particular multiple.

So, what are the companion variables for the most widely used multiples? To arrive at this judgment, we looked at which of the variables listed in table 4.2 was most useful in explaining differences across firms with each multiple and came up with the listing in table 4.4:

Table 4.4: Valuation Mismatches

Multiple	*Companion variable*	*Valuation mismatch*
PE Ratio	Expected Growth	Low PE stock with high expected growth rate in earnings per share
PBV Ratio	ROE	Low PBV stock with high ROE
PS Ratio	Net Margin	Low PS stock with high net profit margin.
EV/EBITDA	Reinvestment Rate	Low EV/EBITDA stock with low reinvestment needs
EV/Capital	Return on Capital	Low EV/Capital stock with high return on capital.
EV/Sales	After-tax operating margin	Low EV/Sales ratio with a high after-tax operating margin.

Relationship

Knowing the fundamentals that determine a multiple is a useful first step, but understanding how the multiple changes as the fundamentals change is just as critical to using the multiple. To illustrate, knowing that higher growth firms have higher PE ratios is not a sufficient insight if we are called upon to analyze whether a firm with a growth rate that is twice as high as the average growth rate for the sector should have a PE ratio that is 1.5 times or 1.8 times or 2 times the average price earnings ratio for the sector. To make this judgment, we need to know how the PE ratio changes as the growth rate changes.

A surprisingly large number of valuation analyses are based upon the assumption that there is a linear relationship between multiples and fundamentals. For instance, the PEG ratio, which is the ratio of the PE to the expected growth rate of a firm and widely used to analyze high growth firms, implicitly assumes that PE ratios and expected growth rates are linearly related.

One of the advantages of deriving the multiples from a discounted cash flow model, as was done in the last section, is that we can analyze the relationship between each fundamental variable and the multiple by keeping everything else constant and changing the value of that variable. When we do this, we will find that there are very few linear relationships in valuation.

Application Tests

When multiples are used, they tend to be used in conjunction with comparable firms to determine the value of a firm or its equity. But what is a comparable firm? While the conventional practice is to look at firms within the same industry or business, this is not necessarily always the correct or the best way of identifying these firms. In addition, no matter how carefully we choose comparable firms, differences will remain between the firm we are valuing and the comparable firms. Figuring out how to control for these differences is a significant part of relative valuation.

What is a comparable firm?

A comparable firm is one with cash flows, growth potential, and risk similar to the firm being valued. It would be ideal if we could value a firm by looking at how an exactly identical firm - in terms of risk, growth and cash flows - is priced. Nowhere in this definition is there a component that relates to the industry or sector to which a firm belongs. Thus, a telecommunications firm can be compared to a software firm, if the two are identical in terms of cash flows, growth and risk. In most analyses, however, analysts define comparable firms to be other firms in the firm's business or businesses. If there are enough firms in the industry to allow for it, this list is pruned further using other criteria; for instance, only firms of similar size may be considered. The implicit assumption being made here is that firms in the same sector have similar risk, growth, and cash flow profiles and therefore can be compared with much more legitimacy. The tradeoff is therefore a simple one. Defining an industry more broadly increases the number of comparable firms, but it also results in a more diverse group of companies.

Controlling for Differences across Firms

No matter how carefully we construct our list of comparable firms, we will end up with firms that are different from the firm we are valuing. The differences may be small on some variables and large on others and we will have to control for these differences in a relative valuation. There are three ways of controlling for these differences:

1. Subjective Adjustments

Relative valuation begins with two choices - the multiple used in the analysis and the group of firms that comprises the comparable firms. In many relative valuation, the

multiple is calculated for each of the comparable firms and the average is computed. To evaluate an individual firm, the analyst then compare the multiple it trades at to the average computed; if it is significantly different, the analyst can make a subjective judgment about whether the firm's individual characteristics (growth, risk or cash flows) may explain the difference. Thus, a firm may have a PE ratio of 22 in a sector where the average PE is only 15, but the analyst may conclude that this difference can be justified because the firm has higher growth potential than the average firm in the industry. If, in the judgment of the analyst, the difference on the multiple cannot be explained by the fundamentals, the firm will be viewed as over valued (if its multiple is higher than the average) or undervalued (if its multiple is lower than the average). The weakness in this approach is not that analysts are called upon to make subjective judgments, but that the judgments are often based upon little more than guesswork. All too often, these judgments confirm their biases about companies.

2. Modified Multiples

In this approach, we modify the multiple to take into account the most important variable determining it – the companion variable. To provide an illustration, analysts who compare PE ratios across companies with very different growth rates often divide the PE ratio by the expected growth rate in EPS to determine a growth-adjusted PE ratio or the PEG ratio. This ratio is then compared across companies with different growth rates to find under and over valued companies.

There are two implicit assumptions that we make when using these modified multiples. The first is that these firms are comparable on all the other measures of value, other than the one being controlled for. In other words, when comparing PEG ratios across companies, we are assuming that they are all of equivalent risk. The other assumption generally made is that that the relationship between the multiples and fundamentals is linear. Again, using PEG ratios to illustrate the point, we are assuming that as growth doubles, the PE ratio will double; if this assumption does not hold up and PE ratios do not increase proportional to growth, companies with high growth rates will look cheap on a PEG ratio basis.

3. Statistical Techniques

Subjective adjustments and modified multiples are difficult to use when the relationship between multiples and the fundamental variables that determine them becomes complex. There are statistical techniques that offer promise, when this happens. In a multiple regression, for instance, we attempt to explain a dependent variable by using independent variables that we believe influence the dependent variable. This mirrors what we are attempting to do in relative valuation, where we try to explain differences across firms on a multiple (PE ratio, EV/EBITDA) using fundamental variables (such as risk, growth and cash flows). Regressions offer three advantages over the subjective approach:

a. The output from the regression gives us a measure of how strong the relationship is between the multiple and the variable being used. Thus, if we are contending that higher growth companies have higher PE ratios, the regression should yield clues to both how growth and PE ratios are related (through the coefficient on growth as an independent variable) and how strong the relationship is (through the t statistics and R squared).
b. If the relationship between a multiple and the fundamental we are using to explain it is non-linear, the regression can be modified to allow for the relationship.
c. Unlike the modified multiple approach, where we were able to control for differences on only one variable, a regression can be extended to allow for more than one variable and even for cross effects across these variables.

In general, regressions seem particularly suited to our task in relative valuation, which is to make sense of voluminous and sometimes contradictory data.

Exhibit 4.1 illustrates all three approaches – the subjective comparison, modified multiples and statistical techniques applied to companies in the beverage sector. Based just on PE ratios, Hansen Natural, Andres Wine and Todhunter look cheap, trading at less than ten times earnings. With PEG ratios, Hansen Natural continues to look cheap but Andres Wine and Todhunter do not. Controlling for differences in risk and growth in a regression, Hansen Natural looks under valued by about 20%, relative to the sector.

Exhibit 4.1: A Relative Valuation of Beverage Companies

Subjective Comparisons
Stocks with the lowest PE ratios (Andres Wine, Hansen Natural and Todhunter look cheap but Andres Wine and Todhunter have low expected growth and Hansen looks riskier than the other firms. Stocks with high PE ratios (Coca Cola and Molson) look expensive but also have high growth.

Company Name	*Trailing PE*	*Expected Growth*	*Standard Deviation*	*PEG*
Andres Wines Ltd. 'A'	8.96	3.50%	24.70%	2.56
Anheuser-Busch	24.31	11.00%	22.92%	2.21
Boston Beer 'A'	10.59	17.13%	39.58%	0.62
Brown-Forman 'B'	10.07	11.50%	29.43%	0.88
Chalone Wine Group Ltd.	21.76	14.00%	24.08%	1.55
Coca-Cola	44.33	19.00%	35.51%	2.33
Coca-Cola Bottling	29.18	9.50%	20.58%	3.07
Coca-Cola Enterprises	37.14	27.00%	51.34%	1.38
Coors (Adolph) 'B'	23.02	10.00%	29.52%	2.30
Corby Distilleries Ltd.	16.24	7.50%	23.66%	2.16
Hansen Natural Corp	9.70	17.00%	62.45%	0.57
Molson Inc. Ltd. 'A'	43.65	15.50%	21.88%	2.82
Mondavi (Robert) 'A'	16.47	14.00%	45.84%	1.18
PepsiCo, Inc.	33.00	10.50%	31.35%	3.14
Todhunter Int'l	8.94	3.00%	25.74%	2.98
Whitman Corp.	25.19	11.50%	44.26%	2.19
Average	*22.66*	*12.60%*	*33.30%*	*2.00*

Modifiied Multiple
Dividing the PE ratio by the expected growth yields the PEG ratio. Firms that have low PEG ratios (Hansen, Boston Beer and Brown Forman) are cheap, given their expected growth potential. Risk, however, remains the wild card.

Statistical Adjusttments
. A regression of PE ratios on the expected growth rate and risk (standard deviation) is presented below:
PE = 20.87 - 63.98 Standard deviation + 183.24 Expected Growth R2 = 51%
(3.01) (2.63) (3.66)
The numbers in brackets are t-statistics. The regression itself can be used to get predicted PE ratios for the companies in the list. The predicted PE ratio for Hansen Natural, based upon its standard deviation of 62.45% expected growth of 17% is:
Predicted PEfor Hansen = 20.87 - 63.98 (0.6245) + 183.24 (0.17) = 12.06
Since the actual PE ratio for Hansen was .70, this would suggest that the stock is under valued by about 20%, given how the rest of the sector is priced.

Reconciling Relative and Intrinsic Valuations

The two approaches to valuation – discounted cash flow valuation and relative valuation – will generally yield different estimates of value for the same firm at the same point in time. It is even possible for one approach to generate the result that the stock is under valued while the other concludes that it is over valued. Furthermore, even within relative valuation, we can arrive at different estimates of value depending upon which multiple we use and what firms we based the relative valuation on.

The differences in value between discounted cash flow valuation and relative valuation come from different views of market efficiency, or put more precisely, market inefficiency. In discounted cash flow valuation, we assume that markets make mistakes, that they correct these mistakes over time, and that these mistakes can often occur across entire sectors or even the entire market. In relative valuation, we assume that while markets make mistakes on individual stocks, they are correct on average. In other words, when we value a new software company relative to other small software companies, we

are assuming that the market has priced these companies correctly, on average, even though it might have made mistakes in the pricing of each of them individually. Thus, a stock may be over valued on a discounted cash flow basis but under valued on a relative basis, if the firms used for comparison in the relative valuation are all overpriced by the market. The reverse would occur, if an entire sector or market were underpriced.

Conclusion

In relative valuation, we estimate the value of an asset by looking at how similar assets are priced. To make this comparison, we begin by converting prices into multiples – standardizing prices – and then comparing these multiples across firms that we define as comparable. Prices can be standardized based upon earnings, book value, revenue or sector-specific variables.

While the allure of multiples remains their simplicity, there are four steps in using them soundly. First, we have to define the multiple consistently and measure it uniformly across the firms being compared. Second, we need to have a sense of how the multiple varies across firms in the market. In other words, we need to know what a high value, a low value and a typical value are for the multiple in question. Third, we need to identify the fundamental variables that determine each multiple and how changes in these fundamentals affect the value of the multiple. Finally, we need to find truly comparable firms and adjust for differences between the firms on fundamental characteristics.

Section 2: Valuing across the life cycle

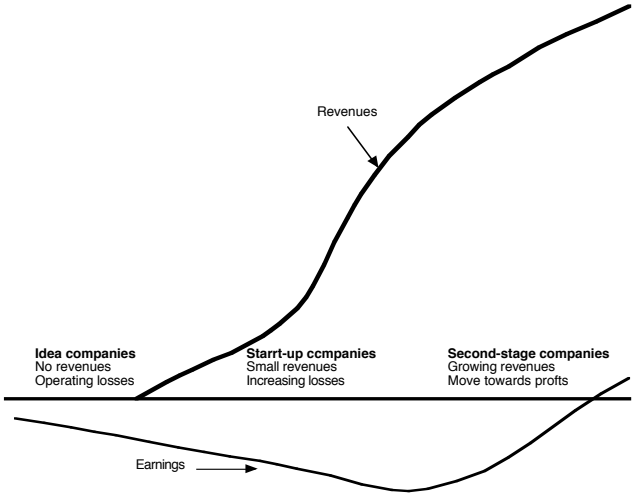
Chapter 5: Valuing young growth companies and businesses

Valuing young companies is difficult to do, partly because of the absence of operating history and partly because most young firms do not make it through these early stages to success. Until recent years, these companies were funded primarily by venture capitalists, but in the last two decades, some of these young companies have become publicly listed, exposing investors to both opportunity and threat. In this chapter, we begin by looking at characteristics shared by young companies and the valuation challenges that they create. We follow up by looking at how intrinsic and relative valuation techniques can be adapted to meet these challenges and draw from these techniques some pointers that can help investors in finding good deals.

What is a young growth company?

If every business starts with an idea, young companies can range the spectrum from idea companies, that have no revenues or products, to start-up companies that are testing out product appeal to second stage companies that are moving on the path to profitability. Figure 5.1 illustrates the diversity of young, growth companies:

Figure 5.1: The Early Stages of the Life Cycle



Most young growth companies tend to be privately owned and funded, either entirely by their founder/owner or by venture capitalists. In the last two decades, though, companies in some sectors such as technology and biotechnology have been able to leap frog the process and go public.

Characteristics of young growth companies

Young companies are diverse, but they generally share some common attributes:

1. No history: At the risk of stating the obvious, young companies have very limited histories. Many of them have only one or two years of data available on operations and financing and some have financials for only a portion of a year, for instance.
2. Small or no revenues, operating losses: The limited history that is available for young companies is rendered even less useful by the fact that there is little operating detail in them. Revenues are small or non-existent for idea companies and the expenses often are associated with getting the business established, rather than generating revenues. In combination, they result in significant operating losses.
3. Dependent on private equity: For the most part, young businesses are dependent upon equity from private sources, rather than public markets. At the earlier stages, the equity is provided almost entirely by the founder (and friends and family). As the promise of future success increases, and with it the need for more capital, venture capitalists become a source of equity capital, in return for a share of the ownership in the firm. In recent years, though, some companies have bypassed the venture capital route and gone directly to public markets.
4. Many don't survive: Most young companies don't survive the test of commercial success and fail. One study of the Bureau of Labor Statistics Quarterly Census of Employment and Wages (QCEW) computes survival statistics across firms.[15] Using a seven-year database from 1998 to 2005, the authors concluded that only

[15] Knaup, Amy E., May 2005,, "Survival and longevity in the Business Employment Dynamics data," *Monthly Labor Review*, pp. 50–56; Knaup, Amy E. and MC. Piazza, September 2007, Business Employment Dynamics Data: Survival and Longevity, Monthly Labor Review, pp 3-10.

44% of all businesses that were founded in 1998 survived at least 4 years and only 31% made it through all seven years. In addition, they find that survival rates vary across sectors, with only 25% of firms in the information sector (which includes technology) surviving 7 years, whereas almost 44% of health service businesses make it through that period.

5. Investments are illiquid: Since equity investments in young firms tend to be privately held and in non-standardized units, they are also much more illiquid than investments in their publicly traded counterparts.

Valuation Issues

The fact that young companies have limited histories and are particularly susceptible to failure all contribute to making them more difficult to value. In this section, we will begin by considering the estimation issues that we run into in discounted cash flow valuations and we will follow up by evaluating why these same issues crop up when we do relative valuation.

Intrinsic (DCF) Valuation

In chapter 3, we laid out the four pieces that make up the intrinsic valuation puzzle – the cash flows form existing assets, the expected growth from both new investments and improved efficiency on existing assets, the discount rates that emerge from our assessments of risk in both the business and its equity, and the assessment of when the firm will become a stable growth firm (allowing us to estimate terminal value). On each of these measures, young firms pose estimation challenges that can be traced back to their common characteristics.

Cash flows from existing assets

When valuing any company, we start by estimating cash flows from existing investments, not only to provide a base figure, but also to learn something about the company's business model. There are three problems with estimating cash flows from existing assets for young firms.

- The fact that existing assets represent a sliver of the potential value of the firm that the revenues and income they generate are of little import in assessing overall firm value.
- The absence of historical data makes it difficult to assess how well the revenues from existing assets will hold up if macro economic conditions become less favorable. In other words, if all you have is one year of financial data, it is more difficult to make a judgment on whether the revenues represent a flash in the pan or are sustainable.
- The expenses that young companies incur to generate future growth are often mixed in with the expenses associated with generating current revenues. For instance, it is not unusual to see the Selling, General and Administrative (S,G&A) expenses at some young companies be three or four times larger than revenues, largely because they include the expenses associated with lining up future customers.

Growth Assets

The bulk of a young company's value comes from growth assets, but estimating their value comes with its own impediments.

- The absence of revenues in some cases, and the lack of history on revenues in others, means that we cannot use past revenue growth as an input into the estimation of future revenues. As a result, we are often dependent upon the firm's own estimates of future revenues, with all the biases associated with these numbers.
- Even if we were able to estimate revenues in future years, we have to also estimate how earnings will evolve in future years, as revenues change. Here again, the fact that young companies tend to report losses and have no history on operating income makes it more difficult to assess what future profit margins will be.
- In chapter 3, we noted that it is not revenue or even earnings growth per se that determines value, but the quality of that growth. To assess the quality of growth, we looked at how much the firm reinvested to generate its expected growth,

noting that value creating growth arises only when a firm generates a return on capital greater than its cost of capital on its growth investments. This intuitive concept is put to the test with young companies, because there is little to base the expected return on capital on new investments. Past data provides little guidance, because the company has made so few investments in the past and these investments have been in existence for short periods. The current return on capital, which is often used as a starting point for estimating future returns, is generally a negative number for young companies.

In summary, we have a tough time estimating future growth in revenues and operating margins for young companies, and the estimation problems are accentuated by the difficulties we face in coming up with reinvestment assumptions that are consistent with our growth estimates.

Discount Rates

The standard approaches for assessing the risk in a company and coming up with discount rates are dependent upon the availability of market prices for the securities issued by the firm. Thus, we estimate the beta for equity by regressing returns on a stock against returns on a market index, and the cost of debt by looking at the current market prices of publicly traded bonds. With young companies, these assumptions are open to challenge. First, most young companies are not publicly traded and have no publicly traded bonds outstanding. Consequently, there is no way in which we can run a regression of past returns, to get an equity beta, or use a market interest rate on debt. To add to the problem, a large proportion of the equity in a young company is often held by investors who are either completely invested in the company (founders) or only partially diversified (venture capitalists). As a result, these investors are unlikely to accept the notion that the only risk that matters is the risk that cannot be diversified away and instead will demand compensation for at least some of the firm specific risk.

Terminal Value

If the terminal value accounts for a large proportion of the overall value of a typical firm, it is an even bigger component of the value of a young company. In fact,

it is not unusual for the terminal value to account for 90%, 100% or even more than 100% of the current value of a young company. Consequently, assumptions about when a firm will reach stable growth, a pre-requisite for estimating terminal value, and its characteristics in stable growth can have a substantial impact on the value that we attach to a young company. Our task, though, is complicated by our inability to answer three questions:

a. Will the firm make it to stable growth? In an earlier section, we noted the high failure rate among young firms. In effect, these firms will never make it to stable growth and the terminal value will not provide the large windfall to value that is does for a going concern. Estimating the probability of survival for a firm, early in the life cycle, is therefore a critical component of value, but not necessarily an easy input to estimate.
b. When will the firm become a stable growth firm? Even if we assume that a firm will make it to stable growth in the future, estimating when that will occur is a difficult exercise. After all, some firms reach steady state in a couple of years, whereas others have a much longer stretch of high growth, before settling into mature growth. The judgment of when a firm will become stable is complicated by the fact that the actions of competitors can play an important role in how growth evolves over time.
c. What will the firm look like in stable growth? In chapter 3, we noted that it is not just the growth rate in the stable growth rate that determines the magnitude of terminal value but the concurrent assumptions we make about risk and excess returns during the stable phase. In effect, assuming that a firm will continue to generate excess returns forever will lead to a higher terminal value than assuming that excess returns will converge on zero or be negative. While this is a judgment that we have to make for any firm, the absence of any historical data on excess returns at young firms does complicate estimation.

Value of Equity Claims

Once the cash flows have been estimated, a discount rate computed and the present value computed, we have estimated the value of the aggregate equity in the

firm. If all equity claims in the firm are equivalent, as is the case with a publicly traded firm with one class of shares, we divide the value of equity proportionately among the claims to get the value per claim. With young firms, there can be large differences across equity claims on cash flows and control rights, with some claimholders getting preferential rights over others. As a final point, the lack of liquidity in equity investments in private business have an effect on how much value we attach to them. In general, we should expect more illiquid investments to have less value than more liquid investments, but measuring and pricing the illiquidity in the equity of private businesses is far more difficult to do than in their publicly traded counterparts.

Relative Valuation

The difficulties that we have outlined in valuing young companies in a discounted cash flow model lead some analysts to consider using relative valuation approaches to value these companies. In effect, they try to value young companies using multiples and comparables. However, this task is also made more difficult by the following factors:

1. <u>What do you scale value to?</u> All valuation multiples have to be scaled to some common measure, and conventional scaling measures include earnings, book value and revenues. With young companies, each of these measures can pose problems. Since most of them report losses early in the life cycle, multiples such as price earnings ratios and EBITDA multiples cannot be computed. Since the firm has been in operation only a short period, the book value is likely to be a very small number and not reflect the true capital invested in the company. Even revenues can be problematic, since they can be non-existent for idea companies and miniscule for companies that have just transitioned into commercial production.
2. <u>What are your comparable companies?</u> When relative valuation is used to value a publicly traded company, the comparable firms are usually publicly traded counterparts in the same sector. With young companies, the comparison would logically be to other young companies in the same business but these companies

are usually not publicly traded and have no market prices (or multiples that can be computed). We could look at the multiples at which publicly traded firms in the same sector trade at, but these firms are likely to have very different risk, cash flow and growth characteristics than the young firm being valued.

3. What is the best proxy for risk? Many of the proxies used for risk, in relative valuation, are market based. Thus, beta or standard deviation of equity returns are often used as measures of equity risk, but these measures cannot be computed for young companies that have been listed for short periods.
4. How do you control for survival? In the context of discounted cash flow valuation, we looked at the problems created by the high failure rate of young companies. This is also an issue with using relative valuation. Intuitively, we would expect the relative value of a young company (the multiple of revenues or earnings that we assign it) to increase with its likelihood of survival. However, putting this intuitive principle into practice is not easy to do.
5. How do you adjust for differences in equity claims? With intrinsic valuation, we noted the effect that differences in cash flows and control claims can have on the value of equity claims. When doing relative valuation, we will have to confront the same issues.

In conclusion, the use of relative valuation may seem like an easy solution, when faced with the estimation challenges posed in intrinsic valuation, but all of the problems that we face in the latter remain problems when we do the former.

Valuation Solutions

While it is understandable that analysts, when confronted with the myriad uncertainties associated with valuing young companies, look for short cuts, there is no reason why young companies cannot be valued systematically. In this section, we will begin by laying out the foundations for estimating the intrinsic value of a young company, move on to consider how best to adapt relative valuation for the special characteristics of young companies and close with a discussion of how real options may be useful, at least for some small businesses.

Discounted Cash Flow Valuation

To applying discounted cash flow models to valuing young companies, we will move systematically through the process of estimation, considering at each stage, how best to deal with the characteristics of young companies.

1. Estimation of future cash flows

To estimate cash flows, we start by estimating the total market for a product or service and derive the rest of the numbers from that top line. In effect, we estimate the revenues first and then consider how much we need as capacity (and capital to create this capacity) to sustain these revenues. The steps involved in the process are the following:

- Potential market for the product/service: The first step in deriving the revenues for the firm is estimating the total potential market for its products and services. There are three challenges we face at this juncture.
    a. Defining the product/service offered by the firm: If the product or service offered by the firm is defined narrowly, the potential market will be circumscribed by that definition and will be smaller. If we use a broader definition, the market will expand to fit that definition. For example, defining Amazon.com as a book retailer, which is what it was in 1998, would have yielded a total market of less than $ 10 billion in that year, representing total book retailing sales in 1998. Categorizing Amazon.com as a general retailer would have yielded a much larger potential market. While that might have been difficult to defend in 1998, it did become more plausible as Amazon expanded its offerings in 1999 and 2000.
    b. Estimating the market size: Having defined the market, we face the challenge of estimating the size of that market. For a product or service that is entering an established market, the best sources of data tend to be trade publications and professional forecasting services.
    c. Evolution in total market over time: Since we have to forecast revenues into the future, it would be useful to get a sense of how the total market is expected to change or grow over time. This information is usually also

usually available from the same sources that provide the numbers for the current market size.

2. Market Share: Once we have a sense of the overall market size and how it will changeover time, we have to estimate the share of that market that will be captured by the firm being analyzed, both in the long term and in the time periods leading up to steady state. There are two other variables that have to be considered. One is the capacity of the management of the young company to deliver on its promises; many entrepreneurs have brilliant ideas but do not have the management and business skills to take it to commercial fruition. That is part of the reason that venture capitalists look for entrepreneurs who have had a track record of success in the past. The other is the resources that the young company can draw on to get its product/service to the desired market share. Optimistic forecasts for market share have to be coupled with large investments in both capacity and marketing; products usually don't produce and sell themselves.

3. Operating expenses/ margins: Revenues may be the top line but as investors, but a firm can have value only if it ultimately delivers earnings. Consequently, the next step is estimating the operating expenses associated with the estimated revenues and we would separate the estimation process into two parts. In the first part, we would focus on estimating the operating margin in steady state, primarily by looking at more established companies in the business. Once we have the target margin, we can then look at how we expect the margin to evolve over time; this "pathway to profitability" can be rockier for some firms than others, with fixed costs and competition playing significant roles in the estimation.

4. Investments for growth: When owners are asked for forecasts of revenues and earnings (step 2 and 3), it is natural that they go for optimistic values: revenues increase at exponential rates and margins quickly move towards target values. In any competitive business, though, neither revenue growth nor margin improvement is delivered for free. Consequently, it is critical that we estimate how much the firm is reinvesting to generate the forecasted growth. With a manufacturing firm, this will take the firm of investments in additional production capacity and with a technology firm, it will include not only investments in R&D and new patents, but

also in human capital (hiring software programmers and researchers). There are two reasons to pay attention to this step in the process. The first is that these investments will require cash outflows and thus affect the final bottom line, which is the cash flow that can be delivered to investors. The second, and this is especially so with young firms, this reinvestment will often result in negative cash flows, which will then have to be covered with new capital infusions. Thus, existing equity investors will see their share of the ownership either reduced (when new equity investors come in) or be called upon to make investments to keep the business going.

2. Estimating Discount Rates

In chapter 2, we laid out the inputs that go into discount rates. Summarizing, we estimated the cost of equity by looking at the beta (or betas) of the company in question, the cost of debt from a measure of default risk (an actual or synthetic rating) and applied the market value weights for debt and equity to come up with the cost of capital. There are both conceptual and estimation issues that make each of these ingredients difficult to deal with, when it comes to young companies.

- Beta and cost of equity: Large portions of equity in even publicly traded young growth companies is held by either undiversified founder owners or by partially diversified venture capitalists. Consequently, it does not make sense to assume that the only risk that should be priced in is the market risk; the cost of equity has to incorporate some (in the case of venture capitalists) or maybe even all (for completely undiversified owners) of the firm specific risk.
- Cost of debt: Young firms almost never have bonds outstanding and are instead dependent on bank loans for debt. C. Even though we may be able to use the process described in chapter 2 to estimate a synthetic rating, the resulting cost of debt may not appropriately capture the interest rates actually paid by these small and risky businesses, since banks may charge them a premium.
- Debt ratio: Since the equity and debt in young companies is not traded, there are no market values that can be used to weight the debt and equity to arrive at the cost of capital.

- Dynamic inputs: Even if we are able to estimate the current costs of debt and equity for a firm, each of the numbers that we use in the cost of capital computation can and will change over time, as the company evolves.

To estimate the cost of capital, we would suggest an approach that looks past the company and at the business the company operates in, and adjusting for key differences. In effect, we use sector averages for discount rates, adjusted for the higher risk and lack of diversification of younger companies. Thus, in the early years, costs of equity and capital will be much higher for young companies than for more mature counterparts in the same business. As we go further into the future, though, we should expect to see the cost of capital move towards sector averages, as the young company grows and matures.

3. Estimating Value today and adjusting for survival

The expected cash flows and discount rates, estimated in the last two steps, are key building blocks towards estimating the value of the business and equity today. However, there are three more components that we have to deal with at this stage in getting to the value of the firm. The first is determining what happens at the end of our forecast period, i.e., the assumptions that lead to the value we assign the business at the end of the period. The second is adjusting for the likelihood that the business will not survive, an issue that has added relevance with young firms, because so many fail early in the process. The third factor that we have to deal with, at least in businesses that are dependent upon a person or a few key people for their success, is how best to incorporate into the value the effects of their loss.

Terminal Value

The "terminal value" estimate represents a big chunk of the value of any business, but is an even bigger component of value for a young firm that has small or negative cash flows in the near years, accounting for 80%, 90% or more than 100% of value today.[16] The basic principles that govern terminal value remain unchanged:

---

[16] The "more than 100%" may strike readers as impossible, but in young companies with substantial negative cash flows in the early years, the terminal value can be more than 100% of value. In effect,

the growth rate used has to be less than the growth rate of the economy, the cost of capital has to converge on that of a mature firm and there has to be enough reinvestment to sustain the "stable" growth. When valuing firms, where success will translate into an initial public offering or sale to a publicly traded firm, the perpetual growth model makes the most sense. For smaller, less ambitious firms, where success will be defined as surviving the forecast period and delivering cash flows beyond, assuming a finite life for the cash flows and estimating a liquidation value will yield the most reasonable value.

Adjust for Survival

Many young firms succumb to the competitive pressures of the market place and don't make it. Rather than try to adjust the discount rate for this likelihood, a difficult exercise, we would suggest a two-step approach. In the first step, we would value the firm on the assumption that it survives and makes it to financial heath. This, in effect, is what we are assuming when we a terminal value and discount cash flows back to today at a risk-adjusted discount rate. In the second step, we would bring in the likelihood that the firm will not survive. The probability of failure can be assessed most simply, by using sector averages. Earlier in the chapter we noted a study that used data from the Bureau of Labor Statistics to estimate the probability of survival for firms in different sectors from 1998 to 2005. We could use the sector averages from this study as the probability of survival for individual firms in the sector. For a software firm that has been in existence for one year, for instance, the likelihood of failure over a five-year period would be assessed at 38% (the difference between the probability of surviving 2 years – 64.85% - and the probability of surviving 7 years – 24.78%). These sector averages can then be adjusted for specifics about the firm being valued: the quality of its management, its access to capital and its cash balances. Once the probability of failure has been assessed, the value of the firm can be written as an expected value of the two

---

to make money on your investment, you have to survive multiple infusions of new equity (dilution) along the way.

scenarios – the intrinsic value (from the discounted cash flows) under the going concern scenario and the distress value under the failure scenario.

Key Person Discounts

Young companies, especially in service businesses, are often dependent upon the owner or a few key people for their success. Consequently, the value we estimate for these businesses can change significantly if one or more of these key people will no longer be associated with the firm. To assess a key person discount in valuations, we would suggest that the firm be first valued, with the status quo (with key people involved in the business) and be valued again, with the loss of these individuals built into revenues, earnings and expected cash flows. To the extent that earnings and cash flows suffer when key people leave, the value of the business will be lower with the loss of these individuals, thus leading to a "key person discount". There is no simple formula that will help in determining how much cash flows will be lost as a result of the loss of key personnel, since it will vary not only across businesses but across the personnel involved. One way to assess it is to survey existing customers to see how they will respond if the key personnel leave and to then build in this impact into operating forecasts.

4. Valuing Equity Claims in the business

To estimate the value of equity per share in a publicly traded company, we divide the estimated value of equity by the number of shares outstanding. This works if all shareholders have the same claims on cash flows and equivalent voting rights. In young growth companies, there can be differences in both cash flow and control claims across shares.

- Cash flow claims: There are two types of preferential cash flow rights that can be embedded in equity claims. The first allows some equity investors, generally holding preferred shares, to claim a share of the operating cash flows, usually in the form of preferential dividends, before other claimholders get paid. The second gives preferred stockholders first claim on the cash flows of the firm on liquidation. Other things remaining equal, the value of common stock in a

company with preferred shares outstanding will be lower than in a company without preferred shares outstanding.

- Control claims: In some young publicly traded growth firms, the voting rights are either entirely or disproportionately held by the founders and other inside stockholders, leaving the shares with lower voting rights to outside investors. Other things remaining equal, voting shares should trade at a premium over non-voting shares and the magnitude of the premium will reflect the value of control. How much are voting rights worth? The earliest studies of voting share premiums were done with companies with different voting share classes in the United States, and concluded found that voting shares in that market trade, on average, at a relatively small premium of 5-10% over non-voting shares.[17] A more updated study looked at 28 companies In with voting and non-voting shares in 1994 and 1999 and concluded that that the median voting share premium increased from 2% in 1994 to 2.8% in 1999.[18]

There are ways in which we can value cash flow and control claims more precisely, but they require substantial information and some degree of sophistication. Even if you decide not to explicitly value these claims, being aware of their existence is a significant step in the right direction.

Exhibit 5.1 provides an illustration of a discounted cash flow valuation for Evergreen Solar, a small battery manufacturer, with significant growth potential and risk in early 2009. The value that we estimated, $0.63 per share, was significantly lower than the stock price of $2.70 per share at the time of the valuation.[19] Note that this value is not adjusted for the risk that the firm may not survive through year 10. If we did adjust for that likelihood, the value per share will be even lower.

---

[17] Lease, R.C., J.J. McConnell and W.H. Mikkelson 1983, *The market value of control in publicly-traded corporations,* Journal of Financial Economics, v11, 439-471.

[18] Reilly, R.F., 2005, *Quantifying the Valuation Discount for Lack of Voting Rights and Premium,* American Bankruptcy Institute Journal.

[19] To get to the present value, the cash flows are discounted back at the cumulated cost of capital, reflecting the changing cost of capital over time. For instance, the cumulated cost of capital in year 7 is computed as follows:

Cumulated cost of capital in year 7 = $(1.1037)^5$ (1.0984) (1.0937) = 1.9672

Exhibit 5.1: Value Drivers for a Young Growth Company - Evergreen Solar in Early 2009

The setting: Evergreen Solar, a manufacturer of solar power cells was founded in 2003 and is a Massachusetts-based company. Revenues at the firm increased from $24 million in 2004 to $70 million in 2007, yielding a compounded growth rate of 43% a year. In the trailing 12 months leading into this valuation, the company generated a pre-tax operating loss of $ 50 million on revenues of $ 90 million.

Revenue Growth
As revenues grow, the growth rate begins to drop off. Total revenues of $1.089 billion still small, relative to overall market.

Taxes
Net operating losses carried forward shield income from taxes until year 9.

Reinvestment
Assume that you need $1 in new investment for every $1.25 in revenue (industry average)

Access to capital
Firm will be able to raise fresh capital to cover its negative FCFF for the next 9 years.

Year	Revenues	Revenue Growth	Operating Margin	Operating Income	After-tax Operating income	Reinvestment	FCFF	Cost of capital	Present Value
Current	$90		-55.31%	-$50	-$50	$267	-$317	10.21%	
1	$126	40.00%	-28.39%	-$36	-$36	$29	-$64	10.21%	-$59
2	$176	40.00%	-12.23%	-$22	-$22	$40	**-$62**	10.21%	-$51
3	$247	40.00%	-2.54%	-$6	-$6	$56	-$63	10.21%	-$47
4	$345	40.00%	3.28%	$11	$11	$79	-$68	10.21%	-$46
5	$483	40.00%	6.77%	$33	$33	$111	-$78	10.21%	-$48
6	$628	30.00%	8.86%	$56	$56	$116	-$60	9.70%	-$34
7	$786	25.00%	10.12%	$79	$73	$126	-$52	9.01%	-$27
8	$943	20.00%	10.87%	$102	$61	$126	-$64	7.69%	-$31
9	$1,037	10.00%	11.32%	$117	$70	$75	-$5	7.33%	-$2
10	$1,089	5.00%	11.59%	$126	$76	$41	$34	7.20%	$14
Beyond	$1,113	2.25%	12.00%	$134	$80	$18	$1,255	7.20%	$521
Value of the operating assets =									$192
(Add) Cash									$285
(Minus) Debt									$374
Value of equity									$103
Value per share									$ 0.63

Survival
Firm will get larger, healthier and survive to generate value well into the future.

Target Operating Margins
As firms becomes larger, its margins converge on margins more typical of mature firms in the sector.

Terminal Value
Computed on the assumption that the earnings grow at 2.25% a year forever, and that the return on capital after year 10 is equal to the cost of capital of 7.23%.

$$\frac{\$134(1-.40)(1-.3112)}{(.0723-.0225)} = \$1,255$$

Cost of capital
High initially, with high beta (1.60) and cost of debt. As firm becomes larger and profitable, the cost of capital moves towards that of mature firms in the business.

Relative Valuation

The essence of relative valuation is that you value a firm, based upon how much the market is paying for similar firms. This premise is clearly more challenging with young firms that often have little to show in terms of operations and face substantial risks in operations and threats to their existence. The issues we face in applying public market multiples to young companies, especially early in the life cycle, are fairly obvious:

a. Life cycle affects fundamentals: If we accept the premise that only those young firms that make it through the early phase of the life cycle and succeed are likely to go public, we also have to accept the reality that public firms will have different fundamentals than private firms. Generally, public firms will be larger, have less potential for growth and have more established markets than private businesses, and these differences will manifest themselves in the multiples investors pay for public companies.

b. Survival: A related point is that there is a high probability of failure in young firms. However, this probability of failure should decrease as firms establish their product offerings and those firms that go public should have a greater chance of surviving than younger private firms. The former should therefore trade at higher market values, for any given variable such as revenues, earnings or book value, holding all else (growth and risk) constant.

c. Scaling variable: Assuming that we are able to obtain a reasonable multiple of revenues or earnings from our public company dataset, we face one final problem. Young firms often have very little revenues to show in the current year and many will be losing money; the book value is usually meaningless. Applying a multiple to any one of these measures will result in strange valuations.

d. Liquidity: Since equity in publicly traded companies is often more liquid than equity in young growth companies, the value obtained by using these multiples may be too high if applied to a young company. Just as we had to adjust for illiquidity in intrinsic valuation, we have to adjust for illiquidity with relative valuation.

There are simple practices that can not only prevent egregious valuation errors but also lead to better valuations:

a. Use forward revenues/ earnings: One of the problems we noted with using multiples on young companies is that the current operations of the company do not provide much in terms of tangible results: revenues are very small and earnings are negative. One solution is to forecast the operating results of the firm further down the life cycle and use these forward revenues and earnings as the basis for valuation. In effect, we will estimate the value of the business in five years, using revenues or earnings from that point in time.

b. Adjust the multiple for your firm's characteristics at time of valuation: If we are valuing the firm five years down the road, we have to estimate a multiple that is appropriate for the firm at that point in time, rather than today. Consider a simple illustration. Assume that you have a company that is expected to generate a compounded revenue growth of 50% a year for the

next five years, as it scales from being a very small firm to a more established enterprise. Assume that revenue growth after year 5 will drop to a more moderate compounded annual rate of 10%. The multiple that we apply to revenues or earnings in year 5 should reflect an expected growth rate of 10% (and not 50%).

c. Adjust for survival: When we estimated the intrinsic value for young firms, we allowed for the possibility of failure by adjusting the value for the probability that the firm would not make it. We should stick with that principle, since the value based upon future revenues/ earnings is implicitly based upon the assumption that the firm survives and succeeds.

d. Adjust for illiquidity: In the last section on intrinsic valuation, we presented different ways of estimating illiquidity discounts for equity in private businesses. We could adopt the same techniques to adjust the public multiple value for illiquidity.

A Relative Valuation of Evergreen Solar

With small revenues, miniscule book value and big operating losses, Evergreen Solar presents more of a challenge when it comes to relative valuation. To begin with, the only multiple that we can use, with current numbers, is a revenue multiple. We will consider two ways in which we can approach this relative valuation and the pluses and minuses of each.

a. Similar companies: The simplest way to approach relative valuation is to find alternative energy companies that are close to Evergreen in terms of current operating statistics – revenues and margins – and compare the multiples of revenues that these firms trade at in comparison to Evergreen's numbers. In January 2009, for instance, we found four companies that had revenues similar to Evergreen (about $ 90 million). Table 5.4 summarizes the Enterprise value/Sales multiples for these firms:

Table 5.4: Enterprise Value to Sales: Comparable Alternative Energy Companies

Company Name	*Market Capitalization*	*Enterprise Value*	*EV/Sales*	*Total Revenue ($mm) [LTM]*

Sumatec Resources	10.7	74.71	1.23	60.7
China Windpower	169.3	174.73	2.62	66.7
Viridis Clean Energy	52.9	290.4	3.31	87.8
Terna Energy SA	580.1	580.1	5.51	105.3
Average			**3.17**	

Note that the average EV/Sales ratio across these firms is 3.17 and applying this to Evergreen Solar's revenues of $ 90 million in 2007 would have yielded an enterprise value today of $285.3 million. In using this sample, we are implicitly assuming that these firms have the same growth and profit potential in steady state. Note also how much variation there is in the EV/Sales ratio across these four firms.

b. Forward values: One reason why we have trouble using multiples with growth companies is that the current numbers are not representative of what we expect the firm to look like in the future. One way around this problem is to use the forecasted revenues and other operating numbers to estimate value. With Evergreen, we draw on our estimates of future revenues and earnings in illustration 10.5. In table 10.6, we forecast Evergreen's revenues in year 5 to be $483 million and its pre-tax operating income to be $27.75 million. Looking across the larger publicly traded firms that are listed under alternative energy, we estimated an average EV/Sales ratio of 1.55. Applying this to the revenues in year 5 yields an expected enterprise value (in year 5) of $749 million.

Expected Enterprise value in year 5 = $483 * 1.55 = $749 million

Discounting this value back at Evergreen's cost of capital of 10.37% for the first 5 years (see table 10.8), we estimate an enterprise value today of $457 million.

Enterprise value today = $749 million/ 1.1037^5 =$457 million

This approach is based on three premises. The first is that the firm will survive and make it to year 5, which may or may not happen, given operating losses and the overlay of debt. The second is that our forecasts of revenues and operating income in year 5 are reasonable numbers; if we have over or under estimated these numbers, the resulting estimates of values will also be affected. The third is

that the firm being valued will look like larger firms in the industry in five years, thus making it all right to use the industry average multiple.

Are we missing something?

In both discounted cash flow and relative valuation, we build in our expectations of what success for a young firm will look like in terms of revenues and earnings. Thus, it can be argued that the potential upside is already reflected in the value. The counter to this argument is that success in one business or market can sometimes be a stepping-stone to success in other businesses/markets:

a. New products: Success with an existing product or service can sometimes provide an opening for a firm to introduce a new product. A classic example would be Microsoft building off the operating system (MSDOS and Windows) it developed for the PC to produce Microsoft Office, an immensely profitable addition to its product line. Apple's introduction of the iPhone to take advantage of the customer base that it had developed with the iPod would be another example. While neither new product (MS Office and the iPhone) could have been predicted at the time of the original product's introduction, the success of the initial product was clearly the launching pad for these offerings.
b. New markets: In some cases, companies that succeed with a product in one market may be able to expand into other markets, with similar success. The most obvious example of this is expanding into foreign markets to build on domestic market success, a pathway adopted by companies like Coca Cola, McDonalds and many retail companies. The more subtle examples are products that are directed at one market that serendipitously find new markets: an ulcer drug that reduces cholesterol would be a good example.

Why cannot we build expectations about new products and new markets into our cash flows and value? We can try, but there are two problems. The first is that our forecasts about these potential product and market extensions will be very hazy at the time of the initial valuation and the cash flows will reflect this uncertainty. In other words, neither Microsoft nor Apple would have been able to visualize the potential markets for Microsoft Office or the iPhone at the time that they were

introducing MSDOS or the iPod. The second is that it is the information gleaned and the lessons learned during the initial product launch and subsequent development that allows firms to take full advantage of the follow-up offerings. It is this learning and adaptive behavior that gives rise to value that adds to the estimated intrinsic value. While we will not explicitly value this "real option", we should at least qualitatively consider the possible upside.

Value Plays

Having laid out the challenges that we face when valuing young growth companies and ways to meet those challenges in both intrinsic and relative valuation, we can extract general lessons that can stand us in good stead when investing in these companies.

Consider what a young growth company has to pull off to become a successful mature company. In the near future, the firm has to post high revenue growth, while margins improve towards target margins at a reasonable clip. During this period, the company will have to raise new capital to sustain its revenue growth, while dealing with uncertainty in its operations. Once it gets past this initial phase successfully, the company will have to be able to defend its margins and profitability against larger players in the sector, while holding on or increasing its market share. Now consider all the things that can go wrong:

a. The expected revenue growth may not materialize, either because the potential market turns out to be smaller than expected or because customers are not as receptive to the product as anticipated.
b. The revenue growth may be spot on, but the target margin may be much lower than anticipated, either because expenses or competition were under estimated. In addition, the pathway to the target may turn out to be rockier, with larger losses for longer periods preceding profitability.
c. The revenue growth and margins may both be in line with expectations, but the firm may be unable to raise the capital it needs to fund its growth. In the face of this constraint, it will either have to shut down or sell itself at a bargain basement price to competitors.

d. Every part of the initial phase may work out, but just as the company hits its target margin and starts making money, competitors may swarm the market and push down margins and returns.
e. If there is a key person or persons that the firm depends on for its success, their absence caused by rifts within the firm or acts of God can put the firm to the test.

While there is no way to insulate yourself against all of these concerns, investors can improve their odds of success in young, growth companies by focusing on those that share the following characteristics

a. Big potential market: The potential market for the company's products and services has to be large enough to absorb high revenue growth for an extended period, without being overwhelmed. In the process, though, you have to be realistic about the reach of the products and services offered.
b. Expense tracking and controls: It is easy for young companies to become undisciplined in tracking and controlling expenses, while chasing high growth. As investors, we should be looking for management that is focused on both growth and margins and results that deliver the same. Thus, it may make sense, when investing in young companies, to set targets for margin improvement and view failure to meet these targets as reasons to exit (sell the stock)
c. Access to capital: To survive and grow, young firms need to be able to access capital in the near future. We would look at three indicators of access. First, we would look at the current cash balance and the rate at with which the company spends its cash; this is termed the cash burn ratio. Thus, a company with $ 100 million in cash that burns through$ 10 million each month has about 10 months of cash on hand, before it has to go back to raise more money. Second, we would look at who owns and trades the listed shares in the company. A company where a large portion of the shares are traded (high float) frequently by institutional investors is in a much better position to raise new capital than a company with low float, light trading and no institutional investors.
d. Dependence on key individuals: Young firms are always dependent upon key individuals to succeed, but those firms where these individuals nurture talent to succeed them (and are

e. The option bonus; One of the payoffs to investing in a young growth company is that success can feed on itself and allow it to enter new markets and introduce new products. While this may seem completely unpredictable, it is more likely to happen if the company has some exclusivity in its business, arising either from a proprietary technology or a patent.

f. Risk/Return: Even successful investments in high growth companies take investors on wild rides, with significant uncertainty along the way. Investors have to be proactive and demand higher returns for investing in young growth companies, before they enter the tumultuous growth period. Put in simpler terms, even the best young, growth company can be a bad investment, at the wrong price. At the minimum, therefore, investors have to make judgments of revenues and margins at a future date (at the end of the initial growth phase, for instance) and check the current values against these forward estimates. If the current market value is too high, relative to these future operating numbers, it makes no sense to invest in the company.

In summary, then, you want to invest in young growth companies that are entering huge markets with proprietary technologies, have management that is focused on keeping expenses under control and have access to enough capital to fund growth. Not easy to do, but done right, it is a high risk, high return proposition.

Conclusion

There can be no denying the fact that young companies pose the most difficult estimation challenges in valuation. A combination of factors – short and not very informative histories, operating losses and the possibility that high probability of failure – all feed into valuation practices that try to avoid dealing with the uncertainty by using a combination of forward multiples and arbitrarily high discount rates. We have laid out processes that can be used to apply conventional valuation models to young companies. While these approaches require us to estimate inputs that are often difficult to nail down, they are still useful insofar as they force us to confront the sources of uncertainty, learn more about them and make our best estimates. While we may be tempted to add premiums to these

values for potential opportunities that we see in the future, the use of real option premiums should be limited to those companies that have some degree of exclusivity in exploiting these opportunities.

Chapter 6: Valuing growth companies

In the last chapter, we looked at the estimation challenges associated with valuing young and idea companies. One of the issues that we confronted was the question of survival, since many young companies fail early in their lives. But what about those firms that make it through the test of competition and become successful businesses? In this chapter, we will look at a sub-set of young firms that become growth companies. Many of the concerns that we had with young companies – short and volatile operating histories, uncertainty about future growth and changing risk profiles – remain problems when we value growth companies, especially in their initial phases, though the data and tools that we have to deal with them do become better. They will be joined by new concerns about how growth rates may change, as companies get larger, and how access and exposure to public capital markets will change financing and investment decisions at the firm.

What is a growth company?

While investors and managers often talk about growth and mature companies as distinct groups, the distinction is more difficult to make in the real world. So, what is a growth company? There are many definitions for growth companies used in practice but they all tend to be both subjective and have significant flaws.

a. Sector based measures: Many analysts categorize companies as growth companies or mature companies, based upon the sector that they operate in. Thus, technology companies in the United States are treated as growth companies, whereas steel companies are considered mature. This definition clearly misses the vast differences in growth prospects across companies within any given sector. Technology companies like Intel and Microsoft may be more mature businesses than growth businesses, at this stage of their corporate evolution.

b. Analyst Growth Estimates/ Growth history: A second categorization of companies into growth and mature companies is based upon expected growth in future earnings, based usually on forecasts by equity research analysts. The limits of this approach are that it is circumscribed by its focus on earnings, as opposed to revenues or units sold. After all, there are many young high growth companies that may have exponential growth in revenues, while losing money. Similarly, mature companies can post healthy earnings increases with improved efficiency and relatively little operating unit growth.

c. Market based measures: Morningstar, as part of its mutual fund tracking service, categorizes mutual funds into those investing in growth stocks and those investing in mature companies. They base their categorization on the market multiples that companies trade at, arguing that companies that are perceived to be growth companies will trade at higher multiples of earnings, revenues and book value than mature companies.

While we do not have a perfect alternative, we would suggest using the financial balance sheet that we introduced in the earlier chapters to make this judgment. In figure 6.1, we focus on the asset side of the financial balance sheet, where assets are broken down into existing investments and growth assets.

Figure 6. 1: A Financial Balance Sheet for a Business

Assets			Liabilities
Existing Investments Generate cashflows today	Investments already made	*Mature companies get most of their value from existing assets.*	Debt
Expected Value that will be created by future investments	Investments yet to be made	*Growth companies derive a significant [proportion of their value from growth assets.*	Equity

Growth firms get a significant portion of their value from growth assets, i.e., investments that they expect to make in the future. While this may seem like a restatement of the growth categorization described earlier, where firms with high

growth rates are treated as growth companies, there is an important difference. As we noted in chapter 3, the value of growth assets is a function of not only how much growth is anticipated but also the quality of that growth, measured in terms of excess returns: returns on the invested capital in these assets, relative to the cost of capital.

Characteristics of growth companies

Growth companies are diverse in size, growth prospects and can be spread out over very different businesses but they share some common characteristics that make an impact on how we value them. In this section, we will look at some of these shared features:

1. Dynamic financials: Much of the information that we use to value companies comes from their financial statements (income statements, balance sheets and statements of cash flows). One feature shared by growth companies is that the numbers in these statements are in a state of flux. Not only can the numbers for the latest year be very different from numbers in the prior year, but can change dramatically even over shorter time periods.
2. Size disconnect: The market values of growth companies, if they are publicly traded, are often much higher than the accounting (or book) values, since the former incorporate the value of growth assets and the latter often do not. In addition, the market values can seem discordant with the operating numbers for the firm – revenues and earnings. Many growth firms that have market values in the hundreds of millions or even in the billions can have small revenues and negative earnings. Again, the reason lies in the fact that the operating numbers reflect the existing investments of the firm and these investments may represent a very small portion of the overall value of the firm.
3. Use of debt: While the usage of debt can vary across sectors, growth firms in any business will tend to carry less debt, relative to their value (intrinsic or market), than more stable firms in the same business, simply because they do not have the cash flows from existing assets to support more debt.

4. Market history is short and shifting: We are dependent upon market price inputs for several key components of valuation and especially so for estimating risk parameters (such as betas). Even if growth companies are publicly traded, they tend to have short and shifting histories.

While the degree to which these factors affect growth firms can vary across firms, they are prevalent in almost every growth firm.

Valuation Issues

The shared characteristics of growth firms – dynamic financials, a mix of public and private equity, disconnects between market value and operating data, a dependence on equity funding and a short and volatile market history – have consequences for both intrinsic and relative valuations.

Intrinsic (DCF) Valuation

If the intrinsic value of a company comes from its cash flows and risk characteristics, there are problems that we will run into while valuing growth companies that can be traced back to where they are in the life cycle. In this section, we will break down the valuation issues specific to growth companies by the key components of intrinsic value – existing asset value, growth asset value, risk (discount rates), terminal value and equity value per share.

Cash flows from Existing Assets

To value existing assets, we start with the cash flows generated by these assets and discount back at an appropriate risk-adjusted rate. There are two considerations that come into play that can make this measurement complicated.

1. Poorly measured earnings: With growth firms, existing assets tend to be a small part of overall value and can be easily swamped by what a firm expends to sustain and nurture its growth assets. Consider, for instance, the standard assumption that we make in discounted cash flow valuation that the existing operating income can be attributed to existing assets and thus be the basis for valuing those assets. With any company, the existing operating income (or loss) will be after selling, advertising and other administrative expenses. While we

assume that these expenses are associated with existing assets, that assumption may not hold up in a growth company. After all, the sales force in a growth company may be less interested in pushing existing products and more focused on cultivating a customer base for future products. By treating all sales expenses as operating expenses, we are understating the earnings from and consequently the value of existing assets.

2. Shifting profitability: If one of the key inputs into value is the measure of the future profitability of the firm, the fact that margins and returns at growth firms change significantly over time can make it difficult to make forecasts. Unlike mature firms, where margins usually move within a narrow range and returns are stable, using past margins and returns to forecast future values for a growth firm may not yield reasonable numbers.

Growth Assets

The bulk of a value for a growth company obviously comes from growth assets, making it imperative that we assess the value right. The challenges we face in attaching a reasonable value to growth assets in a growth company can also be daunting:

1. The scaling effect on growth: One of the biggest questions that we have to answer about expected growth rates for a company is how they will be affected by the changing size of the company. Consider, for instance, a company that has posted a growth rate of 80% over the last 5 years. The company today is obviously much larger (by a factor of 18) than it was five years ago.[20] It is extremely unlikely that it will be able to maintain an 80% growth rate for the next year, given its larger size. In general, delivering a given growth rate will become more difficult as a company gets bigger.
2. Success attracts competition: A small company can operate under the radar and sometimes show exceptional profitability. As the company grows, though, its success will attract attention from larger and more predatory competitors, often

[20] Applying a compounded growth rate of 80% for five years to $ 1 results in an end value of of almost $ 19, an overall increase of 1800% over five years.

possessing more significant resources. This competition, in turn, will result in lower profitability and value for growth.

3. Macro economic effects: While all companies are susceptible to macro economic shocks, small companies are more exposed to economic downturns because their products are often niche products that are discretionary. While customers may be inclined to buy them in good book economic times, they are likely to hold back during recessions or economic slowdowns.

Questions about how quickly growth rates will scale down, how profitability will survive competitive assaults and the effects of overall economic growth will have to be answered if we intend to attach a value to growth assets.

Discount Rates

The two key determinants of discount rates are the risk in the underlying investments of a business and the mix of debt and equity used to fund the business. On both dimensions, growth companies pose a challenge in valuation.

1. Risk of existing assets versus risk of growth assets: Since growth companies derive significant value from both growth assets and existing assets, delineating the risk in each category can make a big difference in how we value them. In other words, if growth assets are riskier than existing assets, we should be using higher discount rates for expected cash flows from the former and lower discount rates for cash flows from the latter.
2. Changing risk for the firm over time: If computing the current risk parameters and debt ratio for a growth firm is difficult, the task is complicated further by a simple fact. On both dimensions, growth firms can be expected to change over time, leading to discount rates that vary across the years. To be more specific, as a firm become larger over time (as it will in future periods, with growth), we should expect existing assets to become a larger proportion of overall value and risk measures to change to reflect the increasing (and more stable) earnings of the firm. Concurrently, the firm's capacity to borrow money will increase and if it exploits this capacity, its debt ratio will change as well. Generally speaking, the

discount rates used to value growth firms should be higher in the earlier periods and decrease in later periods towards mature company levels.

Terminal Value

Two key questions that overhang the valuation of any firm relate to when the firm will become a stable growth firm and the characteristics that it will possess in this phase. The answer to the first question will determine the length of the high growth period, with the terminal value being computed on the assumption that growth beyond that point will be sustained forever. The answers to the second question, especially on risk and the returns generated on new investments, will influence the value that we assign to the firm, for any given level of growth. Again while these are estimation issues that arise in any valuation, they can be more problematic for growth companies for the following reasons:

a. Terminal value is a bigger proportion of value: Since growth companies generate relatively low cash flows from existing assets, the terminal value will comprise a much larger proportion of their overall value. Thus, the assumptions we make about terminal value will matter more in any assessment of current value for a growth firm than at a mature firm.
b. More uncertainty about terminal value assumptions: Concurrent with the terminal value being a larger proportion of the value of a growth company than for a mature firm comes the fact that there is significantly more uncertainty about assessing that value for two reasons. First, we are looking at a young and often untested firm and making our assessments of not only how quickly it will continue to grow but also how it will respond to more aggressive competition. Second, the fact that the firm is evolving makes it difficult to evaluate what market it is aspiring to be in or even who its direct competitors are.
c. Terminal value characteristics: Earlier in this section we noted the difficulties we face in arriving at the current cash flows, returns and discount rates for a growth firm. We will be called upon to estimate all of these numbers again, when we put the firm into stable growth, in ten or fifteen years. If we cannot estimate the current cost of capital for a growth firm, it seems unreasonable to believe that

we can estimate this and other numbers for the same firm ten or fifteen years in the future.

The irony of terminal value estimation for growth firms is that it is both more important that we get it right and that we have far less basis for making the estimate in the first place. The way we resolve this contradiction will play a key role in whether the value we arrive at for a growth firm is a reasonable one.

Relative Valuation

Many analysts, when confronted with the problems with intrinsic valuation outlined in the earlier section, decide that relative valuation is a much easier path to follow with growth companies. The issues that make discounted cash flow valuation difficult also crop up, not surprisingly, when we do relative valuation.

1. Comparable firms: The conventional practice of using other publicly traded companies in the same sector can be dangerous with relative valuation for a few reasons. The first is that a growth company in a mature sector will (and should) bear little or no resemblance on either fundamentals or pricing multiples to the rest of the firms in the sector. The second is that even if every firm in the sector has growth potential, growth firms can vary widely in terms of risk and growth characteristics, thus making it difficult to generalize from industry averages.
2. Base year values and choice of multiples: Most multiples are stated as a function of base year values for revenues, earnings and book value. To estimate the PE ratio, for instance, we divide the stock price today by the earnings per share in the most recent fiscal year or four quarters. If a firm is a growth firm, the current values for these numbers will bear little resemblance to the future potential for the firm. Using PE ratios to illustrate this point, this can lead to either very high PE ratios (since current earnings per share will be small relative to stock prices today) or not meaningful values (because earnings currently are negative and PE ratios cannot be computed), for many growth companies. Moving up the income statement to

EBITDA or revenues offers little solace, since the values for these items will also be low, relative to future potential.

3. Controlling for growth differences: Since growth potential is the key dimension on which these firms vary, it becomes critical that we control for growth when comparing firms or extrapolating from industry averages. Unfortunately, the relationship between growth and value is too complex to lend itself to the simplistic generalizations that make relative valuation so attractive to both analysts and investors. Not only does the level of growth make a difference to value, but so does the length of the growth period and the excess returns that accompany that growth rate. Put another way, two companies with the same expected growth rate in earnings can trade at very different multiples of these earnings, because they vary on other dimensions.
4. Controlling for risk differences: Growth and risk are twin variables, with higher values for one generally going with higher values for the other. Determining how the net trade off will affect value is difficult to do in any valuation but becomes doubly so in relative valuation, where many companies have both high growth and high risk. Furthermore, as risk and growth characteristics change over time, as they inevitably will for any growth company, the multiple that we will apply to the company's operating numbers should also change.

Analysts who use multiples and comparables to value growth firms may feel a false sense of security about their valuations, since their assumptions are often implicit rather than explicit. The reality, though, is that relative valuations yield valuations that are just as subject to error as discounted cash flow valuations.

Valuation Solutions

While growth companies raise thorny estimation problems, we can navigate our way through these problems to arrive at values for these firms that are less likely to be contaminated by internal inconsistencies. In this section, we will lay out the steps to follow in discounted cash flow and relative valuations of growth companies.

Discounted Cash Flow Valuation

Given that the objective in discounted cash flow valuation is to arrive at reasonable estimates of the cash flows and the discount rates, we will list out some of the considerations that should enter into the process when we are valuing growth companies.

Choice of Model

In chapter 2, we listed the choices we face with discounted cash flow models, noting that we can either value the entire business (by discounting cash flows to the firm at the cost of capital) or value the equity directly (by discounting cash flows to equity at the cost of equity). While both approaches should yield the same value for the equity, estimating cash flows to equity, if we expect the debt ratio to change over time, is much more difficult than estimating the cost of capital. The former requires us to forecast new debt issues, debt repayments and interest payments each period, as the dollar debt changes, whereas the latter is based upon changing debt ratios.

Since many growth companies have little or no debt in their capital structure, analysts often fall back on equity valuation models, using the absence of debt as justification. However, this assumes that growth companies will continue with their policy of not using debt in perpetuity, even as growth decreases and the companies become more mature. If we make the more reasonable assumption that growth companies will become mature companies over time and adopt the financing practices of the latter, firm valuation models provide analysts with more flexibility to reflect these changes.

Needless to say, the discounted cash flow models used need to allow for high growth and even changing operating margins over time. As a general rule, rigid models that lock in the current characteristics of the company do not perform as well as more flexible models, where analysts can change the inputs over time, in valuing growth companies.

Valuing the operating assets

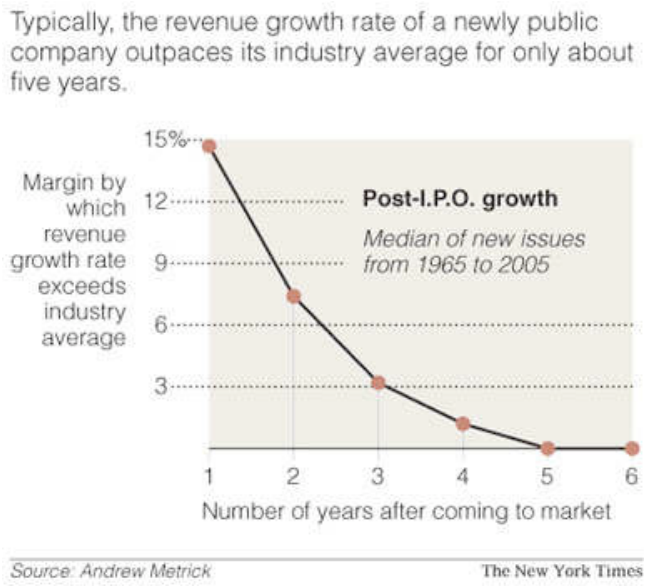
If we accept the premise that firm valuation models work better than equity valuation models, when valuing growth companies, the first step in the process is

valuing the operating assets of the firm, incorporating both existing assets and growth assets.

Revenue growth rates

The valuation process starts with estimating future revenues. In making these estimates, many of the considerations that we raised in chapter 9, for young companies, come into play. The biggest issue, and one that we have emphasized repeatedly in this chapter, is the scaling factor. Revenue growth rates will decrease as companies get larger, and every growth company will get larger over time, if our forecasts of growth come to fruition. In a test of how growth rates change as firms get larger, a study examined the revenue growth rate for high growth firms, relative to growth rate in revenues for the sector in which they operate, in the immediate aftermath of their initial public offerings.[21] The results are reported in Figure 6.2 below:

Figure 6.2: Revenue growth in years after initial public offering



[21] Metrick, A., 2006, Venture Capital and the Finance of Innovation, John Wiley & Sons.

At the time of going public, firms have growth rates that are much higher than the industry average. Note how quickly the revenue growth at these high growth firms moves towards the industry average – from a 15% higher revenue growth (then the industry average) one year after the IPO to 7% higher in year 2 to 1% higher in year 4 to the industry average in year 5. While we are not suggesting that this will happen at every high growth firm, the aggregate evidence suggests that growth firms that are able to maintain high growth rates for extended periods remain the exception rather than the rule.

The question of how quickly revenue growth rates will decline at a given company can generally be addressed by looking at the company's specifics – the size of the overall market for its products and services, the strength of the competition and quality of both its products and management. Companies in larger markets with less aggressive competition (or protection from competition) and better management can maintain high revenue growth rates for longer periods.[22]

Current margins versus target margins

To get from revenues to operating income, we need operating margins over time. The easiest and most convenient scenario is the one where the current margins of the firm being valued are sustainable and can be used as the expected margins over time. In fact, if this is the case, we can dispense with forecasting revenue growth and instead focus on operating income growth, since the two will be the equivalent. In most growth firms, though, it is more likely that the current margin is likely to change over time.

Let us start with the most likely case first, which is that the current margin is either negative or too low, relative to the sustainable long-term margin. There are three reasons why this can happen. One is that the firm has up-front fixed costs that have to be incurred in the initial phases of growth, with the payoff in terms of revenue and growth in later periods. This is often the case with infrastructure companies such as energy, telecommunications and cable firms. The second is the

---

[22] For an extended discussion of this issue, see Damodaran, A., 2008, The Origins of Growth, Working Paper, SSRN.

mingling of expenses incurred to generate growth with operating expenses; we noted earlier that selling expenses at growth firms are often directed towards future growth rather than current sales but are included with other operating expenses. As the firm matures, this problem will get smaller, leading to higher margins and profits. The third is that there might be a lag between expenses being incurred and revenues being generated; if the expenses incurred this year are directed towards much higher revenues in 3 years, earnings and margins will be low today.

The other possibility, where the current margin is too high and will decrease over time, is less likely but can occur, especially with growth companies that have a niche product in a small market. In fact, the market may be too small to attract the attention of larger, better-capitalized competitors, thus allowing the firms to operate under the radar for the moment, charging high prices to a captive market. As the firm grows, this will change and margins will decrease. In other cases, the high margins may come from owning a patent or other legal protection against competitors, and as this protection lapses, margins will decrease.

In both of the latter two scenarios – low margins converging to a higher value or high margins dropping back to more sustainable levels – we have to make judgment calls on what the target margin should be and how the current margin will change over time towards this target. The answer to the first question can be usually be found by looking at both the average operating margin for the industry in which the firm operates and the margins commanded by larger, more stable firms in that industry. The answer to the second will depend upon the reason for the divergence between the current and the target margin. With infrastructure companies, for instance, it will reflect how long it will take for the investment to be operational and capacity to be fully utilized.

Reinvest to sustain growth

A constant theme in the earlier chapters has been the insistence that growth is not free and that firms will have to reinvest to growth. As we noted earlier in the chapter, it is dangerous to base reinvestment assumptions on a growth company's history of reinvestment. In other words, taking the net capital expenditures and

working capital changes from the most recent year and assuming that these items will grow at the same rate as revenues can result in reinvestment numbers that are both unrealistic and inconsistent with our assumptions about growth.

To estimate reinvestment for a growth firm, we will follow one of three paths, depending largely upon the characteristics of the firm in question:

1. For growth firms earlier in the life cycle, we will adopt the same roadmap we used for young growth companies, where we estimated reinvestment based upon the change in revenues and the sales to capital ratio. Thus, assuming a sales to capital ratio of 2.5, in conjunction with a revenue increase of $ 250 million will result in reinvestment of $ 100 million. We can build in lags between the reinvestment and revenue change into the computation, by using revenues in a future period to estimate reinvestment in the current one.
2. With a growth firm that has a more established track record of earnings and reinvestment, we can estimate the growth rate as a product of the reinvestment rate and the return on capital on these investments. Thus, a firm that reinvests 80% of its after-tax operating income back into the business and earn a 25% return on capital on its investments, can be expected to grow about 20% a year.
   Expected growth rate in operating income = 0.80*25% = 20%
3. Growth firms that have already invested in capacity for future years are in the unusual position of being able to grow with little or no reinvestment for the near term. For these firms, we can forecast capacity usage to determine how long the investment holiday will last and when the firm will have to reinvest again. During the investment holiday, reinvestment can be minimal or even zero, accompanied by healthy growth in revenues and operating income.

With all three classes of firms, though, the leeway that we have in estimating reinvestment needs during the high growth phase should disappear, once the firm has reached its mature phase. The reinvestment in the mature phase should hew strictly to fundamentals:

Reinvestment rate in mature phase = $\frac{\text{Expected Growth}}{\text{Stable ROC}} = \frac{3.00\%}{6.76\%} = 44.40\%$

In fact, even in cases where reinvestment is estimated independently of the operating income during the growth period, and without recourse to the return on capital, we should keep track of the imputed return on capital (based on our forecasts of operating income and capital invested) to ensure that it stays within reasonable bounds. The process for doing so is described in chapter 9.

Risk profile consistent with growth and operating numbers

While the components of the cost of capital – the beta(s) and the cost of equity, the cost of debt and the debt ratio – are the same for a growth company as they are for a mature company, what sets growth companies apart is that their risk profiles will shift over time. The key to maintaining balance in growth company valuations is to adjust the discount rates over time to keep them consistent with the growth and margin assumptions that we are making in each period. As general rules:

- Growth firms should have high costs for equity and debt when revenue growth is highest, but the costs of debt and equity should decline as revenue growth moderates and margins improve.
- As earnings improve and growth drops, another phenomenon will come into play, which is that the firm will generate more cash flows than it needs, which it can use to not only pay dividends but also to service debt financing. While firms are not required to use this debt capacity, and some of them do not, the tax advantages of debt will lead some firms to borrow, causing debt ratios to increase over time.

In summary, the cost of capital for a growth company should almost never be a number that remains unchanged over the entire time horizon. Instead, it should be a year-specific number that keeps track with the rest of the changes that we are forecasting at the firm.

In terms of estimating risk parameters (betas), we would steer as far as we can from using the limited price data that is available on growth companies: the standard errors on the estimates are likely to be huge. Instead, we would use estimates of betas obtained by looking at other publicly traded firms that share the

same risk, growth and cash flow characteristics as the firm being valued. If the case for using these bottom up betas (industry average as opposed to a regression beta) is strong with any firm, it is even stronger with growth firms.

Stable growth assumptions: When and what will the firm look like?

The assumptions we make about terminal value loom large with a growth company, since it will comprise a much larger portion of the firm's current value than is the case with a mature firm. When will a growth firm become a stable growth, mature firm? While we have a little more information than we did with young companies, in making this assessment, it is difficult to do, and akin to looking at a teenager and wondering what he or she will look like or be doing, in middle age.

While no one answer or approach will work with every growth company, we will draw on the discussion both in chapter 2 and this one to develop the following general propositions:

- Do not wait too long to put a firm into stable growth: As we noted in the section on the dark side of valuing growth companies, analysts often allow for very long growth periods for growth firms and justify the assumption by pointing to past growth. As we noted in figure 6.2, both scale and competition conspire to lower growth rates quickly at even the most promising growth companies. Growth periods that exceed 10 years, especially when accompanied by high growth rates over these periods, are difficult to defend, since only a handful of companies have been able to accomplish this over time.
- When you put your firm into stable growth, give it the characteristics of a stable growth firm: In keeping with the emphasis on preserving internal consistency, we should change the characteristics of the company to reflect stable growth. With discount rates, as we noted in the last section, this will take the form of using lower costs of debt and equity and a higher debt ratio. With reinvestment, the key assumption will be the return on capital that we assume for the stable growth phase. While some analysts believe that the return on capital should be set equal to the cost of capital in stable growth, we would preserve some company-specific flexibility and suggest that the difference between return on

capital and cost of capital should narrow during stable growth to a sustainable level (less than 4 or 5%).

The nature of the cash flows at growth companies – low or negative in the early years and higher later – will ensure that the terminal value is a high proportion of value. Some analysts use this as ammunition against using discounted cash flow valuations, suggesting that assumptions about the high growth phase will be drowned out by the terminal value assumptions. This is not true, since the base year value for the terminal value calculation (earnings and cash flows in year 5 or 10) is a function of the assumptions during the high growth phase; changing these assumptions will have dramatic effects (as it should) on value.

From operating asset value to equity value per share

Navigating the way from operating asset value to equity value per share for growth companies can be fraught with dangers, many of which we outlined in earlier sections. In this section, we will outline precautions that can protect, at least partially, against some of these dangers.

Cash and non-operating assets

Earlier in this chapter, we noted how quickly growth firms can burn through cash balances and how using the cash balance from the most recent financial statements can lead to misleading values. At least in theory, it would be useful to know what the cash balance is today, when valuing a firm. While investors in public equity markets have no way of accessing this information, an acquirer (or at least a friendly acquirer) should be able to get this information from the target firm and use it estimate an updated value. Even public investors can make judgments about current cash balances by using two pieces of public information – the cash flows of the firm and any new financing during the period since the last financial statement. For instance, assume that the last cash balance (from three months ago) is $ 100 million for a firm that reported EBITDA of negative $80 million in the most recent twelve-month period. If the firm has not raised any new financing in the last three months (through either equity issues or new debt), the current cash balance for the

firm is likely to be closer to $ 80 million than $ 100 million. (We are reducing the cash balance by the estimated EBITDA of -$20 million – one quarter of -$80 million.)

Debt and other Non-equity Claims

If convertible debt is the preferred mode of borrowing for a growth firm, we should treat it for what it is – a hybrid security that is part debt and part equity. Since the conversion option is equity and the rest is debt, the easiest way to decompose convertible debt into debt and equity is to value the convertible debt as if it were straight debt and to treat the resulting number as debt. For instance, assume that a growth firm has 5-year convertible debt outstanding, with a face value of $ 50 million and a coupon rate of 4%, and that the current pre-tax cost of debt for the firm, assuming it used conventional debt, is 10%. The value of the convertible bond, treated like a conventional debt would be:[23]

$$\text{Debt Value of convertible bond} = (50 * .04)\left(\frac{1-(1.10)^{-5}}{.10}\right) + \frac{50}{(1.10)^5} = \$38.63 \text{ million}$$

Subtracting this value from the market value of convertible bond will yield the value of the conversion option. Thus, if the convertible debt is trading at $52 million, the conversion option (equity) will be valued at $13.37 million.[24] When valuing this company, we would treat this part as equity and rest as debt in computing the current cost of capital. We would subtract out only the debt portion from the value of the firm to arrive at the overall value of equity.

Another aspect of debt that is potentially problematic is that the debt ratios will change over time, and with those changes will come changes in the debt outstanding at the firm. Since the value of equity is the value of the firm, net of debt, analysts often get caught up in the question of whether they should subtract the debt outstanding today (which may be negligible) or the expected debt outstanding in the future (which may be very large). The answer, when valuing the firm, is that we should subtract out only the current debt outstanding, even though that value may be miniscule relative to future debt issues.

---

[23] For simplicity, we have assumed annual interest payments. This equation can be easily modified to allow for semi-annual payments.

[24] If the market value of the debt is not available, an approximation would be to use the face value.

Post-valuation corrections

Once we have derived the value of equity in a growth firm, the final step is to allocate the value of equity across the shares outstanding in the firm. In making this final judgment, there are three considerations to keep in mind, all of which we raised and addressed in the chapter on young growth companies:

- While concerns about whether a firm will survive may abate as the company ages, they will not disappear. Even publicly traded growth firms have to shut down operations, especially if they run out of cash, and/or access to the market is restricted.
- While a few growth firms have large market capitalizations and consequently are heavily traded, most firms in this phase remain small, lightly held by institutions and relatively illiquid.
- Many growth firms continue to be controlled by their founder(s), who try to maintain that control by holding on to shares with disproportionate voting rights.

The corrections we would make for survival, illiquidity and different voting rights mirror those that we described for young growth firms.

Illustration: A DCF valuation of Underarmour

Under Armour, a company with a very successful line of microfiber apparel for athletes, was founded by Kevin Plank in 1996, is headquartered in Baltimore, and capitalized on its success by going public in 2006. Revenues at the firm tripled from $205 million in 2004 to $607 million in 2007; over the three-year period, the company had a compounded growth rate in revenues of 44% a year. Table 6.1 summarizes the most recent twelve months of data for Under Armour (at least as of January 2009):

Table 6.1: Updated twelve-month numbers: Under Armour

	Last 10K	*Last year thru Q*	*This year's thru Q*	*Trailing 12 month*
Current EBIT =	$86.27	$58.01	$54.05	$82.31
EBIT (adjusted for leases)				$88.28
Current Interest Expense =	$0.80	$0.53	$0.86	$1.13
Current Capital Spending	$33.96	$26.24	$30.85	$38.57

Current Depreciation and Amortization =	$14.62	$10.00	$15.48	$20.10
Current Revenues =	$606.56	$431.72	$545.97	$720.81

The updated numbers reflect the higher revenues and lower operating income at the firm over the last twelve months, relative to the last annual report.

Table 6.2 summarizes our estimates of cash flows and cost of capital for Underarmour over the next 10 years:

Table 6.2: Cash flows and Costs of capital – Underarmour

Year	*Revenues*	*Revenue growth*	*Pre-tax Margin*	*Operating income*	*After-tax Operating Income*	*Reinvestment*	*FCFF*	*Cost of capital*
Trailing 12 month	$721		12.25%	$88		$53		
1	$973	35.00%	12.46%	$121	$73	$138	-$65	9.27%
2	$1,216	25.00%	12.57%	$153	$92	$133	-$41	9.27%
3	$1,460	20.00%	12.64%	$184	$111	$133	-$22	9.27%
4	$1,679	15.00%	12.67%	$213	$128	$120	$8	9.27%
5	$1,846	10.00%	12.69%	$234	$141	$92	$49	9.27%
6	$1,994	8.00%	12.71%	$253	$152	$81	$71	8.87%
7	$2,114	6.00%	12.71%	$269	$161	$65	$96	8.62%
8	$2,209	4.50%	12.72%	$281	$169	$52	$117	8.34%
9	$2,275	3.00%	12.72%	$289	$174	$36	$137	7.98%
10	$2,343	3.00%	12.72%	$298	$179	$37	$142	7.28%
Terminal year	$2,396	2.25%	12.72%	$305	$183			

The key assumptions underlying these numbers are as follows:

a. Revenue Growth: We expect revenue growth to remain healthy for the next 5 years, but lower than past growth of 44%. As the company becomes larger, we expect growth rates to decrease reflecting both the scaling effect and competition.

b. Operating margin: We expect margins to improve slightly from the current level of 12.25% to the industry average of 12.72%. The tax rate is assumed to be 40%.

c. Reinvestment: The reinvestment each period is estimated by assuming that every dollar of new capital invested generates $1.83 in additional revenues (again an industry average).

d. Cost of capital: The cost of capital decreases as the firm matures, from the starting value of 9.27% to a stable period level of 7.28%.

As a final step, we assess what Under Armour will look like as a stable growth firm. As a slightly less risky business, with more debt in its capital structure, it should have a lower cost of capital (7.28%). We will assume that the firm will be able to use the magic of its brand name to generate returns on capital of 9% in perpetuity on new investments, above the cost of capital. The reinvestment rate and terminal value for the firm are computed below:

Reinvestment rate (after year 10) = Stable growth rate/ Stable Return on capital

= 2.25%/ 9% = 25%

$$\text{Terminal value} = \frac{EBIT_{11}(1-t)(1-\text{Reinvestment Rate})}{(\text{Cost of capital}_{stable} - g_{stable})}$$

$$= \frac{304.79(1-.40)(1-.25)}{(.0728-.0225)} = \$2729.50 \text{ million}$$

As the final step, we pull together the expected free cash flows to the firm and the costs of capital in table 6.3, with the intent of computing the value of the operating assets of the firm:

Table 6.3: Cash flows, discount rates and value

Year	Cost of capital	Cumulated Cost of Capital	FCFF	Terminal value	Present value
1	9.27%	1.0927	-$65.13		-$59.60
2	9.27%	1.1939	-$41.17		-$34.48
3	9.27%	1.3046	-$22.25		-$17.05
4	9.27%	1.4254	$8.01		$5.62
5	9.27%	1.5575	$48.92		$31.41
6	8.87%	1.6957	$71.31		$42.05
7	8.62%	1.8418	$95.85		$52.04
8	8.34%	1.9953	$116.55		$58.41
9	7.98%	2.1545	$137.40		$63.77
10	7.28%	2.3112	$141.54	$2,729.50	$1,242.22
		Sum of PV of cash flows =			$1,384.40

The value that we estimate for the operating assets of the firm is $1.384 billion.

To get from operating asset value to equity value for Under Armour poses fewer challenges than it did at Evergreen Solar. The cash balance of $40.15 million reported by the firm in its most recent financial statements in September 2008 is very similar to the cash balance of $40.59 million it reported in December 2007.

Since the firm is generating positive earnings, we feel comfortable assuming that the cash balance is intact at the time of the valuation in January 2009. With debt, the only confounding detail is that the bulk of the debt takes the form of leases and sponsorship commitments; of the total debt of $133 million, $95.5 million represents the present value of future commitments. From a valuation perspective, though, we see no need to separate the two types of debt. Consequently, the value of equity at Under Armour can be estimated as follows:

Value of Equity = Value of operating assets + Cash – Debt

= $1,384 million + $40 million - $133 million = $1.292 million

To get from this value to the value of equity per share, there are two final details that we have to take care of. The first is to consider the overhang of equity options that have been granted in the past to management; we estimated the value of the 2.13 million options outstanding at the firm, with an average strike price of $8.26 and an average expiration of 3.7 years, to be $ 23 million after taxes.[25] (We will return to this issue in more depth in a later chapter, when we look at technology firms) The second is that the firm has two classes of shares – 36.791 million class A shares that are held by the investing public and are traded and 12.5 million class B shares that are held by Kevin Plank, the founder of the firm; class B shares have 10 times the voting rights of class A shares. If we attribute no value to voting rights the value per share that we estimate is $25.73.

$$\text{Value per share} = \frac{\text{Value of Equity}}{(\#\ \text{Class A shares} + \#\ \text{Class B shares})} = \frac{(\$1{,}292 - \$23)}{(36.791 + 12.5)} = \$25.73$$

Note that control component should give class B shares a premium but the fact that the shares are not traded will result in an illiquidity discount. If we assume that class B shares have a 10% premium on class A shares, we can estimate the value of equity per share for class A and class B shares as follows:

[25] We used the standard Black Scholes model and adjusted for the potential dilution and the tax savings that will accrue to the firm when the options are exercised.

$$\text{Value per class A share} = \frac{\text{Value of Equity}}{(\#\ \text{Class A shares} + \text{Value premium} * \#\ \text{Class B shares})}$$

$$= \frac{(\$1{,}292 - 23)}{(36.791 + 1.1 * 12.5)} = \$25.09/share$$

Value per class B share = Value of class A share * 1.1 = $27.60

Relative Valuation

There is no reason why relative valuation cannot be used to arrive at an independent estimate of the value of equity in a growth firm as long as we keep two key factors in mind. The first is that using multiples and comparables cannot reduce the uncertainty inherent in valuing growth companies. The second is that relative valuation techniques have to be adapted to meet the limitations of growth companies – the paucity and unreliability of current operating numbers and the shifting risk/growth characteristics over time.

Comparable firms

Optimally, we would like to assess how a growth firm is valued by the market by comparing its pricing with that of otherwise similar growth firms. In a business like software, where there are growth firms aplenty, this can be accomplished by staying within the traditional framework of defining comparable firms as those in the same industry. In businesses like retailing or automotive parts, a growth firm may be the exception in a sector where the bulk of the firms are either mature or in decline. In these cases, we may have to abandon the conventional practice and define growth firms in terms of fundamentals rather than business. The pricing of a retail firm with growth prospects should be compared to how the market is pricing growth firms in other sectors rather than more mature firms in its own industry; there is no reason why the PE ratio for a high growth retail firm should not be comparable to the PE ratio for a high growth software firm.

Choice of multiples and base year

As we noted in the section on the dark side as it relates to relative valuation, analysts valuing growth companies tend to use either revenues in the current year

or estimates of operating performance in future years (forward earnings or revenues) to compute multiples. Each carries some danger:

- Revenue multiples are troubling simply because they gloss over the fact that the company being valued could be losing significant amounts of money. Consequently, we would suggest bringing in the expected future profit margins (which will be estimates) into the discussion of what comprises a reasonable multiple of revenues. Other things held constant, we would expect firms with higher expected profit margins (in mature phase) to trade at higher multiples of current revenues than firms with lower expected profit margins.
- Forward earnings multiples implicitly assume that the firm being valued will survive to the forward year and that the estimates of earnings for that year are reasonable. If forward multiples are used, controlling for survival becomes a critical component of the analysis. Firms that have a greater chance of surviving to the forward year, should trade at a higher multiple of earnings than firms that have a greater chance of failure.

As a general rule, we would steer away from multiples of either current book value or current earnings with growth companies early in the growth cycle, simply because these numbers are likely to be small and unstable.

Adjusting for differences in growth and risk

With growth firms, no matter how careful we are about constructing a set of comparable firms and picking the right multiple, there will be significant differences across the firms on both the level and the quality of expected growth. In chapter 4, we made note of the three ways in which analysts adjust for these differences: story telling, adjusted multiples and statistical approaches. Each has its use with growth companies, but each comes with limitations:

a. The growth story: When comparing the pricing of growth firms, analysts often try to explain why a company trades at a higher multiple than comparables by pointing to its "higher growth potential".
b. Adjusted multiples: In the chapter 4, we introduced the PEG ratio, the PE ratio divided by expected growth in the future, as a growth-adjusted version

of the PE ratio. In effect, a firm that trades at a lower PEG ratio is cheaper than one that trades at a higher PEG ratio.

c. Statistical approaches: When firms vary not only on expected growth, but also on the quality of that growth and risk, the first two approaches become difficult to apply. A multiple regression, with the multiple as the dependent variable, and risk and growth as independent variables, allow us to control for differences across firms on these dimensions.

Illustration 10.6: A Relative Valuation of Under Armour

To evaluate how Under Armour is being priced, relative to other firms in the sector, we extracted information on 34 publicly traded apparel firms in the United States, with earnings growth rates estimates available for the next 5 years from analysts following the companies.[26] Table 6.4 provides the PE ratios, expected growth rates and two-year betas for these companies:

Table 6.4: PE, Growth and Risk – Apparel Companies

Company Name	*Market Capitalization*	*PE*	*Expected growth rate in EPS - Next 5 years*	*Beta*	*PEG Ratio*
Nike Inc. (NYSE:NKE)	$22,102.80	11.91	13.10%	1.04	0.91
Kimberly-Clark Corporation (NYSE:KMB)	$21,830.00	12.64	7.33%	0.603	1.72
Avon Products Inc. (NYSE:AVP)	$8,896.80	11.53	13.00%	0.966	0.89
VF Corp. (NYSE:VFC)	$5,996.90	9.21	10.60%	1.02	0.87
Polo Ralph Lauren Corp. (NYSE:RL)	$3,781.30	8.00	13.60%	1.25	0.59
Cintas Corp. (NasdaqGS:CTAS)	$3,627.20	11.26	10.80%	0.999	1.04
Guess? Inc. (NYSE:GES)	$1,472.40	6.67	16.00%	1.25	0.42
Gildan Activewear Inc. (NYSE:GIL)	$1,291.90	8.93	12.50%	1.1	0.71
Tupperware Brands Corporation (NYSE:TUP)	$1,285.10	8.54	12.00%	1.45	0.71
Columbia Sportswear Company (NasdaqGS:COLM)	$1,083.70	8.87	9.70%	0.664	0.91
Warnaco Group Inc. (NYSE:WRC)	$969.80	11.21	20.00%	1.91	0.56
Under Armour, Inc. (NYSE:UA)	$969.10	20.71	20.90%	1.44	0.99
Phillips-Van Heusen Corp. (NYSE:PVH)	$968.30	6.05	14.00%	1.25	0.43
Carters Inc. (NYSE:CRI)	$964.70	12.64	15.00%	1.19	0.84
Wolverine World Wide Inc. (NYSE:WWW)	$929.80	9.56	12.60%	0.693	0.76
Hanesbrands Inc. (NYSE:HBI)	$918.00	5.77	15.30%	1.44	0.38
Fossil Inc. (NasdaqGS:FOSL)	$814.70	5.62	16.50%	1.35	0.34
Polaris Industries, Inc. (NYSE:PII)	$772.50	6.50	11.20%	1.5	0.58
Timberland Co. (NYSE:TBL)	$642.50	11.92	12.00%	1.06	0.99

[26] There were 84 companies in the overall sample; only 34 had positive earnings (to compute PE) and expected growth rates in earnings for the next 5 years.

UniFirst Corp. (NYSE:UNF)	$551.20	8.69	13.00%	0.906	0.67
Jos. A Bank Clothiers Inc. (NasdaqGS:JOSB)	$529.10	9.71	15.00%	1.11	0.65
Iconix Brand Group, Inc. (NasdaqGS:ICON)	$511.20	7.08	17.00%	1.74	0.42
Bebe Stores, Inc. (NasdaqGS:BEBE)	$507.30	8.61	13.00%	1.55	0.66
Raven Industries Inc. (NasdaqGS:RAVN)	$415.40	12.94	14.50%	1.08	0.89
K-Swiss Inc. (NasdaqGS:KSWS)	$382.00	11.24	19.00%	1.17	0.59
G&K Services Inc. (NasdaqGS:GKSR)	$347.10	9.86	10.70%	0.922	0.92
Ennis Inc. (NYSE:EBF)	$301.40	7.32	10.00%	1.21	0.73
True Religion Apparel Inc. (NasdaqGS:TRLG)	$284.30	7.00	17.00%	1.3	0.41
Volcom Inc. (NasdaqGS:VLCM)	$225.20	5.99	20.00%	1.9	0.30
Maidenform Brands Inc. (NYSE:MFB)	$211.70	7.40	14.50%	1.45	0.51
American Apparel, Inc. (AMEX:APP)	$157.10	18.33	30.00%	0	0.61
Cherokee Inc. (NasdaqGS:CHKE)	$133.70	9.03	12.00%	1.27	0.75
Oxford Industries Inc. (NYSE:OXM)	$104.90	20.61	13.00%	1.54	1.59
G-III Apparel Group, Ltd. (NasdaqGS:GIII)	$85.80	4.47	18.30%	1.32	0.24
Lacrosse Footwear Inc. (NasdaqGM:BOOT)	$64.90	8.78	13.50%	0.283	0.65
Perry Ellis International Inc. (NasdaqGS:PERY)	$60.80	3.27	13.50%	1.57	0.24
Sector Average	**2288.25**	**9.70**	**15.00%**	**1.15**	**0.70**

Let us consider first the two most common approaches used by analysts, following growth companies.

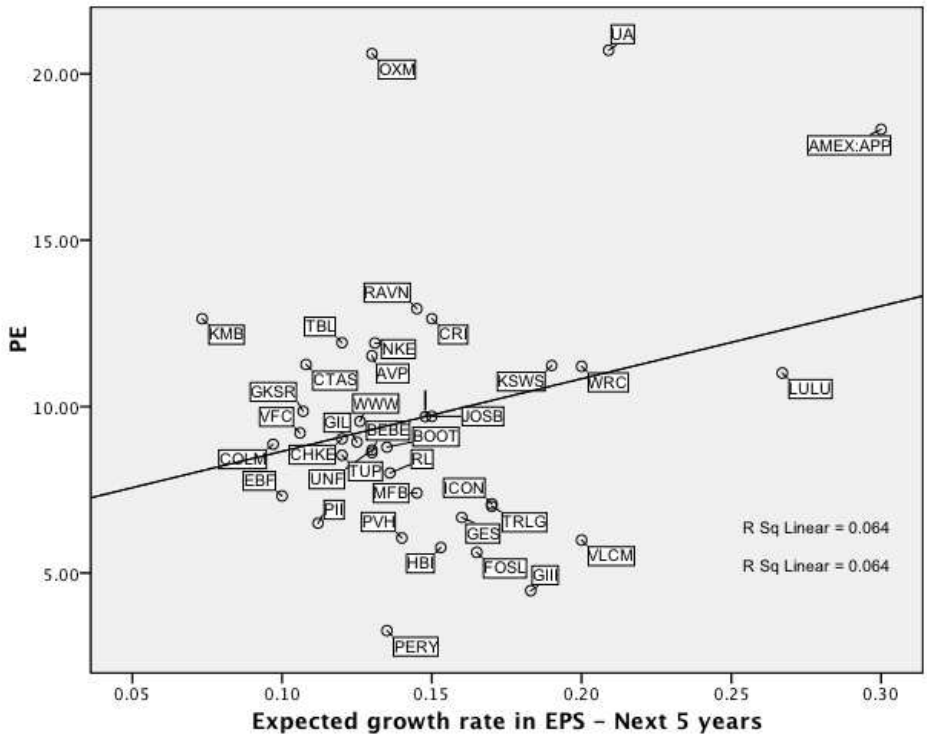
- Subjective: The PE ratio for Under Armour in January 2009 is 20.71, well above the average PE ratio for the sector (9.70). An optimistic analyst would undoubtedly point to Under Armour's higher expected growth in earnings (20.90% versus the industry average of 15%) as justification for a higher PE, though a pessimistic analyst would draw attention to Under Armour's higher risk (a beta of 1.44 versus the industry average of 1.15) to suggest that the stock was over valued.
- PEG ratio: A simple way to control for differences is growth is to compute the PEG ratio that we described earlier in the chapter; lower PEG ratios are considered indicative of an under valued company. The PEG ratio for Under Armour is a shade under 1.00:

$$PEG_{\text{Under Armour}} = \frac{\text{PE}}{\text{Expected growth rate in EPS}} = \frac{20.71}{20.90} = 0.99$$

Since the average PEG ratio for the sector is much lower at 0.70, this would seem to indicate that Under Armour is overvalued.

Since neither approach captures the effects of both growth and risk satisfactorily, and the PEG ratio assumes that PE increases in lock step with growth, we first graphed the PE ratios of apparel firms against expected EPS growth in the next 5 years in figure 6.3:

Figure 6.3: PE versus Expected Growth – Apparel companies



Note that Under Armour (UA) has the highest PE ratio in the group, but it also has a high growth rate. We regressed PE ratios against the expected growth rate and betas, weighting each firm, by market capitalization. The results are as follows (with t statistics in brackets below the coefficients):

PE = 13.78 + 32.04 (Expected Growth Rate) - 6.60 Beta R^2 = 25.9%

(10.88) (3.49) (1.91)

In keeping with intuition, higher growth firms have higher PE ratios and higher risk pushes down the PE, and both relationships are statistically significant. Using the

expected growth in earnings per share of 20.9% that analysts are forecasting for Under Armour for the next 5 years and the 2-year beta estimate of 1.44, we can estimate the expected PE ratio for the firm:

Expected $PE_{\text{Under Armour}}$ = 13.78 + 32.04 (.209) - 6.60 (1.44) = 10.98

At its existing PE ratio of 20.71, Under Armour looks overvalued, on a relative valuation basis, by almost 89%.

Value Plays

Investing successfully in growth companies requires skillful navigation through dangerous currents – unsustainable or value destroying growth and high growth. Consider what you need for a growth company to succeed:

a. The capacity to scale up growth, i.e., be able to maintain high revenue growth as it gets bigger,. There are a few tools that we can use to assess whether the assumptions we are making about revenue growth rates in the future, for an individual company, are reasonable:

- Absolute revenue changes: Even experienced analysts often under estimate the compounding effect of growth and how much revenues can balloon out over time with high growth rates. Computing the absolute change in revenues, given a growth rate in revenues, can be a sobering antidote to irrational exuberance when it comes to growth.
- Past history: Looking at past revenue growth rates for the firm in question should give us a sense of how growth rates have changed as the company size changed in the past. To those who are mathematically inclined, there are clues in the relationship that can be used for forecasting future growth.
- Sector data: The final tool is to look at revenue growth rates of more mature firms in the business, to get a sense of what a reasonable growth rate will be as the firm becomes larger.

In summary, expected revenue growth rates will tend to drop over time for all growth companies but the pace of the drop off will vary across companies.

b. The ability to generate returns on capital invested that exceed the cost of capital on new growth investments. It is not growth per se that creates value, but

growth with excess returns. Firms that are able to sustain margins and returns on capital, as they grow, are much better investments that firms that have to trade off lower margins and returns for higher growth. As a generalization, firms that are able to maintain organic growth have better odds of preserving excess returns than firms that have to rely on acquisitions to deliver high growth.

c. Increasing stability in earnings as the firm grows and matures, accompanied by a decline in risk: Firms that are able to diversify their product offerings and cater to a wider customer base, as they grow, will see their earnings become more stable as they scale up.

Finally, as with any investment, a growth company is worth investing in, only at the right price. While multiples such as PEG ratios have their limitations, they can be used to screen growth firms for likely bargains. There is one factor that can work in your favor, in this search process. Every growth company (including those that pass all of the screens above) will disappoint its investors at some point, delivering earnings that do not match up to lofty expectations. When that happens, there will be some growth investors who over react, dumping their shares and embarking on their search for the next great growth story. The resulting drop in price may offer you an opportunity to pick up the company at the right price.

Conclusion

When valuing a growth company, we confront many of the issues that we faced with young and idea companies, albeit on a lesser scale. There are three key components to valuing growth companies well, in a discounted cash flow framework. The first is to ensure that the assumptions we are making about growth and margins reflect not only market potential and competition, but also change to reflect the firm's changing size over time. The second is to reinvest enough back into the business to sustain the forecasted growth rates. The third is modify the risk profile of the firm to match its growth characteristics – the costs of equity, debt and capital are all likely to decrease as the firm goes from high growth to stable growth.

With relative valuation, controlling for differences in growth and risk when comparing companies is essential.

Chapter 7: Valuing Mature Companies

Mature companies should be the easiest ones to value. They have long periods of operating and market history, allowing us to estimate most of the inputs for valuation from historical data. They have also settled into established patterns of investment and financing, resulting in fundamentals (risk and returns) that are stable over time, giving us more confidence in our estimates of these numbers. It is, however, these established patterns that may present a problem, since not all long standing practice is good. Put another way, there are mature firms that make financing and investment choices that are not optimal or sensible, and have been doing so for long periods. It is possible, therefore, that these firms, with new management in place, could be run differently (and better) and have higher values. Investors valuing mature companies have to juggle two values – the value with existing management in place and a higher value form new or different management – and how they deal with them will in large part determine the quality of the valuation.

What is a mature company?

One way to categorize companies as growth and mature companies is to look at the growth rate, with lower growth companies being treated as mature. There are two problems with this approach. First, given that growth is a continuum, any growth rate that we adopt as a cut off point will be subjective –we will find more mature companies, if we adopt a 6% growth rate cut off than a 4% growth rate. Second, not all operating measures grow at the same rate; we have to decide whether the growth rate that we use for the categorization will be growth in revenues, units or earnings. It is conceivable for a company with low growth in revenues to deliver high earnings growth, at least over short periods.

A better way of thinking about growth is to use the financial balance sheet construct that we developed in the last chapter. Rather than focus on operating measures such as revenue or earnings growth, we can look at the proportion of a

firm's value that comes from existing investments as opposed to growth assets. If growth companies get the bulk of their value from growth as growth assets, mature companies must get the bulk of their value from existing investments (see figure 7.1 below):

Figure 7.1: A Financial Balance Sheet for a Business

Assets			Liabilities
Existing Investments Generate cashflows today	Investments already made	*Mature companies get most of their value from existing assets.*	Debt
Expected Value that will be created by future investments	Investments yet to be made	*Growth companies derive a significant [proportion of their value from growth assets.*	Equity

We can use the distribution, across all companies, of the proportion of value that comes from mature assets to determine our threshold for mature companies. Thus, if we define mature companies as the top 20% of all companies, in terms of proportion of value from mature assets, the threshold for being a mature company will vary across markets (it will be lower in growth economies like India an China, than in the US or Western Europe) and across time (the threshold will be higher, when economies slow down as they did in 2008 and 2009, and lower, when economies are booming).

Characteristics of Mature Companies

There are clear differences across mature companies in different businesses, but there are some common characteristics that they share. In this section, we will look at what they have in common, with an eye on the consequences for valuation.

1. Revenue growth is approaching growth rate in economy: In the last section, we noted that there can be a wide divergence between growth rate in revenues and earnings in many companies. While the growth rate for earnings for mature firms can be high, as a result of improved efficiencies, the revenue growth is more difficult to alter. For the most part, mature firms will register growth rates

in revenues that, if not equal to, will converge on the nominal growth rate for the economy.

2. Margins are established: Another feature shared by growth companies is that they tend to have stable margins, with the exceptions being commodity and cyclical firms, where margins will vary as a function of the overall economy. While we will return to take a closer look at this sub-group later in the book, event these firms will have stable margins across the economic or commodity price cycle.
3. Competitive advantages? The dimension on which mature firms reveal the most variation is in the competitive advantages that they hold on to, manifested by the excess returns that they generate on their investments. While some mature firms see excess returns go to zero or become negative, with the advent of competition, other mature firms retain significant competitive advantages (and excess returns). Since value is determined by excess returns, the latter will retain higher values, relative to the former, even as growth rates become anemic.
4. Debt capacity: As firms mature, profit margins and earnings improve, reinvestment needs drop off and more cash is available for servicing debt. As a consequence, debt ratios should increase for all mature firms, though there can be big differences in how firms react to this surge in debt capacity. Some will choose not to exploit any or most of the debt capacity and stick with financing policies that they established as growth companies. Others will over react and not just borrow, but borrow more than they can comfortably handle, given current earnings and cash flows. Still others will take a more reasoned middle ground, and borrow money to reflect their improved financial status, while preserving their financial health.
5. Cash build up and return? As earnings improve and reinvestment needs drop off, mature companies will be generating more cash from their operations than they need. If these companies do not alter their debt or dividend policies, cash balances will start accumulating in these firms. The question of whether a company has too much cash, and, if so, how it should return this cash to stock holders becomes a standard one at almost every mature company.

6. Acquisition driven growth: The transition from a growth company to a mature company is not an easy one for most companies (and the managers involved). As companies get larger and investment opportunities internally do not provide the growth boost that they used to, it should not be surprising that many growth companies look for quick fixes that will allow them to continue to maintain high growth. One option, albeit an expensive one, is to buy growth: acquisitions of other companies can provide boosts to revenues and earnings.

One final point that needs to be made is that not all mature companies are large companies. Many small companies reach their growth ceiling quickly and essentially stay as small, mature firms. A few growth companies have extended periods of growth before they reach stable growth and these companies tend to be the large companies that we find used as illustrations of typical mature companies: Coca Cola, IBM and Verizon are all good examples.

Valuation Issues

As with young businesses and growth firms, the characteristics of mature companies can create estimation challenges, during valuations. In this section, we will first focus on the valuation issues in the discounted or intrinsic valuation of mature companies and then look at manifestations of the same problems, when we do relative valuation.

Intrinsic (DCF) Valuation

If the intrinsic value of a firm is the present value of the expected cash flows from its investments, discounted back at a risk adjusted rate, it would seem that mature firms should be easiest to value on that basis. While this is generally true, there are still problems that can lurk under the surface of the long and seemingly stable histories of these firms.

Existing Assets

We categorized mature companies as those that get the bulk of their value from existing assets. Consequently, measuring the value of these assets correctly becomes far more critical with mature firms than it was with the growth firms that

we analyzed in the last two chapters. Since a key input into valuing existing assets is estimating the cash flows that they generate, there are two issues that we encounter when we value mature companies.

a. Managed Earnings: Mature companies are particularly adept at using the discretionary power offered in accounting rules to manage earnings. While they are not necessarily committing accounting fraud or even being deceptive, it does imply that the earnings reported from existing assets by companies that aggressively approach accounting choices will be much higher than the earnings reported by otherwise similar conservative companies. Failing to factor in the differences in "accounting" mindset can lead us to over value the existing assets of the aggressive companies and under value them for conservative companies.
b. Management inefficiencies: When valuing mature companies, we are often lulled by the fact that they have long periods of stable operating history into believing that the numbers from the past (operating margins, returns on capital) are reasonable estimates of what existing assets will continue to generate in the future. However, past earnings reflect how the firm was managed over the period. To the extent that managers may not have made the right investment or financing choices, the reported earnings may be lower than what the existing assets would be able to produce under better or optimal management. If there is the possibility of such a management change on the horizon, we will under value existing assets using reported earnings.

In summary, the notion that existing assets can be easily valued at a mature company, because of its long operating history, is defensible only at well-managed companies or at companies where existing management is so entrenched that there is no chance of a management change.

Growth Assets

There are two ways in which companies can create growth assets. One is to invest in new assets and projects that generate excess returns: this is generally termed organic growth. The other is to acquire established businesses and

companies and thus short circuit the process: this is inorganic or acquisition driven growth. While both options are available to companies at any stage in the life cycle, mature companies are far more likely to take the "acquired growth" route for three reasons. The first is that as companies mature, internal investments start to become scarce, relative to what the firm has available to invest. The second is that as companies get larger, the new investments that they make also have to grow in size to have any impact on overall growth. While it is difficult to find multi-billion dollar internal projects, it is easier to find acquisitions that are of that size and affect the growth rate almost immediately. The third applies in businesses where there is a long lead-time between investment and payoff. In these businesses, there will be a lag between the initial investment in a new asset and the growth generated by that investment. With an acquisition, we are in effect speeding up the payoffs.

So what are the consequences for intrinsic valuation? As a general rule, the value of acquisition driven growth is much more difficult to assess than the value of organic growth. Unlike organic growth, where firms take several small investments each period, acquisitions tend to be infrequent and lumpy: a multi-billion dollar acquisition in one year may be followed by two years of no investments at all and then followed by another acquisition. The consequences of this lumpiness can be seen ifwe relate growth to fundamentals:

Expected Growth Rate = Reinvestment Rate * Return on Capital

Since reinvestment and returns on capital should reflect both organic and acquisition driven growth, we think it is far more difficult to estimate these numbers for acquisitive companies. If we follow the standard practice of using the reinvestment numbers from the most recent financial statement, we risk overstating the value (if there was a large acquisition during the period) or understating it (if it was a period between acquisitions). Computing the return on capital on investments is much more difficult with acquisitions, partly because of the accounting treatment of the price paid and its allocation to goodwill and partly because we have far fewer observations to base our judgments on.

Discount Rates

When estimating discount rates, we start from a position of more strength, when analyzing mature companies, because we have more data to work with. Most mature companies have been publicly traded for extended periods, giving us access to more historical price data, and have settled risk profiles, which stabilizes the data. Thus, estimating equity risk parameters from historical data is more defensible with this group of companies than it was with the growth companies that we analyzed in the last two chapters. In addition, many mature companies, at least in the United States, use the corporate bond route to raise debt, which yields two benefits. The first is that we can get updated market prices and yields on these bonds, which are an input into the cost of debt. The second is that the bonds are accompanied by bond ratings, which not only provide measures of default risk but pathways to default spreads and costs of debt.

There are, however, three estimation issues that can affect discount rate estimates. The first is that mature companies accumulate debt from multiple places, leading to a complex mix of debt – fixed and floating rate, in multiple currencies, senior and subordinated, and with different maturities. Since they often carry different interest rates (and even different ratings), analysts are left with the challenge of how to deal with this complexity, when computing debt ratios and costs of debt. The second is that discount rates (costs of debt, equity and capital) are affected by the firm's mix of debt and equity and the estimates that we obtain from the current price data and ratings are reflective of the current financing mix of the firm. If that mix is altered, the discount rate will have to be re-estimated. The third factor comes into play for those firms that follow the acquisitive route to growth. Acquiring a firm in a different business or with a different risk profile can alter the discount rate for the firm.

Terminal Value

As in any intrinsic valuation, the terminal value accounts for a large share of the overall value of a mature firm. Since mature firms have growth rates that are close to that of the economy, the computation of terminal value may seem both

more imminent and simpler with a mature company than a growth company. While this may be true, there are two factors that can still cause distortions in the computation.

a. Stable growth rates, unstable risk and investment profile: While many mature companies have growth rates low enough to qualify for stable growth (by being less than the growth rate of the economy and the riskfree rate), the other inputs into the valuation may not reflect this maturity. Thus, a firm with a 2% growth rate in revenues and earnings would qualify as a stable firm, based upon its growth rate, but not if its beta is 2.00 and it is reinvesting 90% of its after-tax operating income back into the business. To qualify as a stable growth firm that can be valued using the terminal value equation, the firm should not only have a sustainable growth rate but also have a risk profile of a stable firm (close to average risk) and behave like a stable firm (in terms of reinvestment).
b. Lock in inefficiencies in perpetuity: The cash flows from existing assets and the discount rates that we obtain from past data will reflect the choices made by the firm. To the extent that the firm is not managed optimally, the cash flows may be lower and the discount rate higher than it would have been for the same firm with a different management. If we lock in current values (margins, returns on investment and discount rates), when estimating terminal value, and the firm is poorly run, we are in effect under valuing the firm by assuming that the current practices will continue forever.

The assumption that a firm is in stable growth and can be valued using a terminal value equation cannot be made easily, even for mature firms.

Relative Valuation

With mature companies, with positive revenues and earnings and book values that are meaningful, we have a luxury of riches when it comes to relative valuation. We can estimate revenues, earnings and book value multiples and compare how a company is priced, relative to other companies like it.

a. Too many choices? The problem, though, is that while finding a multiple that works and comparable companies is easier with mature firms than with the growth firms that we analyzed in the last two chapters, the fact that each multiple that we use gives us a different estimate of value can be problematic. Put another way, relative valuation is a subjective process, where the same company can be assigned very different values, depending upon whether we are using a firm or equity multiple, whether that multiple is stated as a function of revenues, earnings and book value and the companies we pick to be its comparables. With mature firms, the problem we face is not that we cannot estimate a relative value but that there are too many values to pick between.

b. Management change: The multiples that we compute of revenues, earnings and book value reflect the mature firm as it is managed today. To the extent that changing the management of the firm could change all of these numbers, we are faced with the same problem as we were with discounted cash flow valuation. How do we best reflect, in a relative valuation, the potential for management change and the consequent increase in value? The problem is magnified, though, because the same issue of how a different management can affect operating numbers also affects all of the other companies that are used as a comparable firm.

c. Acquisition Noise: Acquisition driven growth, a source of intrinsic valuation angst, contaminates relative valuation. The accounting aftermath of acquisitions – the creation of goodwill as an asset and its subsequent treatment – can affect both earnings and book value, making multiples based on either number dicey.

d. Changing financial leverage: The other factor that can throw a wrench into relative valuations is changing financial leverage. Mature companies are capable of making large changes to their debt ratios overnight – debt for equity swaps, recapitalizations – and some multiples can be affected dramatically by such actions. In general, equity multiples, such as PE and Price to book ratios, will change more as financial leverage changes than enterprise value or firm multiples, that are based upon the collective value of both debt and equity. A stock buyback, using borrowed funds, can reduce market capitalization dramatically (by reducing the shares outstanding), but will have a much smaller impact on enterprise value (since we are

replacing equity with debt). For the same reason, equity earnings (earnings per share, net income) will change when firms alter debt ratios.

Valuation Solutions

Looking at both the estimation issues related to valuation and the dark side as it manifests itself in the valuation of mature companies, there are two aspects of mature companies that we have to be able to deal with well, in order to value them correctly. The first is how best to value growth at mature companies, especially when the growth is generated by acquisitions. The other is how to assess the impact on value of changing the way a mature company is run – as we noted earlier, there can be significant operating and financial inefficiencies in mature firms that have endured for long periods, under existing management.

Growth and Acquisitions

Some firms grow primarily through acquisitions and valuing these firms poses challenges. If acquisitions are in the company's past and are unlikely to occur in the future, their effect is already embedded in value and there is no future impact. If acquisitions are part of the long-term strategy of the firm, the task becomes more difficult since we have to assess how much the firm will spend on acquisitions in the future and what return on capital it will generate on these acquisitions. To the extent that acquisitions generate a return on capital that is greater (lower) than the cost of capital, they will add to (detract from) value.

Changing management

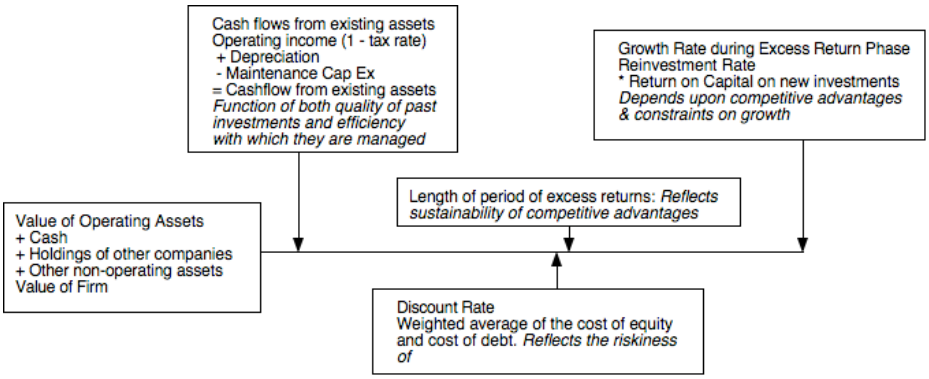
If the key to valuing mature companies is assessing the potential change in value from changing the way they are run, it is critical that we come up with better ways of assessing that effect. In this section, we will begin by looking at the potential for enhancing value at any firm, first by focusing on changes in operations and then examining how financing policy and strategy can alter value. We will extend the discussion by bringing both the potential for value change and the possibility of making that change into a measure of the expected value of control. We will close the section by evaluating the implications of the expected value of control not only

in valuing mature companies but also in estimating the premium that we should pay for voting rights.

Operating Restructuring

When valuing a company, our forecasts of earnings and cash flows are built on assumptions about how the company will be run. If these numbers are based upon existing financial statements, we are, in effect, assuming that the firm will continue to be run the way it is now. In this section, we will look at how changes in operations manifest themselves in valuation, using the intrinsic value framework that we have already used extensively in this book. In this approach, the value of a firm is a function of five key inputs. The first is the cash flow from assets in place or investments already made, the second is the expected growth rate in the cash flows during what we can term a period of both high growth and excess returns (where the firm earns more than its cost of capital on its investments), the third is the length of time before the firm becomes a stable growth firm and the fourth is the discount rate reflecting both the risk of the investment and the financing mix used to fund it. The final element represents cash, cross holdings and other non-operating assets that the firm may hold that augment the value of operating assets. Figure 7.2 captures all five elements:

Figure 7.2: Determinants of Value



A firm can increase its value by increasing cash flows from current operations, increasing expected growth and the period of high growth, by reducing its composite cost of financing and managing its non-operating assets better. In this section, we will focus on all of these inputs, other than the cost of financing.

1. Increase Cash Flows From Assets In Place

The first place to look for value is in the existing assets of the firm. These assets reflect investments that have already been made and that generate the firm's current operating income. To the extent that these investments earn less than the cost of capital, or are earning less than they could, if optimally managed, there is potential for value enhancement. In general, actions taken to increase cash flows from assets in place can be categorized into the following groups:

- *Asset Redeployment*: To the extent that the assets of a business are poorly invested, you can increase the cash flows and value of the firm by divesting poorly performing assets[27] or by moving assets from their existing uses to ones that generate higher value. One example would be a retail firm that owns its stores deciding that the store spaces would be worth more developed as commercial real estate instead of being used in retailing.
- *Improved operating efficiency*: When a firm's operations are riddled with inefficiencies, reducing or eliminating these inefficiencies will translate into an increase in operating cash flows and value. Thus, a telecommunications firm that is overstaffed should be able to generate value by reducing the size of its workforce. A steel company that is losing money because of outdated equipment in its plants may be able to increase its value by replacing them with newer, more efficient equipment. In recent years, manufacturing companies in developed markets like the United States and Western Europe have been able to generate substantial savings in costs by moving their operations to emerging markets where labor costs are lower.

---

[27] At first sight, divesting businesses that are earning poor returns or losing money may seem like the ticket to value creation. However, the real test is whether the divestiture value exceeds the value of continuing in the business; if it is, divestiture makes sense. After all, when a business is earning poor returns, it is unlikely that a potential buyer will pay a premium price for it.

- *Reduce tax burden*: It is every firm's obligation to pay its rightful due in taxes but not to pay more than its fair share. If a firm can legally reduce its tax burden, it should do so. A multinational firm may be able to reduce its taxes by moving more of its operations (and the ensuing earnings) to lower tax locales. Risk management can also play a role in reducing taxes by smoothing out earnings over periods; spikes in income can subject a firm to higher taxes.
- *Reduce capital maintenance and working capital investments*: A significant portion of after-tax operating income is often reinvested in the firm not to generate future growth but to maintain existing operations. This reinvestment includes capital maintenance (which is capital expenditure designed to maintain and replace existing assets) and investments in inventory or accounts receivable. Much of this reinvestment may be unavoidable, because assets age and firms need working capital to generate sales. In some firms, though, there may be potential for savings, especially in working capital. A retail firm that maintains inventory at 10% of sales, when the average for the sector is only 5%, can increase cash flows substantially if it can bring its inventory levels down to industry standards.

2. Increase Expected Growth

A firm with low current cash flows can still have high value if it is able to grow quickly during the high growth period. As noted earlier, higher growth can come either from new investments or from more efficiently utilizing existing assets.

- With new investments, higher growth has to come from either a <u>higher reinvestment rate or a higher return on capital on new investments or both</u>. Higher growth does not always translate into higher value, since the growth effect can be offset by changes elsewhere in the valuation. Thus, higher reinvestment rates usually result in higher expected growth but at the expense of lower cash flows, since more reinvestment reduces free cash

flows at least in the near term.[28] To the extent that the return on capital on the new investments is higher (lower) than the cost of capital, the value of the business will increase (decrease) as the reinvestment rate rises. Similarly, higher returns on capital also cause expected growth to increase, but value can still go down if the new investments are in riskier businesses and there is a more than proportionate increase in the cost of capital.

- With existing assets, the effect is more unambiguous, with higher returns on capital translating into higher growth and higher value. A firm that is able to increase its return on capital on existing assets from 2% to 8% over the next 5 years will report healthy growth and higher value.

Which of these two avenues offers the most promise for value creation? The answer will depend upon the firm in question. For mature firms with low returns on capital (especially when less than the cost of capital), extracting more growth from existing assets is likely to yield quicker results, at least in the short term. For smaller firms with relatively few assets in place, generating reasonable returns, growth has to come from new investments that generate healthy returns.

3. Lengthen the Period of High Growth

As noted above, every firm, at some point in the future, will become a stable growth firm, growing at a rate equal to or less than the economy in which it operates. In addition, growth creates value only if the return on investments exceeds the cost of capital. Clearly, the longer high growth and excess returns last, other things remaining equal, the greater the value of the firm. Note, however, that no firm should be able to earn excess returns for any length of period in a competitive product market, since competitors will be attracted by the excess returns into the business. Thus, implicit in the assumption that there will be high growth, in conjunction with excess returns, is also the assumption that there exist some barriers to entry that prevent firms from earning excess returns for extended time periods.

---

[28] Acquisitions have to be considered as part of capital expenditures for reinvestment. Thus, it is relatively easy for firms to increase their reinvestment rates but very difficult for these firms to maintain high returns on capital as they do so.

Given this relationship between how long firms can grow at above-average rates and the existence of barriers to entry, one way firms can increase value is by augmenting existing barriers to entry and coming up with new barriers to entry. Another way of saying the same thing is to note that companies that earn excess returns have significant competitive advantages. Nurturing these advantages can increase value.

4. Manage non-operating Assets

In the first three components of value creation, we have focused on ways in which a firm can increase its value from operating assets. A significant chunk of a firm's value can derive from its non-operating assets – cash and marketable securities and holdings in other companies. To the extent that these assets are sometimes mismanaged, there is potential for value enhancement here.

- Cash: While cash and marketable securities are by themselves neutral investments, earning a fair rate of return (a low one, but a fair one given the risk and liquidity of the investments), there are two scenarios where it can be value destructive. The first is when cash is invested at below market rates. A firm with $ 2 billion in a cash balance held in a non-interest bearing checking account is clearly hurting its stockholders. The second arises if investors are concerned that cash will be misused by management to make poor investments or acquisitions. In either case, there will be a discount applied to cash to reflect the likelihood that management will misuse the case the consequences of such misuse. If cash is being discounted, returning some or all of this cash to stockholders in the form of dividends or stock buybacks will make stockholders better off.
- Cross holdings are difficult to value, especially when they are in subsidiary firms with different risk and growth profiles than the parent company. It is not surprising that firms with substantial cross holdings in diverse businesses often find these holdings being undervalued by the market. In some cases, this undervaluation can be blamed on information gaps, caused by the failure to convey important details on growth, risk and cash flows on cross holdings to the

markets. In other cases, the undervaluation may reflect market skepticism about the parent company's capacity to manage its cross holding portfolio; consider this a conglomerate discount.[29] If such a discount applies, the prescription for increased value is simple. Spinning off or divesting the cross holdings and thus exposing their true value should make stockholders in the parent company better off.

Financial Restructuring

The one element of value that we did not address in the last chapter was the cost of financing. In this chapter, we will look at two aspects of financing that affect the cost of capital, and through it, the value that we derive for a firm. First, we will look at how best to reflect changes the mix of debt and equity used to fund operations in the cost of capital. Second, we will look at how the choices of financing (in terms of seniority, maturity, currency and other add-on features) may affect the cost of funding and value.

Changing financial mix

The question of whether changing the mix of debt and equity can alter the value of a business has long been debated in finance. While the answer to some may seem obvious – debt after all is always less expensive than equity – the choice is not that simple. In this section, we will first lay out the trade off on debt versus equity in qualitative terms and then consider three tools that can be used to assess the effect of financing mix on value.

Debt versus Equity: The trade off

Debt has two key benefits, relative to equity, as a mode of financing. First, the interest paid on debt financing is tax deductible, whereas cash flows to equity (such as dividends) are generally not.[30] Therefore, the higher the tax rate, the greater the tax benefit of using debt. This is absolutely true in the United States and partially

---

[29] Studies looking at conglomerates conclude that they trade at a discount of between 5 and 10% on the value of the pieces that they are composed of.

[30] This is clearly the case in the United States. There are other markets, such as Brazil, where equity cash flows also provide tax advantages. Even in those markets, the tax advantages for debt tend to be higher than the tax advantages for equity.

true in most parts of the world. The second benefit of debt financing is more subtle. The use of debt, it can be argued, induces managers to be more disciplined in project selection. That is, the managers of a company funded entirely by equity, and with strong cash flows, have a tendency to become lazy. For example, if a project turns sour, the managers can hide evidence of their failure under large operating cash flows, and few investors notice the effect in the aggregate. But if those same managers had to use debt to fund projects, then bad projects are less likely to go unnoticed. Since debt requires the company to make interest payments, investing in too many bad projects can lead to financial distress or even bankruptcy, and managers may lose their jobs.

Relative to equity, the use of debt has three disadvantages—an expected bankruptcy cost, an agency cost, and the loss of future financing flexibility.

- The expected bankruptcy cost has two components. One is simply that as debt increases, so does the probability of bankruptcy. The other component is the cost of bankruptcy, which can be separated into two parts. One is the direct cost of going bankrupt, such as legal fees and court costs, which can eat up to a significant portion of the value of the assets of a bankrupt firm. The other (and more devastating) cost is the effect on operations of being perceived as being in financial trouble. Thus, when customers learn that a company is in financial trouble, they tend to stop buying the company's products. Suppliers stop extending credit, and employees start looking for more reliable employment elsewhere. Borrowing too much money can create a downward spiral that ends in bankruptcy.
- Agency costs arise from the different and competing interests of equity investors and lenders in a firm. Equity investors see more upside from risky investments than lenders to. Consequently, left to their own devices, equity investors will tend to take more risk in investments than lenders would want them to and to alter financing and dividend policies to serve their interests as well. As lenders become aware of this potential, they alter the terms of loan agreements to protect themselves in two ways. One is by adding covenants to these agreements, restricting investing, financing and dividend policies in the future;

these covenants create legal and monitoring costs. The other is by assuming that there will be some game playing by equity investors and by charging higher interest rates to compensate for expected future losses. In both instances, the borrower bears the agency costs.

- As firms borrow more money today, they lose the capacity to tap this borrowing capacity in the future. The loss of future financing flexibility implies that the firm may be unable to make investments that it otherwise would have liked to make, simply because it will be unable to line up financing for these investments.

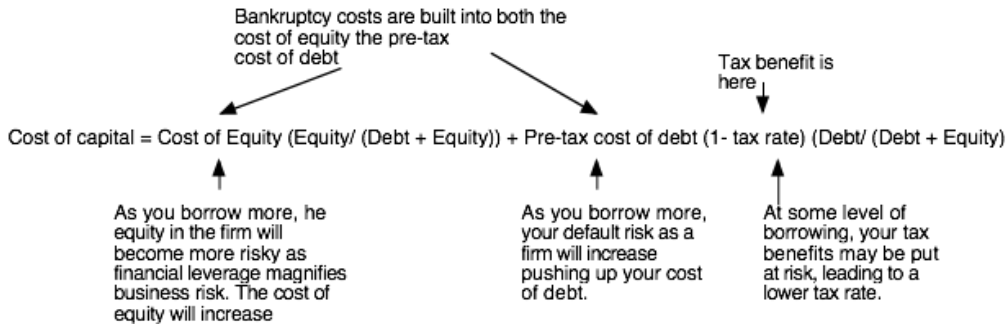
The effect of financing mix on value: The cost of capital approach

In the cost of capital approach, the optimal financing mix is the one that minimizes a company's cost of capital. In effect, we keep operating cash flows fixed and assume that changing debt changes only the cost of capital. By minimizing the cost of capital, we are maximizing firm value. This approach has its roots in the discounted cash flow model for valuing a firm, where expected cash flows to the firm (prior to debt payments but after taxes and reinvestment needs) are discounted back at the cost of capital. If a company can keep its cash flows unchanged and lower its cost of capital, it will increase its present value. Therefore, the optimal debt ratio is the one at which the cost of capital is minimized.

At first sight, the answer to what will happen to the cost of capital as the debt ratio is increased seems trivial, given that the cost of debt is almost always lower than the cost of equity for a business. However, that solution misses the dynamic effects of introducing debt into a business. To see these effects, consider the two components that drive the cost of capital— the cost of equity and cost of debt. As the company borrows more money, its equity will become riskier. Even though it has the same operating assets (and income), it now has to make interest payments, and financial leverage magnifies the risk in equity earnings. Thus, the cost of equity will be an increasing function of the debt ratio. Furthermore, as borrowing increases, so does default risk, which, in turn, increases the cost of debt. The trade off on debt's effect on the cost of capital can be summarized as follows: replacing equity with debt has the positive effect of replacing a more expensive mode of funding with a

less expensive one but in the process the increased risk in both debt and equity will push up the costs of both components, creating a negative effect. Whether the cost of capital increases or decreases will be a function of which effect dominates. Figure 7.3 captures the trade off:

Figure 7.3: Trade off on Cost of Capital and Debt



The cost of capital approach relies on sustainable cash flow to determine the optimal debt ratio. The more stable and predictable a company's cash flow and the greater the magnitude of these cash flows—as a percentage of enterprise value—the higher the company's optimal debt ratio can be. Furthermore, the most significant benefit of debt is the tax benefit. Higher tax rates should lead to higher debt ratios. Based on the insights offered so far, the best candidates for large amounts of financial leverage will be mature or declining companies that have large, reliable cash flows. Growth companies—companies with their best days ahead of them—are not good candidates for high financial leverage because such companies have high market values, relative to cash flows, and usually need to plough these cash flows back into the business (rather than pay interest expenses) to generate future growth.

Illustration 7.1: The Cost of Capital Approach – Hormel Foods

In January 2009, we looked at Hormel Foods and computed a cost of capital of 6.79%, based upon the existing debt ratio of 10.39%. The cost of capital approach, in conjunction with the default spreads in January 2009, was used to derive the costs of debt and equity for Hormel at different debt ratios, with the results summarized in Table 7.1:

Table 7.1: Costs of Equity, Debt and Capital: Hormel Foods

Debt Ratio	Beta	Cost of Equity	Bond Rating	Interest rate on debt	Tax Rate	Cost of Debt (after-tax)	WACC	Firm Value (G)
0%	0.78	7.00%	AAA	3.60%	40.00%	2.16%	7.00%	$4,523
10%	0.83	7.31%	AAA	3.60%	40.00%	2.16%	6.80%	$4,665
20%	0.89	7.70%	AAA	3.60%	40.00%	2.16%	6.59%	$4,815
30%	0.97	8.20%	A+	4.60%	40.00%	2.76%	6.57%	$4,834
40%	1.09	8.86%	A-	5.35%	40.00%	3.21%	6.60%	$4,808
50%	1.24	9.79%	B+	8.35%	40.00%	5.01%	7.40%	$4,271
60%	1.47	11.19%	B-	10.85%	40.00%	6.51%	8.38%	$3,757
70%	1.86	13.52%	CCC	12.35%	40.00%	7.41%	9.24%	$3,398
80%	2.70	18.53%	CC	14.35%	38.07%	8.89%	10.81%	$2,892
90%	5.39	34.70%	CC	14.35%	33.84%	9.49%	12.01%	$2,597

We use the unlevered beta of 0.78 to estimate the levered beta at 10% increments on the debt ratio, up to 90 percent debt. . The table also shows the effect of the rising debt ratio on the company's bond ratings, interest rate on debt, tax rate, cost of debt, weighted average cost of capital (WACC), and firm value.

The cost of equity and cost of debt both rise as debt increases, but the cost of capital drops and the firm value increases, at least initially. The benefits of debt exceed its costs, until the debt reaches 30 percent, at which point, the cost of capital starts climbing again and the firm value begins to drop. To minimize the cost of capital for Hormel, the optimal debt ratio would be about 30 percent, or $1.4 billion in debt.

Financing Type

The fundamental principle in designing the financing of a firm is to ensure that the cash flows on the debt match as closely as possible the cash flows on the asset. Firms that mismatch cash flows on debt and cash flows on assets (by using short term debt to finance long term assets, debt in one currency to finance assets in a different currency or floating rate debt to finance assets whose cash flows tend to be adversely impacted by higher inflation) will end up with higher default risk, higher costs of capital and lower firm values. To the extent that firms can use derivatives and swaps to reduce these mismatches, firm value can be increased.

Converting this intuitive statement about matching financing to assets and its effect on default risk into specifics can be difficult. In many cases, mismatching financing to assets shows up only after it has created a crisis: a firm that has used

short term funding to finance long term assets is unable to refinance its debt and has to put its assets up for sale. We would suggest that there is a much simpler lesson embedded in the financing matching principle. Companies often use a bewildering array of debt and justify this complexity on the basis of cheapness, defined purely in terms of interest payments. Note that, if we follow this path, short term debt will be cheaper than long term debt in most periods as will borrowing money in lower inflation currencies (yen, dollar, euro) will be cheaper than borrowing money in high inflation currencies (peso, ruble). Rather than trying to assign different costs to each layer of debt, we would recommend a consolidation of all debt with a composite cost of debt attached to it; this composite cost should reflect the overall default risk of the firm (rather than the default risk of a specific bond or debt) and the cost of borrowing long term (even if the firm uses short term debt).

The Expected Value of Control

By considering the effects of operating and financing changes on value explicitly, rather than attaching an arbitrary control premium, we can get a much better handle on the value of control. In this section, we will first consolidate and summarize the effects of changing management on value and then look at the likelihood that we can make this change. The product of these analyses will be the expected value of control, which we then use to examine a wide array of valuation issues.

The Value of Changing Management

If we consider value to be the end result of the investment, financing and dividend decisions made by a firm, the value of firm will be a function of how optimal (or sub-optimal) we consider a firm's management to be. If we estimate a value for the firm, assuming that existing management practices continue, and call this a status quo value and re-estimate the value of the same firm, assuming that it is optimally managed, and call this estimate the optimal value, the value of changing management can be written as:

Value of management change = Optimal firm value – Status quo value

The value of changing management will be a direct consequence of how much we can improve the way the firm is run. The value of changing management will be zero in a firm that is already optimally managed and substantial for a firm that is badly managed.

Retracing the steps through value, it should also be quite clear that the pathway to value enhancement will vary for different firms. Sub-optimal management can manifest itself in different ways for different firms. For firms where existing assets are poorly managed, the increase in value will be primarily from managing those assets more efficiently – higher cash flows from these assets and efficiency growth. For firms where investment policy is sound but financing policy is not, the increase in value will come from changing the mix of debt and equity and a lower cost of capital. Table 7.2 considers potential problems in existing management, fixes to these problems and the value consequences:

Table 7.2: Ways of Increasing Value

Potential Problem	Manifestations	Possible fixes	Value Consequence
Existing assets are poorly managed	Operating margins are lower than peer group and return on capital is lower than the cost of capital	Manage existing assets better. This may require divesting some poorly performing assets.	Higher operating margin and return on capital on existing assets -> Higher operating income Efficiency growth -> in near term as return on capital improves
Management is under investing (It is too conservative in exploiting growth opportunities)	Low reinvestment rate and high return on capital in high growth period	Reinvest more in new investments, even if it means lower return on capital (albeit > cost of capital)	Higher growth rate and higher reinvestment rate during high growth period -> Higher value because growth is value creating.
Management is over investing (It is investing in value destroying new investments)	High reinvestment rate and return on capital that is lower than cost of capital	Reduce reinvestment rate until marginal return on capital is at	Lower growth rate and lower reinvestment rate during high growth period -> Higher

		least equal to cost of capital	value because growth is no longer value destroying
Management is not exploiting possible strategic advantages	Short or non-existent high growth period with low or no excess returns.	Build on competitive advantages	Longer high growth period, with larger excess returns -> Higher value
Management is too conservative in its use of debt	Debt ratio is lower than optimal (or industry average)	Increase debt financing	Higher debt ratio and lower cost of capital -> Higher firm value
Management is over using debt	Debt ratio is higher than optimal	Reduce debt financing	Lower debt ratio and lower cost of capital -> Higher firm value
Management is using wrong type of financing	Cost of debt is higher than it should be, given the firm's earning power	Match debt up to assets, using swaps, derivatives or refinancing	Lower cost of debt and cost of capital -> Higher firm value
Management holds excess cash and is not trusted by the market with the cash.	Cash and marketable securities are a large percent of firm value; Firm has poor track record on investments.	Return cash to stockholders, either as dividends or stock buybacks	Firm value is reduced by cash paid out, but stockholders gain because the cash was discounted in the firm's hands.
Management has made investments in unrelated companies.	Substantial cross holdings in other companies that are being undervalued by the market.	As a first step, try to be more transparent about cross holdings. If that is not sufficient, divest cross holdings	Firm value is reduced by divested cross holdings but increased by cash received from divestitures. When cross holdings are under valued, the latter should exceed the former.

Illustration 7.2: A Valuation of Hormel Foods- Status Quo versus Optimal

To value control at Hormel, we valued the firm twice: the first time with existing management continuing to run the firm and once with an optimal management team in place. For the status quo value, we assumed that the firm

would stay with its existing financing mix (10.5% debt) and that they will stay with their existing investment policy, thus preserving current reinvestment rates and returns on capital for the next 3 years. At the end of the third year, we assumed that the firm would be in stable growth and that excess returns would fade to zero. Table 7.3 summarizes the assumptions used to value Hormel Foods under the status quo:

Table 7.3: Inputs for Valuation: Hormel Foods – Status Quo

	High Growth Period	Stable growth phase
Length of High Growth Period =	3	After year 3
Growth Rate =	2.75%	2.35%
Debt Ratio used in Cost of Capital Calculation=	10.39%	10.39%
Beta used for stock =	0.83	0.90
Riskfree rate =	2.35%	2.35%
Risk Premium =	6.00%	6.00%
Cost of Debt =	3.60%	4.50%
Tax Rate =	40.00%	40.00%
Cost of capital =	6.79%	7.23%
Return on Capital =	14.34%	7.23%
Reinvestment Rate =	19.14%	32.52%

There are two things to note about the high growth phase. The first is that the high growth period is short and the growth rate during the period is anemic (2.75%), reflecting the conservative reinvestment policy of the existing management; the returns on capital on existing investments is healthy (14.34%) but the firm reinvests only 19.14% of its after-tax operating income back. The second is that the low growth rate that we have estimated, by itself, is not sufficient to allow us to use a stable growth model because it is higher than the risk free rate and the excess returns generated currently (almost 7.45%) are not compatible with a stable firm. Table 11.13 summarizes our estimates of cash flows for the first 3 years and the present value of these cash flows, for Hormel Foods:

Table 11.13: Cash Flows and Value Today

	Current	1	2	3
EBIT * (1 - tax rate)	$315	$324	$333	$342
- Reinvestment	$60	$62	$64	$65
Free Cashflow to Firm	$255	$262	$269	$276
Cost of capital		6.79%	6.79%	6.79%

Present value		**$245**	**$236**	**$227**

In stable growth, we not only move the growth rate down to the riskfree rate but also assume that the return on capital drops to equal the cost of capital of 7.23%; the cost of capital increases marginally because we increased the beta in stable growth.

Reinvestment rate = $\frac{g_{stable}}{ROC_{stable}} = \frac{2.35\%}{7.23\%} = 32.52\%$

Terminal Value $= \frac{\text{After - tax Operating Income}_5\,(1+g_{stable})(1-\text{Reinvestement Rate})}{(\text{Cost of capital}_{stable} - g_{stable})}$

$= \frac{342(1.0235)(1-.3252)}{(.0723-.0235)} = \$4{,}840 \text{ million}$

Adding the present value of the terminal value, discounted back 3 years at the current cost of capital of 6.79%, to the present value of the cash flows for the first 3 years gives us a value for the operating assets of $4,682 million. Adding the cash holdings of the firm ($155 million), subtracting out debt ($491 million) and the value of management options outstanding ($53 million) yields the value of equity in common stock, which when divided by the number of shares outstanding (134.53 million) generates a value per share of $31.91.

Value per share = $\frac{4682+155-491-53}{134.53} = \$31.91/share$

To value the firm under optimal management, we made three key changes.

a. <u>More debt based financing</u>: Based upon our analysis of Hormel's financing mix in the last section, we increased the debt ratio from 10.56% to 20%. Even allowing for the higher risk in equity (beta goes up to 0.90), the cost of capital for the firm decreases to 6.63% in the high growth phase and to 6.74% in stable growth.

b. <u>Higher reinvestment rate</u>: We assumed that the firm would be more aggressive in seeking out new investments, using a higher reinvestment rate of 40%, but assumed that the return on capital would drop as a result to 14%.

c. Longer growth period: Hormel has several key brand names in its stable and we will assume that it can exploit these brand names to generate excess returns for a longer time periods – 5 years instead of 3.

The resulting valuation inputs are summarized in table 11.14:

Table 11.14: Valuation Inputs: Hormel Foods – Optimal Management

	High Growth	*Stable growth*
Length of High Growth Period =	5	Forever
Growth Rate =	5.60%	2.35%
Debt Ratio used in Cost of Capital Calculation=	20.00%	20.00%
Beta used for stock =	0.90	0.90
Riskfree rate =	2.35%	2.35%
Risk Premium =	6.00%	6.00%
Cost of Debt =	3.60%	4.50%
Tax Rate =	40.00%	40.00%
Cost of capital	0.10%	6.74%
Return on Capital =	14.00%	6.74%
Reinvestment Rate =	40.00%	34.87%

The cash flows, terminal value and the value of the firm today that emerge from these assumptions are shown in table 11.15:

Table 11.15: Cashflows and Value today: Optimal Management

	Current	1	2	3	4	5
EBIT * (1 - tax rate)	$315	$333	$351	$371	$392	$414
- Reinvestment	$131	$133	$141	$148	$157	$165
Free Cashflow to Firm	$184	$200	$211	$223	$235	$248
Terminal value						$6282
Cost of capital		6.63%	6.63%	6.63%	6.63%	6.63%
Present value		$187	$185	$184	$182	$5,655
Value of operating assets	$5,474					

To complete the story, we make the same adjustments for cash, debt and management options that we did in the status quo valuation to arrive at a value per share.

$$\text{Value per share} = \frac{5474 + 155 - 491 - 53}{134.53} = \$37.80/\text{share}$$

The value per share that we obtain for Hormel Foods, with a different management team in place, is $37.80, an increase of $ 5.89 over the status quo value per share. That would represent the overall value of control at Hormel Foods.

The Probability of Changing Management

While the value of changing management in a badly managed firm can be substantial, the increased value will be created only if management policies are changed. While this change can sometimes be accomplished by convincing existing managers to modify their ways, all too often it requires replacing the managers themselves. If the likelihood of management change happening is low, the expected value of control will also be low. In this section, we first consider the mechanisms for changing management, and then some of the factors that determine the likelihood of management change.

Mechanisms for changing management

It is difficult to change the way a company is run, but in general, there are four ways in which it can be done.

1. Activist investors: The first is a variation of moral or at least economic suasion, where one or more large institutional investors introduce shareholder proposals designed primarily to improve corporate governance, holding the threat of more extreme action over the heads of managers. A mix of pension funds and private investors has shown a willingness to confront incumbent managers. These activist investors, with the weight of their large stockholdings, are able to present proposals to stockholders to change policies that they feel are inimical to shareholder interests. Often, these proposals are centered on corporate governance; changing the way the board of directors is chosen and removing anti-takeover clauses in the corporate charter are common examples.
2. Proxy contests: The second is a proxy contest, where incumbent managers are challenged by an investor who is unhappy with the way the firm is run, for proxy votes; with sufficient votes, the investor can get representation on the board and may be able to change management policy. In most companies, investors vote with their feet – selling their stockholdings when dissatisfied – and thus concede power to incumbent managers. In some companies, however, activist investors compete with incumbent managers for the proxies of individual investors, with the intent of getting their nominees for the board elected. While they may not

always succeed at winning majority votes, they do put managers on notice that they are accountable to stockholders. There is evidence that proxy contests occur more often in companies that are poorly run, and that they create significant changes in management policy and improvements in operating performance.

3. Replacing Management The third is to try to replace the existing managers in the firm with more competent manager. In publicly traded firms, this will require a board of directors that is willing to challenge management. Top management turnover at most firms is usually a consequence of retirement or death and the successor usually follows in the incumbent's footsteps. In some cases, though, top managers are forced out by the board, because of displeasure over their performance, and a new management is brought in to head the firm. This provides an opening for a reassessment of the firm's current management policies and for significant changes. While forced management turnover was uncommon outside the United States until recent years, it is becoming more frequent.
4. Hostile acquisition: The fourth and most extreme is a hostile acquisition of the firm by an investor or another firm; the incumbent management is usually replaced after the acquisition and management policy is revamped. Investor pressure, CEO turnover and proxy contests represent internal processes for management discipline. When these fail, the only weapon that stockholders have left is to hope that the firm will become the target of a hostile acquisition, where the acquirer will take over the company and change the way it is run. For hostile acquisitions to be effective as management disciplining mechanism, several pieces have to fall into place. First, firms that are badly managed and run should be targeted for acquisitions. Second, the system should give potential hostile acquirers a reasonable chance of success; the bias towards incumbency should be negligible or small. Third, the acquirer has to change both the managers and the management policies of the target company after the acquisition.

Determinants of Management Change

There is a strong bias towards preserving incumbent management at firms, even when there is widespread agreement that the management is incompetent or does not have the interests of stockholders at heart. Some of the difficulties arise from the institutional tilt towards incumbency and others are put in place to make management change difficult, if not impossible. In general, there are four determinants of whether management will be changed at a firm:

1. Institutional concerns: The first group of constraints on challenging incumbent management in companies that are perceived to be badly managed and badly run is institutional. Some of these constraints can be traced to difficulties associated with raising the capital needed to fund the challenge, some to state restrictions on takeovers and some to inertia.
2. Firm-specific constraints: There are some firms where incumbent managers, no matter how incompetent, are protected from stockholder pressure by actions taken by these firms. This protection can take the form of anti-takeover amendments to the corporate charter, elaborate cross holding structures and the creation of shares with different voting rights. In some cases, the incumbent managers may own large enough stakes in the firm to stifle any challenge to their leadership. The time-honored way for protecting incumbent management is to issue shares with different voting rights. In its most extreme form, the incumbent managers hold all of the shares with voting rights and issue only non-voting shares to the public. In effect, this allows the insiders in these firms to control their destiny with a small percentage of all outstanding stock. More generally, firms can accomplish the same objective by issuing shares with different voting rights.
3. Corporate Holding Structures: Control can be maintained over firms with a variety of corporate structures including pyramids and cross holdings. In a pyramid structure, an investor uses control in one company to establish control in other companies. For instance, company X can own 50% of company Y and use the assets of company Y to buy 50% of company Z. In effect, the investor

who controls company X will end up controlling companies Y and Z, as well. Studies indicate that pyramids are a common approach to consolidating control in family run companies in Asia and Europe. In a cross holding structure, companies own shares in each other, thus allowing the group's controlling stockholders to run all of the companies with less than 50% of the outstanding stock. The vast majority of Japanese companies (keiretsus) and Korean companies (chaebols) in the 1990s were structured as cross holdings, immunizing management at these companies from stockholder pressure.

4. Large Shareholder/Managers: In some firms, the presence of a large stockholder as a manager is a significant impediment to a hostile acquisition or a management change. Consider, a firm like Oracle, where the founder/CEO, Larry Ellison, owns almost 30% of the outstanding stock. Even without a dispersion of voting rights, he can effectively stymie hostile acquirers. Why would such a stockholder/manager mismanage a firm when it costs him or her a significant portion of market value? The first reason can be traced to hubris and ego. Founder CEOs, with little to fear from outside investors, tend to centralize power and can make serious mistakes. The second is that what is good for the inside stockholder, who often has all of his or her wealth invested in the firm may not be good for the other investors in the firm.

While the determinants of management change can be listed, it is far more difficult to quantitatively estimate the probability that it will occur. We can however make subjective judgments of this probability by looking at firms where the top manager (CEO) is forced out by the board. In recent years, researchers have examined when forced CEO turnover is most likely to occur.

- The first factor is stock price and earnings performance, with forced turnover more likely in firms that have performed poorly relative to their peer group and to expectations.[31] One manifestation of poor management is overpaying

[31] Warner, J., R. Watts and K. Wruck, 1988, *Stock Prices and Top Management Changes*, Journal o Financial Economics, v20, 461-492; Murphy, K and J. Zimmerman, 1993, *Financial Performance surrounding CEO Turnover*, Journal of Accounting and Economics, v16, 273-316; Puffer, S. and J.B.

on acquisitions, and there is evidence that CEOs of acquiring firms that pay too much on acquisitions are far more likely to be replaced than CEOs who do not do such acquisitions.[32]

- The second factor is the structure of the board, with forced CEO changes more likely to occur when the board is small[33], is composed of outsiders[34] and when the CEO is not also the chairman of the board of directors.[35]
- The third and related factor is the ownership structure; forced CEO changes are more common in companies with high institutional and low insider holdings.[36] They also seem to occur more frequently in firms that are more dependent upon equity markets for new capital.[37]
- The final factor is industry structure, with CEOs more likely to be replaced in competitive industries.[38]

In summary, firms where you see forced CEO change share some characteristics with firms that are targets of hostile acquisitions – they are poorly managed and run – but they tend to have much more effective boards of directors and more activist investors who are able to change management without turning over the firm to a hostile acquirer.

Weintrop, 1991, *Corporate Performance and CEO Turnover: The Role of Performance Expectations*, Administrative Science Quarterly, v36, 1-19;

[32] Lehn, K. and M. Zhao, 2004, *CEO Turnover after Acquisitions*: *Do Bad Bidders get fired?*, Working Paper, University of Pittsburgh.

[33] Faleye, O., 2003, *Are large boards poor monitors? Evidence from CEO turnover*, Working Paper, SSRN. Using a proportional hazard model, he finds that every additional director on the board reduces the probability of a forced CEO change by 13%.

[34] Weisbach, M., 1988, *Outside Directors and CEO Turnover*, Journal of Financial Economics, v20, 431-460.

[35] Goyal. V.K., and C.W. Park, 2001, *Board Leadership Structure and CEO Turnover*, Journal of Corporate Finanee, v8, 49--66.

[36] Dennis, D.J., D.K. Dennis and A. Sarin, 1997, *Ownership Structure and Top Executive Turnover*, Journal of Financial Economics, v45, 193-221.

[37] Hillier, D., S. Linn and P. McColgan, 2003, *Equity Issuance, Corporate Governance Reform and CEO Turnover in the UK*, Working Paper, SSRN. They find that CEO are more likely to be forced out just before new equity issues or placings.

[38] DeFondt, M.L. and C.W. Park, 1999, *The effect of competition on CEO Turnover*, Journal of Accounting and Economics, v27, 35-56.

Illustration 11.7: The Probability of Control Changing – Hormel Foods

While making a precise estimate of the probability of control changing may be difficult with Hormel Foods, the fact that the Hormel Foundation holds 47.4% of the outstanding stock in the company is a key factor. While the foundation is run by independent trustees, it retains strong links with the incumbent managers and is unlikely to acquiese to a hostile acquisition that will change key parts of the company. Management change, if it does come, will have to be made with the agreement of the foundation. Consequently, we will estimate a probability of only 10% of the change occurring; in effect, the firm has to be under extreme duress before the foundation will step in and agree to a change.

Implications

Once we have a measure of the expected value of control, it is useful not just to acquirers who are to trying to buy a firm but to any investor in the firm. The market price that we observe for a publicly traded stock should reflect the expected value of control, as should the premium that we observe for voting shares, relative to non-voting shares.

Expectations and Stock Prices

To see how the expected value of control shows up in stock prices, assume that you live in a world where management change never happens and that the market is reasonably efficient about assessing the values of the firms that it prices. In this scenario, every company will trade at its status quo value, reflecting both the strengths and weaknesses of existing management. Now assume that you introduce the likelihood of management change into this market, either in the form of hostile acquisitions or CEO changes. If the market remains reasonably efficient, the stock price of every firm should rise to reflect this likelihood:

Market value = Status quo value + (Optimal value – Status quo value)* Probability of management changing

The degree to which this will affect stock prices will vary widely across firms, with the expected value of control being greatest for badly managed firms where there is

a high likelihood of management turnover and lowest for well managed firms and for firms where there is little or no chance of management change.

There are many who will be skeptical about the capacity of markets to make these assessments with any degree of accuracy and whether investors actually try to estimate the expected value of control. The evidence indicates that while markets may not use sophisticated models to make these assessments, they do try to value and price in control.

To the extent that the expected value of control is already built into the market value, there are important implications for acquirers, investors and researchers:

a. *Paying a premium over the market price can result in over payment*: If the current market price incorporates some of all of the value of control, the effect of management change on market value (as opposed to status quo value) will be small or non-existent. In a firm where the market already assumes that management will be changed and builds it into the stock price, acquirers should be wary of paying a premium on the current market price even for a badly managed firm. Consider an extreme example. Assume that you have a firm with a status quo value of $ 100 million and an optimally managed value of $ 150 million and that the market is already building in a 90% chance that the management of the firm will change in the near future. The market value of this company will be $ 145 million. If an acquirer decides to pay a substantial premium (say $ 40 million) for this firm, based upon the fact that the company is badly managed, he will overpay substantially; in this example, he will pay $ 185 million for a company with a value of $ 150 million.

b. *Anything that causes market perception of the likelihood of management change to shift can have large effects on all stocks*. A hostile acquisition of one company, for instance, may lead investors to change their assessments of the likelihood of management change for all companies and to an increase in stock prices. Since hostile acquisitions often are clustered in a particular sector – oil companies in the 1980s, for instance – it is not surprising that a hostile acquisition of a single company often leads to increases in stock prices of companies in its peer group.

c. *Poor corporate governance = Lower stock prices*: The price of poor corporate governance can be seen in stock prices. After all, the essence of good corporate governance is that it gives stockholders the power to change the management of badly managed companies. Consequently, stock prices in a market where corporate governance is effective will reflect a high likelihood of change for bad management and a higher expected value for control. In contrast, it is difficult, if not impossible, to dislodge managers in markets where corporate governance is weak. Stock prices in these markets will therefore incorporate lower expected values for control. The differences in corporate governance are likely to manifest themselves most in the worst managed firms in the market.

Illustration 7.3: Market Prices and the Expected Value of Control

Consider the valuation of Hormel Foods in illustration 11.7. We estimated both the status quo and the optimal value of the equity in the company and arrived at the following results:

	Value of Equity	Value per share
Status Quo	$ 4,293 million	$ 31.91 per share
Optimally managed	$ 5,085 million	$ 37.80 per share

In illustration 11.*, we estimated the probability of management change happening at only 10%. If we assume that these are all reasonable estimates, the expected value per share for Hormel will be:

Expected Value per share = $31.91 (.90) + #37.80 (.10) = $32.51

If our assessments are correct, the stock should be trading at $32.51. The actual market price at the time of this valuation was about $32.25. Assuming that both the market price and our values per share are correct, the market price can be written in terms of a probability of control changing and the expected value of control:

Expected value per share = Status Quo Value + Probability of control changing * (Optimal Value – Status Quo Value)

$ 32.25 = $ 31.91 + Probability of control changing ($37.80 - $31.91)

The market is attaching a probability of 5.6% that management policies can be changed.

Value Plays

There are two ways in which investors can profit from investing in mature companies. One is to adopt the classic value approach and look for mature companies that continue to deliver high returns and solid earnings, but which investors have turned sour on, either because they are not the flavor of the moment in the market or they are boring. These are the companies that you hope to find with traditional value screens – they trade at low multiples of earnings and book value, have good management and still deliver reasonable earnings.

The other is a more perverse strategy. You try to find mature companies that are poorly managed and run, that are perceived to be so by the market, but where you perceive the possibility of a change in management that can generate substantially higher value. The indicator variables for these factors include:

a. Poor performance: If poor management is manifested in poor investment and financing choices, it should be revealed in a company's profitability measures and its financing mix. In particular, you could look for firms that have low operating margins relative to the sector, low returns on capital relative to cost of capital and very low debt ratios that are not consistent with their strong operating cash flows and mature company status.
b. Market perception matches performance: The rest of the market has to not only agree with your assessment of current management quality but also see little chance of changing the management. These two factors will contribute to keeping the market value low.
c. Potential for large increase in value with change in management: There has to be potential for a significant increase in value if the firm were managed differently. While this may seem a given, if the firm is currently badly managed, it requires two other pieces to fall into place.
    - The assets that are currently deployed poorly (and earnings low margins) can either be redeployed quickly and put to more profitable use or can be sold to buyers who can put it to that use. For example, a retail firm that has

stores occupying valuable real estate is in a better position to redeploy those assets than a manufacturing company with inefficient factories.

- To the extent that the firm has too little debt, having ready access to fairly priced debt can make changing the financing mix much easier. Thus, firms in countries with well developed capital markets may be able to tap their debt potential quicker and more cheaply than firms in some emerging markets.

d. Change on the horizon: If you wait for the announcement of a hostile acquisition or a management change, it will be too late, since the market price will respond and the stock will no longer be a bargain. Thus, you have to see the potential or possibility of management change before most other investors do. To make this difficult judgment, you will want to look at companies that bear the following characteristics:

- The game is not tilted unfairly in favor of incumbent management. This can happen if the state restricts takeovers or management change, incumbent managers hold shares with disproportionate voting rights or the company has anti takeover amendments on the books.
- If the company has been badly managed for a while, there has to be a catalyst for management change. This catalyst can be an ageing CEO with long tenure, a founding family that wishes to exit the company and/or interest from an activist investor.

Assuming that you can find a poorly managed company, priced as such, with potential for change and a catalyst, your payoff will occur when the company's management does change. Note that you may not be right on each one of your picks, but if you are right on even a small proportion, your portfolio will generate high returns.

Conclusion

In general, it is easier to value mature companies than high growth companies, because we have more relevant historical data on earnings, cash flows and revenues. That does not mean, however, that mature companies pose no challenges. In this chapter, we looked at two aspects of mature companies that may

create problems in valuations, the shift that some mature firms make to acquisitions to jump start growth and the possibility that a change in management could create a change in value.

Valuing acquisitive companies requires us to deal with two estimation issues – the lumpiness of acquisitions, where a big acquisition in one year is followed by inactivity in the other and the accounting treatment of acquisitions, which is inconsistent with the treatment of internal investment, and skews key measures such as return on capital. Rather than treat acquisitions separately, we included them with capital expenditures, when estimating reinvestment, and used averages over extended periods to overcome the year-to-year volatility. When measuring returns, we removed goodwill from invested capital, assuming that it represents a premium for growth assets.

To deal with the possibility of management change and the consequences for value, we value a firm twice, once with the incumbent management (status quo) and once with a different and better management in place (optimal). The difference between the two numbers is the overall value of control and when multiplied by the probability of management changing, yields the expected value of control. This has consequences not only in acquisitions, but also in valuing any publicly traded company and in assigning premiums to voting shares.

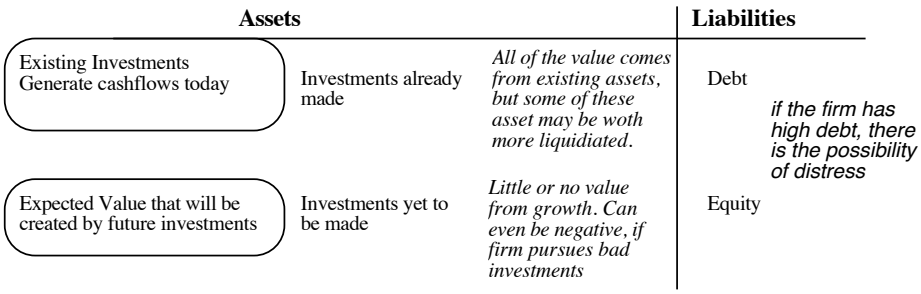
Chapter 8: Valuing Declining and Distressed Companies

In chapter 5, we examined firms at the earliest stages in the life cycle and wrestled with how best to build in the reality that most young, idea firms do not survive to become healthy business. In chapter 6, we moved forward in the life cycle to look at growth firms, and our biggest challenge became estimating growth rates, as firms became larger and competition entered the business. In chapter 7, we continued further up the life cycle to look at mature companies, a grouping that most growth companies seek to avoid but inevitably join, and evaluated the valuation consequences of acquisitions and management changes. In this chapter, we will turn to the final phase of the life cycle, which is decline and examine the key questions that drive the value of firms that enter this phase.

What is a declining company?

As we noted in the chapter on mature companies, growth companies do not want to become mature companies and mature companies constantly try to rediscover their growth roots. By the same token, no mature company wants to go into decline, with the accompanying loss of earnings and value. So, how do would we differentiate between mature firms and firms in decline? We will use the financial balance sheet, as we did in the earlier chapters in figure 8.1, to illustrate the difference:

Figure 8.1: A Financial Balance Sheet for a Declining Business



There are two key areas where mature companies are different from companies in decline. The first is on the asset side of the balance sheet. If mature companies get

the bulk of their value from existing assets and less from growth assets, declining companies get none (or close to none) of their value from growth assets. In fact, it is not uncommon for declining companies to actually lose value from growth investments, especially if they decide to reinvest at rates well below their cost of capital. Existing assets not only represent all of the value of declining firms but some firms may actually get more from liquidating or divesting these assets than from continuing operations. On the liability side, declining firms face much more dire consequences from being over levered, since they cannot count on higher earnings in the future to cover debt obligation. In other words, decline and distress often go hand in hand.

Characteristics of Declining Companies

In this section, we will look at characteristics that declining companies tend to share, with an eye towards the problems that they create for analysts trying to value these firms. Note again that not every declining company possesses all of these characteristics but they do share enough of them to make these generalizations.

1. Stagnant or declining revenues: Perhaps the most telling sign of a company in decline is the inability to increase revenues over extended periods, even when times are good. Flat revenues or revenues that grow at less than the inflation rate is an indicator of operating weakness. It is even more telling if these patterns in revenues apply not only to the company being analyzed but to the overall sector, thus eliminating the explanation that the revenue weakness is due to poor management (and can thus be fixed by bringing in a new management team).
2. Shrinking or negative margins: The stagnant revenues at declining firms are often accompanied by shrinking operating margins, partly because firms are losing pricing power and partly because they are dropping prices to keep revenues from falling further. This combination results in deteriorating or negative operating income at these firms, with occasional spurts in profits generated by asset sales or one time profits.

3. Asset divestitures: If one of the features of a declining firm is that existing assets are sometimes worth more to others, who intend to put them to different and better uses, it stands to reason that asset divestitures will be more frequent at declining firms than at firms earlier in the life cycle. If the declining firm has substantial debt obligations, the need to divest will become stronger, driven by the desire to avoid default or to pay down debt.
4. Big payouts – dividends and stock buybacks: Declining firms have few or any growth investments that generate value, existing assets that may be generating positive cashflows and asset divestitures that result in cash inflows. If the firm does not have enough debt for distress to be a concern, it stands to reason that declining firms not only pay out large dividends, sometimes exceeding their earnings, but also buy back stock.
5. Financial leverage – the downside: If debt is a double-edged sword, declining firms often are exposed to the wrong edge. With stagnant and declining earnings from existing assets and little potential for earnings growth, it is not surprising that many declining firms face debt burdens that are overwhelming. Note that much of this debt was probably acquired when the firm was in a healthier phase of the life cycle and at terms that cannot be matched today. In addition to difficulties these firms face in meeting the obligations that they have committed to meet, they will face additional trouble in refinancing the debt, since lenders will demand more stringent terms.

Valuation Issues

The issues that we face in valuing declining companies come from their common characteristics. Most of the valuation techniques we use for businesses, whether intrinsic or relative, are built for healthy firms with positive growth and they sometimes break down when a firm is expected to shrink over time or if distress is imminent.

Intrinsic (DCF) Valuation

The intrinsic value of a company is the present value of the expected cashflows of the company over its lifetime. While that principle does not change with declining firms, there are practical problems that can impede valuations.

Existing Assets

When valuing the existing assets of the firm, we estimate the expected cash flows from these assets and discount them back at a risk-adjusted discount rate. While this is standard valuation practice in most valuations, there are two aspects of declining companies that may throw a wrench in the process.

- Earning less than cost of capital: In many declining firms, existing assets, even if profitable, earn less than the cost of capital. The natural consequence is that discounting the cash flows back at the cost of capital yields a value that is less than the capital invested in the firm. From a valuation perspective, this is neither surprising nor unexpected: assets that generate sub-par returns can be value destroying.
- Divestiture effects: If existing assets earn less than the cost of capital, the logical response is to sell or divest these assets and hope that the best buyer will pay a high price for them. From a valuation perspective, divestitures of assets create discontinuities in past data and make forecasts more difficult to make. To see how divestitures can affect past numbers, consider a firm that divested a significant portion of its assets midway through last year. All of the operating numbers from last year – revenues, margins and reinvestment – will be affected by the divestiture, but the numbers for the year will reflect the operating results from the portion of the year prior to the divestiture. Similarly, risk parameters such as betas, where we use past prices or returns, can be skewed by divestitures of assets midway through the time period. For the forecasting consequences, try estimating the revenues and earnings numbers for a firm that is expected to divest a large portion of its assets over the next few years. Not only do we have to pinpoint the assets that will be divested and the effects of the divestiture on operating revenues and earnings, but we also have to estimate the

proceeds from the divestitures. Put another way, a divestiture, by itself, does not affect value, but what we expect to receive in comparison for the divested assets can affect value.

Thus, what makes the valuation of existing assets of a declining firm difficult is that the value you derive from these assets in cash flows may be lower than that value you obtain from divesting the assets.

Growth Assets

Declining firms derive little from growth assets, and the valuation of these assets should therefore not have a significant impact on value. While this is generally true, we have to leave open the possibility that some declining firms are in denial about their status and continue to invest in new assets, as it they had growth potential. If these assets earn less than the cost of capital, the value of adding new assets will be negative and reinvestment will lower the value of the firm.

We can actually go further. If we viewed divestitures as reductions in capital invested, the reinvestment rate for a declining firm can be negative in future years, which will lead to negative growth rates, at least for the foreseeable future. Analysts who have learned their valuation fundamentals at healthier companies often are uncomfortable with the notion of negative earnings growth rates and cash flows that exceed earnings, but that combination will characterize many declining firms.

Discount Rates

If the cost of capital is a weighted average of the costs of debt and equity, what is it about declining firms that makes it difficult to estimate these numbers? First, the large dividends and buybacks that characterize declining firms can have an effect on the overall value of equity and on the debt ratios that we use in the computation. In particular, returning large amounts of cash to stockholders will reduce the market value of equity, through the market price, with dividends, and the number of shares, with stock buybacks. If debt is not repaid proportionately, the debt ratio will increase, which will then affects of costs of debt, equity and capital.

Second, the presence of distress can have significant effects on both the cost of equity and debt. The cost of debt will increase as default risk increases and some

rated firms will see their ratings drop to junk status – BB, B or lower. If operating earnings drop below interest expenses, the tax benefits of debt will also dissipate, leading to further pressure upwards on the after-tax cost of debt. As debt to equity ratios climb, the cost of equity should also increase, as equity investors will see much more volatility in earnings. From a measurement standpoint, analysts who use regression betas, which reflect changes in equity risk on a lagged basis, may find themselves facing the unusual scenario of a cost of equity that is lower than the pre-tax cost of debt.[39]

Terminal Value

The standard procedures for estimating terminal value have been examined in detail in prior chapters. We first estimate a growth rate that a firm can sustain forever, with the caveat that the growth cannot exceed the growth rate of the economy, with the riskfree rate acting as a proxy. We follow up by making reasonable assumptions about what a firm can generate as excess returns in perpetuity and use this number to forecast a reinvestment rate for the firm. We complete the process by estimating a discount rate for the terminal value computation, with the qualifier that the risk parameters used should reflect the fact that the company will be a more stable one.

At each stage of this process, declining and distressed firms pose special challenges. At the first stage, we have to consider the possibility, which will be significant, that the firm being valued will not make it to stable growth; many distressed firms will default and go out of business or be liquidated. Even if a firm is expected to survive to reach steady state, the expected growth rate in perpetuity will not only be well below the growth rate of the economy and inflation, but in some cases, it can even be negative. Essentially, the firm will continue to exist but

---

[39] There are two reasons why regression betas may not adequately capture the risk in the equity of a firm in financial distress. The first is that they are computed using a long period of historical returns; to the extent that the firm was healthy (or healthier than it is today) over some of that period, the regression beta will understate the true beta. The second is that during periods of distress, the stock prices of companies tend to be volatile but often with no relation to the market; they may move up an down as a function of debt restructuring talks or rumors of impending bankruptcy. Since the regression beta captures how a stock moves with the market, it may actually decrease during periods of financial distress.

get progressively smaller over time, as its market shrinks. At the second step, the biggest estimation issues we face will arise with declining firms that are earning well below their cost of capital currently, with no reason for optimism about the future. In effect, the most reasonable assumption to make about this firm may be that it will continue to earn a return on its capital that is below the cost of capital in perpetuity. This will have consequences for both reinvestment and the terminal value. Finally, the problems that we mentioned in the previous section relating to discount rates can spill over into the terminal value computation. In other words, a distressed firm can have sky-high costs of equity and debt at the moment and leaving these numbers at or even close to current levels can cause terminal values to implode.

Relative Valuation

Analysts who fall back on relative valuation as a solution to the problems of valuing declining or distressed firms, using intrinsic valuation, will find themselves confronting the estimation issues that we listed in the earlier sections either explicitly or implicitly when they use multiples and comparables.

a. Scaling Variable: All multiples have to be scaled to common variables, which can be broadly categorized into revenues, earnings, book value or sector specific measures. With distressed companies, earnings and book values can become inoperative very quickly, the former because many firms in decline have negative earnings and the latter because repeated losses can drive the book value of equity down and into negative territory. We can scale value to revenues, but we are then implicitly assuming that the firm will be able to turn its operations around and deliver positive earnings.
b. Comparable firms: There are two possible scenarios that we can face when valuing declining firms. One is when we are valuing a declining firm in a business where the remaining firms are all healthy and growing. Since markets value declining firms very differently from healthy firms, the challenge in this case is working out how much of a discount the declining firm should trade at, relative to the values being attached to healthy firms.

We face the second scenario when we are valuing a declining/distressed firm in a sector where many or even all of the firms share the same characteristic. In this case, not only do our choices of what multiple to use become more limited, but we have to consider how best to adjust for the degree of decline in a firm. For instance, in early 2009, Ford, GM and Chrysler all showed signs of distress but GM was in the worst shape, followed by Chrysler and Ford.

c. Incorporating Distress: While analysts often come up with creative solutions to the first two problems – using multiples of future earnings and controlling for differences in decline, for instance – the presence of distress puts a wild card in the comparison. Put another way, when firms are not only in decline but are viewed as distressed, we should expect those firms that have a higher likelihood of distress to trade at lower values (and hence at lower multiples) than firms that are more likely to make it. Unless we explicitly control for distress, we will find ourselves concluding, based on relative valuation, that the first group of firms are under valued and the second growth over valued.

By now, the message should be clear. Any issues that skew intrinsic valuations also skew relative valuations. The symptoms of decline – negative growth rates, poor or negative margins, flat revenues- and the potential for failure – caused by too much debt and declining earnings – will not disappear as issues just because we base our value on a revenue multiple.

Valuation Solutions

Flat revenues, declining margins and the potential for distress makes valuing distressed companies tricky. In this section, we will look at how best to navigate the challenges in both the intrinsic and relative valuation framework.

Intrinsic Valuation

We will build our analysis of declining firms around two key questions. The first is whether the decline that we are observing in a firm's operations is reversible or permanent. In some cases, a firm may be in a tailspin but can pull out of it, with a

new management team in place. The second relates to whether the firm faces a significant possibility of distress; not all declining firms are distressed.

To assess the question of reversibility, we can look at a firm's own history as well as the state of other companies in the sector. A firm that has gone through cycles of good and bad times, as is often the case with cyclical and commodity companies, is more likely to be able to move back to health than a firm that has not been subject to these cycles. Similarly, it can be argued that a firm that is doing badly in a sector filled with healthy firms has problems that are more attributable to poor management than to fundamentals; with better management, the firm may be able to revert back to health, if not growth. In contrast, a firm that is doing badly in a sector filled with poorly performing firms, with no obvious macro economic reasons for the problems, has problems that will not be remedied by changing managers.

To evaluate distress, the place to start is the debt load that the firm may have accumulated over time. Declining firms with significant debt obligations are more likely to face default, resulting in cessation of operations and liquidation by creditors. If the firm is rated by a bond ratings agency, we would expect to see a low credit rating, generally below investment grade. Declining firms that do not face these fixed obligations should be able to survive even with poor earnings and no or even negative growth. The way we estimate value will be different, depending upon the combination of reversibility and decline that we observe in a given firm. We will consider four possible combinations below:

1. <u>Reversible decline, Low distress</u>: If a firm has flat revenues and declining margins, but the problems are fixable, we would follow the framework we developed in chapter 11 for valuing control. We would first value the firm, run by existing management, with continuing decline in operations; the resulting status quo value will be low. We would then revalue the firm, assuming that better management is in place and that the decline is reversed; this optimal value should be much higher. Finally, we would estimate a probability of management changing and compute an expected value, based up the status quo and optimal values.

   Expected Value = Status Quo Value (1 – Probability of Management Change)

+ Optimal Value (Probability of Management Change)

2. Irreversible decline, Low distress: If a firm's poor performance cannot be attributed to poor management and is not easily fixable, we cannot revalue the firm with operating improvements. However, the assets that are deployed by the firm may be put to better use by others (in other businesses) and thus be worth more in a divestiture. Since the firm is under no pressure to sell its assets to meet fixed obligations, it can liquidate the assets in an orderly manner, waiting for both the best time and the highest bidder for each asset. The expected proceeds from this orderly liquidation will provide an alternate estimate of value for the firm. The final value that we would attach to the firm would be the higher of the two numbers.[40]

   Expected Value = Maximum (Status Quo Value, Orderly Liquidation Value)

3. Reversible decline, High distress: For firms where there is a high probability of distress, we will consider two courses of action. In the first one, we will try to bring in the probability of distress into the expected cash flows and discount rates, and derive distress-adjusted values. In the second one, we will compute the expected values just as we did in the low distress scenario and then estimate the probability of distress separately. Following up this with an estimate of the proceeds that we can expect to receive in a distress sale will yield the distress-adjusted value of the firm. The fact that decline is reversible, though, may give equity investors in this firm the possibility of large payoffs, if distress is avoided and the firm recovers its bearings, that has the characteristics of an option and can add to equity value.

4. Irreversible decline, High distress: When decline is inevitable and is overlaid with distress, we have a toxic combination for value. As with the previous case, we have to adjust expected values for distress, by either changing the inputs to discounted cash flow or by adjusting the expected no-distress value for the probability of distress. There are two significant differences, from the reversible

---

[40] We are assuming that since managers in this firm will take the right action and liquidate, if that is the course that will deliver higher value. If they don't, we will have to estimate probabilities, just as we did for firms with management deficiencies.

decline scenario, that will depress value. The first is that, if distress occurs, the proceeds from a distress sale with irreversible decline will be lower, both because the pool of buyers is thin (if most firms in the sector are troubled) and because buyers don't see much potential upside. The second is that equity investors have less to gain from the option to liquidate, since the best case value is constrained by the poor quality of the assets.

We summarize the four scenarios in table 8.1 below.

Table 8.1: A framework for dealing with decline and/or distress

	No or low Distress (Not much debt, investment grade rating)	High Distress (High debt commitments, low rating)
Irreversible (Sector in trouble)	1. Value the firm with existing management and expected decline (Going concern value) 2. Value the firm, assuming orderly liquidation of all of its assets. 3. Expected Value = Maximum (Going concern value, Orderly liquidation value)	1. Start with the expected value (irreversible, no distress) 2. Estimate the probability of distress and proceeds from forced liquidation of firm. 3. Re-compute the expected value, adjusting for distress.
Reversible (Firm outlier n healthy sector)	a. Value the firm with existing management and expected decline. b. Value the firm with better management and recovery. c. Expected Value = Status Quo Value (Probability of no management change) + Optimum Value (Probability of management change)	1. Start with the expected value (reversible, no distress) 2. Estimate the probability of distress and proceeds from distress sale of firm. 3. Re-compute the expected value, adjusting for distress. 4. If equity investors run the firm, value the option to liquidate.

I. Irreversible Decline, Low distress

With some firms, the symptoms of operating decline – flat revenues and declining margins –have deep roots that cannot be easily remedied for three reasons. The first is that the overall market for the firm's products and services is shrinking and is expected to keep doing so over time. The second is that all of the firms in the sector share some or many of the same symptoms as the firm being valued. The third is that there is no macroeconomic factor that can be pointed to as the reason for the decline, as is the case with an economic recession for a cyclical firm or the price cycle for a commodity company. A good example would the airline business in much of the developed world since deregulation; in this business, the healthy company has been the exception, rather than the rule, and most companies seem to teeter on the edge of bankruptcy, even in good economic times.

The first step in valuing such companies is to estimate the value as going concerns, notwithstanding the fact that the assets, as invested currently, may be earning less than the cost of capital. In effect, we are valuing the businesses on the assumption that they will continue to be operated, while destroying value for existing investors. This status quo value may be well below the book value of the firm but that is not surprising given the negative excess returns that we are projecting on existing assets. The value will be depressed further, if management insists on adding to its asset base with new investments in the same business.

The second step is to consider a logical alternative to continuing in business. If the assets deployed by the firm can be used elsewhere (in other businesses or by other firms) to generate higher returns, we can consider divesting these assets and liquidating the business over time. Since distress is not a concern, the firm can wait until the right time and the right bidder, and extract the maximum value from divestiture; hence, we will term this an <u>orderly liquidation</u>, which can occur over many years. As for estimating what the divestiture proceeds will be, the answer will vary across sectors and for different assets, but there are a few general propositions that we can develop. The first is that the expected proceeds should be higher than the present value of the cash flows that would have been generated by the assets in the existing use; if this is not the case, the divestiture would not make sense in the first place. Thus, if the asset is expected to generate $ 50 million in perpetuity, in its existing use, with a risk-adjusted discount rate of 10%, the divestiture proceeds should exceed $ 500 million. The second is that there should be little or no illiquidity discount applied to the divestiture value, since there is no urgency associated with the sale.

Once we have both values in had – the value of the firm as a going concern and the value from an orderly liquidation of the firm over time – we would expect the firm to trade at the higher of the two values. In fact, there is an intermediate solution, where a portion of the firm, composed of assets that are more valuable to others, is liquidated, but the rest of the firm continues as a going concern.

Illustration 8.1: Valuing a company in irreversible decline: Sears in September 2008

Sears, a retailer with a long and distinguished history, has seen its fortunes fade over the last 10 years, as its core customer base has abandoned it to go to more upscale retailers like Target, at one end of the spectrum, and to discount retailers like Walmart, at the other end. In the fiscal year ending in February 2008, Sears reported revenues of $50.7 billion, just 3.2% higher than its revenues in 2006, and operating income of $1.54 billion, about 17.7% lower than operating income in 2006. During the 2008 fiscal year, Sears reduce the number of stores it operated and bought back almost $2.9 billion of its own stock. Finally, Sears generated a return on capital of 4.99% on its retail operations for the year, well below its cost of capital of 7.50%. In summary, Sears has all the hallmarks of a company in decline.

We will assume that the firm, while not under any significant threat of default, will shut down less profitable stores over time, liquidating some of its real estate holdings in the process, and become a smaller firm, with higher quality assets.

- Revenues will decrease 5% a year, each year for the next 5 years, from the current level of $50.7 billion to $39.2 billion in the fifth year, as the firm shuts down stores. After year 5, we will assume that the shrinkage will stop and that revenues will grow 2% a year forever.
- The pre-tax operating margin will improve from 3.05% to 4%, in linear increments over the next five years, reflecting both the cost savings from shutting down unprofitable stores and a reversal back to health at the other stores. The tax rate is assumed to remain unchanged at 38%.
- For the first five years, we will use the current cost of capital for Sears, which we estimate to be 7.50%. During the five-year period, as stores are being closed and assets divested, Sears will be reducing its capital invested and collecting proceeds from the divestitures. To estimate the proceeds, we first assumed that the return on capital at Sears will increase from the current level of 4.99% to 7.50%, in linear increments, over the next 5 years, and then backed out the capital invested based upon the after-tax operating income and the return on capital estimate. Using the change in book capital each year as the basis, we then

estimated the divestiture proceeds as a proportion of the book value. Table 8.2 reports the numbers by year:

Table 8.2: Divestiture Proceeds by year

	Current	1	2	3	4	5
Growth rate		-5%	-5%	-5%	-5%	-5%
Revenues	$50,703	$48,168	$45,759	$43,471	$41,298	$39,233
Operating margin	3.05%	3.24%	3.43%	3.62%	3.81%	4.00%
EBIT	$1,548	$1,562	$1,570	$1,574	$1,574	$1,569
EBIT * (1 - tax rate)	$960	$968	$974	$976	$976	$973
Return on capital	4.99%	5.50%	6.00%	6.50%	7.00%	7.50%
Capital Invested	$19,234	$17,606	$16,227	$15,015	$13,939	$12,973
Change in book capital		-$1,628	-$1,379	-$1,212	-$1,077	-$965
Divestiture Proceeds/ Capital		54.08%	58.9%	61.9%	64.7%	67.2%
Divestiture Proceeds ($)		$880	$811	$751	$697	$649

Since we are assuming that the most unprofitable stores will be closed first, we also assume that the divestitures proceeds, as a percentage of book value, will be lower in the earlier years and increase over time. Note, though, that we receive well below 100% of capital invested in every year, ranging from 54% of capital in year 1 to 67% of capital in year 5, reflecting the fact that the investments being sold are delivering poor returns.[41]

- To apply closure in the valuation, we assume that the divestitures end in year 5 and that Sears will revert back to a more traditional stable growth firm after year 5. In keeping with this assumption, the cost of capital after year drops to 7.13% and the return on capital will stay at 7.50% in perpetuity. Finally, we assume that the after-tax operating income will grow 2% a year in perpetuity, thus allowing us to estimate a reinvestment rate and terminal value.

Reinvestment rate = $\frac{g_{stable}}{ROC_{stable}} = \frac{2\%}{7.5\%} = 26.7\%$

Terminal Value $= \frac{\text{After - tax Operating Income}_5\ (1+g_{stable})(1\text{ - Reinvestement Rate})}{(\text{Cost of capital}_{stable}\text{ - }g_{stable})}$

[41] It is possible that we are being too conservative in this estimate, since the buyers of these assets may have no intention of preserving them as retail stores.

$$= \frac{973\ (1.02)(1 - .267)}{(.0713 - .02)} = \$14{,}187$$

Table 8.3 summarizes the cash flows for the next 5 years and the terminal value at the end of the fifth year and the present values of these cash flows (discounted back at the cost of capital of 7.50%).

Table 8.3: Expected Cash flows and Value Today

	Current	1	2	3	4	5	Terminal year
EBIT * (1 - tax rate)	$960	$968	$974	$976	$976	$973	$992
- Reinvestment		-$880	-$811	-$751	-$697	-$649	$265
Free Cashflow to Firm		$1,849	$1,785	$1,727	$1,673	$1,622	$728
Terminal Value						$14,187	
Present Value		$1,720	$1,545	$1,390	$1,252	$11,012	
Cost of capital		7.50%	7.50%	7.50%	7.50%	7.50%	7.13%

Summing up the present values over time yields a value of $16,918 million for the operating assets. Adding cash ($1,622 million), subtracting out debt ($7,728 million) and dividing by the number of shares outstanding (132.01) yields a value per share of $81.91, about 7% higher than the prevailing market price ($76.25).

$$\text{Value per share} = \frac{16{,}918 + 1{,}622 - 7728}{132.01} = \$81.91/\text{share}$$

II. Reversible Decline, Low distress

The history of the corporate world has its share of rebirths, where firms that were viewed as in decline reversed the process and returned to growth or mature status. One example was Harley Davidson, the manufacturer of a cult classic motorcycle. The firm saw sales slip to 32,400 motorcycles in 1982 and reported a loss of about $ 30 million in that year. While many analysts were writing its epitaph,, a new management team devised a strategy built around an loyal customer base and an iconic brand and Harley rebounded to profitability and financial health.

It is worth noting, however, that most troubled firms never turn around and that righting the ship is not easy. Consequently, we should be realistic in our assessments of when decline is reversible and when it is not. Decline is more likely to be reversible if one or more of the following conditions hold:

- The company being analyzed has a history of operating ups and downs, and has come back from decline before. Cycles in revenue growth and margins may then be part of the company's make up and should be considered when valuing it.
- The sector or industry to which the company belongs is, for the most part, healthy and the firm being analyzed is more the exception than the rule. When most of a firm's competitors are growing and making reasonable returns, and the firm is not, it seems reasonable to conclude that decline is the result of choices made by the firm, and that new management could conceivably turn the company around.
- The company is in a business that can benefit from macro economic trends. Even declining companies in a very cyclical business can see their operating results improve if the economy booms.

Note that while we may conclude that decline is likely to be reversed, there are no guarantees.

The first step in valuing declining companies, where reversal is possible, is to estimate a value with existing policies and strategies, notwithstanding past failures. Since we are assuming continuing decline, it is entirely possible that the firm's revenues will stay flat, margins will decrease over time and the returns on capital invested on both existing assets and new investments will be less than the cost of capital, i.e., are value destroying. In effect, we are valuing the firm under the status quo, assuming that it is unable to turn itself around.

The second step is to assume that the firm's fortunes can be turned around either by new management/ownership or existing management changing its policies. Assuming that the firm will revert back to financial health, if not immediately, but in the near future, we can re-estimate the firm's value, with the operating improvements built into the cash flows. If the rest of the sector is healthy, we could assume that the firm's margins and returns on invested capital will revert to industry averages. If not, we can look at the company's own history to get a sense of what it will look like, if it reverts back to health. With these improvements in place, we should derive a higher value for the firm, under new or optimal management, than we did with the status quo.

The third step is estimating a probability of change occurring, using some of the techniques that we described in chapter 11. Using a mix of subjective judgment and quantitative techniques, we can compute the likelihood that management will change. This estimate, though, will change as a function of the external environment, with the entrance of an activist investor into the mix changing the assessment. The expected value for the firm will then reflect the weighted average of the status quo and optimal values, based upon the probability of change.

III. Distress

Growth is not inevitable and firms may not remain as going concerns. In fact, even large publicly traded firms sometimes become distressed for one reason or the other and the consequences for value can be serious. In this section, we will consider first how often firms become distressed and follow up by looking at the costs they face as a consequence. We will close the section by examining why, given the frequency with which firms face distress, that we have historically not paid attention to distress in valuation.

Distress in Discounted Cashflow Valuation

Financial distress is far more common in the real world that most of us assume it to be. In fact, even casual empirical observation suggests that a very large number of firms do not survive and go out of business. Some will fail because they borrow money to fund their operations and then are unable to make these debt payments. Other will fail because they do not have the cash to cover their operating needs.

Consider how we value a firm in a discounted cash flow world. We begin by projecting expected cash flows for a period, we estimate a terminal value at the end of the period that captures what we believe the firm will be worth at that point in time and we then discount the cash flows back at a discount rate that reflects the riskiness of the firm's cash flows. This approach is an extraordinarily flexible one and can be stretched to value firms ranging from those with predictable earnings and little growth to those in high growth with negative earnings and cash flows. Implicit in this approach, though, is the assumption that a firm is a going concern,

with potentially an infinite life. The terminal value is usually estimated by assuming that earnings grow at a constant rate forever (a perpetual growth rate). Even when the terminal value is estimated using a multiple of revenues or earnings, this multiple is derived by looking at publicly traded firms (usually healthy ones). The failure to explicitly consider distress in discounted cash flow valuation will not have a material impact in value if any the following conditions hold:

1. There is no possibility of bankruptcy, either because of the firm's size and standing or because of a government guarantee.
2. Easy access to capital markets allows firms with good investments to raise debt or equity capital to sustain themselves through bad times, thus ensuring that these firms will never be forced into a distress sale.
3. We use expected cash flows that incorporate the likelihood of distress and a discount rate that is adjusted for the higher risk associated with distress. In addition, we have to assume that the firm will receive sale proceeds that are equal to the present value of expected future cash flows as a going concern in the event of a distress sale.

If these conditions do not hold, and it is easy to make an argument that they will not for some firms at some points in time, discounted cash flow valuations will overstate firm value.

Discounted Cash flow Valuation

When will the failure to consider distress in discounted cash flow valuation have a material impact on value? If the likelihood of distress is high, access to capital is constrained (by internal or external factors) and distress sale proceeds are significantly lower than going concern values, discounted cash flow valuations will overstate firm and equity value for distressed firms, even if the cash flows and the discount rates are correctly estimated. An alternative to the standard discounted cash flow model is to separate the going concern assumptions and the value that emerges from it from the effects of distress. To value the effects of distress, we estimate the cumulative probability that the firm will become distressed over the

forecast period, and the proceeds that we estimate we will get from the distress sale. The value of the firm can then be written as:

Firm Value = Going concern value * $(1-\pi_{Distress})$+ Distress sale value * $\pi_{Distress}$

where $\pi_{distress}$ is the cumulative probability of distress over the valuation period. In addition to making valuation simpler, it also allows us to make consistent assumptions within each valuation.

You may wonder about the differences between this approach and the far more conventional one of estimating liquidation value for deeply distressed firms. You can consider the distress sale value to be a version of liquidation value, and if you assume that the probability of distress is one, the firm value will, in fact, converge on liquidation value. The advantage of this approach is that it allows us to consider the possibility that even distressed firms have a chance of becoming going concerns.

Going Concern DCF

To value a firm as a going concern, we consider only those scenarios where the firm survives. The expected cash flow is estimated only across these scenarios and thus should be higher than the expected cash flow estimated in the modified discounted cash flow model. When estimating discount rates, we make the assumption that debt ratios will, in fact, decrease over time, if the firm is over levered, and that the firm will derive tax benefits from debt as it turns the corner on profitability. This is consistent with the assumption that the firm will remain a going concern. Most discounted cash flow valuations that we observe in practice are going concern valuations, though they may not come with the tag attached.

A less precise albeit easier alternative is to value the company as if it were a healthy company today. This would require estimating the cashflows that the firm would have generated if it were a healthy firm, a task most easily accomplished by replacing the firm's operating margin by the average operating margin of healthy firms in the business. The cost of capital for the distressed firm can be set to the average cost of capital for the industry and the value of the firm can be computed.

The danger with this approach is that it will overstate firm value by assuming that the return to financial health is both painless and imminent.

Estimating the Probability of Distress

A key input to this approach is the estimate of the cumulative probability of distress over the valuation period. In this section, we will consider two ways in which we can estimate this probability. The first is a statistical approach, where we relate the probability of distress to a firm's observable characteristics – firm size, leverage and profitability, for instance – by contrasting firms that have gone bankrupt in prior years with firms that did not. This approach has been used effectively to compute credit scores (like the Altman Z score) that predict bankruptcy.

The second is a less data intensive approach, where we use the bond rating for a firm, and the empirical default rates of firms in that rating class to estimate the probability of distress. Altman (2007) has estimated the cumulative probabilities of default for bonds in different ratings classes over five and ten-year periods, following issuance, and the estimates are reproduced in table 12.5 below[42]:

Table 12.5: Bond Rating and Probability of Default – 1971 - 2007

Rating	Cumulative Probability of Distress	
	5 years	10 years
AAA	0.04%	0.07%
AA	0.44%	0.51%
A+	0.47%	0.57%
A	0.20%	0.66%
A-	3.00%	5.00%
BBB	6.44%	7.54%
BB	11.9%	19.63%
B+	19.25%	28.25%
B	27.50%	36.80%
B-	31.10%	42.12%
CCC	46.26%	59.02%
CC	54.15%	66.6%
C+	65.15%	75.16%
C	72.15%	81.03%

[42] Altman, E.I.[43] The reported operating income of $ 163 million was after an impairment charge for disposal of assets of $46 million. The adjusted operating income is $ 209 million.

Unknown
Deleted:

C-	80.00%	87.16%

As elaboration, the cumulative default probability for a bond rated BB at the start of the period is 19.63% over the next 10 years. We can illustrate the use of this approach with Delta Airlines and Las Vegas Sands, two operating companies with significant probability of default at the start of 2009:

Company	Bond Rating	Estimated probability of distress
Delta Airlines	BBB-	13.58%
Las Vegas Sands	B+	28.25%

Estimating Distress Sale Proceeds

Once we have estimated the probability that the firm will be unable to make its debt payments and cease to exist, we have to consider the logical follow-up question. What happens then? As noted earlier in the chapter, it is not distress per se that is the problem but the fact that firms in distress have to sell their assets for less than the present value of the expected future cash flows from existing assets and expected future investments. Often, they may be unable to claim even the present value of the cash flows generated even by existing investments. Consequently, a key input that we need to estimate is the expected proceeds in the event of a distress sale. We have three choices:

a. Estimate the present value of the expected cash flows in a discounted cash flow model, and assume that the distress sale will generate only a percentage (less than 100%) of this value. Thus, if the discounted cash flow valuation yields $ 5 billion as the value of the assets, we may assume that the value will only be $ 3 billion in the event of a distress sale.
b. Estimate the present value of expected cash flows only from existing investments as the distress sale value. Essentially, we are assuming that a buyer will not pay for future investments in a distress sale. In practical terms, we would estimate the distress sale value by considering the cash flows from assets in place as a perpetuity (with no growth).

c. The most practical way of estimating distress sale proceeds is to consider the distress sale proceeds as a percent of book value of assets, based upon the experience of other distressed firms.

Note that many of the issues that come up when estimating distress sale proceeds – the need to sell at below fair value, the urgency of the need to sell – are issues that are relevant when estimating liquidation value.

Illustration 8.2: Valuing Las Vegas Sands with Distress valued separately

To value Las Vegas Sands with distress valued separately, we began with a going concern valuation of Las Vegas Sands, assuming that the firm survives and reverts back to financial health.

- For the fiscal year ended December 2008, Las Vegas Sands reported revenues of $4,390 million and pre-tax operating income of $209 million, yielding pre-tax operating margin of 4.76%.[43] The capital invested in the company at the start of the year was $9,832 million, yielding an abysmal after-tax return on capital of 1.44% (assuming the firm's effective tax rate of 26%)

  Return on capital = $\frac{209(1-.26)}{8975} = 1.72\%$

- To map out a path to recovery, we have to first estimate what we believe to be reasonable profitability measures, if Las Vegas Sands can turn things around. To make these estimates, we first looked at the operating margins and returns on capital reported by the firm over the last 5 years in table 8.4:

Table 8.4: Revenues, Margins and Return on Capital: LVS 2003-2008

Year	Revenues	Operating Income	Pre-tax Margin	Capital invested	ROC
2004	1197	233	19.47%	1575	9.17%
2005	1741	491	28.20%	1810	16.82%
2006	2237	577	25.79%	2791	12.82%
2007	2951	331	11.22%	2049	10.02%
2008	4390	209	4.76%	8974	1.44%

Both the margins and returns earned in 2008 represent a break from a generally profitable past for Las Vegas Sands. We followed by estimating the average pre-tax operating margin (16.96%) and after-tax return on capital (approximately 10%) for casino firms in the United States at the start of 2009. Based on these

numbers, we will assume that Las Vegas Sands, assuming it makes it as a healthy firm, will have a pre-tax operating margin of 17% and earn an after-tax return on capital of 10%.

- To project operating results into the future, we will assume that revenues will grow only 1% next year and 2% the year after, we will assume that the revenue growth rate will pick up, especially as two new casinos in development come online, picking up to 20% in years 3-5, before dropping back to 5% a year from years 6-10. We will also assume that the recovery to the targeted margin will occur gradually over the next 10 years; with pre-tax operating margins improving to 10% by year 5 and then posting a further increase to 17% by year 10; the changes in each period occur in linear increments. Table 8.5 summarizes our forecasts of revenues, margins and operating income each year for the next 10 years; we used a 26% effective tax rate to estimate the after-tax operating income for the first 5 years but gradually move that number up to the marginal tax rate of 38% by year 10.

Table 8.5: Expected Revenues and Operating Income – LVS

Year	Revenues	Revenue growth	Operating margin	Operating income	After-tax Operating income	Reinvestment	FCFF	Debt ratio	Cost of capital
Current	$4,390		4.76%	$209	$155				
1	$4,434	1%	5.81%	$258	$191	-$19	$210	73.50%	9.88%
2	$4,523	2%	6.86%	$310	$229	-$11	$241	73.50%	9.88%
3	$5,427	20%	7.90%	$429	$317	$0	$317	73.50%	9.88%
4	$6,513	20%	8.95%	$583	$431	$22	$410	73.50%	9.88%
5	$7,815	20%	10.00%	$782	$578	$58	$520	73.50%	9.88%
6	$8,206	5%	11.40%	$935	$670	$67	$603	68.80%	9.79%
7	$8,616	5%	12.80%	$1,103	$763	$153	$611	64.10%	9.50%
8	$9,047	5%	14.20%	$1,285	$858	$215	$644	59.40%	9.01%
9	$9,499	5%	15.60%	$1,482	$954	$286	$668	54.70%	8.32%

10	$9,974	5%	17.00%	$1,696	$1,051	$350	$701	50.00%	7.43%

Since much of the capital for the new casinos has already been invested, we will hold capital expenditures down for much of the high growth period; in effect, the company is living off past investments.[44] As a consequence, the reinvestment rate will be negative for the next two years, since there will be significant cash inflows from depreciation charges, before increasing in the rest of the high growth period.[45]

We will begin the valuation with a cost of capital for Las Vegas Sands that reflects its tenuous hold on going concern status. Using the current debt ratio of 73.50%, we estimate a cost of capital of 9.88% for Las Vegas Sands:

Cost of capital = Cost of equity (1- Debt ratio) + After-tax cost of debt (Debt ratio)

= 21.82% (.265) + 5.58% (.735) = 9.88%

However, we will assume that as the firm becomes healthier, its debt ratio will converge on the casino industry average of 50% and that its cost of capital will also move down to 7.43% to reflect the return to financial health. To complete the valuation, we will assume that Las Vegas Sands will be in stable growth after year 10, growing at 3% a year (set equal to the riskfree rate cap) forever. We will also assume that the return on capital will be 10% in perpetuity and that the stable period cost of capital is 7.43% (from table 12.9). The terminal value can then be computed.

$$\text{Reinvestment rate} = \frac{g_{stable}}{ROC_{stable}} = \frac{3\%}{10\%} = 30\%$$

$$\text{Terminal Value} = \frac{\text{After - tax Operating Income}_5\ (1+g_{\text{stable}})(1-\text{Reinvestement Rate})}{(\text{Cost of capital}_{\text{stable}} - g_{\text{stable}})}$$

$$= \frac{1051\ (1.03)(1-.30)}{(.0743-.03)} = \$17{,}129$$

[44] LVS had invested almost $3 billion in new developments in January 2009 that had still not commenced operations.

Bringing together the free cash flows, the terminal value above and the cost of capital, we can compute the value of the operating assets in table 8.6:[46]

Table 8.6: Value of operating assets – LVS

Year	FCFF	Terminal value	Cost of capital	Cumulated cost of capital	PV
1	$210		9.88%	1.0988	$190.79
2	$241		9.88%	1.2075	$199.54
3	$317		9.88%	1.3268	$239.25
4	$410		9.88%	1.4579	$281.12
5	$520		9.88%	1.6021	$324.88
6	$603		9.79%	1.7590	$342.71
7	$611		9.50%	1.9261	$316.98
8	$644		9.01%	2.0997	$306.52
9	$668		8.32%	2.2744	$293.72
10	$701	$17,129.27	7.43%	2.4433	$7,297.83
			Value of operating assets =		$9,793.34

Adding cash ($3,040 million), subtracting out the market value of debt ($7,565 million) and dividing by the number of shares outstanding (641.839 million) yields a value per share of $8.21.

$$\text{Value per share} = \frac{9793 + 3040 - 7565}{641.839} = \$8.21/\text{share}$$

Note that the market value of debt is significantly lower than the face value of almost $ 10.47 billion. However, it is consistent with our assumption that LVS will make it as a going concern.

We can now bring in the probability of distress and the consequences into our final estimate of equity value. We estimated the probability of distress from probability of distress, based upon the B+ rating, of 28.25%, and concluded that the distress sale value of the assets would be lower than the debt outstanding (making equity worthless). The expected value of equity per share in LVS can then be computed:

Expected Value per share = Value per share as going concern (1- Probability of distress) + Value per share in distress (Probability of distress) = $8.21 (1-.7175) + $0.00 (.2825) = $5.89

[46] Since the cost of capital changes over time, we have to compute a cumulated cost. For instance, the cost of capital in year 7 = 1.0988^5*1.979*1.0950=1.9261

If we adjust for the possibility of distress, the value per share is only $5.89, about 35% higher than the stock price of $4.25 in February 2009.

Equity as an Option

In most publicly traded firms, equity has two features. The first is that the equity investors run the firm and can choose to liquidate its assets and pay off other claim holders at any time. The second is that the liability of equity investors in some private firms and almost all publicly traded firms is restricted to their equity investments in these firms. This combination of the option to liquidate and limited liability allows equity to have the features of a call option. In firms with substantial debt and a significant potential for bankruptcy, the option value of equity may be in excess of the discounted cash flow value of equity. When the equity in a firm takes on the characteristics of a call option, we have to change the way we think about its value and what determines its value. There are a number of potential implications for equity investors and bondholders in the firm.

- In discounted cash flow valuation, we argue that equity is worthless if what we own (the value of the firm) is less than what we owe. The first implication of viewing equity as a call option is that equity will have value, even if the value of the firm falls well below the face value of the outstanding debt. While the firm will be viewed as troubled by investors, accountants and analysts, its equity is not worthless. In fact, just as deep out-of-the-money traded call options command value because of the possibility that the value of the underlying asset may increase above the strike price in the remaining lifetime of the option, equity commands value because of the time premium on the option (the time until the bonds mature and come due) and the possibility that the value of the assets may increase above the face value of the bonds before they come due.
- In traditional discounted cash flow valuation, higher risk almost always translates into lower value for equity investors. When equity takes on the characteristics of a call option, we should not expect this relationship to continue to hold. Risk can become

our ally, when we are equity investors in a troubled firm. In essence, we have little to lose and much to gain from swings in firm value.

Relative Valuation

Most valuations in practice, including those of distressed firms, are relative valuations. In particular, firms are valued using multiples and groups of comparable firms. An open question then becomes whether the effects of distress are reflected in relative valuations and, if not, how best to do so.

Distress in Relative Valuation

It is not clear how distress is incorporated into an estimate of relative value. Consider how relative valuation is most often done. We choose a group of firms that we believe are comparable to the firm that we are valuing. Usually, we pick firms in the same business that our firm is in. We then standardize prices by computing a multiple – price earnings, price to book, enterprise value to sales or enterprise value to EBITDA. Finally, we examine how our firm measures up on this multiple, relative to the comparable firms. While this time honored approach is used for distressed firms as well, the issues listed below generally are unique to distressed firms:

1. Revenue and EBITDA multiples are used more often to value distressed firms than healthy firms. The reasons are pragmatic. Multiple such as price earnings or price to book value often cannot even be computed for a distressed firm. Analysts therefore move up the income statement looking for a positive number. For firms that make heavy infrastructure investments, where depreciation and amortization is a significant charge against operating income and there are substantial interest expenses, the EBITDA is often positive while net income is negative. For some firms, even EBITDA is negative and revenue multiples are only multiples that yield positive values.

2. Analysts who are aware of the possibility of distress often consider them subjectively when they compare the multiple for the firm they are analyzing to the industry average. For example, assume that the average telecomm firm trades at 2 times revenues and that the firm we are analyzing trades at 1.25 times revenues. Assume also that the firm has substantially higher default risk than the average

telecomm firm. We may conclude that the firm is not undervalued even though it trades at a significant discount on the average, because of the potential for default. The perils of subjective adjustment are obvious. Barring the most egregious misvaluations, analysts will find a way to justify their prior biases about firms.

Adapting Relative Valuation to Distress

Is there a way in which relative valuation can be adapted to cover distressed firms? We believe so, though the adjustments tend to be much more approximate than those described in the discounted cash flow section. We consider two ways of building distress explicitly into relative valuations. In the first, we compare a distressed company's valuation to the valuations of other distressed companies. In the second, we use healthy companies as comparable companies, but find a way to adjust for the distress that the firm we are valuing is facing.

Choosing the Comparables

To value a distressed firm, we can find a group of distressed firms in the same business and look at how much the market is willing to pay for them. For instance, we

could value a troubled telecomm firm by looking at the enterprise value to sales (or book capital) multiples at which other troubled telecomm firms trade. While there is promise in this approach, it works only if a large number of firms in a sector slip into financial trouble at the same time. In addition, by categorizing firms as distressed or not distressed firms, we run the risk of lumping together firms that are distressed to different degrees.

One possible way to expand this approach is to look at distressed firms across the whole market, rather than just the sector in which the firm operates. This will allow for a larger sample though there is the possible disadvantage that a troubled grocery store may be in a better position (in terms of generating distress sale proceeds) than a troubled technology company.

Illustration 8.3: Choosing distressed comparables

To value Las Vegas Sands, we considered only casino firms with high financial leverage (market debt to capital ratios that exceed 60%). Our objective was to arrive at a sample of casino firms that have a significant likelihood of distress. We used the EBITDA in the most recent trailing 12 months and computed the enterprise value by adding the market value of equity to the book value of debt and subtracting out cash. Table 8.7 summarizes the resulting EV/EBITDA ratios for these firms:

Table 8.7: Distressed Casino Firms

Company	Market Cap	Total Debt	Cash	EV	EBITDA	EV/EBITDA
Codere, S.A.	516.6	1072.8	144.1	1445.3	295.2	4.90
Ameristar Casinos Inc.	561.1	1615.7	68.2	2108.6	289.4	7.29
Las Vegas Sands Corp.	2729	10470	1276	11628.9	812.5	14.31
Groupe Partouche SA	139.1	675	146.9	667.2	178.2	3.74
Boyd Gaming Corp.	431.2	2624.1	123.6	2931.7	372.5	7.87
MGM Mirage (NYSE:MGM)	1548.4	13288.3	250.1	14586.6	1959.6	7.44
Wynn Resorts Ltd.	2747	4917.7	1713.7	5951	714.4	8.33
				Average across firms =		7.70
				Average (without LVS) =		6.60

The average EV/EBITDA multiple, not including LVS, across these firms is 6.60 and Las Vegas Sands looks clearly over valued, relative to the rest of the group. There are, however, a couple of caveats that we would offer. The first is that the use of book value of debt clearly overstates the enterprise value computations, and more so for companies that are more distressed. For instance, using the estimated market value of debt of $7.57 billion instead of the book value would lower LVS's multiple to about 10 times EBITDA.

Considering the Possibility of Distress Explicitly

One of the adaptations that we suggested for discounted cash flow valuation was an explicit assessment of default risk and a firm value that was a weighted estimate of a going concern value and a distress sale value. For a distressed firm in a sector where the average firm is healthy, this approach offers promise. We can estimate the value of the distressed firm using the comparable firms and consider it the going concern value. For instance, if healthy firms in the business trade at 2

times revenues, we would multiple the firm's revenues by 2 to arrive at the going concern value. We could then estimate the firm value as follows:

Firm Value = Going concern relative value * $(1-\pi_{Distress})$+ Distress sale value * $\pi_{Distress}$

The probability of distress and the distress sale value would be estimated just as they were in the last section. This approach makes the most sense when valuing a firm that is distressed in a sector containing mostly healthy firms, since the prior two approaches could not be used here.

In some cases, we may have to use forecasted values for revenues and operating income to arrive at the going concern value, especially if current revenues and operating income are adversely impacted by the overhang of distress.

Illustration 8.4: Forward Multiples and Distress

Consider the forecasts of revenues and EBIT made in table 12.7 for Las Vegas Sands. While the firm is not generating a large operating income currently, we are anticipating an improvement in margins and growth in revenues, resulting in an expected operating income of $1,696 million in year 10; adding the expected depreciation charge of $572 million to this value yields an expected EBITDA of 2,268 million.[47] Using the average enterprise value/EBITDA multiple of 8.25 at which healthy casino firms trade, we can estimate an expected enterprise value in year 10.[48]

Expected Enterprise value in year 10 = $EBITDA_{10}$ * EV/ $EBITDA_{Current\ for\ healthy\ casino\ firms}$

= 2,268 * 8.25 = $18,711 million

We can estimate the present value of this estimated value by discounting back at Las Vegas Sand's cumulated cost of capital, computed earlier in illustration 12.6 to be 2.4433.

Enterprise value today = 18,711/2.4433 = $7,658 million

---

[47] While we have not explicitly forecast depreciation, the depreciation in the current year is $ 509 million and the book capital invested in year 10 is about 25% higher than today's book capital. We scaled up depreciation by the same proportion.

[48] We defined healthy casino firms as those with market debt ratios lower than 50%.

This, of course, is based upon the assumption that Las Vegas Sands will become a healthy firm. Using the probability of survival (71.75%) and distress (28.25%) estimated earlier as well as the distress sale proceeds ($2,769 million) we can value LVS's operating assets today:

Estimated Enterprise Value
= Going Concern Value ($\pi_{\text{Going Concern}}$) + Distress Sale Value (1 - $\pi_{\text{Going Concern}}$)

= 7,658(.7175) + 2769 (.2825) = $ 6,277 million

By ignoring the cash flows over the next 10 years, we are also significantly understating the value of LVS, as a going concern.[49] Adding back the cash balance of the firm ($ 3,040 million) and subtracting out debt ($ 7.565 million) yields a value for the equity:

Enterprise Value	= $ 6,277 million
+ Cash & Marketable Securities	= $ 3,040 million
- Debt	= $7,565 million
Value of Equity	= $1,751 million

Dividing by the 641.839 million shares outstanding yields a value of equity per share of $2.73.

Value Plays

Most investors steer instinctively away from declining firms, and are even more averse when these companies carry large debt burdens. While this may be a sound strategy for a risk averse investor, there is potential for investment gold in these companies, for investors with longer time horizons and strong stomachs.

The first strategy is to focus on declining firms that have low debt burdens and thus face little risk of bankruptcy. So, what would make these firms attractive investments?

- Low price: If the decline in the firm's fortunes is visible to the market (flat revenues and declining margins are tough to hide), the market price will

[49] We could always add back the present value of cash flows, but if we do, we are mixing up intrinsic and relative valuations.

generally decline as well. Consequently, these firms will trade at low multiples of earnings and at below book value.

- Aware management: A low market price alone does not make for a good investment, especially if a company in decline persists in trying to expand and make more bad investments. Consequently, you need managers at these companies who are aware that they have reached the end of the growth road and are willing to not just scale down new investment but liquidate old ones.
- Divestiture potential: As declining firms try to liquidate or unload poorly performing assets, it does help to have buyers who are willing to pay higher values for these assets (than could be generated by the firm). These higher values may be the result of better uses that the assets can be put to or reflect irrational buyers.
- High cash return: As the management holds back on new investment and liquidates old ones, it will see its cash balances rise. This cash has to find its way back to investors either as high dividends or in the form of stock buybacks.

The payoff on this portfolio, assuming that you make the right choices, will be in the form of high cash yield on the portfolio and relatively low price appreciation.

The second and riskier strategy is to direct your money to declining firms with large debt burdens. The additional risk in this strategy comes from the distress potential created by the debt. To succeed with this "vulture" investing strategy, you have to find firms where distress is either more distant or less likely than the market assessment in the equity market. This is a tall order but there are potential indicators:

- Look for distressed companies that have both bonds and stocks that are traded, where the two markets are generating different signals about likely distress. If you are investing in equity, you want to invest in companies where stock prices have collapsed because of the fear of default but bond prices have remained relatively intact.
- If an external entity, such as the government or an agency, can provide backing for the debt, the company may find the breathing room to pay down its debt and escape default. This will require pinpointing the types of companies where this

assistance will be forthcoming, perhaps larger companies with lots of employees, where the social costs of corporate failure are viewed as too large to allow the failure.

If you do invest in distressed companies, it is important that recognize that some or many of these companies will fail and that you may very well lose everything you invested in their equity. Your hope is that those companies that manage to turn themselves around will offer high enough returns to cover your losses.

Conclusion

Looking across the life cycle, it is the firms at either end of the life cycle that seem to pose the most valuation challenges, and sometimes for the same reasons. If the question with young firms is whether they will survive to become profitable businesses, a key issue with declining firms is whether they will survive deteriorating operations and large debt obligations and emerge as going concerns.

In this chapter, we looked at the interplay between decline and distress to develop a framework for valuing declining companies. When decline is irreversible, but distress is not imminent, we argued for valuing the firm twice, once as a going concern and once in an orderly liquidation, and using the higher of the two numbers. When the decline is attributable to poor management and thus reversible, and no distress overhanging the firm, we again argued for two valuations, once with existing management and once with better management, and estimated an expected value, based upon the probability of management change.

When distress is a distinct possibility, we have three choices. In the first, we can develop probability distributions for key variables and run simulations, with distress built into the process. In the second, we can try to adjust the expected cash flows and discount rates in a valuation to reflect the probability of distress occurring and the resulting cash flows. In the third, we value the firm as a going concern and then adjust for the likelihood of distress separately. The contrast here between reversible and irreversible decline shows up in two places with distressed firms. One is that the proceeds from a distress sale are likely to be higher for the first group, where buyers see a potential for turnaround for the assets, than for the

second group. The other is that equity in distressed firms with the potential for a turnaround can have option characteristics.

Section 3: Special Situations in Valuation

Chapter 9: Valuing Financial Service Companies

Banks, insurance companies and other financial service firms pose special challenges for an analyst attempting to value them, for three reasons. The first is the nature of their businesses makes it difficult to define both debt and reinvestment, making the estimation of cash flows much more difficult. The second is that they tend to be heavily regulated and changes in regulatory requirements can have significant effect on value. The third is that the accounting rules that govern bank accounting have historically been very different from the accounting rules for other firms, with assets being marked to market more frequently for financial service firms.

What is a financial service firm?

Any firm that provides financial products and services to individuals or other firms can be categorized as a financial service firm. We would categorize financial service businesses into four groups from the perspective of how they make their money. A *bank* makes money on the spread between the interest it pays to those from whom it raises funds and the interest it charges those who borrow from it, and from other services it offers it depositors and its lenders. *Insurance companies* make their income in two ways. One is through the premiums they receive from those who buy insurance protection from them and the other is income from the investment portfolios that they maintain to service the claims. An *investment bank* provides advice and supporting products for other firms to raise capital from financial markets or to consummate deals such as acquisitions or divestitures. *Investment firms* provide investment advice or manage portfolios for clients. Their income comes from advisory fees for the advice and management and sales fees for investment portfolios. With the consolidation in the financial services sector, an increasing number of firms operate in more than one of these businesses. For example, Citigroup, created by the merger of Travelers and Citicorp operates in all four businesses. At the same time, however, there remain a large number of small banks, boutique investment banks and specialized insurance firms that still derive the bulk of their income from one source.

Characteristics of financial service firms

There are many dimensions on which financial service firms differ from other firms in the market. In this section, we will focus on four key differences and look at why these differences can create estimation issues in valuation. The first is that many categories (albeit not all) of financial service firms operate under strict regulatory constraints on how they run their businesses and how much capital they need to set aside to keep operating. The second is that accounting rules for recording earnings and asset value at financial service firms are at variance with accounting rules for the rest of the market. The third is that debt for a financial service firm is more akin to raw material than to a source of capital; the notion of cost of capital and enterprise value may be meaningless as a consequence. The final factor is that the defining reinvestment (net capital expenditures and working capital) for a bank or insurance company may be not just difficult, but impossible, and cash flows cannot be computed.

The Regulatory Overlay

Financial service firms are heavily regulated all over the world, though the extent of the regulation varies from country to country. In general, these regulations take three forms. First, banks and insurance companies are required to maintain regulatory capital ratios, computed based upon the book value of equity and their operations, to ensure that they do not expand beyond their means and put their claimholders or depositors at risk. Second, financial service firms are often constrained in terms of where they can invest their funds. For instance, until a decade ago, the Glass-Steagall Act in the United States restricted commercial banks from investment banking activities as well as from taking active equity positions in non-financial service firms. Third, the entry of new firms into the business is often controlled by the regulatory authorities, as are mergers between existing firms.

Why does this matter? From a valuation perspective, assumptions about growth are linked to assumptions about reinvestment. With financial service firms, these assumptions have to be scrutinized to ensure that they pass regulatory constraints. There might also be implications for how we measure risk at financial service firms. If

regulatory restrictions are changing or are expected to change, it adds a layer of uncertainty (risk) to the future, which can have an effect on value. Put more simply, to value banks, insurance companies and investment banks, we have to be aware of the regulatory structure that governs them.

Differences in Accounting Rules

The accounting rules used to measure earnings and record book value are different for financial service firms than the rest of the market, for two reasons.

a. Mark to Market: The first is that the assets of financial service firms tend to be financial instruments (bonds, securitized obligations). Since the market price is observable for many of these investments, accounting rules have tilted towards using market value (actual of estimated) for these assets. To the extent that some or a significant portion of the assets of a financial service firms are marked to market, and the assets of most non-financial service firms are not, we fact two problems. The first is in comparing ratios based upon book value (both market to book ratios like price to book and accounting ratios like return on equity) across financial and non-financial service firms. The second is in interpreting these ratios, once computed. While the return on equity for a non-financial service firm can be considered a measure of return earned on equity invested originally in assets, the same cannot be said about return on equity at financial service firms, where the book equity measures not what was originally invested in assets but an updated market value. The second is that the nature of operations for a financial service firm is such that long periods of profitability are interspersed with short periods of large losses; accounting standards have been developed to counter this tendency and create smoother earnings.

a. Loss Provisions and smoothing out earnings: Consider a bank that makes money the old fashioned way – by taking in funds from depositors and lending these funds out to individuals and corporations at higher rates. While the rate charged to lenders will be higher than that promised to depositors, the risk that the bank faces is that lenders may default, and the rate at which they default will vary widely over time – low during good economic times and high during economic

downturns. Rather than write off the bad loans, as they occur, banks usually create provisions for losses that average out losses over time and charge this amount against earnings every year. Though this practice is logical, there is a catch, insofar as the bank is given the responsibility of making the loan loss assessment. A conservative bank will set aside more for loan losses, given a loan portfolio, than a more aggressive bank, and this will lead to the latter reporting higher profits during good times.

Valuation Issues

When valuing financial service firms, there are two primary challenges that undercut any valuation. The first is that drawing a distinction between debt and equity, one that we have made for all other firms, is difficult to do for financial service firms. The second is that estimating cash flows is complicated because capital expenditures and working capital are not easily measured at financial service firms.

Debt and Equity

In the financial balance sheet that we used to describe firms, there are only two ways to raise funds to finance a business – debt and equity. While this is true for both all firms, financial service firms differ from non-financial service firms on three dimensions:

a. Debt is raw material, not capital: When we talk about capital for non-financial service firms, we tend to talk about both debt and equity. A firm raises funds from both equity investor and bondholders (and banks) and uses these funds to make its investments. When we value the firm, we value the value of the assets owned by the firm, rather than just the value of its equity. With a financial service firm, debt has a different connotation. Rather than view debt as a source of capital, most financial service firms seem to view it as a raw material. In other words, debt is to a bank what steel is to a manufacturing company, something to be molded into other products which can then be sold at a higher price and yield a profit. Consequently, capital at financial service firms seems to be narrowly defined as including only equity capital. This definition of capital is reinforced by the regulatory authorities, who evaluate the equity capital ratios of banks and insurance firms.

b. Defining Debt: The definition of what comprises debt also is murkier with a financial service firm than it is with a non-financial service firm. For instance, should deposits made by customers into their checking accounts at a bank be treated as debt by that bank? Especially on interest-bearing checking accounts, there is little distinction between a deposit and debt issued by the bank. If we do categorize this as debt, the operating income for a bank should be measured prior to interest paid to depositors, which would be problematic since interest expenses are usually the biggest single expense item for a bank.

c. Degree of financial leverage: Even if we can define debt as a source of capital and can measure it precisely, there is a final dimension on which financial service firms differ from other firms. They tend to use more debt in funding their businesses and thus have higher financial leverage than most other firms. While there are good reasons that can be offered for why they have been able to do this historically - more predictable earnings and the regulatory framework are two that are commonly cited – there are consequences for valuation. Since equity is a sliver of the overall value of a financial service firm, small changes in the value of the firm's assets can translate into big swings in equity value.

Estimating cash flows is difficult

We noted earlier that financial service firms are constrained by regulation in both where they invest their funds and how much they invest. If, as we have so far in this book, define reinvestment as necessary for future growth, there are problems associated with measuring reinvestment with financial service firms. Note that, we consider two items in reinvestment – net capital expenditures and working capital. Unfortunately, measuring either of these items at a financial service firm can be problematic.

Consider net capital expenditures first. Unlike manufacturing firms that invest in plant, equipment and other fixed assets, financial service firms invest primarily in intangible assets such as brand name and human capital. Consequently, their investments for future growth often are categorized as operating expenses in accounting statements. Not surprisingly, the statement of cash flows to a bank show little or no capital expenditures and correspondingly low depreciation. With working capital, we run into a different problem. If we define working capital as the difference between current assets and current liabilities, a large proportion of a bank's balance sheet would fall into one or

the other of these categories. Changes in this number can be both large and volatile and may have no relationship to reinvestment for future growth.

As a result of this difficulty in measuring reinvestment, we run into two practical problems in valuing these firms. The first is that we cannot estimate cash flows without estimating reinvestment. In other words, if we cannot identify how much a company is reinvesting for future growth, we cannot identify cash flows either. The second is that estimating expected future growth becomes more difficult, if the reinvestment rate cannot be measured.

Valuation Solutions

If you cannot clearly delineate how much a financial service firm owes and what its cash flows are, how can you ever get an estimate of value? That is the question we will address in this section.

Discounted Cash Flow Models

In a discounted cash flow model, we consider the value of an asset to be the present value of the expected cash flows generated by that asset. In this section, we will first lay out the argument that financial service firms should be valued on an equity basis, rather than as on a firm basis, and that dividends, for better or worse, are often the only tangible cash flow that we can observe or estimate. Consequently, our focus will be on variants of the dividend discount model and how they can best be used in valuing banks and insurance companies.

Equity versus Firm Valuation

Early in this book, we noted the distinction between valuing a firm and valuing the equity in the firm. We value firms by discounting expected after tax cash flows prior to debt payments at the weighted average cost of capital. We value equity by discounting cash flows to equity investors at the cost of equity. Estimating cash flows prior to debt payments at a weighted average cost of capital is problematic, when debt and debt payments cannot be easily identified, which, as we argued earlier, is the case with financial service firms. Equity can be valued directly, however, by discounting cashflows

to equity at the cost of equity. Consequently, we would argue for the latter approach for financial service firms.

Even with equity valuation, we have a secondary problem. To value the equity in a firm, we normally estimate the free cashflow to equity. In Chapter 2, we defined the free cash flow to equity.

Free Cashflow to Equity = Net Income – Net Capital Expenditures – Change in non-cash working capital – (Debt repaid – New debt issued)

If we cannot estimate net capital expenditures or non-cash working capital, we clearly cannot estimate the free cashflow to equity. Since this is the case with financial service firms, we have three choices. The first is to use dividends as cash flows to equity and assume that firms over time pay out their free cash flows to equity as dividends. Since dividends are observable, we therefore do not have to confront the question of how much firms reinvest. The second is to adapt the free cashflow to equity measure to allow for the types of reinvestment that financial service firms make. For instance, given that banks operate under a regulatory capital ratio constraint, it can be argued that these firms have to increase regulatory capital in order to make more loans in the future. The third is to keep the focus on excess returns, rather than on earnings, dividends and growth rates, and to value these excess returns.

Dividend Discount Models

In the basic dividend discount model, the value of a stock is the present value of the expected dividends on that stock. While many analysts view the model as old fashioned, it retains a strong following among analysts who value financial service companies, because of the difficulties we face in estimating cash flows. In this section, we will begin by laying out the basic model and then consider ways in which we can streamline its usage, when valuing financial service companies.

The standard model

If we start with the assumption that equity in a publicly traded firm has an infinite life, we arrive at the most general version of the dividend discount model:

$$\text{Value per share of equity} = \sum_{t=1}^{t=\infty} \frac{DPS_t}{(1+k_e)^t}$$

where

DPS_t = Expected dividend per share in period t

k_e = Cost of equity

In the special case where the expected growth rate in dividends is constant forever, this model collapses into the Gordon Growth model.

$$\text{Value per share of equity in stable growth} = \frac{DPS_1}{(k_e - g)}$$

In this equation, g is the expected growth rate in perpetuity and DPS_1 is the expected dividends per share next year. In the more general case, where dividends are growing at a rate which is not expected to be sustainable or constant forever during a period (called the extraordinary growth period), we can still assume that the growth rate will be constant forever at some point in the future. This allows us to then estimate the value of a stock, in the dividend discount model, as the sum of the present values of the dividends over the extraordinary growth period and the present value of the terminal price, which itself is estimated using the Gordon growth model.

$$\text{Value per share of equity in extraordinary growth} = \sum_{t=1}^{t=n} \frac{DPS_t}{(1+k_{e,hg})^t} + \frac{DPS_{n+1}}{(k_{e,st} - g_n)(1+k_{e,hg})^n}$$

The extraordinary growth is expected to last n years, g_n is the expected growth rate after n years and k_e is the cost of equity (hg: high growth period and st: stable growth period).

While the dividend discount model is intuitive and has deep roots in equity valuation, there are dangers in using the model blindly. As we noted in the section on the dark side, there are many analysts who start with the current dividends of the bank as a base, apply a growth rate to these earnings, based on either history or analyst forecasts, and compute a present value. For the model to yield a value that is reasonable, the assumptions have to be internally consistent, with the expected growth rate numbers jelling with the dividend forecasts and risk measures.

A Consistent Dividend Discount Model

Looking at the inputs into the dividend discount model, there are three sets of inputs that determine the value of equity. The first is the cost of equity that we

use to discount cash flows, with the possibility that the cost may vary across time, at least for some firms. The second is the proportion of the earnings that we assume will be paid out in dividends; this is the dividend payout ratio and higher payout ratios will translate into more dividends for any given level of earnings. The third is the expected growth rate in dividends over time, which will be a function of the earnings growth rate and the accompanying payout ratio. In addition to estimating each set of inputs well, we also need to ensure that the inputs are consistent with each other.

Risk and Cost of Equity

In keeping with the way we have estimated the cost of equity for firms so far in this book, the cost of equity for a financial service firm has to reflect the portion of the risk in the equity that cannot be diversified away by the marginal investor in the stock. This risk is estimated using a beta (in the capital asset pricing model) or betas (in a multi-factor or arbitrage pricing model). There are three estimation notes that we need to keep in mind, when making estimates of the cost of equity for a financial service firm:

a. Use bottom-up betas: In our earlier discussions of betas, we argued against the use of regression betas because of the noise in the estimates (standard errors) and the possibility that the firm has changed over the period of the regression. We will continue to hold to that proposition, when valuing financial service firms. In fact, the large numbers of publicly traded firm in this domain should make estimating bottom up betas much easier.
b. Do not adjust for financial leverage: When estimating betas for non-financial service firms, we emphasized the importance of unlevering betas (whether they be historical or sector averages) and then relevering them, using a firm's current debt to equity ratio. With financial service firms, we would skip this step for two reasons. First, financial service firms tend to be much more homogeneous in terms of capital structure – they tend to have similar financial leverage primarily due to regulations. Second, and this is a point made earlier, debt is difficult to measure for financial service firms. In practical terms, this will mean that we will

use the average levered beta for comparable firms as the bottom-up beta for the firm being analyzed.

c. Adjust for regulatory and business risk: If we use sector betas and do not adjust for financial leverage, we are in effect using the same beta for every company in the sector. As we noted earlier, there can be significant regulatory differences across markets, and even within a market, across different classes of financial service firms. To reflect this, we would define the sector narrowly; thus, we would look the average beta across large money center banks, when valuing a large money center bank, and across small regional banks, when valuing one of these. We would also argue that financial service firms that expand into riskier businesses – securitization, trading and investment banking – should have different (and higher betas) for these segments, and that the beta for the company should be a weighted average. Table 9.1 summarizes the betas for different groups of financial service companies, categorized by region, in February 2009.

Table 9.1: Betas for financial service businesses

Category	US	Europe	Emerging Markets
Large Money Center Banks	0.71	0.80	0.9
Small/Regional Banks	0.91	0.98	1.05
Thrifts	0.66	0.75	0.85
Brokerage Houses	1.37	1.25	1.5
Investment Banks	1.50	1.55	1.9
Life Insurance	1.17	1.20	1.1
Property and Casualty Insurance Companies	0.91	0.95	0.9

d. Consider the relationship between risk and growth: Through the book, we have emphasized the importance of modifying a company's risk profile to reflect changes that we are assuming to its growth rate. As growth companies mature, betas should move towards one. We see no need to abandon that principle, when valuing banks. We would expect high growth banks to have higher betas (and costs of equity) than mature banks. In valuing such banks, we would therefore start with higher costs of equity but as we reduce growth, we would also reduce betas and costs of equity.

There is one final point that bears emphasizing here. The average betas that we get across financial service firms reflect the regulatory constraints that they operated under during

that period. When significant changes are expected to regulation, we should consider the potential impact on betas across the board. For instance, the crisis of 2008 will cause banking regulations to be tightened globally and may very well push up the betas for all banks at least for the foreseeable future.

Growth and Payout

There is an inherent trade off between dividends and growth. When a company pays a larger segment of its earnings as dividends, it is reinvesting less and should thus grow more slowly. With financial service firms, this link is reinforced by the fact that the activities of these firms are subject to regulatory capital constraints; banks and insurance companies have to maintain equity (in book value terms) at specified percentages of their activities. When a company is paying out more in dividends, it is retaining less in earnings; the book value of equity increases by the retained earnings. In recent years, in keeping with a trend that is visible in other sectors as well, financial service firms have increased stock buybacks as a way of returning cash to stockholders. In this context, focusing purely on dividends paid can provide a misleading picture of the cash returned to stockholders. An obvious solution is to add the stock buybacks each year to the dividends paid and to compute the composite payout ratio. If we do so, however, we should look at the number over several years, since stock buybacks vary widely across time – a buyback of billions in one year may be followed by three years of relatively meager buybacks, for instance.

To ensure that assumptions about dividends, earnings and growth are internally consistent, we have to bring in a measure of how well the retained equity is reinvested; the return on equity is the variable that ties together payout ratios and expected growth. In chapter 2, we introduced a fundamental growth measure for earnings:

Expected growth in earnings = Return on equity * (1 – Dividend Payout ratio)

For instance, a bank that payout out 60% of its earnings as dividends and earns a return on equity of 12% will have an expected growth rate in earnings of 4.8%. When we introduced the fundamental equation in chapter 2, we also noted that

firms can deliver growth rates that deviate from this expectation, if the return on equity is changing.

$$\text{Expected Growth}_{EPS} = (1 - \text{ Payout Ratio})(ROE_{t+1}) + \frac{ROE_{t+1} - ROE_t}{ROE_t}$$

Thus, if the bank is able to improve the return on equity on existing assets from 10% to 12%, the efficiency growth rate in that year will be 20%. However, efficiency growth is temporary and all firms ultimately will revert back to the fundamental growth relationship.

The linkage between return on equity, growth and dividends is therefore critical in determining value in a financial service firm. At the risk of hyperbole, the key number in valuing a bank is not dividends, earnings or growth rate, but what we believe it will earn as <u>return on equity in the long term</u>. That number, in conjunction with payout ratios, will help in determining growth. Alternatively, the return on equity, together with expected growth rates, can be used to estimate dividends. This linkage is particularly useful, when we get to stable growth, where growth rates can be very different from the initial growth rates. To preserve consistency in the valuation, the payout ratio that we use in stable growth, to estimate the terminal value, should be:

$$\text{Payout ratio in stable growth} = 1 - \frac{g}{ROE_{\text{stable growth}}}$$

The risk of the firm should also adjust to reflect the stable growth assumption. In particular, if betas are used to estimate the cost of equity, they should converge towards one in stable growth.

Illustration 9.1: Wells Fargo Banks – February 2009

In illustration 14.1, we examined the effects of leaving dividends unchanged and using historical dividend growth to value Wells Fargo in early 2009 and concluded that we would over value the firm for two reasons. First, we are overstating the expected dividends in the future by basing it on the dividends paid in 2008. Second, the growth rate we were assuming for the future (4%) may not be consistent with the payout ratio that we were assuming in the valuation. Based on the 2008 numbers, where dividends per share were $1.30 per share and earnings

per share was $1.71, the payout ratio is 76%. To deliver a growth rate of 4% a year forever, the return on equity that Wells Fargo would have to deliver on it's new investment is 16.67%.

$$\text{Implied Return on Equity} = \frac{g}{(1 - \text{Payout ratio})} = \frac{4\%}{(1 - .76)} = 16..67\%$$

If we believe that the return on equity, in the future, at Wells Fargo will be lower than 16.67%, we have to either lower growth or reduce dividends.

Rather than base the valuation on the 2008 dividend and earnings numbers, which are unstable and reflect the market crisis, we chose a different path. We started with the book value of equity of $47,628 million that Wells Fargo reported at the end of 2008, and estimated what earnings and dividends would be at a normalized return on equity. For instance, take the most the optimistic scenario, where the return on equity at Wells Fargo reverts back quickly to 18.91%, the average ROE from 2001 to 2007. The normalized net income for next year would be as follows:

Normalized net income = Book value of equity * Normalized ROE

= $47,628 million * .1891 = $9,006 million

Assuming that these earnings would grow at a stable rate of 3% a year in perpetuity, we next estimated the dividend payout ratio:

$$\text{Dividend Payout ratio} = 1 - \frac{g}{\text{ROE}} = 1 - \frac{.03}{.1891} = .8414 \text{ or } 84.14\%$$

If we assume that the cost of equity of 9% that we estimated earlier is a reasonable value, we can estimate the value of equity in Wells Fargo:[50]

$$\text{Value of equity} = \frac{\text{Expected Dividends next year}}{(\text{Cost of equity - Stable growth rate})}$$

$$= \frac{\text{Net Income} * \text{Payout ratio}}{(\text{Cost of equity - Stable growth rate})} = \frac{9006 * (0.8414)}{(.09 - .03)} = \$126{,}293 \text{ mil}$$

Under the most optimistic scenario, Wells Fargo is significantly under valued in February 2009 at its existing market value for equity of $66,640 million.

---

[50] To get to this cost of equity, we assumed a beta of one and an equity risk premium of 6%. With a riskfree rate of 3%, we obtain a cost of equity = 3% + 6% = 9%.

The two inputs that will determine the value of equity at Wells Fargo are the return on equity and the cost of equity. As we lower the return on equity, the normalized net income will decrease and the payout out ratio will decrease as well (for the given growth rate of 3%). The cost of equity can also change, if we perceive that banks have become riskier. Following the same procedure that we did for the most optimistic scenario, we valued equity at Wells Fargo under two other scenario – an intermediate scenario where the normalized return on equity drops to 15% and the cost of equity increases to 10% and a pessimistic scenario, where the return on equity reverts to 12% and the cost of equity increases to 11%. Table 9.2 summarizes our findings under each scenario:

Table 9.2: Value of Wells Fargo Equity: February 2009

	Net Income	ROE	Payout ratio	Cost of equity	Value of equity
Quick bounce back to normalcy	$9,006.45	18.91%	84.14%	9%	126293.58
Slow bounce back to normalcy	$7,144.20	15.00%	80.00%	10%	$81,648.00
Long term change to lower profitability and higher risk	$5,715.36	12.00%	75.00%	11%	$53,581.50
Market Cap (2/2009)					$66,643.00

While Wells Fargo continues to look under valued, if we assume a slow bounce back to normalcy, it does not look cheap if we assume that banks will be riskier and less profitable from this point on.

Cashflow to Equity Models

At the beginning of this discussion, we noted the difficulty in estimating cashflows when net capital expenditures and non-cash working capital cannot be easily identified. It is possible, however, to estimate cashflows to equity for financial service firms if we define reinvestment differently. The cashflow to equity is the cashflow left over for equity investors after debt payments have been made and reinvestment needs met. With financial service firms, the reinvestment generally does not take the form of plant, equipment or other fixed assets. Instead, the investment is in regulatory capital; this

is the capital as defined by the regulatory authorities, which, in turn, determines the limits on future growth.

$$FCFE_{\text{Financial Service Firm}} = \text{Net Income} - \text{Reinvestment in Regulatory Capital}$$

To estimating the reinvestment in regulatory capital, we have to define two parameters. The first is the book equity capital ratio that will determine the investment; this will be heavily influenced by regulatory requirements but will also reflect the choices made by a bank. Conservative banks may choose to maintain a higher capital ratio than required by regulatory authorities whereas aggressive banks may push towards the regulatory constraints. For instance, a bank that has a 5% equity capital ratio can make $100 in loans for every $5 in equity capital. When this bank reports net income of $15 million and pays out only $5 million, it is increasing its equity capital by $10 million. This, in turn, will allow it to make $200 million in additional loans and presumably increase its growth rate in future periods. The second is the profitability of the activity, defined in terms of net income. Staying with the bank example, we have to specify how much net income the bank will generate with the additional loans; a 0.5% profitability ratio will translate into additional net income of $1 million on the additional loans.

Excess Return Models

The third approach to valuing financial service firms is to use an excess return model. In such a model, the value of a firm can be written as the sum of capital invested currently in the firm and the present value of excess returns that the firm expects to make in the future. In this section, we will consider how this model can be applied to valuing equity in a financial service firm.

Basic Model

Given the difficulty associated with defining total capital in a financial service firm, it makes far more sense to focus on just equity when using an excess return model to value a financial service firm. The value of equity in a firm can be written as the sum of the equity invested in a firm's current investments and the expected excess returns to equity investors from these and future investments.

Value of Equity = Equity Capital invested currently + Present Value of Expected Excess Returns to Equity investors

The most interesting aspect of this model is its focus on excess returns. A firm that invests its equity and earns just the fair-market rate of return on these investments should see the market value of its equity converge on the equity capital currently invested in it. A firm that earns a below-market return on its equity investments will see its equity market value dip below the equity capital currently invested.

The other point that has to be emphasized is that this model considers expected future investments as well. Thus, it is up to the analyst using the model to forecast not only where the financial service firm will direct its future investments but also the returns it will make on those investments.

Inputs to Model

There are two inputs needed to value equity in the excess return model. The first is a measure of equity capital currently invested in the firm. The second and more difficult input is the expected excess returns to equity investors in future periods.

The equity capital invested currently in a firm is usually measured as the book value of equity in the firm. While the book value of equity is an accounting measure and is affected by accounting decisions, it should be a much more reliable measure of equity invested in a financial service firm than in a manufacturing firm for two reasons. The first is that the assets of a financial service firm are often financial assets that are marked up to market; the assets of manufacturing firms are real assets and deviations between book and market value are usually much larger. The second is that depreciation, which can be a big factor in determining book value for manufacturing firms, is often negligible at financial service firms. Notwithstanding this, the book value of equity can be affected by stock buybacks and extraordinary or one-time charges. The book value of equity for financial service firms that have one or both may understate the equity capital invested in the firm.

The excess returns, defined in equity terms, can be stated in terms of the return on equity and the cost of equity.

Excess Equity return = (Return on equity – Cost of equity) (Equity capital invested)

Here again, we are assuming that the return on equity is a good measure of the economic return earned on equity investments. When analyzing a financial service firm, we can obtain the return on equity from the current and past periods, but the return on equity that

is required is the expected future return. This requires an analysis of the firm's strengths and weaknesses as well as the competition faced by the firm

Illustration 9.2: Excess Return Valuation – Goldman Sachs

In February 2009, Goldman Sachs, perhaps the best-regarded investment bank in the world, was trading at a market capitalization for equity of $48.7 billion, well below its book value of equity of $60.6 billion. A significant factor underlying the stock price collapse was the decline in profitability at the firm, which reported $2,322 million in net income in 2008, well below the $11,599 million it reported as profits in the previous year. Goldman paid out $850 million in dividends during 2008.

To value Goldman Sachs, we begin with the current cost of equity. Using the average beta of 1.50, reported by investment banks in 2008, in conjunction with a treasury bond rate of 3% and an equity risk premium of 6%, yields a cost of equity of 12% for the firm:

$$\text{Cost of equity} = 3\% + 1.5\ (6\%) = 12\%$$

While the return on equity at Goldman Sachs has ranged from 16 to 20% between 2001 and 2007, the expected return on equity, looking forward, will be much lower. For the next 5 years, we will assume that the return on equity at Goldman will be 9%, well below not only the historical average return on equity but also its own cost of equity. The resulting negative excess returns and present value are summarized in table 9.3:

Table 9.3: Excess Returns – High Growth Period

	1	*2*	*3*	*4*	*5*
Net Income	$5,941.08	$6,384.60	$6,861.23	$7,373.44	$7,923.89
- Equity Cost (see below)	$7,921.44	$8,512.80	$9,148.30	$9,831.25	$10,565.18
Excess Equity Return	-$1,980.36	-$2,128.20	-$2,287.08	-$2,457.81	-$2,641.30
Cumulated Cost of Equity	1.12000	1.25440	1.40493	1.57352	1.76234
Present Value	-$1,768.18	-$1,696.59	-$1,627.90	-$1,561.98	-$1,498.74
Beginning BV of Equity	$66,012.00	$70,939.98	$76,235.86	$81,927.08	$88,043.17
Cost of Equity	12.00%	12.00%	12.00%	12.00%	12.00%
Equity Cost	$7,921.44	$8,512.80	$9,148.30	$9,831.25	$10,565.18
Return on Equity	9.00%	9.00%	9.00%	9.00%	9.00%
Net Income	$5,941.08	$6,384.60	$6,861.23	$7,373.44	$7,923.89
Dividend Payout Ratio	17.05%	17.05%	17.05%	17.05%	17.05%
Dividends paid	$1,013.10	$1,088.73	$1,170.00	$1,257.35	$1,351.21
Retained Earnings	$4,927.98	$5,295.87	$5,691.22	$6,116.09	$6,572.67

The net income each year is computed by multiplying the return on equity each year by the beginning book value of equity. The book value of equity each year is augmented by the portion of earnings that is not paid out as dividends; the dividend payout ratio is based upon current dividends and normalized earnings.

To put closure on this valuation, we have to make assumptions about excess returns after year 5. We assumed that the net income would grow 3% a year beyond year 5 and that the beta for the stock would decline to 1.20. For Goldman Sachs, we will assume that the return on equity after year 5 will be 10.20%, set equal to the cost of equity in stable growth:

Cost of equity in stable growth period = 3% + 1.2(6%) = 10.20%

Net Income_6 = Book value of equity at start of year 6* Stable ROE

= ($88,043*1.03) *.102 = $9249.82 million

Note that the net income in year 6 is significantly higher than the net income in year 5, as the return on equity bounces back from 9% to 10.20%. The terminal value of excess returns to equity investors can then be computed.

$$\text{Terminal value of excess returns} = \frac{\text{Net Income}_6 - (\text{Cost of equity}_6)(\text{BV of Equity}_6)}{\text{Cost of equity} - \text{Expected growth rate}}$$

$$= \frac{9{,}249.82 - (90684.47)(0.102)}{0.102 - 0.03}$$

$$= \$0$$

Since the firm earns its cost of equity after year 5, there is no value gained or lost after that year. The value of equity can then be computed as the sum of the three components – the book value of equity invested today, the present value of excess equity returns over the next 5 years and the present value of the terminal value of equity.

Book value of Equity Invested currently	=	$66,012
PV of Equity Excess Return – next 5 years	=	- $8,154
PV of terminal value of excess returns	=	0
Value of Equity	=	$57,859
Number of shares	=	461.874
Value Per Share	=	$125.29

At the time of this valuation in February 2009, Goldman Sachs was trading at $ 96.45 a share.

Relative Valuation

In our chapters on relative valuation, we examined a series of multiples that are used to value firms, ranging from earnings multiples to book value multiples to revenue multiples. In this section, we consider how relative valuation can be used for financial service firms.

Choices in Multiples

Firm value multiples such as Value to EBITDA or Value to EBIT cannot be easily adapted to value financial service firms, because neither value nor operating income can be easily estimated for banks or insurance companies. In keeping with our emphasis on equity valuation for financial service firms, the multiples that we will work with to analyze financial service firms are equity multiples. The three most widely used equity multiples are price earnings ratios, price to book value ratios and price to sales ratios. Since sales or revenues are not really measurable for financial service firms, price to sales ratios cannot be estimated or used for these firms. We will look, in this section, at the use of price earnings and price to book value ratios for valuing financial service firms.

Price Earnings Ratios

The price earnings ratio for a bank or insurance companies is measured much the same as it is for any other firm.

$$\text{Price Earnings Ratio} = \frac{\text{Price per share}}{\text{Earnings per share}}$$

In Chapter 4, we noted that the price earnings ratio is a function of three variables – the expected growth rate in earnings, the payout ratio and the cost of equity. As with other firms, the price earnings ratio should be higher for financial service firms with higher expected growth rates in earnings, higher payout ratios and lower costs of equity.

An issue that is specific to financial service firms is the use of provisions for expected expenses. For instance, banks routinely set aside provisions for bad loans. These provisions reduce the reported income and affect the reported price earnings ratio. Consequently, banks that are more conservative about categorizing bad loans will report lower earnings and have higher price earnings ratios, whereas banks that are less conservative will report higher earnings and lower price earnings ratios.

Another consideration in the use of earnings multiples is the diversification of financial service firms into multiple businesses. The multiple that an investor is willing to pay for a dollar in earnings from commercial lending should be very different than the multiple that the same investor is will to pay for a dollar in earnings from trading. When a firm is in multiple businesses with different risk, growth and return characteristics, it is very difficult to find truly comparable firms and to compare the multiples of earnings paid across firms. In such a case, it makes far more sense to break the firm's earnings down by business and assess the value of each business separately.

Illustration 9.3: Comparing PE ratios: Insurance Companies

In Table 9.4, we compare the current price earnings ratios of life insurance companies in February 2009.

Table 9.4: PE Ratios and Expected Growth Rates – Insurance Companies

Company Name	PE Ratio	Expected growth in EPS	Beta
Torchmark Corp. (NYSE:TMK)	4.11	3.60%	1.87
Odyssey Re Holdings Corp. (NYSE:ORH)	5.15	4.00%	1.53
Manulife Financial Corporation (TSX:MFC)	5.4	5.20%	2.41
MetLife, Inc. (NYSE:MET)	5.45	4.50%	1.96
Assurant Inc. (NYSE:AIZ)	5.56	5.00%	2.16
Principal Financial Group Inc. (NYSE:PFG)	5.85	5.50%	2.15
AFLAC Inc. (NYSE:AFL)	6.01	6.40%	2.4
Unum Group (NYSE:UNM)	6.33	6.00%	1.47
Aon Corporation (NYSE:AOC)	7.04	6.20%	1.7
The Travelers Companies, Inc. (NYSE:TRV)	7.58	6.00%	1.87
HCC Insurance Holdings Inc. (NYSE:HCC)	7.75	7.00%	2.05
The Chubb Corporation (NYSE:CB)	7.94	10.50%	1.67
American Financial Group Inc. (NYSE:AFG)	9.41	11.00%	1.31
ProAssurance Corporation (NYSE:PRA)	10.74	10.30%	0.89
Reinsurance Group of America Inc. (NYSE:RGA)	11.71	11.50%	1.24
W.R. Berkley Corporation (NYSE:WRB)	12.3	12.50%	1.98
Sun Life Financial Inc. (TSX:SLF)	12.8	10.00%	1.16
RLI Corp. (NYSE:RLI)	13.48	13.00%	1.62
Brown & Brown Inc. (NYSE:BRO)	14.36	13.70%	1.44
Arthur J Gallagher & Co. (NYSE:AJG)	20.21	12.67%	1.21
Transatlantic Holdings Inc. (NYSE:TRH)	20.36	15.00%	1.22
Lincoln National Corp. (NYSE:LNC)	30.5	10.20%	0.86
The Hanover Insurance Group Inc. (NYSE:THG)	35.52	15.00%	0.98

The PE ratios vary widely and range from 4.11 for Nationwide Financial to 35.52 for the Hanover Insurance Group. We also report the consensus estimates by analysts of the growth rate in earnings per share over the next 5 years and the equity beta for each of these firms, as a proxy for risk. Some of the variation in PE ratios can be explained by differences in the expected growth rate – higher growth firms tend to have higher PE ratios - and some of it is due to differences in risk – more risky firms have lower PE ratios. Regressing PE ratios against the expected growth rate and the standard deviation yields the following:

PE Ratio = 12.41 + 109.95 Expected Growth Rate – 6.60 Beta R^2=59%

(1.61) (2.86) (-2.14)

The regression confirms the intuition that higher growth and lower risk firms have higher PE ratios than other firms. Table 9.5 uses this regression to estimate predicted PE ratios for the companies in the table and reports on whether the firms are under or over valued.

Table 9.5: Predicted and Actual PE ratios: Insurance Companies in February 2009

Company Name	PE Ratio	Predicted PE	% Under or over valued
American Financial Group Inc. (NYSE:AFG)	9.41	15.86	-40.66%
ProAssurance Corporation (NYSE:PRA)	10.74	17.86	-39.87%
The Chubb Corporation (NYSE:CB)	7.94	12.93	-38.61%
Unum Group (NYSE:UNM)	6.33	9.31	-31.97%
Reinsurance Group of America Inc. (NYSE:RGA)	11.71	16.87	-30.59%
Odyssey Re Holdings Corp. (NYSE:ORH)	5.15	6.71	-23.25%
Brown & Brown Inc. (NYSE:BRO)	14.36	17.97	-20.09%
Sun Life Financial Inc. (TSX:SLF)	12.8	15.75	-18.72%
RLI Corp. (NYSE:RLI)	13.48	16.01	-15.81%
Aon Corporation (NYSE:AOC)	7.04	8.01	-12.08%
W.R. Berkley Corporation (NYSE:WRB)	12.3	13.09	-6.00%
Transatlantic Holdings Inc. (NYSE:TRH)	20.36	20.85	-2.35%
Torchmark Corp. (NYSE:TMK)	4.11	4.03	2.08%
Arthur J Gallagher & Co. (NYSE:AJG)	20.21	18.35	10.11%
The Travelers Companies, Inc. (NYSE:TRV)	7.58	6.67	13.73%
HCC Insurance Holdings Inc. (NYSE:HCC)	7.75	6.58	17.84%
MetLife, Inc. (NYSE:MET)	5.45	4.42	23.25%
Principal Financial Group Inc. (NYSE:PFG)	5.85	4.27	37.09%
Assurant Inc. (NYSE:AIZ)	5.56	3.65	52.27%

The Hanover Insurance Group Inc. (NYSE:THG)	35.52	22.43	58.33%
AFLAC Inc. (NYSE:AFL)	6.01	3.61	66.63%
Lincoln National Corp. (NYSE:LNC)	30.5	17.95	69.93%
Manulife Financial Corporation (TSX:MFC)	5.4	2.22	143.09%

Based upon this regression, Manulife Financial looks significantly overvalued while American Financial and ProAssurance look significantly undervalued.

Price to Book Value Ratios

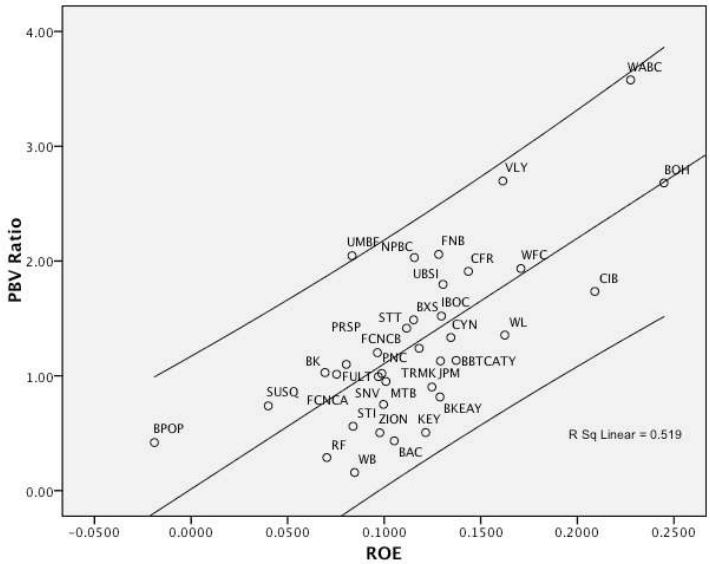
The price to book value ratio for a financial service firm is the ratio of the price per share to the book value of equity per share.

$$\text{Price to Book Ratio} = \frac{\text{Price per share}}{\text{Book value of equity per share}}$$

Other things remaining equal, higher growth rates in earnings, higher payout ratios, lower costs of equity and higher returns on equity should all result in higher price to book ratios. Of these four variable, the return on equity has the biggest impact on the price to book ratio, leading us to identify it as the companion variable for the ratio.

If anything, the strength of the relationship between price to book ratios and returns on equity should be stronger for financial service firms than for other firms, because the book value of equity is much more likely to track the market value of equity invested in existing assets. Similarly, the return on equity is less likely to be affected by accounting decisions. The strength of the relationship between price to book ratios and returns on equity can be seen when we plot the two on a scatter plot for U.S. commercial banks with market capitalization exceeding $ 1 billion, in the United States in February 2009, in figure 9.1.

Figure 9.1: Price to Book Ratios and Returns on Equity: Banks



[a]Regression line, with 90% confidence range on estimate.

Note that these numbers were extracted in the midst of the biggest crisis in banking since the Great Depression, and in an environment where most analysts have come to the conclusion that investors are in crisis mode and that equity values in banks reflect the panic and irrationality. It is therefore astounding how close the link is between price to book ratios for banks in February 2009 and the returns on equity, based upon trailing 12-month earnings. Banks such as Valley National (VLY) and WestAmerica Bancorp (WABC) that have high price to book value ratios tend to have high returns on equity. Banks such as Banco Popular (BPOP) and Wachovia (WB) that have low returns on equity trade at low price to book value ratios. The correlation between price to book ratios and returns on equity is in excess of 0.70. Put another way, there seems to be a fundamental order to the chaos that has undercut the banking sector.

While emphasizing the relationship between price to book ratios and returns on equity, we should not ignore the other fundamentals. For instance, banks vary in terms of risk, and we would expect for any given return on equity that riskier banks should have lower price to book value ratios. Similarly, banks with much greater potential for growth

should have much higher price to book ratios, for any given level of the other fundamentals. In February 2009, one factor that should make a difference is the exposure that different banks have to toxic securities – mortgage backed bonds and collateralized debt obligations (CDOs) – on their balance sheets.

Illustration 9.4: Price to Book Value Ratios: Small commercial banks

In figure 9.1, we noted the strong relationship between price to book ratios and returns on equity at large banks. Does the same apply relationship apply to smaller banks? To answer to this question, we looked at banks with market capitalizations between $ 500 million and $ 1 billion in table 9.6:

Table 9.6: Price to Book Ratios and Returns on Equity: Small Commercial Banks

Company Name	PBV Ratio	Expected Growth in EPS: next 5 years	Std deviation in stock prices	ROE
East West Bancorp	0.76	-2.50%	57.75%	13.76%
Webster Fin'l	0.37	2.00%	31.06%	6.44%
NBT Bancorp	2.13	5.00%	32.72%	12.66%
PacWest Bancorp	0.60	5.00%	40.09%	7.93%
WesBanco	1.08	5.00%	41.77%	7.70%
Chemical Financial	1.12	5.00%	33.98%	7.67%
CVB Financial	2.05	6.33%	33.02%	14.26%
First Commonwealth	1.52	6.50%	30.81%	8.14%
Pacific Cap. Bancorp	1.13	6.50%	42.12%	13.26%
Community Bank Sys.	1.43	7.30%	24.10%	8.96%
First Busey Corp	1.17	8.00%	30.34%	5.95%
Tompkins Financial Corp	2.75	8.00%	27.89%	13.39%
S & T Bancorp	2.70	9.00%	23.69%	16.62%
Umpqua Holdings Corporation	0.68	10.00%	30.42%	5.11%
MB Finl Inc	1.07	12.00%	25.50%	7.19%
PrivateBancorp Inc	2.17	15.60%	41.03%	2.57%
Pinnacle Financial Prtners Inc	1.34	16.00%	33.69%	4.93%
UCBH Hldgs Inc	0.61	24.33%	77.25%	11.35%

While the relationship between price to book ratios and returns on equity is weaker for this sample than it is for commercial banks, higher price to book value ratios tend to go with higher returns on equity. Since the assumption about all banks being equally risky was put to the test during this period, we used the standard deviation in stock price as a

proxy for this risk. Regressing the price to book ratios against the return on equity and standard deviation yields the following:[51]

$$\text{Price to Book Ratio} = \underset{(2.94)}{1.527} + \underset{(1.93)}{8.63}\,(\text{Return on Equity}) - \underset{(2.36)}{2.63}\,(\sigma_{\text{Stock price}}) \quad R^2 = 31\%$$

Using this regression yields predicted price to book ratios for any firm in the sample. For instance, the predicted price to book ratio for Tompkins Financial, which at 2.75 times book value of equity looks expensive, would be:

Predicted P/BV for Tompkins Financial = 1.527 + 8.63 (0.1338) – 2.63 (0.2789)= 1.95

Based on how other small banks are priced, Tompkins looks over valued by about 30%.

Value Plays

Investing in financial service companies has historically been viewed as a conservative strategy for investors who want high dividends and prefer price stability. In effect, investors made a bargain, where they accepted the absence of transparency and difficulties in estimating cash flows in return for a regulatory overlay that provided constraints on where these firms invested and how much risk they took. Implicitly, they assumed that regulators would prevent banks and insurance companies from taking on too much risk in their operations. The last few years have stripped this illusion away, exposing investors to significant risks even when investing in large banks, investment

When investing in financial service firms today, success requires a strategy that goes beyond looking at the dividend yield and the current earnings. While the actual loans and investments made by these firms are still beyond our purview, we can look at the following:

1. Regulatory buffers: While regulatory authorities impose capital requirements for financial service firms, these requirements represent a floor and banks and insurance companies can set their actual or target ratios above this floor. One measure of safety in a bank is how much of a buffer that bank has in its capital ratios. Thus, a bank with a regulatory capital ratio of 12% can be viewed as safer than one with a regulatory capital ratio of 8%.

[51] With 18 firms in the sample, we are pushing the limits of allowable independent variables, with two. A larger sample will provide more precision.

2. Risk in assets: While capital ratios are one half of the risk equation, the types of assets that are held by the financial service firm represent the other half. There is more risk in investing in the equity of a bank that specializes in lending to entities with poor credit histories or an insurance company that seeks out and insures high risk individuals.
3. Transparency: Some financial service firms are more forthcoming in providing information both about their regulatory capital holdings and the risk in their assets than others. To the extent that withheld information is likely to contain more bad news than good, investing in more transparent companies is a good strategy.
4. Significant restrictions on new entrants into the business: Like companies in any other business, banks, investment banks and insurance companies generate higher returns on equity for their investors when there are significant barriers to entry into that business. For instance, restrictions on foreign banks inflate the earnings at many emerging market banks and make them more attractive investments.

In summary, then, you would like to invest in financial service firms that not only deliver high dividends, but also generate significant returns on equity from an asset base that is not abnormally risky, while preserving a large regulatory capital buffer.

Conclusion

The basic principles of valuation apply just as much for financial service firms as they do for other firms. There are, however, a few aspects relating to financial service firms that can affect how they are valued. The first is that debt, for a financial service firm, is difficult to define and measure, making it difficult to estimate firm value or costs of capital. Consequently, it is far easier to value the equity directly in a financial service firm, by discounting cash flows to equity at the cost of equity. The second is that capital expenditures and working capital, which are required inputs to estimating cash flows, are often not easily estimated at financial service firms. In fact, much of the reinvestment that occurs at these firms is categorized under operating expenses. To estimate cashflows to

equity, therefore, we either have to use dividends (and assume that what is not paid out as dividend is the reinvestment) or modify our definition of reinvestment.

Even if we choose to use multiples, we run into many of the same issues. The difficulties associated with defining debt make equity multiples such as price earnings or price to book value ratios better suited for comparing financial service firms than value multiples. In making these comparisons, we have to control for differences in fundamentals – risk, growth, cash flows, loan quality – that affect value.

Finally, regulatory considerations and constraints overlay financial firm valuations. In some cases, regulatory restrictions on competition allow financial service firms to earn excess returns and increase value. In other case, the same regulatory authorities may restrict the potential excess returns that a firm may be able to make by preventing the firm from entering a business.

Chapter 10: Valuing Cyclical and Commodity Companies

Uncertainty and volatility are endemic to valuation, but cyclical and commodity companies have volatility thrust upon them by external factors – the ups and downs of the economy with cyclical companies, and movements in commodity prices with commodity companies. As a consequence, even mature cyclical and commodity companies have volatile earnings and cash flows. When valuing these companies, the danger of focusing on the most recent fiscal year is that the resulting valuation will depend in great part on where in the cycle (economic or commodity price) that year fell. If the most recent year was a boom (down) year, the value will be high (low).

What makes commodity and cyclical companies different?

There are two groups of companies that we look at in this chapter. The first group includes cyclical companies, i.e., companies whose fortunes rest in large part on how the economy is doing. There are two ways of identifying these firms. The first is to categorize industry sectors into cyclical and non-cyclical, based on historical performance, and to assume that all firms in the sector share the same characteristics. For instance, the housing and automobile sectors have historically been considered to be cyclical, and all firms in these sectors will share that label. While the approach is low-cost and simple, we run the risk of tarring all firms in a sector with the same brush; thus Walmart and Abercombie & Fitch would both be categorized as cyclical firms because they are in the retailing business. The second is to look at a company's own history, in conjunction with overall economic performance, to make a categorization. Thus, a company that has historically reported lower earnings/revenues during economic downturns and higher earnings/revenues during economic boom times would be viewed as cyclical.

The second group of companies are commodity companies that derive their earnings from producing commodities that may become inputs to other companies in the economy (oil, iron ore) or be desired as investments in their own right (gold,

platinum, diamonds). We can categorize commodity companies into three groups. The first group has products that are inputs to other businesses, but are not consumed by the general public; included in this group would be mining companies like Vale, Rio Tinto and BHP Billiton. The second group generates output that is marketed to consumers, though there may other intermediaries involved in the process; in this group would be most of the food and grains companies. The third group includes firms whose output serves both other businesses and consumers; the oil and natural gas businesses come to mind but gold mining companies can also be considered part of this group.

Characteristics of commodity & cyclical firms

While commodity companies can range the spectrum from food grains to precious metals and cyclical firms can be in diverse businesses, they do share some common factors that can affect both how we view them and the values we assign to them.

1. The Economic/Commodity price cycle: Cyclical companies are at the mercy of the economic cycle. While it is true that good management and the right strategic and business choices can make some cyclical firms less exposed to movements in the economy, the odds are high that all cyclical companies will see revenues decrease in the face of a significant economic downturn. Unlike firms in many other businesses, commodity companies are, for the most part, price takers. In other words, even the largest oil companies have to sell their output at the prevailing market price. Not surprisingly, the revenues of commodity companies will be heavily impacted by the commodity price. In fact, as commodity companies mature and output levels off, almost all of the variance in revenues can be traced to where we are in the commodity price cycle. When commodity prices are on the upswing, all companies that produce that commodity benefit, whereas during a downturn, even the best companies in the business will see the effects on operations.
2. Finite resources: With commodity companies, there is one final shared characteristic. There is a finite quantity of natural resources on the planet; if

oil prices increase, we can explore for more oil but we cannot create oil. When valuing commodity companies, this will not only play a role in what our forecasts of future commodity prices will be but may also operate as a constraint on our normal practice of assuming perpetual growth (in our terminal value computations).

In summary, then, when valuing commodity and cyclical companies, we have to grapple with the consequences of economic and commodity price cycles and how shifts in these cycles will affect revenues and earnings. We also have to come up with ways of dealing with the possibility of distress, induced not by bad management decisions or firm specific choices, but by macro economic forces.

Valuation Issues

1. Volatile earnings and cash flows: The volatility in revenues at cyclical and commodity companies will be magnified at the operating income level because these companies tend to have high operating leverage (high fixed costs). Thus, commodity companies may have to keep mines (mining), reserves (oil) and fields (agricultural) operating even during low points in price cycles, because the costs of shutting down and reopening operations can be prohibitive.
2. Volatility in earnings flows into volatility in equity values and debt ratios: While this does not have to apply for all cyclical and commodity companies, the large infrastructure investments that are needed to get these firms started has led many of them to be significant users of debt financing. Thus, the volatility in operating income that we referenced earlier, manifests itself in even greater swing in net income.
3. Even the healthiest firms can be put at risk if macro move is very negative: Building on the theme that cyclical and commodity companies are exposed to cyclical risk over which they have little control and that this risk can be magnified as we move down the income statement, resulting in high volatility in net income, even for the healthiest and most mature firms in the sector, it is easy to see why we have to be more concerned about distress and survival

with cyclical and commodity firms than with most others. An extended economic downturn or a lengthy phase of low commodity prices can put most of these companies at risk.

Valuation Solutions

If volatility in earnings is a given at cyclical and commodity companies, and forecasting the cycles that the cause the volatility is often impossible to do, how can we value such companies? In this section, we will examine healthy responses to the volatility in the valuation of these companies.

Intrinsic Valuation

In chapter 2, we noted that the discounted cash flow value of a company rests on four inputs – earnings and cash flows from existing assets, the growth in these cash flows in the near term, a judgment on when the company will become mature and a discount rate to apply to the cash flows. Using this framework, we will develop two ways of adapting discounted cash flow valuations for cyclical and commodity companies. In the first, we will normalize our estimates for all four of these inputs, using normalized cash flows, growth rates and discount rates to estimate a normalized value for a firm. In the second, we will try to adjust the growth rate in the cash flows to reflect where we are in the cycle – setting it to low or even negative values at the peak of a cycle (reflecting the expectation that earnings will decline in the future) and high values at the bottom of a cycle.

Normalized Valuations

The easiest way to value cyclical and commodity companies is to look past the year-to-year swings in earnings and cash flows and to look for a smoothed out numberunderneath. In this section, we will begin by defining what comprises a normal value first and then consider different techniques that can be used to estimate this number.

What are normal numbers?

If the current financial statements of a company answer the questions we have about how much a company earned, reinvested and generated as cash flows in

the most recent period, the normalized versions of these numbers would answer a different question: How much earnings, reinvestment and cash flow would this company have generated in a normal year?

If we are talking about cyclical companies, a normal year would be one that represents the mid-point of the cycle, where the numbers are neither puffed up nor deflated by economic conditions. With commodity companies, a normal year would be one where commodity prices reflect the intrinsic price of the commodity, reflecting the underlying demand and supply. Each of these definitions conveys the subjective component to this process, since two analysts looking at the same economy or commodity can make very different judgments on what is normal.

Measuring normalized values for cyclical companies

If we accept the proposition that normalized earnings and cash flows have a subjective component to them, we can begin to lay out procedures for estimating them for individual companies. With cyclical companies, there are usually three standard techniques that are employed for normalizing earnings and cash flows:

1.Absolute average over time: The most common approach used to normalize numbers is to average them over time, though over what period remains in dispute. At least in theory, the averaging should occur over a period long enough to cover an entire cycle. In chapter 8, we noted that economic cycles, even in mature economies like the United States, can range from short periods (2-3 years) to very long ones (more than 10 years). The advantage of the approach is its simplicity. The disadvantage is that the use of absolute numbers over time can lead to normalized values being misestimated for any firm that changed its size over the normalization period. In other words, using the average earnings over the last 5 years as the normalized earnings for a firm that doubled its revenues over that period will understate the true earnings.

2. Relative average over time: A simple solution to the scaling problem is to compute averages for a scaled version of the variable over time. In effect, we can average profit margins over time, instead of net profits, and apply the average profit margin to revenues in the most recent period to estimate normalized earnings. We can

employ the same tactics with capital expenditures and working capital, by looking at ratios of revenue or book capital over time, rather than the absolute values.

3. Sector averages: In the first two approaches to normalization, we are dependent upon the company having a long history. For cyclical firms with limited history or a history of operating changes, it may make more sense to look at sector averages to normalize. Thus, we will compute operating margins for all steel companies across the cycle and use the average margin to estimate operating income for an individual steel company. The biggest advantage of the approach is that sector margins tend to be less volatile than individual company margins, but this approach will also fail to incorporate the characteristics (operating efficiencies or inefficiencies) that may lead a firm to be different from the rest of the sector.

Illustration 10.1: Valuing Toyota – Normalized Earnings

By most accounts in early 2009, Toyota was considered the best-run automobile company in the world. However, the firm was not immune to the ebbs and flows of the global economy and reported a loss in the last quarter of 2008, a precursor to much lower and perhaps negative earnings in its 2008-2009 fiscal year (stretching from April 2008 to March 2009).

To normalize Toyota's operating income, we look at its operating performance from 1998 to 2008 in table 10.1:

Table 10.1: Toyota's Operating Performance – 1998-2009 (in millions of Yen)

Year	Revenues	Operating Income	EBITDA	Operating Margin	EBITDA/ Revenues
FY1 1998	¥11,678,400	¥779,800	¥1,382,950	6.68%	11.84%
FY1 1999	¥12,749,010	¥774,947	¥1,415,997	6.08%	11.11%
FY1 2000	¥12,879,560	¥775,982	¥1,430,982	6.02%	11.11%
FY1 2001	¥13,424,420	¥870,131	¥1,542,631	6.48%	11.49%
FY1 2002	¥15,106,300	¥1,123,475	¥1,822,975	7.44%	12.07%
FY1 2003	¥16,054,290	¥1,363,680	¥2,101,780	8.49%	13.09%
FY1 2004	¥17,294,760	¥1,666,894	¥2,454,994	9.64%	14.20%
FY1 2005	¥18,551,530	¥1,672,187	¥2,447,987	9.01%	13.20%
FY1 2006	¥21,036,910	¥1,878,342	¥2,769,742	8.93%	13.17%
FY1 2007	¥23,948,090	¥2,238,683	¥3,185,683	9.35%	13.30%
FY1 2008	¥26,289,240	¥2,270,375	¥3,312,775	8.64%	12.60%
FY 2009 (Est)	¥22,661,325	¥267,904	¥1,310,304	1.18%	5.78%

Average		¥1,306,867		7.33%	

Each year, we report the operating income or loss, the EBITDA and the margins relative to revenues. We considered three different normalization techniques:

- Average income: Averaging the operating income from 1998 to 2009 yields an value of 1,332.9 billion yen. Since the revenues over the period more than doubled, this will understate the normalized operating income for the firm.
- Industry average margin: The average pre-tax operating margin of automobile firms (global) over the same time period (1998-2008) is about 6%. In 2009, however, many of these firms were in far worse shape than Toyota and many are likely to report large losses. While we could apply the industry average margin to Toyota's 2009 revenues to estimate a normalized operating income (6% of 22,661 billion yen=1,360 billion yen), this will also understate the normalized operating income, since it will not reflect the fact that Toyota has been among the most profitable firms in the sector.
- Historical margin: Averaging the pre-tax operating margin from 1998 to 2009 yields an average operating margin of 7.33%. Applying this margin to the revenues in 2009 yields a normalized operating income of 1,660.7 billion yen (7.33% of 22,661 billion yen), an estimate that captures both the larger scale of the firm today and its success in this business. We will use this value as our normalized operating income.

To value the firm, we will also make the following assumptions.

- To estimate Toyota's cost of equity, we will use a bottom up beta (estimated from the automobile sector) of 1.10. Using the ten-year Japanese yen government bond rate of 1.50% as the riskfree rate and an equity risk premium of 6.5%, we compute a cost of equity of 8.65%. [52]

  Cost of equity = Riskfree rate + Beta * Equity Risk Premium

  = 1.50% + 1.10 (6.5%) = 8.65%

[52] We are using a mature market equity risk premium of 6.5% for Toyota. An argument can be made that we should be adding a country risk premium to reflect Toyota's sales exposure in emerging markets in Asia and Latin America.

- In early 2009, Toyota had 11,862 billion yen in debt outstanding and the market value of equity for the firm was 10,551 billion (3.448 billion shares outstanding at 3060 Yen/share). Using a rating of AA and an associated default spread of 1.75% over the riskfree rate, we estimated a pre-tax cost of debt of 3.25%. Assuming that the current debt ratio is a sustainable one, we estimate a cost of capital of 5.09%; the marginal tax rate for Japan in 2009 was 40.7%.

  Debt Ratio = 11,862/ (11,862+ 10,551) = 52.9%

  Cost of capital = 8.65% (.471) + 3.25% (1-.407) (.529) = 5.09%

  We did examine the cost of capital for Toyota over time, and since neither the debt ratio nor the cost of capital has moved substantially over time, we will use this as the normalized cost of capital.
- Since Toyota is already the largest automobile firm in the world, in terms of market share, we will assume that the firm is in stable growth, growing at 1.50% (capped at the riskfree rate) in perpetuity. We will also assume that the firm will be able to generate a return on capital equal to its cost of capital on its investments.[53] The reinvestment rate that emerges from these two assumptions is 29.46%:

  Stable period reinvestment rate = $\frac{g}{ROC} = \frac{.015}{.0509} = .2946$

Bringing together the normalized operating income (1,660.7 billion yen), the marginal tax rate for Japan (40.7%), the reinvestment rate (29.46%), the stable growth rate of 1.5% and the cost of capital of 5.09%, we can estimate the value of the operating assets at Toyota:

$$\text{Value}_{\text{Operating Assets}} = \frac{\text{Operating Income } (1+g)\ (1-\ \text{tax rate})\ (1-\ \text{Reinvestment Rate})}{(\text{Cost of capital } -\ g)}$$

$$= \frac{1660.7\ (1.015)\ (1-\ .407)\ (1-\ .2946)}{(.0509\ -\ .015)} = 19{,}640 \text{ billion Yen}$$

[53] Our reasoning was as follows. By most indicators, Toyota is the most efficiently run automobile firm. We are assuming that it will not generate excess returns but will be able to break even. In fact, the return on capital that we computed based on the normalized income and the capital invested at the end of 2008 was 4.98%, very close to the estimated value of 5.09%.

Adding in cash (2,288 billion Yen) and non-operating assets (6,845 billion Yen), subtracting out debt (11,862 billion Yen) and minority interests in consolidated subsidiaries (583 billion Yen), and dividing by the number of shares (3.448 billion) yields a value per share of 4735 yen/share.[54]

Value per share

$$= \frac{\text{Operating Assets} + \text{Cash} + \text{Non-operating Assets} - \text{Debt} - \text{Minority Interest}}{\text{Number of shares}}$$

$$= \frac{19640 + 2288 + 6845 - 11862 - 583}{3.448} = 4735 \text{ Yen/share}$$

Based on the normalized income, Toyota looks significantly undervalued at its stock price of 3060 yen per share in early 2009.

Measuring normalized earnings for commodity companies

With commodity companies, the variable that causes the volatility is the price of the commodity. As it moves up and down, it not only impacts revenues and earnings but also reinvestment and financing costs. Consequently, normalization with commodity companies has to be built around a normalized commodity price.

Normalized commodity prices

What is a normalized price for oil? Or gold? There are two ways of answering this question.

- One is to look at history. Commodities have a long trading history and we can use the historical price data to come up with an average, which we can then adjust for inflation. Implicitly, we are assuming that the average inflation-adjusted price over a long period of history is the best estimate of the normalized price.
- The other approach is more complicated. Since the price of a commodity is a function of demand and supply for that commodity, we can assess (or at least

[54] The non-operating assets include marketable securities and holdings in other companies. Absent detailed information, we are assuming that the book value of these assets is the market value. The minority interests are also taken at book value, but the amount is small enough that using a market value would have made little difference in our final value per share.

try to assess the determinants of that demand and supply) and try to come up with an intrinsic value for the commodity.

Once we have normalized the price of the commodity, we can then assess what the revenues, earnings and cashflows would have been for the company being valued at that normalized price. With revenues and earnings, this may just require multiplying the number of units sold at the normalized price and making reasonable assumptions about costs. With reinvestment and cost of financing, it will require some subjective judgments on how much (if any) the reinvestment and cost of funding numbers would have changed at the normalized price.

Market-based forecasts

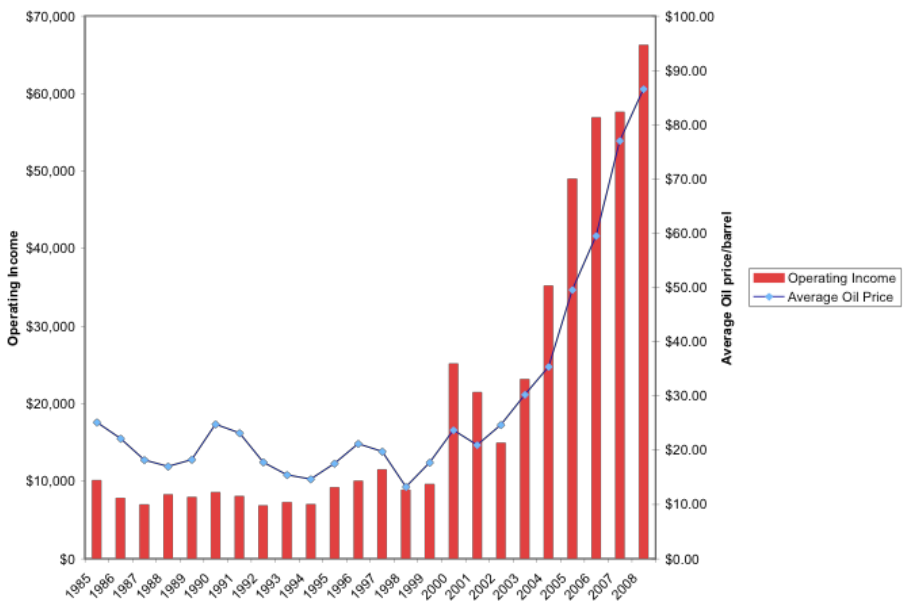
Using a normalized commodity price to value a commodity company does expose us to the critique that the valuations we obtain will reflect our commodity price views as much as they do our views on the company. For instance, assume that the current oil price is $45 and that we use a normalized oil price of $100 to value an oil company. We are likely to find the company to be undervalued, simply because of our view about the normalized oil price. If we want to remove our views of commodity prices from valuations of commodity companies, the safest way to do this is to use market-based prices for the commodity in our forecasts. Since most commodities have forward and futures markets, we can use the prices for these markets to estimate cash flows in the next few years. For an oil company, then, we will use today's oil prices to estimate cash flows for the current year and the expected oil prices (from the forward and futures markets) to estimate expected cash flows in future periods.

The advantage of this approach is that it comes with a built-in mechanism for hedging against commodity price risk. An investor who believes that a company is under valued but is shaky on what will happen to commodity prices in the future can buy stock in the company and sell oil price futures to protect herself against adverse price movements.

Illustration 10.2: Valuing Exxon Mobil – Normalized commodity prices

Exxon Mobil may be the largest of the oil companies, with diversified operations in multiple locations, but it is as dependent upon oil prices as the rest of the companies in its sector. In figure 10.1, we graph Exxon's operating income as a function of the average oil price each year from 1985 to 2008.

Figure 10.1: Operating Income versus Oil Prices for Exxon Mobil: 1985-2008



The operating income clearly increases (decreases) as the oil price increases (decreases). We regressed the operating income against the oil price per barrel over the period and obtained the following:

Operating Income = -6,395 + 911.32 (Average Oil Price) R^2 = 90.2%

(2.95) (14.59)

Put another way, Exxon Mobil's operating income increases about $9.11 billion for every $ 10 increase in the price per barrel of oil and 90% of the variation in Exxon's earnings over time comes from movements in oil prices.[55]

To get from operating income to equity value at Exxon, we made the following assumptions:

- We estimated a bottom-up beta of 0.90 for Exxon Mobil, and then used the treasury bond rate of 2.5% and an equity risk premium of 6.5% to estimate a cost of equity.

  Cost of equity = 2.5% + 0.90 (6.5%) = 8.35%

  Exxon has $9.4 billion of debt outstanding and a market capitalization of $320.4 billion (4941.63 million shares, trading at $64.83/share), resulting in a debt ratio of 2.85%. As a AAA rated company, its cost of debt is expected to be 3.75%, reflecting a default spread of 1.25% over the risk free rate. Using a marginal tax rate of 38% (rather than the effective tax rate), we estimate a cost of capital of 8.18% for the firm.

  Cost of capital = 8.35% (.9715) + 3.75% (1-.38) (.0285) = 8.18%
- Exxon Mobil is in stable growth with the operating income growing at 2% a year in perpetuity. New investments are expected to generate a return on capital that reflects the normalized operating income and current capital invested; this return on capital is used to compute a reinvestment rate.

Exxon reported pre-tax operating income in excess of $60 billion in 2008, but that reflects the fact that the average oil price during the year was $86.55. By March 2009, the price per barrel of oil had dropped to $ 45 and the operating income for the coming year will be much lower. Using the regression results, the expected operating income at this oil price is $34,614 billion:

Normalized Operating Income = -6,395 + 911.32 ($45) = $34,614

This operating income translates into a return on capital of approximately 21% and a reinvestment rate of 9.52%, based upon a 2% growth rate. [56]

---

[55] The relationship is very strong at Exxon because it has been a large and stable firm for decades. It is likely that the relationship between earnings and oil prices will be weaker at smaller, evolving oil companies.

Reinvestment Rate = g/ ROC = 2/21% = 9.52%

Value of Operating Assets

$$= \frac{\text{Operating Income } (1+g)\ (1-\ \text{tax rate})\ (1-\ \frac{g}{\text{ROC}})}{(\text{Cost of capital}\ -\ g)}$$

$$= \frac{34614\ (1.02)\ (1-\ .38)\ (1-\ \frac{2\%}{21\%})}{(.0818\ -\ .02)} = \$320{,}472 \text{ million}$$

Adding the current cash balance ($32,007 million), subtracting out debt ($9,400 million) and dividing by the number of shares (4,941.63 million) yields the value per share.

$$\text{Value per share} = \frac{\text{Operating Assets}\ +\ \text{Cash}\ -\ \text{Debt}}{\text{Number of shares}}$$

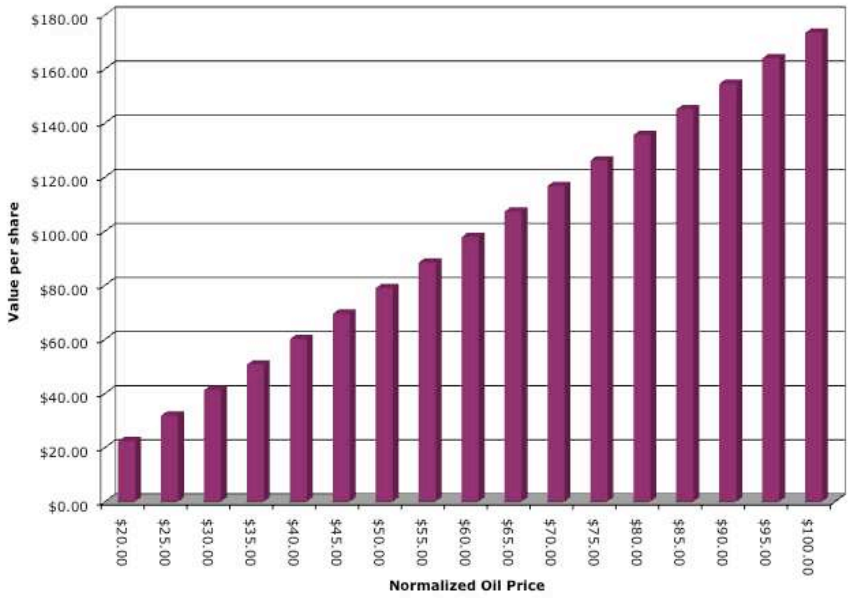
$$= \frac{320472\ +\ 32007\ -\ 9400}{4941.63} = \$69.43/share$$

At its current stock price of $64.83, the stock looks slightly under valued. However, that reflects the assumption that the current oil price (of $45) is the normalized price. In figure 10.2, we graph out the value of Exxon Mobil as a function of the normalized oil price:

---

[56] To compute the return on capital, we aggregated the book value of equity ($126,044 million), the book value of debt ($9,566 million) and netted out cash ($33,981 million) from the end of 2007, to arrive at an invested capital value of $101,629 million. The return on capital is computed as follows: Return on capital = Operating Income (1-tax rate)/ Invested Capital = 34614 (1-.38)/101629 = 21.1%

Figure 10.2: Normalized Oil price and Value per Share –Exxon Mobil



As the oil price changes, the operating income and the return on capital change; we keep the capital invested number fixed at $10,629 million and re-estimate the return on capital with the estimated operating income. If the normalized oil price is $42.52, the value per share is $64.83, equal to the current stock price. Put another way, any investor who believes that the oil price will stabilize above this level will find Exxon Mobil to be under valued.

Adaptive Growth

One of the perils of normalization, no matter what approach you use, is that we are replacing the current numbers of a company with what we believe the company will generate as earnings and cash flows, if the cycle rights itself. Since cycles can last for long periods, the danger is that normalization, even if warranted, may be a long time coming. One compromise solution is to assume normalization in the long term, but to allow earnings to follow the current cycle for the short term, and to use the growth rate as a mechanism to bring us back to normalcy.

Consider first the case of a cyclical company, with the economy mired in recession, or a commodity company, when the price is at the low point on the cycle. The earnings will be negative or low in the most recent time period and may get worse before it gets better. We can allow for the deterioration, by lowering revenues, earnings and cash flows in the near term (the first year) and for the improvement by allowing for higher revenue growth and improved margins in the medium term, as the company takes advantage of the economic cycle. With a cyclical company at the peak of the economic cycle or a commodity company when commodity companies have peaked, we reverse the process, allowing for short-term prosperity from the cycle, before reducing revenues and profit margins as the cycle reverts back to historic norms.

In effect, we are splitting the difference between normalization and forecasting the cycle. We are assuming that we have enough information to forecast how the economic or commodity price cycle will play out in the short term (next 6 months to a couple of years) but that we do not have to capacity to forecast it in the long term. Using the normalized numbers as our long-term targets, we estimate the rest of the numbers.

Illustration 10.3: Valuing Toyota with adaptive growth

In illustration 13.3, we valued Toyota, using normalized earnings and arrived at a value per share of 4735 Yen/share. Implicit in this valuation, however, is the assumption that while Toyota's earnings have been hurt by the economic slowdown that began in 2008, they will bounce back very quickly to pre-recession levels. To the extent that the recession that started in September 2008 was viewed as deeper and potentially longer-lasting than other recessions, we will over value the equity as a consequence.

To generate a more realistic estimate of the value of equity, we started with the assumption that the revenues in the next financial year (April 2009 – March 2010) would decline 10% and be accompanied by operating losses, and that the recovery would gradually begin the following year before picking up steam in the third year. In year 4, we will assume that Toyota will reach the stable state that we

assumed in illustration 10.1 – earning its historical average operating margin of 7.33% on revenues and generating a return on capital of 5.09% (equal to the cost of capital). Table 10.2 summarizes the year-by-year estimates of revenues, operating income and cash flows for the next 3 years and for the terminal year (year 4):

Table 10.2: Expected Free Cash Flow to Firm – Toyota

	Current	1	2	3	Terminal year
Revenue growth rate		-10%	4%	8%	1.50%
Revenues	¥22,661	¥20,395	¥21,211	¥22,908	¥23,251
Operating Margin	1.18%	-3%	1%	4%	7.33%
Operating Income	¥268	-¥612	¥212	¥916	¥1,704
Taxes	¥93	¥0	¥0	¥203	¥694
After-tax Operating Income	¥175	-¥612	¥212	¥714	¥1,011
- Reinvestment	-¥79	-¥200	¥300	¥400	¥298
FCFF	¥254	-¥412	-¥88	¥314	¥713
Terminal value				¥19,856	
Present value		-¥392	-¥80	¥17,378	

Capital Invested	¥14,945	¥14,745	¥15,045	¥15,445	
Return on Capital	1.79%	-4.15%	1.41%	5.93%	5.09%
Tax rate	34.73%	36.22%	37.72%	39.21%	40.70%
NOL			¥611.85	¥399.74	
Cost of capital	5.09%	5.09%	5.09%	5.09%	5.09%

Value of Operating Assets =	¥16,907
+ Cash & Other non-operating assets	¥9,133
- Debt	¥11,862
- Minority Interest	¥583
Value of Equity =	¥13,595
Value per share =	¥3,943

Capital invested$_t$ = Capital invested$_{t-1}$ + Reinvestment$_t$

There are several things to note about the projections. The first is that revenue growth is negative in year 1, but bounces back sharply in years 2 and 3, reflecting the climb back to normalcy. The second is that the operating loss that we forecast for year 1 creates a Net Operating Loss (NOL) carry forward that shelters the firm entirely from taxes in year 2 and partially in year 3; the tax rate also climbs from the current effective rate of 34.73% to the marginal rate of 40.7% in year 4. The third is that Toyota pulls back from reinvesting in the first year, but returns strongly to reinvest (again making up for lost ground) in years 2 and 3, before settling into its steady state reinvestment rate of 29.46%.

Stable period reinvestment rate = $\frac{g}{ROC} = \frac{.015}{.0509} = .2946$

We keep the cost of capital unchanged at 5.09% over the period, and the present value of the cash flows over the next 3 years and the terminal value yields a value for the operating assets of 16,907 billion Yen. Making the identical adjustments for cash, non-operating assets, debt and minority interests that we used in illustration 10.1, we estimate a value of equity per share of 3,943 Yen. While this is lower than 4735 Yen per share we estimated, with instant normalization, it is still significantly higher than the price per share of 3,060 Yen in February 2009.

Relative Valuation

The two basic approaches that we developed in the discounted cash flow approach –using normalized earnings or adapting the growth rate – are also the approaches we have for making relative valuation work with cyclical and commodity companies.

Normalized Earnings Multiples

If the normalized earnings for a cyclical or commodity firm reflect what it can make in a normal year, there has to be consistency in the way the market values companies relative to these normalized earnings. In the extreme case, where there are no growth and risk differences across firms, all firms should trade at the same multiple of normalized earnings. In effect, the PE ratios for these firms, with normalized earnings per share, should be identical across firms.

In the more general case, where growth and risk differences persist even after normalization, we would expect to see differences in the multiples that companies trade at. In particular, we should expect to see firms that have more risky earnings trade at lower multiples of normalized earnings than firms with more stable earnings. We would also expect to see firms that have higher growth potential trade at higher multiples of normalized earnings than firms with lower growth potential. To provide a concrete illustration, Petrobras and Exxon Mobil are both oil companies whose earnings are affected by the price of oil. Even if we normalize earnings, thus controlling for the price of oil, Petrobras should trade at a different

multiple of earnings than Exxon Mobil, because its earnings are riskier (because they are derived almost entirely from Brazilian reserves) and also because it has higher growth potential.

Adaptive fundamentals

For those analysts who are reluctant to replace the current operating numbers of a company with normalized values, the multiples at which cyclical and commodity firms trade at will change as we move through the cycle. In particular, the multiples of earnings for cyclical and commodity firms will bottom out at the peak of the cycle and be highest at the bottom of the cycle. While this may seem counter intuitive, it reflects the fact that markets have to value these companies for the long term,

If the earnings of all companies in a sector (cyclical and commodity) move in lock step, there are no serious consequences to comparing the multiples of current earnings that firms trade at. In effect, we may conclude that a steel company with a PE ratio of 6 is fairly valued at the peak of the cycle, when steel companies collectively report high earnings (and low PE). The same firm will be fairly valued at 15 times earnings at an economic trough, where the earnings of other steel companies are also down.

As with normalized earnings, the primary concern is that we control for other factors that affect the PE. When the cycle is working in your favor (strong economy and high commodity prices), all firms in a sector may report high earnings, but some firms may have better long-term prospects and should trade at higher multiples. By the same token, all oil companies may report lower earnings, when oil prices are down, but some of these companies may have more predictable earnings and therefore trade at higher multiples of earnings.

Illustration 10.3: PE ratios for oil companies

In February 2009, oil companies that had benefited over the prior five years of rising oil prices were shaken by the sudden drop in the price per barrel of oil, from $ 140 a barrel a year prior to $45 a barrel. While the market prices of oil companies tumbled to reflect the lower oil prices, the earnings reported by these

companies for the previous year reflected the high oil prices over that period. In table 10.3, we report on the stock prices of oil companies, in conjunction with four measures of earnings per share – earnings in the most recent (reported) fiscal year, earnings in the last four quarters, expected earnings in the next four quarters and a measure of normalized earnings obtained by averaging earnings per share over the previous five years. The PE ratios are estimated using each measure of earnings.

Table 10.3: PE Ratios – Oil Companies in February 2009

Company Name	Stock Price	Current EPS	EPS Trail 12 Mo	EPS Next 4 quarters	Average EPS _ Last 5 years	Current PE	Trailing PE	Forward PE	Normalized PE
BP PLC ADR	$37.21	$3.84	$8.18	$4.25	$6.20	9.69	4.55	8.76	6.00
Chevron Corp.	$61.22	$5.24	$11.67	$4.00	$7.30	11.68	5.25	15.31	8.39
ConocoPhillips	$37.98	$4.78	$10.69	$4.75	$6.25	7.95	3.55	8.00	6.08
Exxon Mobil Corp.	$65.77	$5.15	$8.66	$5.00	$6.50	12.77	7.59	13.15	10.12
Frontier Oil	$13.97	$0.21	$0.77	$1.35	$1.90	66.52	18.14	10.35	7.35
Hess Corp.	$57.17	$0.42	$7.24	$1.05	$3.40	136.12	7.90	54.45	16.81
Holly Corp.	$22.03	$3.06	$2.41	$2.75	$3.50	7.20	9.14	8.01	6.29
Marathon Oil Corp.	$22.59	$2.04	$4.94	$2.90	$4.20	11.07	4.57	7.79	5.38
Murphy Oil Corp.	$41.00	$2.88	$8.73	$2.85	$5.50	14.24	4.70	14.39	7.45
Occidental Petroleum	$55.59	$3.18	$8.97	$3.05	$5.50	17.48	6.20	18.23	10.11
Petroleo Brasileiro ADR	$30.47	$4.05	$4.44	$4.05	$4.15	7.52	6.86	7.52	7.34
Repsol-YPF ADR	$15.76	$1.48	$3.49	$2.45	$3.70	10.65	4.52	6.43	4.26
Royal Dutch Shell 'A'	$43.32	$5.42	$10.15	$5.10	$6.40	7.99	4.27	8.49	6.77
Sunoco Inc.	$28.33	$5.68	$7.48	$3.65	$4.30	4.99	3.79	7.76	6.59
Tesoro Corp.	$13.67	$2.60	$1.76	$2.10	$2.80	5.26	7.77	6.51	4.88
Total ADR	$49.85	$5.84	$9.16	$5.65	$7.15	8.54	5.44	8.82	6.97

As can be seen from the table, each version of the PE ratio tells a different story. With current PE (based on earnings per share in the most recent fiscal year), the cheapest stock is Sunoco, with a PE of 4.99 and Hess is off the charts with its PE ratio of 136, but the fact that the most recent fiscal year is different for different

firms – 2007 for some, midway through 2008 for others and the end of 2008 for a handful – gives us pause. With trailing PE, the cheapest stock is ConocoPhillips and the most expensive is Frontier Oil, and there are relatively few outliers. If we assume that all oil companies benefited equally from the oil price boom in the last four quarter and that there are no significant differences in growth and risk across oil companies, this would suggest that Conoco Phillips is cheap. However, perusing the expected growth rates in earnings per share, we find that Conoco has an expected growth rate of only 4% for the next 5 years, whereas analysts are forecasting growth of 8.5% a year for Petrobras. With forward PE ratios, there are no stocks that trade at PE ratios less than 6, but Repsol does have the lowest PE with 6.43. Finally, with normalized EPS, the cheapest stock remains Repsol with a PE of 4.26 and the most expensive is Hess; our assumption that the average earnings per share over the last 5 years is normal can be contested.

What are we to make of this mishmash of recommendations? First, it is critical that we stay consistent in how we measure earnings with commodity and cylcial companies. If we decide to use trailing earnings, we should do so for all companies. Second, the fundamentals that determine multiples – cash flows, growth and risk – apply just as much to commodity companies as they do to the rest of the market. To the extent that commodity companies are becoming more diverse, with large differences in growth potential and risk (especially in emerging markets), we should try to factor in these differences into our analyses.

The Real Option argument for undeveloped reserves

One critique of conventional valuation approaches is that they fail to consider adequately the interrelationship between the commodity price and the investment and financing actions of commodity companies. In other words, oil companies behave very differently (in terms of exploration and financing) when oil prices are $100 a barrel than they do when oil prices are only $20 a barrel. Since the managers of commodity companies get to observe the commodity price before they act, it can be argued that the learning and adaptive behavior that follows gives at least the semblance of a real options argument in these firms. If we accept this argument, the

upshot in valuation is that we should be adding a premium to conventional discounted cash flow valuations, to reflect this optionality, and the premium should become larger as commodity prices become more volatile.

Even if we never explicitly use option pricing models to value natural resource reserves or firms, there are implications for other valuation approaches:

a. Price volatility affects value: The value of a commodity company is a function of not only the price of the commodity but also the expected volatility in that price. The price matters for obvious reasons – higher commodity prices translate into higher revenues, earnings and cash flows. The variance in that price can affect value by altering the option values of undeveloped reserves. Thus, if the price of oil goes from $25 a barrel to $40 a barrel, you would expect all oil companies to become more valuable. If the price drops back to $25, the values of oil companies may not decline to their old levels, since the perceived volatility in oil prices may have changed.
b. Mature versus Growth commodity companies: As commodity prices become more volatile, commodity companies that derive more of their value from undeveloped reserves will gain in value, relative to more mature companies that generate cash flows from developed reserves. In the example used above, where oil price volatility is perceived to have changed even though the price itself has not changed, we would expect Petrobras to gain in value, relative to Exxon Mobil.
c. Development of reserves: As commodity price volatility increases, commodity companies will become more reluctant to develop their reserves. If we treat undeveloped reserves as options, and developing those reserves as the equivalent of exercising those options, higher volatility in the underlying commodity price will make exercise less likely (since we will lose the time premium on the option).
d. Optionality increases as commodity price decreases: The time premium on an option becomes smaller (as a percent of the option value) as it becomes in-the-money. In the context of natural resource options, this would imply that the option premium is greatest when commodity prices are low (and the reserves

are either marginally viable or not viable) and should decrease as commodity prices increases.

In closing, if we regard undeveloped reserves as options, discounted cash flow valuation will generally under estimate the value of natural resource companies, because the expected price of the commodity is used to estimate revenues and operating profits. As a consequence, we miss the option component of value. Again, the difference will be greatest for firms with significant undeveloped reserves and with commodities where price volatility is highest.

Value Plays

When investing in a commodity company, you are also investing in the commodity. Investing in an under valued oil company will not pay off, if oil prices drop 50% over the next few years. There are two ways you can incorporate this reality into your investing strategy. In the first, you take a stand on where commodity prices are going and then invest in those companies where you can exploit this price view. In the second, you find the best commodity companies to invest in, from a pure value perspective, and then use commodity futures or options to hedge commodity price risk. To make a commodity price play, you have to adopt one of two strategies. In the first, you play off price momentum and assume that if a commodity's price has risen in the recent past, it will continue to rise in the future. In the second, you adopt a more contrarian view and assume that long periods of price upswings will followed by declining prices. There is empirical evidence that supports both strategies, with momentum strategies generating more success for short term investing and reversal strategies more successful in the long term. If commodity prices are low right now, and you believe that they will increase significantly in the future, the value payoff will be highest in companies with the following characteristics:

- Significant undeveloped reserves of the commodity: The fact that these reserves are undeveloped gives these companies that option of waiting until the commodity price increases to develop them.

- ➔ Funding or capital to survive the near term commodity price cycle: To the extent that low commodity prices depress current earnings, you want to avoid companies that may be pushed into distress or have to sell their reserves to raise funding.

If you do not want to bring a commodity view into your investment strategy, your choice becomes a more conventional one. You want to invest in those commodity companies that are priced attractively and have significant low cost reserves of the commodity.

When investing in a cyclical company, you are making a play on overall economic growth. In other words, investing in a good cyclical company will not pay off, if economic growth is anemic. As with commodity companies, there are two investment strategies you can adopt. In the first, you put your faith in your forecasts of overall economic growth. If you believe that overall economic growth will be stronger than the rest of the market thinks it will be, you should buy strong cyclical companies that will benefit from the economic upswing, with strong defined to include the following components:

- A product or service mix that is not only unique but attracts customer loyalty.
- Normalized returns on investments that exceed the cost of capital; normalized would require looking across the economic cycle.
- Low financial leverage and a strong balance sheet that will provide protection in the event of an extended economic downturn.

This strategy is most likely to work in periods of economic malaise, where investors are over reacting to current economic indicators – inflation, GDP growth – and selling cyclical stocks. In the second, you accept your inability to forecast economic cycles and focus on buying the best bargains in each cyclical sector. The bargains will be defined as companies that trade at low prices, relative to their normalized fundamentals; low price to book ratios and high normalized ROE, low enterprise value to sales ratios with high normalized margins.

Conclusion

Cyclical and commodity companies have volatile earnings, with the volatility coming from macro economic factors that are not in the control of these companies. As the economy weakens and strengthens, cyclical companies will see their earnings go up and down, and commodity companies will see their earnings and cash flows track the commodity price.

When valuing these companies, analysts make one of two mistakes. They either ignore the economic and commodity price cycles, and assume that the current year's earnings and cashflows (which are a function of where we are in the cycle) will continue forever, or they expend resources trying to forecast the cycle in the long term. We presented two ways of valuing these firms. In the first, we look past the cycle at the normalized earnings, growth and cash flow for the firm. In effect, we are assuming that while cycles can cause big swings in the numbers, we cannot forecast the year-to-year shifts in cycles. In the second, we still assume normalization, but only in the long term. In the near term, we forecast revenues, earnings and cash flows, based on where we are in the cycle. While the two approaches will converge when firms are in the middle of a cycle, they will diverge at the top or bottom of a cycle.

In the final section of this chapter, we considered the possibility that the undeveloped reserves at commodity companies could be considered options, insofar as the company has the rights to develop these reserves but does not have to develop them. We argued that commodity companies, especially when the commodity price is volatile, can trade at a premium on their discounted cash flow values.

Chapter 12: Valuing companies with intangible assets: Technology, Brand name and Patents

In the last twenty years, we have seen a shift away from manufacturing firms to service and technology firms in the global economy, with the magnitude of the change greatest in the United States. As we value more and more pharmaceutical, technology and service companies, we are faced with two realities. The first is that the assets of these firms are often intangible and invisible – patents, know-how and human capital. The second is that the way in which accounting has dealt with investments in these assets is inconsistent with its treatment of investments in tangible assets at manufacturing firms. As a result, many of the basic inputs that we use in valuation – earnings, cash flows and return on capital – are contaminated.

What are intangible assets?

Looking at publicly traded firms, it is obvious that many firms derive the bulk of their value from intangible assets. From consumer product companies, dependent upon brand names, to pharmaceutical companies, with blockbuster drugs protected by patent, to technology companies that draw on their skilled technicians and know-how, these firms range the spectrum. In this section, we will begin by looking at their place in the market and how it has shifted over time and follow up by identifying characteristics that they share.

The simplest measure of how much intangible assets represent of the economy comes from the market values of firms that derive the bulk of their value from these assets as a proportion of the overall market. While technology firms have fallen back from their peak numbers in 2000, they still represented 14% of the overall S&P 500 index at the end of 2008. If we add pharmaceutical and consumer product companies to this mix, the proportion becomes even higher.

There have been other attempts to capture the importance of intangible assets in th economy. In one study, Leonard Nakamura of the Federal Reserve Bank of Philadelphia provided three different measures of the magnitude of intangible

assets in today's economy – an accounting estimate of the value of the investments in R&D, software, brand development and other intangibles; the wages and salaries paid to the researchers, technicians and other creative workers who generate these intangible assets; and the improvement in operating margins that he attributes to improvements to intangible factors.[57] With all three approaches, he estimated the investments in intangible assets to be in excess of $ 1 trillion in 2000 and the capitalized value of these intangible assets to be in excess of $ 6 trillion in the same year.

Characteristics of firms with intangible assets

While firms with intangible assets are diverse, there are some characteristics that they do have in common. In this section, we will highlight those shared factors, with the intent of expanding on the consequences for valuation in the next section.

1. Inconsistent accounting for investments made in intangible assets: Accounting first principles suggests a simple rule to separate capital expenses from operating expenses. Any expense that creates benefits over many years is a capital expense whereas expenses that generate benefits only in the current year are operating expenses. Accountants hew to this distinction with manufacturing firms, putting investments in plant, equipment and buildings in the capital expense column and labor and raw material expenses in the operating expense column. However, they seem to ignore these first principles when it comes to firms with intangible assets. The most significant capital expenditures made by technology and pharmaceutical firms is in R&D, by consumer product companies in brand name advertising and by consulting firms in training and recruiting personnel. Using the argument that the benefits are too uncertain, accountants have treated these expenses as operating expenses. As a consequence, firms with intangible assets report small capital expenditures, relative to both their size and growth potential.

---

[57] Nakamura, L., 1999,. Intangibles: What put the new in the new economy? Federal Reserve Bank of Philadephia Business Review July/August: 3—16.

2. Generally borrow less money: While this may be a generalization that does not hold up for some sub-categories of firms with intangible assets, many of them tend to use debt sparingly and have low debt ratios, relative to firms in other sectors with similar earnings and cash flows. Some of the low financial leverage can be attributed to the bias that bankers have towards lending against tangible assets and some of it may reflect the fact that technology and pharmaceutical firms are either in or have just emerged from the growth phase in the life cycle.
3. Equity Options: While the use of equity options in management compensation is not unique to firms with intangible assets, they seem to be much heavier users of options and other forms of equity compensation. Again, some of this behavior can be attributed to where these firms are in the life cycle (closer to growth than mature), but some of it has to be related to how dependent these firms are on retaining human capital.

Valuation Issues

The miscategorization of capital expenses, the sparing use of debt and equity-based compensation (options and restricted stocks) can create problems when we value these firms. In this section, we will lay out some of the issues that arise in both discounted cash flow and relative valuation.

- We generally draw on the current earnings and current book value of a firm to derive a value for existing assets. The flawed accounting treatment of intangible assets renders both numbers unreliable, since the reported earnings for a technology firm represent the earnings after reinvestment in R&D, rather than true operating earnings and the book value of assets (and equity) will be understated because the biggest assts for these firms are off the books; if you expense an item, you cannot show it as an asset. This has consequences not only for discounted cash flows valuation, where these numbers become the base from which we forecast, but also in relative valuation, where we compare multiples of accounting earnings and book values across companies.
- If growth is a function of how much firms reinvest and the quality of that reinvestment, the accounting treatment of expenditures on intangible assets

makes it difficult to gauge either number. The reinvestment made by the firm is often buried in the operating expenses (rather than showing up separately as capital expenditures) and the failure to record the book values of intangible assets makes measures like return on equity and capital, widely used to determine the quality of a firm's investments, unreliable.

- In addition to all of the standard variables that affect risk in a company, firms with intangible assets are susceptible to an additional risk. Lenders are wary of lending to firms with intangible assets, since monitoring these assets can be difficult to do. In addition, the values of some intangible assets, like human capital, can dissipate overnight, if a firm gets into trouble or has its reputation besmirched.
- Estimating when a firm with intangible assets gets to steady state can range from simple to complex. Consider the simple scenario first: a biotechnology firm that derives almost all of its growth from a single blockbuster drug, with a patent expiring in 7 years. Having a competitive advantage that comes with a time expiration stamp does make the judgment on when the company will hit stable growth very simple. A more complex scenario is a firm with a well-regarded brand name. Given the durability of consumer brand name as a competitive advantage, analysts face a much tougher task estimating when to put the firm into stable growth. The final and most difficult scenario is a firm, whose biggest intangible asset is human capital – consultants at McKinsey or traders at a private equity fund. Since it is very difficult to lock in human capital, these firms can lose their best assets overnight to the highest bidder. Figuring out how or why these firms manage to hold on to their best personnel is a central component to valuing them correctly.

The defense offered by some analysts is that the rules, flawed though they might be, are the same for all firms within a sector. As we will see in the next section, that does not neutralize the problem.

Valuation Solutions

To value firms with intangible assets, it would seem to us that we have to deal with the two big problems that they share. First, we have to clean up the financial statements (income statement and balance sheet) and re-categorize operating and capital expenses. The intent is not just to get a better measure of earnings, though that is a side benefit, but also to get a clearer sense of what the firm is investing to generate future growth. Second, we need to deal more effectively with equity options – the ones that have been granted in the past as well the ones that we expect to be granted in the future.

Regaining Accounting Consistency

While, in theory, income is not computed after capital expenses, the reality is that there are a number of capital expenses that are treated as operating expenses. A significant shortcoming of accounting statements is the way in which they treat research and development expenses. Under the rationale that the products of research are too uncertain and difficult to quantify, accounting standards have generally required that all R&D expenses to be expensed in the period in which they occur. This has several consequences, but one of the most profound is that the value of the assets created by research does not show up on the balance sheet as part of the total assets of the firm. This, in turn, creates ripple effects for the measurement of capital and profitability ratios for the firm. We will consider how to capitalize R&D expenses in the first part of the section and extend the argument to other capital expenses in the second part of the section.

Capitalizing R&D Expenses

Research expenses, notwithstanding the uncertainty about future benefits, should be capitalized. To capitalize and value research assets, we have to make an assumption about how long it takes for research and development to be converted, on average, into commercial products. This is called the *amortizable life* of these assets. This life will vary across firms and reflect the commercial life of the products that emerge from the research. To illustrate, research and development expenses at a pharmaceutical company should have fairly long amortizable lives, since the approval process for new drugs is long. In contrast, research and development expenses at a software firm, where products

tend to emerge from research much more quickly should be amortized over a shorter period.

Once the amortizable life of research and development expenses has been estimated, the next step is to collect data on R&D expenses over past years ranging back to the amortizable life of the research asset. Thus, if the research asset has an amortizable life of 5 years, the R&D expenses in each of the five years prior to the current one have to be obtained. For simplicity, it can be assumed that the amortization is uniform over time, which leads to the following estimate of the residual value of research asset today.

$$\text{Value of the Research Asset} = \sum_{t=-(n-1)}^{t=0} R\,\&\,D_t \frac{(n+t)}{n}$$

Thus, in the case of the research asset with a five-year life, you cumulate 1/5 of the R&D expenses from four years ago, 2/5 of the R & D expenses from three years ago, 3/5 of the R&D expenses from two years ago, 4/5 of the R&D expenses from last year and this year's entire R&D expense to arrive at the *value of the research asset*. This augments the value of the assets of the firm, and by extension, the book value of equity.

Adjusted Book Value of Equity = Book Value of Equity + Value of the Research Asset

Finally, the operating income is adjusted to reflect the capitalization of R&D expenses. First, the R&D expenses that were subtracted out to arrive at the operating income are added back to the operating income, reflecting their re-categorization as capital expenses. Next, the amortization of the research asset is treated the same way that depreciation is and netted out to arrive at the adjusted operating income.

Adjusted Operating Income = Operating Income + R & D expenses – Amortization of Research Asset

The adjusted operating income will generally increase for firms that have R&D expenses that are growing over time. The net income will also be affected by this adjustment:

Adjusted Net Income = Net Income + R & D expenses – Amortization of Research Asset

While we would normally consider only the after-tax portion of this amount, the fact that R&D is entirely tax deductible eliminates the need for this adjustment.[58]

---

[58] If only amortization were tax deductible, the tax benefit from R&D expenses would be:
Amortization * tax rate
This extra tax benefit we get from the entire R&D being tax deductible is as follows:
(R&D – Amortization) * tax rate

Illustration 11.1: Capitalizing R&D expenses: Amgen in February 2009

Amgen is a biotechnology/ pharmaceutical firm. Like most such firms, it has a substantial amount of R&D expenses and we will attempt to capitalize it in this example. The first step in this conversion is determining an amortizable life for R & D expenses. How long will it take, on an expected basis, for research to pay off at Amgen? Given the length of the approval process for new drugs by the Food and Drugs Administration, we will assume that this amortizable life is 10 years.

The second step in the analysis is collecting research and development expenses from prior years, with the number of years of historical data being a function of the amortizable life. Table 11.1 provides this information for the firm.

Table 11.1: Historical R& D Expenses (in millions)

Year	R& D Expenses
Current	3030.00
-1	3266.00
-2	3366.00
-3	2314.00
-4	2028.00
-5	1655.00
-6	1117.00
-7	864.00
-8	845.00
-9	823.00
-10	663.00

The current year's information reflects the R&D in the most recent financial year (which was calendar year 2008 in this example).

The portion of the expenses in prior years that would have been amortized already and the amortization this year from each of these expenses is considered. To make estimation simpler, these expenses are amortized linearly over time; with a 10-year life, 10% is amortized each year. This allows us to estimate the value of the research asset created at each of these firms and the amortization of R&D expenses in the current year. The procedure is illustrated in table 11.2:

If we subtract out (R&D – Amortization) (1- tax rate) and add the differential tax benefit, which is computed above, (1- tax rate) drops out of the equation.

Table 11.2: Value of Research Asset

Year	*R&D Expense*	*Unamortized portion*		*Amortization this year*
Current	3030.00	1.00	3030.00	
-1	3266.00	0.90	2939.40	$326.60
-2	3366.00	0.80	2692.80	$336.60
-3	2314.00	0.70	1619.80	$231.40
-4	2028.00	0.60	1216.80	$202.80
-5	1655.00	0.50	827.50	$165.50
-6	1117.00	0.40	446.80	$111.70
-7	864.00	0.30	259.20	$86.40
-8	845.00	0.20	169.00	$84.50
-9	823.00	0.10	82.30	$82.30
-10	663.00	0.00	0.00	$66.30
			$13283.60	$1,694.10

Note that none of the current year's expenditure has been amortized because it is assumed to occur at the end of the most recent year (which effectively makes it today). The sum of the dollar values of unamortized R&D from prior years is $13.284 billion. This can be viewed as the value of Amgen's research asset and would be also added to the book value of equity for computing return on equity and capital measures. The sum of the amortization in the current year for all prior year expenses is $1,694 million.

The final step in the process is the adjustment of the operating income to reflect the capitalization of research and development expenses. We make the adjustment by adding back R&D expenses to the operating income (to reflect its reclassification as a capital expense) and subtracting out the amortization of the research asset, estimated in the last step. For Amgen, which reported operating income of $5,594 million in its income statement for 2008, the adjusted operating earnings would be:

Adjusted Operating Earnings

= Operating Earnings + Current year's R&D expense – Amortization of Research Asset

= 5,594 + 3030 – 1694 = $ 6.930 million

The stated net income of $4,196 million can be adjusted similarly.

Adjusted Net Income

= Net Income + Current year's R&D expense – Amortization of Research Asset

= 4,196 + 3030 – 1694 = $ 5,532 million

Both the book value of equity and capital are augmented by the value of the research asset. Since measures of return on capital and equity are based upon the prior year's values, we computed the value of the research asset at the end of 2007, using the same approach that we used in 2008 and obtained a value of $ 11,948 million.[59]

Value of Research Asset_{2007} = $ 11,948 million

Adjusted Book Value of Equity_{2007}

= Book Value of Equity_{2007} + Value of Research Asset_{2007}

= 17,869 million + 11,948 million = $29,817 million

Adjusted Book Value of Capital_{2007}

= Book Value of Capital_{2007} + Value of Research Asset_{2007}

= $ 21,985 million + 11,948 million = $ 33,933 million

The returns on equity and capital are estimated by dividing the earnings in 2008 by the capital invested at the end of 2007 and are reported with both the unadjusted and adjusted numbers below:

	Unadjusted	Adjusted for R&D
Return on Equity	$\frac{4,196}{17,869} = 23.48\%$	$\frac{5,532}{29,817} = 18.55\%$
Pre-tax Return on Capital	$\frac{5,594}{21,985} = 25.44\%$	$\frac{6,930}{33,933} = 20.42\%$

While the profitability ratios for Amgen remain impressive even after the adjustment, they decline significantly from the unadjusted numbers.

Capitalizing Other Operating Expenses

While R&D expenses are the most prominent example of capital expenses being treated as operating expenses, there are other operating expenses that arguably should be treated as capital expenses. Consumer product companies such as Gillette and Coca Cola could make a case that a portion of advertising expenses should be treated as capital expenses, since they are designed to augment brand name value. For a consulting firm like KPMG or McKinsey, the cost of recruiting and training its employees could be

[59] Note that you can arrive at this value using the table above and shifting the amortization numbers by one row. Thus, $ 822.80 million will become the current year's R&D, $ 663.3 million will become the R&D for year –1 and 90% of it will be unamortized and so on.

considered a capital expense, since the consultants who emerge are likely to be the heart of the firm's assets and provide benefits over many years. For many new technology firms, including online retailers such as Amazon.com, the biggest operating expense item is selling, general and administrative expenses (SG&A). These firms could argue that a portion of these expenses should be treated as capital expenses since they are designed to increase brand name awareness and bring in new presumably long term customers.

While this argument has some merit, we should remain wary about using it to justify capitalizing these expenses. For an operating expense to be capitalized, there should be substantial evidence that the benefits from the expense accrue over multiple periods. Does a customer who is enticed to buy from Amazon, based upon an advertisement or promotion, continue as a customer for the long term? There are some analysts who claim that this is indeed the case and attribute significant value added to each new customer. It would be logical, under those circumstances, to capitalize these expenses using a procedure similar to that used to capitalize R&D expenses.

- Determine the period over which the benefits from the operating expense (such as SG&A) will flow.
- Estimate the value of the asset (similar to the research asset) created by these expenses. This amount will be added to the book value of equity/capital and used to estimate the returns on equity and capital.
- Adjust the operating income for the expense and the amortization of the created asset.

The net effects of the capitalization will be seen most visibly in the reinvestment rates and returns on capital that we estimate for these firms.

Illustration 11.2: Capitalizing Brand Name Advertising – Coca Cola in 2009

Coca Cola is widely regarded as possessing one of the most valuable brand names in the world. We know that the company has always spent liberally on advertising, partly directed at building up the brand name. In table 11.3, we report on selling and advertising expenditures at Coca Cola every year for the last 25 years, which we will assume is the amortizable life for brand name. (In truth, we should be going back a lot longer, but data limitations get in the way).

Table 11.3: Advertising Expenditures at Coca Cola: 1984-2008

Year	Year	SG&A Expense	Selling and Advertising	Brand Name Advertising	Amortization this year	Unamortized Expense
1984	1	$2,314	$1,543	$771	$30.85	$0.00
1985	2	$2,368	$1,579	$789	$31.57	$31.57
1986	3	$2,446	$1,631	$815	$32.61	$65.23
1987	4	$2,665	$1,777	$888	$35.53	$106.60
1988	5	$3,038	$2,025	$1,013	$40.51	$162.03
1989	6	$3,348	$2,232	$1,116	$44.64	$223.20
1990	7	$4,076	$2,717	$1,359	$54.35	$326.08
1991	8	$4,604	$3,069	$1,535	$61.39	$429.71
1992	9	$5,249	$3,499	$1,750	$69.99	$559.89
1993	10	$5,695	$3,797	$1,898	$75.93	$683.40
1994	11	$6,297	$4,198	$2,099	$83.96	$839.60
1995	12	$6,986	$4,657	$2,329	$93.15	$1,024.61
1996	13	$8,020	$5,347	$2,673	$106.93	$1,283.20
1997	14	$7,852	$5,235	$2,617	$104.69	$1,361.01
1998	15	$8,284	$5,523	$2,761	$110.45	$1,546.35
1999	16	$9,814	$6,543	$3,271	$130.85	$1,962.80
2000	17	$8,551	$5,701	$2,850	$114.01	$1,824.21
2001	18	$6,149	$4,099	$2,050	$81.99	$1,393.77
2002	19	$7,001	$4,667	$2,334	$93.35	$1,680.24
2003	20	$7,488	$4,992	$2,496	$99.84	$1,896.96
2004	21	$8,146	$5,431	$2,715	$108.61	$2,172.27
2005	22	$8,739	$5,826	$2,913	$116.52	$2,446.92
2006	23	$9,431	$6,287	$3,144	$125.75	$2,766.43
2007	24	$10,945	$7,297	$3,648	$145.93	$3,356.47
2008	25	$11,774	$7,849	$3,925	$156.99	$3,767.68
Total					$2,150.40	$31,910.23

We assume that two-thirds of the S,G and A expenses are for selling and advertising and that 50% of the selling and advertising expenses each year are associated with building up brand name, with the balance used to generate revenues in the current year. In the second-to-last column, we compute the amortization this year of prior year's expenditure, using straight-line amortization over 25 years. In the last column, we keep track of the unamortized portion of prior year's expenditures. The cumulated value of this column ($31.9 billion) can be considered the capital invested in the brand name.

There are potential refinements that will improve this estimate. One is to use a longer amortizable life and to go back further in time to obtain advertising expenses. The other is to convert the past expenditures into current dollar expenditures, based upon inflation. In other words, an expenditure of $ 771 million

in 1984 is really much larger if stated in 2008 dollars.[60] Both of these will increase the capital value of the brand name.

The adjustments to operating income, net income and capital invested, in table 11.4, mirror those made for Amgen for R&D expenses:

Table 11.4: Capitalizing Brand Name Advertising- Coca Cola

	Conventional Accounting	Capitalized Brand Name
Operating Income	$8,446	$10,220
Net Income	$5,807	$7,581
Equity invested	$21,744	$53,654
Capital Invested	$31,073	$62,983
ROE	26.71%	14.13%
Pre-tax ROC	27.18%	16.23%

Capitalizing brand name advertising substantially decreases both the return on equity and capital invested for Coca Cola.

Illustration 11.3: Capitalizing Recruitment and Training Expenses: Cyber Health Consulting

Cyber Health Consulting (CHC) is a firm that specializes in offering management consulting services to health care firms. CHC reported operating income (EBIT) of $51.5 million and net income of $23 million in the most recent year. However, the firm's expenses include the cost of recruiting new consultants ($ 5.5 million) and the cost of training ($8.5 million). A consultant who joins CHC stays with the firm, on average, 4 years.

To capitalize the cost of recruiting and training, we obtained these costs from each of the prior four years. Table 11.5 reports on these expenses and amortizes each of these expenses over four years.

Table 11.5: Human Capital Expenses: CHC

Year	Training & Recruiting Expenses	Unamortized Portion		Amortization this year
Current	$ 14.00	100%	$ 14.00	
-1	$ 12.00	75%	$ 9.00	$ 3.00
-2	$ 10.40	50%	$ 5.20	$ 2.60
-3	$ 9.10	25%	$ 2.28	$ 2.28
-4	$ 8.30	-	$ 0.00	$ 2.08

[60] When we use inflation adjusted values, the value of brand name increases to almost $ 40 billion.

Value of Human Capital Asset =	$ 30.48	
Amortization this year =		$9.95

The adjustments to operating and net income are as follows:

Adjusted Operating Income = Operating Income + Training and Recruiting expenses – Amortization of Expense this year = $ 51.5 + $ 14 - $ 9.95 = $ 55.55 million

Net Income = Net Income + + Training and Recruiting expenses – Amortization of Expense this year = $ 23 million + $ 14 million - $ 9.95 million = $ 27.05 million

These adjusted earnings numbers in conjunction with the value of the human capital asset, estimated in table 11.6, are used to compute the returns on equity and capital.

Table 11.6: Returns on Equity and Capital – Conventional versus Adjusted

	Conventional accounting	Capitalized Training Expenses
Net Income	$23.00	$27.05
Operating Income	$51.50	$55.55
Book Equity	$125.00	$155.48
Book Capital	$250.00	$280.48
ROE	18.40%	17.40%
Pre-tax ROC	20.60%	19.81%

As with Amgen and Coca Cola, capitalizing training expenses decreases the returns on equity and capital for the company.

Consequences for valuation

When we capitalize R&D, brand name advertising and training expenses, there are significant consequences for both discounted cash flow and relative valuation. In discounted cash flow valuation, our estimates of cash flows and growth can be dramatically altered by the use of the adjusted numbers. In relative valuation, comparisons of firms within the same sector can be skewed by where they are in the life cycle.

Intrinsic Valuation

When we capitalize the expenses associated with creating intangible assets, we are in effect redoing the financial statements of the firm and restating numbers that are fundamental inputs into valuation – earnings, reinvestment and measures of returns.

a. Earnings: As we have noted with all three examples of capitalization (R&D, brand name advertising and training/recruiting expenses), the operating and net income of a firm will change as a consequence. Since the adjustment involves adding back the current year's expense and subtracting out the amortization of past expenses, the effect on earnings will be non-existent if the expenses have been unchanged over time, and positive, if expenses have risen over time. With Amgen, for instance, where R&D expenses increased from $663 million at the start of the amortization period to $3.03 billion in the current year, the earnings increased by more than $1.3 billion as a result of the R&D adjustment.
b. Reinvestment: The effect on reinvestment is identical to the effect on earnings, with reinvestment increasing or decreasing by exactly the same amount as earnings.
c. Free Cash flow to the equity(firm): Since free cash flow is computed by netting reinvestment from earnings, and the two items change by the same magnitude, there will be no effect on free cash flows.
d. Reinvestment Rate: While the free cash flow is unaffected by capitalization of these expenses, the reinvestment rate will change. In general, if earnings and reinvestment both increase as a consequence of the capitalization of R&D or advertising expenses, the reinvestment rate will increase.
e. Capital Invested: Since the unamortized portion of prior year's expenses is treated as an asset, it adds to the estimated equity or capital invested in the firm. The effect will increase with the amortizable life and should thererfore be higher for pharmaceutical firms (where amortizable lives tend to be longer) than for software firms (where research pays off far more quickly as commercial products).
f. Return on equity (capital): Since both earnings and capital invested are both affected by capitalization, the net effects on return on equity and capital are unpredictable. If the return on equity (capital) increases after the recapitalization, it can be considered a rough indicator that the returns

earned by the firm on its R&D or advertising investments is greater than its returns on traditional investments.

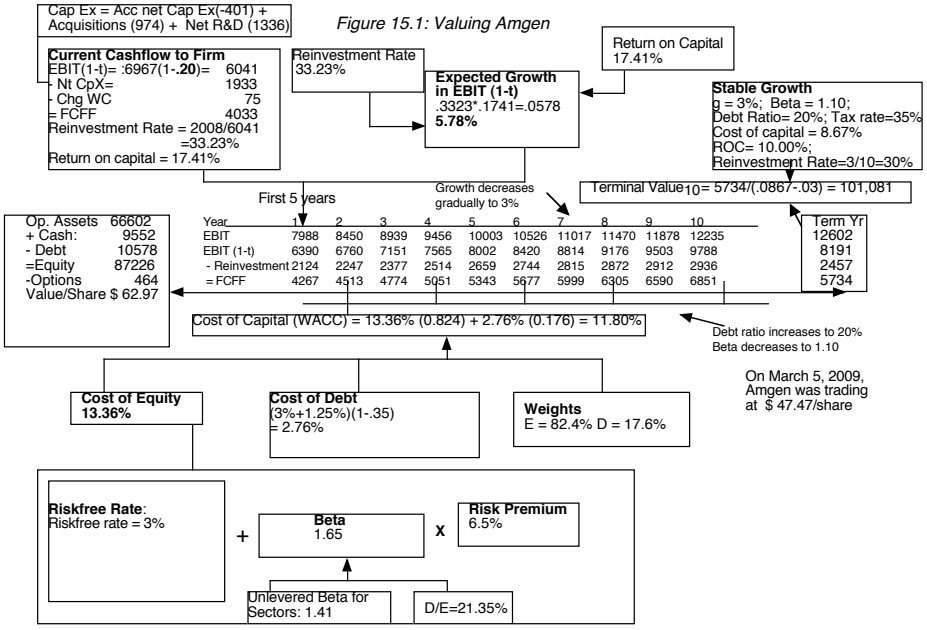
g. Expected growth rates: Since the expected growth rate is a function of the reinvestment rate and the return on capital, and both change as a result of capitalization, the expected growth rate will also change. While the higher reinvestment rate will work in favor of higher growth, it may be more than offset by a drop in the return on equity or capital.

In summary, the variables that are most noticeably affected by capitalization are the return on equity/capital and the reinvestment rate. Since the cost of equity/capital is unaffected by capitalization, any change in the return on capital will translate into a change in excess returns at the firm, a key variable determining the value of growth. In addition to providing us with more realistic estimates of what these firms are investing in their growth assets and the quality of these assets, the capitalization process also restores consistency to valuations by ensuring that growth rates are in line with reinvestment and return on capital assumptions. Thus, technology or pharmaceutical firms that want to continue to grow have to keep investing in R&D, while ensuring that these investments, at least collectively, generate high returns for the firm.

Illustration 11.4: Valuing Amgen

In illustration 11.1, we capitalized R&D expenses for Amgen and computed the adjusted operating income, reinvestment and return on capital at the firm. We used the restated numbers to estimate the value of the firm and equity per share. The valuation, where we assume ten years of high growth, is summarized in Figure 11.1:

Figure 11.1: Valuing Amgen – March 2009



Our estimate of value of equity per share is $62.97 a share, well above the prevailing stock price of $ 47.47.

An intriguing question is how the capitalization of R&D expenses affected value. To investigate, we compared the valuation fundamentals for Amgen, with conventional accounting, and with R&D treated as capital expenses in table 11.7:

Table 11.7: Valuation Fundamentals – With and Without R&D Capitalization

	Conventional	Capitalized R&D
After-tax ROC	14.91%	17.41%
Reinvestment Rate	19.79%	33.23%
Growth Rate	2.95%	5.78%
Value per share	$43.63	$62.97

We then revalued the firm, using both sets of fundamentals. As the table indicates, the value per share would have been $43.63, if we had used conventional accounting numbers. Clearly, capitalization matters and the degree to which it matters will vary across firms. In general, the effect will be negative for firms that invest large amounts in R&D, with little to show (yet) in terms of earnings and cash flows in

subsequent periods. It can be positive for firms that reinvest large amounts in R&D and report large increases in earnings in subsequent periods. In the case of Amgen, capitalizing R&D has a positive effect on value per share, because of its track record of successful R&D.

Relative Valuation

It is true that all technology and pharmaceutical companies operate under the same flawed accounting rules, expensing R&D, rather than capitalizing it. That does not mean, though, that there are no consequences for relative valuation. As we noted in the last section, the effect of capitalizing R&D on earnings and book value can vary widely across firms and will depend upon the following:

a. Age of the firm and stage in life cycle: Generally speaking, the effects of capitalization will be much greater at young firms than at more mature firms. Consider, for instance, the capitalization of R&D expenses. Capitalizing these expenses will increase earnings far more at young firms for two reasons: (a) R&D expenses will comprise a much larger proportion of the total expenses at these firms and (b) R&D expenses are more likely to have increased significantly over time.
b. Amortizable life: The effect of capitalizing expenses will be much greater as we extend the amortizable life of R&D, especially on capital invested. If we assume that all firms in a sector share the same amortizable life for R&D, this will not be an issue, but to the extent that different firms within the same business may convert research into commercial products at different speeds, the effect on earnings of capitalizing R&D can vary across firms

If we ignore accounting inconsistencies and use the reported earnings and book values of firms in the computation of multiples, we are likely to find that younger firms or firms that have R&D with longer gestation periods are overvalued. Their earnings and book value will be understated, leading to much higher PE, EV/EBITDA and book value multiples for these firms.

There are two ways we can incorporate these factors into relative valuation. The first is to capitalize the expenses associated with investing in intangible assets

for each firm and to compute consistent measures of earnings and book value to use in multiples. This approach, while yielding the most precision, is also the most time and data intensive. The second is to stick with the reported accounting values for earnings and book value, which controlling for the factors listed above.

Illustration 11.5: Valuing large pharmaceutical firms with PE ratios

To examine the effect of R&D, we estimated the PE ratios, in February 2009, for pharmaceutical firms, using several measures of net income in table 11.8:

Table 11.8: PE Ratios for Pharmaceutical Companies – February 2009

Company Name	*Market Cap $ (Mil)*	*Net Income*	*R&D expense*	*R&D net of amortization*	*PE*	*P/(E+R&D)*	*P/(E+ Net R&D)*
Merck & Co.	$46,702	$7,804	$4,805	$302	5.98	3.70	5.76
AstraZeneca PLC	$44,366	$6,130	$5,179	$650	7.24	3.92	6.54
GlaxoSmithKline ADR	$77,596	$10,619	$6,707	$225	7.31	4.48	7.16
Lilly (Eli)	$31,232	$3,863	$3,840	$410	8.08	4.05	7.31
Sanofi-Aventis	$67,924	$7,068	$4,575	$450	9.61	5.83	9.03
Novartis AG ADR	$79,954	$8,163	$1,834	$76	9.79	8.00	9.70
Pfizer Inc.	$85,433	$8,104	$7,945	$550	10.54	5.32	9.87
Biogen Idec Inc.	$12,732	$783	$1,072	$415	16.26	6.86	10.63
Wyeth	$54,391	$4,417	$3,373	$155	12.31	6.98	11.90
Bristol-Myers Squibb	$35,019	$2,165	$3,585	$710	16.18	6.09	12.18
Schering-Plough	$26,475	$1,903	$850	$135	13.91	9.62	12.99
Allergan Inc.	$10,901	$577	$798	$255	18.89	7.93	13.10
Teva Pharmac. (ADR)	$34,279	$2,374	$786	$221	14.44	10.85	13.21
Genzyme Corp.	$14,348	$421	$1,308	$622	34.08	8.30	13.76
Novo Nordisk	$28,165	$1,681	$1,368	$355	16.76	9.24	13.83
Abbott Labs.	$71,357	$4,881	$2,689	$250	14.62	9.43	13.91
Gilead Sciences	$40,310	$2,011	$721	$375	20.04	14.75	16.89
Celgene Corp.	$18,302	$226	$399	$215	80.84	29.26	41.46

To contrast with the conventional PE ratio, which is based on reported net income, we computed two alternative measures of earnings. In the first, we use the simplistic adjustment of adding back R&D expenses to net income to arrive at a multiple of the market price to earnings before R&D expenses. In the second, we make the full adjustment for R&D, adding back the R&D and subtracting out the amortization of R&D to arrive at an adjusted net income.

The results are revealing. On all three measures of PE, Merck looks like it is the most undervalued company in the group. As we add back R&D, the differences between the earnings multiples decreases, with Celgene remaining the outlier. Finally, when we compute the multiple of earnings with net R&D added back, the more mature pharmaceutical companies with less attractive growth prospects emerge with lower PE ratios, whereas the smaller, higher growth companies trade at higher multiples of earnings.

Dealing with Equity Options

In the last two decades, firms have increasing turned to compensating managers using equity, with options being a key component, for several reasons. The first is to align management interests with stockholders, i.e, to make managers think like stockholders by giving them an equity stake. The second is it allowed cash-poor firms with significant growth prospects to compete for employee talent against deep-pocketed rivals; young technology firms are prime users of options. The third is that the accounting for options woefully understated the true cost of these options, allowing these firms to report much positive earnings, even as they gave away big chunks of equity to managers.

Firms that pay managers and others with equity options create a second claim on the equity on top of the claim that common stockholders have. Since we are called up to estimate the value of equity per common share, we have to consider how to allocate the aggregate equity value across the two claimholders. In this section, we will first examine how to deal with options that a firm may have granted to managers in the past, that have not been exercised yet; this is the option overhang. In the second section, we will extend the analysis to look at how best to deal with options that may be granted in the future to employees and how to bring the consequences of such grants into the value of equity per share today.

The Option Overhang

There are three approaches that are widely used to deal with outstanding options, issued in prior periods. The crudest way is to assume that all or some of the options will be exercised in the future, adjust the number of shares outstanding and

divide the value of equity by this number to arrive at value per share; this is the diluted shares approach. The second and slightly more tempered approach is to incorporate the exercise proceeds from the options in the numerator and then divide by the number of shares that would be outstanding after exercise; this is the treasury stock approach. The third and preferred approach for dealing with options is to estimate the value of the options today, given today's value per share and the time premium on the option. Once this value has been estimated, it is subtracted from the estimated equity value, and the remaining amount is divided by the number of shares outstanding to arrive at value per share.

I. Use fully diluted number of shares to estimate per-share value

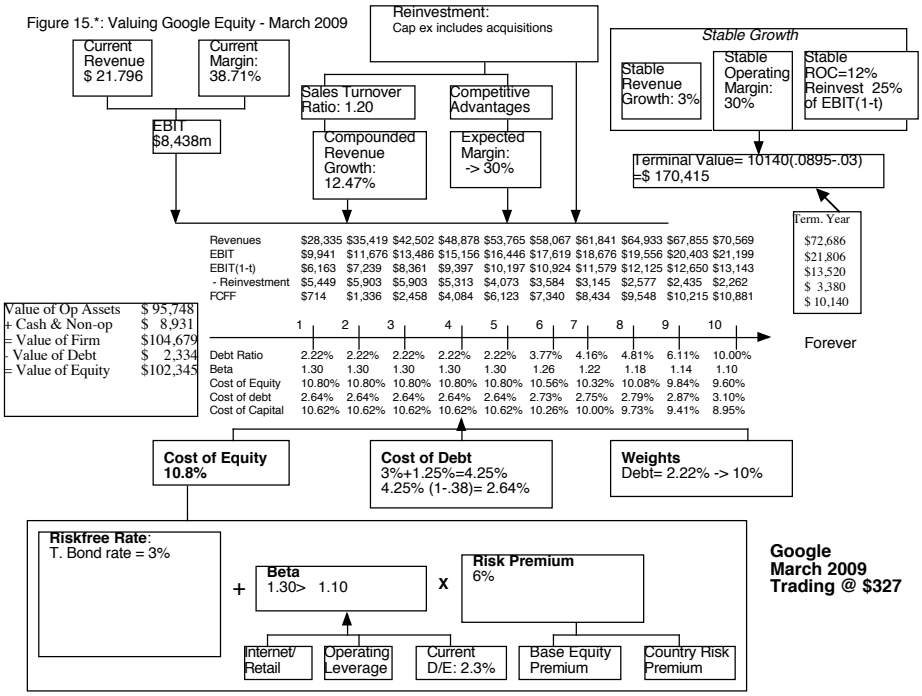
The simplest way to incorporate the effect of outstanding options on value per share is to divide the estimated value of equity from a discounted cash flow model by the number of shares that will be outstanding if all options are exercised today – the fully diluted number of shares. While this approach has the virtue of simplicity, it will lead to too low of an estimate of value per share for three reasons:

- It considers all options outstanding, not just ones that are in the money and vested. To be fair, there are variants of this approach where the shares outstanding are adjusted to reflect only in-the-money and vested options.
- It does not incorporate the expected proceeds from exercise, which will comprise a cash inflow to the firm.
- Finally, this approach does not build in the time premium on the options into the valuation.

Illustration 11.6: Fully Diluted Approach to estimating Value per Share

To apply the fully diluted approach to estimate the per share value, we will value a company with a significant option overhang –Google. We begin by valuing equity in the aggregate, capitalizing R&D along the way (we used a four year amortizable life for Google's R&D) and using a ten-year high growth period. Figure 11.2 summarizes the value of equity.

Figure 11.2 Valuing Google's Equity – March 2009



In February 2009, Google had 315.29 million shares outstanding and 13.97 millioin in options outstanding. To estimate the value of equity per share, we divide the aggregate value of equity estimated in figure 11.2 by the total number of shares outstanding.

Diluted value of equity per share

$$= \frac{\text{Aggregate Value of Equity}}{\text{Fully diluted number of shares}} = \frac{102345}{(315.29 + 13.97)} = \$310.83/share$$

This value, however, ignores both the proceeds from the exercise of the options as well as the time value inherent in the options. At Google, for example, a significant number of the options issued in past years are out-of-the-money and may never be exercised.

A modified version of this approach counts only in-the-money options when computing diluted shares. Of Google's 13.97 options outstanding, 4.75 million were

in-the-money, with an exercise price < stock price. If we count only these shares outstanding, the value of equity per share is $319.79:

Partially diluted valued of equity per share

$$= \frac{\text{Aggregate Value of Equity}}{\text{Fully diluted number of shares}} = \frac{102345}{(315.29 + 4.75)} = \$319.79/share$$

II. Treasury Stock Approach

This approach is a variant of the fully diluted approach. Here, the number of shares is adjusted to reflect options that are outstanding, but the expected proceeds from the exercise (the product of the exercise price and the number of options) are added to the value of equity. The limitations of this approach are that, like the fully diluted approach, it does not consider the time premium on the options and there is no effective way of dealing with vesting. Generally, this approach, by under estimating the value of options granted, will over estimate the value of equity per share.

The biggest advantage of this approach is that it does not require a value per share (or stock price) to incorporate the option value into per-share value. As we will see with the last (and recommended) approach, there is a circularity that is created when the stock price is an input into the process of estimating option value which, in turn, is needed to obtain the value per share.

Illustration 11.7: Treasury Stock Approach

To use the treasury stock approach with Google, we first estimated the average exercise price across all options outstanding and added the exercise proceeds to the estimated value of equity, before dividing by the fully diluted number of shares outstanding. (We used the average exercise price of $391.40 across all options in making this estimate.)

Treasury stock value of equity per share

$$= \frac{\text{Value of Equity} + \text{Options outstanding} * \text{Average Exercise Price}}{\text{Fully diluted number of shares}}$$

$$= \frac{\$102,345 + 13.97 * \$391.40}{(315.29 + 13.97)} = \$327.44/share$$

As with the diluted approach, there are modified versions of this approach where only in-the-money options are considered. This will reduce the value per share for Google considerably since the average exercise price for the in-the-money options, at $185, is much lower than the weighted average exercise price of $391.40.

Treasury stock value of equity per share (based on in-the-money options)

$$= \frac{\text{Value of Equity} + \text{Options outstanding} * \text{Average Exercise Price}}{\text{Fully diluted number of shares}}$$

$$= \frac{\$102{,}345 + 4.75 * \$185}{(315.29 + 4.75)} = \$314.45/share$$

III. Valuing Options

The problem with both the diluted stock and the treasury stock approaches is that they miss the essence of options. After all, the value of an option should include not only the current exercise value (recognized by the treasury stock approach) but also the time premium, reflecting the fact that the option still has life and the underlying stock is volatile. Much of the debate on dealing with options has raged around how well option pricing models work in valuing employee options. In this section, we will review some of this discussion and examine how to adapt conventional option pricing models to value these options.

Measurement Issues

Option pricing models have been widely used, to good effect, for almost four decades now for valuing listed and traded options on the option exchanges. In valuing employee options, however, there are five measurement issues that we have to confront.

a. Vesting: Firms granting employee options usually require that the employee receiving the options stay with the firm for a specified period, to be able to exercise the option (at which point they are vested). When we examine the options outstanding at a firm, we are looking at a mix of vested and non-vested options. The non-vested options should be worth less than the vested options, but the probability of vesting will depend upon how in-the-money the options are and the period left for an employee to vest.

b. Illiquidity: Employee options cannot be traded. As a result, employee options are often exercised before maturity, making them less valuable than otherwise similar traded options that are marketable. In a comprehensive study of 262,931 option exercises of employee options between 1996 and 2003 by U.S. companies, Brooks, Chance and Cline (cited above) note that 92.3% exercise early. On average, they find that exercise takes place 2.69 years after vesting, with 4.71 years left to expiration. Put another way, an employee option with a stated maturity of 10 years is usually exercised in 5.29 years.

c. Stock price or stock value: While conventional option pricing models are built around using the current market price as a key input, we do come up with estimates of value per share when we value companies, and these estimates can be very different from current stock prices. We have to consider whether we want to use our estimates of value per share, rather than the market prices, to preserve valuation consistency.

d. Dilution: Unlike listed options on exchanges, where the exercise of the option has no impact on the number of shares outstanding or the share price, the exercise of employee options can alter both.

e. Tax consequences: Firms are allowed to deduct the difference between the stock and the exercise price of an option at exercise and there is potential tax saving at the time of option exercise. This potential tax benefit reduces the drain on value created by having options outstanding.

f. Unobservable inputs: The final issue relates to options granted at private firms or firms on the verge of a public offering. Key inputs to the option-pricing model, including the stock price and the variance, cannot be obtained for these firms, but the options have to be valued nevertheless.

Modifying Option Pricing Models

With all of these issues affecting valuation, how do we adapt conventional option pricing models to value employee options? This question has been addressed both by academics who value options and by FASB, in its attempts to give guidance to firms that have to value these options for expensing.

Modified Black Scholes

The conventional Black Scholes model is designed to value European options on traded assets and does not explicitly factor in the dilution inherent in employee options or the illiquidity/vesting issues specific to these options. However, adaptations of the model provide reasonable estimates of value:

1. Build in expected dilution into the stock price: One of the inputs into the Black Scholes is the current stock price. To the extent that the exercise of options increases the number of shares outstanding (at a price less than the current stock price), the stock price will drop on exercise. A simple adjustment to the stock price can incorporate this effect:

   $$\text{Adjusted Stock Price} = \text{Current Stock Price}\left[\frac{n_{\text{shares outstanding}}}{(n_{\text{shares outstanding}} + n_{\text{options}})}\right]$$

   The resulting lower adjusted stock price will also reduce the option value.[61]

2. Reduce the life of the option to reflect illiquidity and early exercise: Earlier in this paper, we noted that employees often exercise options well before maturity because these options are illiquid. Typically, options are exercised about half way through their stated lives. Using a reduced life for the option will reduce its value.
3. Adjust option value for probability of vesting: The vesting adjustment can be made in the process of calculating of the option value. If we can assess the probability of vesting, multiplying this probability by the option value will yield an expected value for the option.

While purist would still resist, the model has provided remarkably resilient even in environments where its basic assumptions are violated.

---

[61] A modified version of the adjustment allocates the overall value of equity across all potential shares outstanding:

$$\text{Adjusted Stock Price} = \left[\frac{\text{Share Price} * n_{\text{shares outstanding}} + \text{Value per Option} * n_{\text{options}}}{(n_{\text{shares outstanding}} + n_{\text{options}})}\right]$$

Binomial Models

The possibility of early exercise and non-vesting, which is substantial in employee options, leads many practitioners to argue for the use of Binomial lattice models to value employee options. Unlike the Black-Scholes, these models not only can model for early exercise, but can be modified to allow for other special features specific to employee options, including vesting. In addition, binomial models allow for more flexibility on inputs, with volatility changing from period to period rather than remaining constant (which is the assumption in the Black-Scholes model). The limitation of the binomial models is that they are more information intensive, requiring the user to input prices at each branch of the binomial model. In any realistic version of the model, where the time intervals are short, this could translate into hundreds of potential prices.

The primary benefit of binomial models comes from the flexibility that they offer users to model the interaction between the stock price and early exercise. One example is the Hull-White Model, which proposes reducing the life used to value employee options to a more realistic level.[62] This model take into account the employee exit rate during the vesting period (thus taking into account the probability that options will end up unvested and worthless) and the expected life of the option after they get vested. To estimate the latter, the model assumes that there will be exercise if the stock price reaches a pre-specified multiple of the exercise price, thus making exercise an endogenous component of the model, rather than an exogenous component. The resulting option values are usually lower than those estimated using the Black-Scholes model.

Simulation Models

The third choice for valuing employee options is Monte Carlo simulation models. These models begin with a distribution for stock prices and a pre-specified exercise strategy. The stock prices are then simulated to arrive at the probabilities that employee options will be exercised and an expected value for the options based

---

[62] J. Hull and A. White, How to Value Employee Stock Options, Financial Analysts Journal 60 (1) (2004), 114{119.

upon the exercise. The advantage of simulations is that they offer the most flexibility for building in the conditions that may affect the value of employee options. In particular, the interplay between vesting, the stock price and early exercise can all be built into the simulation rather than specified as assumptions. The disadvantage is that simulations require far more information than other models.

How much does the model matter?

How much does the model used to value employee options matter? Are there significant differences in values when we use alternative models to value employee options? For the most part, the biggest single component determining employee option value is the life of the option. Using the stated life of employee options in the Black-Scholes models yields too high a value for these options. If we use an expected life for the option (which takes into account early exercise and vesting probabilities), the values that we arrive at are not dissimilar using different models. Ammann and Seiz (2003) show that the employee option pricing models in use (the binomial, Black Scholes with adjusted life and Hull White) all yield similar values.[63] As a consequence, they argue we should steer away from models that require difficult to estimate inputs (such as risk aversion coefficients) and towards simpler models.

Illustration 11.8: Option Value Approach

In Table 11.9, we begin by estimating the value of the options outstanding at Google, using the Black-Scholes model, adjusted for dilution and using half the stated maturity (to allow for early exercise). To estimate the value of the options, we first estimate the standard deviation of 50% in stock prices[64] over the previous 2 years. Weekly stock prices are used to make this estimate, and this estimate is annualized[65]. All options, vested as well as non-vested, are valued and there is no adjustment for non-vesting.

---

[63] Ammann, M., and R. Seiz, 2003, *Does the Model Matter? A Valuation Analysis of Employee Stock Options*, Working Ppaer, SSRN.

[64] The variance estimate is actually on the natural log of the stock prices. This allows you us to cling to at least the possibility of a normal distribution. Neither stock prices nor stock returns can be normally distributed since prices cannot fall below zero and returns cannot be lower than –100%.

[65] All of the inputs to the Black Scholes model have to be in annual terms. To annualize a weekly variance, we multiply by 52.

Table 11.9: Estimated Value of Options Outstanding

	Google
Number of Options Outstanding	13.97
Average Exercise Price	$391.41
Estimated Standard Deviation (Volatility)	50%
Average stated maturity	7.00
Maturity adjusted for early exercise	3.50
Stock Price at time of analysis	$326.6
Value per option	$ 103.6
Value of options outstanding	$ 1,447
Tax Rate	38.00%
After-tax Value of options outstanding	$ 897

In estimating the after-tax value of the options at these companies, we have used the marginal tax rate of 38%. Since the tax law allows for tax deductions only at exercise and only for the exercise value, we are potentially overstating the possible tax benefits (and understating the costs).

The value per share is computed by subtracting the value of the options outstanding from the value of equity and then dividing by the primary number of shares outstanding:

$$\text{Value of equity per share} = \frac{\text{Value of equity - Value of options}}{\text{Primary shares outstanding}} = \frac{102{,}345 - 897}{315.29} = \$321.76$$

The inconsistency averred to earlier is clear when we compare the value per share that we have estimated in this table to the price per share that we used in the previous one to estimate the value of the options. For instance, Google's value per share is $321.76, whereas the price per share used in the option valuation is $ 326.60. If we choose to iterate, we would revalue the options using the estimated value, which would lower the value of the options (to $1,406 million) and increase the value per share, leading to a second iteration and a third one and so on. The values converge to yield a consistent estimate of $321.84, close to our original estimate. That is because we estimated a value

per share close to the current price; as the difference widens, the effect of doing the iterative process on value per share will also increase.

Future Option Grants and Effect on Value

Just as options outstanding represent potential dilution or cash outflows to existing equity investors, expected option grants in the future will affect value per share by increasing the number of shares outstanding in future periods. The simplest way of considering why future option grants affect value is to treat them as employee compensation. The resulting increase in operating expenses will decrease operating income and after-tax cash flows in future years, thus reducing the value that we would attach to the firm today.

There are two things to note here. The first is that this process is on top of the adjustment made to equity value per share for the option overhang. It does not represent double counting, because it captures two different drains on equity value per share, one from past option grants and one from expected future grants. However, if we do this, we should not also increase the number of shares outstanding to reflect future option exercise. That would be double counting. The second is that making this estimate has become immeasurably easier, now that the accounting rules have changed to require firms to show option grants as expenses. The operating and net income for most firms now should be after the option expense, and if we forecast future values based on these numbers we are incorporating the expenses associated with future grants into our cash flows. The only note of caution that we would add is that as firms become larger, the option grants as a percent of revenues or value will tend to become smaller. Thus, we should move option grants for firms towards industry averages or mature firm practices as we forecast out further into the future. [66]

---

[66] If the firm is not expensing options, the current earnings of the firm may already include the expenses associated with option exercises in the current period. If the effect on operating income of option exercise in the current period is less than the expected value of new option issues, we have to allow for an additional expense associated with option issues. Conversely, if a disproportionately large number of options were exercised in the last period, we have to reduce the operating expenses to allow for the fact that the expected effect of option issues in future periods will be smaller.

Illustration 11.9: Valuing with expected option issues

When valuing Google, the current operating income was a key input. The way the firm has dealt with employee option expenses will play a key role in what operating income we will use in valuation. Over the past three years, the firm has shifted to expensing employee options. In its 2008 annual report, for instance, the firm highlights employee option expenses as a proportion of total revenues and table 11.10 summarizes the numbers:

Table 11.10: Employee Option Expenses – Google

Year	Value of Employee options granted	As % of Revenues
2006	$458.10	4.30%
2007	$868.60	5.20%
2008	$1,119.80	5.10%

Note that the expense associated with employee options is a significant drain on income and shows no signs of abating as Google becomes larger as a company.

Relative Valuation

Just as options affect intrinsic valuations, they also affect relative valuations. In particular, comparing multiples across companies is complicated by the fact that firms often have varying numbers of employee options outstanding and these options can have very different values. A failure to explicitly factor these options into analysis will result in companies with unusually large or small (relative to the peer group) numbers of options outstanding looking misvalued on a relative basis.

To see the effect of options on earnings multiples, consider the most widely used one, which is the PE ratio. The numerator is usually the current price per share and the denominator is earnings per share. Analysts who use primary earnings per share are clearly biasing their analysis towards finding companies with higher option overhang to be undervalued. To see why, note that the price per share should incorporate the effect of options outstanding – the market price will be lower when there are more employee options outstanding, but the denominator does not since it reflects actual shares outstanding and does not capture potential dilution. Note that this bias will not disappear when firms switch to expensing options.

To counter this, analysts often use fully diluted earnings per share to incorporate the effect of outstanding options, thus penalizing companies with large numbers of options outstanding. The problem with this approach is that it treats all options equivalently, with the number of shares increasing by the same unit whether the option is out-of-the-money and has three weeks left to expiration or deep in-the-money and has five years left to maturity. Clearly, firms that have more of the latter should trade at lower market values (for any given level of earnings) and will look cheaper on a diluted basis.

What is the solution? The only way to incorporate the effect of options into earnings multiples is to value the options at fair value, using the current stock price as the basis, and add this value on to the market capitalization to arrive at the total market value of equity.[67] This total market value of equity can be divided by aggregate net income to arrive at a PE ratio that incorporates (correctly) the existence of options. This will allow analysts to consider all options outstanding and incorporate their characteristics into the value.

$$\text{Option corrected PE} = \frac{(\text{Market Capitalization} + \text{Estimated value of options oustanding})}{\text{Net Income}}$$

The net income used should be the earnings estimated on the assumption that employee options are compensation and operating expenses. With the adoption of 123R, this has become a little easier to do, though many companies still reports net income before and after these expenses.

Everything that we have said about earnings multiples can also be said about book value multiples. Failing to incorporate the value of equity options into the market value of equity will make option-heavy companies look cheaper, relative to companies that have fewer options outstanding. The solution is the same as it was for earnings multiples. Estimating the value of employee options and adding them to market capitalization will almost always eliminate the bias in the comparison process.

---

[67] Harking back to the last section, the value of options used should be calculated based upon the current stock price (rather than an estimated value) and on a pre-tax basis.

Illustration 11.10: Adjusting PE ratio for options outstanding

To examine the effects of options outstanding on relative valuation, we will compare Google and Cisco, two technology firms with a history of using employee options. In table 11.11, we estimate the conventional PE ratio and contrast it with the adjusted PE ratio, using the approach described above:

Table 11.11: PE ratio versus Adjusted PE ratio: Google and Cisco

	Google	Cisco
Stock price	$326.60	$16.23
Primary Shares outstanding	315.29	5986.00
Number of options outstanding	13.97	1199.00
Primary EPS	$13.40	$1.47
Diluted EPS	$12.83	$1.23
Primary PE	24.37	11.04
Diluted PE	25.45	13.25
Market Capitalization	$102,975	$97,153
Value of Options	$1,406	$3,477
Market Value of Equity (Market capitalization + Value of options)	$104,381	$100,630
Net Income before option expensing	$5,347	$8,802
Net Income after option expensing	$4,227	$8,052
Adjusted PE (Market value of Equity/ Net income after option exp)	24.69	12.50

Note that for Google, the effects of incorporating options into the market value of equity and using net income after option expensing does not have a material impact on the PE ratio. For Cisco, the effects are much stronger with the PE ranging from 11 to 13.25, depending on how we deal with options.

Value Plays

The biggest hurdle when investing in firms with intangible assets is that the accounting numbers, at least as stated, are deceptive. Technology and pharmaceutical companies that look good on paper – have high earnings growth and exceptional returns on capital – may not only be much less attractive, when we

correct for accounting miscategorization but may be actually be poorly run and poor long term investments.

As investors in these companies, we have to correct for these accounting problems and focus on companies that have the following characteristics:

a. Intangible assets should generate high margins and returns: The intangible asset possessed by the firm has to be unique and difficult to replicate. We are, in effect, applying the same test to investments in R&D and brand name that we apply to investments in plant and equipment. If a consulting firm claims that its consultants are the best in the business and that this is the source of their excess returns, ask the question: How do you hold on to these consultants?
b. Reasonable prices for "true" earnings: It is undeniable that many firms with intangible assets have high growth potential and prices that reflect that. Using the mismatch test that we have used on other companies, we want to invest in technology companies that have high growth in earnings and trade at low price earnings ratios. The one caveat we would add is that growth in accounting earnings at these firms can be misleading, because it is after R&D has been expensed. Thus, you may want to narrow your search to find firms that have high growth in earnings, prior to R&D expensing, and trade at low multiples of pre-R&D earnings.
c. Company is spending to preserve and augment these intangible assets: If a firm is being valued highly because of its intangible assets, you want to keep tabs on how much it is spending to grow the value of it assets over time. In other words, you want pharmaceutical firms that are continuing to invest in R&D and consumer product companies that are spending money on brand name advertising.
d. The spending is efficient: While spending money on intangible assets is key to success, just as critical is the requirement that this spending be efficient. If the investment is R&D, you want to see the research pay off in new patents, which are turned into successful commercial products quickly in the pipeline. If the investment is brand name advertising, you want to see sustained high

revenue growth and/or high margins from both existing and new brand names.

e. Watch for other equity claims: Firms with intangible assets tend to be the biggest users of equity options as compensation. To the extent that the recepient of these options are managers who now lay a c

In effect, you want to invest in companies that make good investments in intangible assets and are able to leverage these assets to generate high returns, while selling at low market prices.

Conclusion

In this chapter, we examine the two key issues that we face when valuing firms with substantial intangible assets. The first is that the accounting treatment of what comprises capital expenditures at these firms is inconsistent with the accounting treatment of capital expenditures at manufacturing firms. R&D expenses, brand name advertising and employee recruitment and training expenses are treated as operating, rather than capital expenses. As a result, both the earnings and book value numbers at these firms are skewed and using them in valuation can lead to poor estimates of value. We examined ways of correcting for this accounting inconsistency and the resulting effect on value. In general, firms that can convert R&D expenditures more efficiently and profitably into commercial products will see their estimated values increase, as a result of the correction, whereas firms that spend significant amounts on acquiring intangible assets with little to show for it in terms of higher earnings will see their estimated values decrease.

The second issue that we consider is the use of equity options to compensate employees. We look at two traditional approaches for dealing with these options – the diluted stock and treasury stock approaches – and discard them. Instead, we argue for valuing these options using modified option pricing models and adjusting the value of common shares today both for options that have been granted in the past (the option overhang) and expected future option grants.

Section 4: Mopping up

Chapter 12: Valuation Lessons

Through this book, we have emphasized the importance of first principles in valuation and how they should guide us, when faced with questions. The dark side of valuation, as we have described it, takes different forms with different types of firms and the remedies we offer have also varied with each type. In this chapter, we pull together some of the core ideas that can allow us to combat the pull of the dark side. In the process, we will very quickly review the foundations of what we would like to portray as the right "light" side of valuation.

Enlightening Propositions

When confronted by uncertainty or missing information, we are tempted to adopt loosed-backed rules of thumb and make inconsistent assumptions about growth, risk and cash flows. In this section, we will outline a few propositions that can guide us in making better judgments, when challenged, and result in better valuations.

Proposition 1: First principles matter

There are a few basic principles in valuation that we should never compromise on, no matter what the counter arguments are. An analyst who argues that firms can grow forever without reinvesting is violating a first principle, as is one who says that risk does not matter in determining value. Lest this be seen as a sign of rigidity, we would hasten to add that we should always be willing to compromise on and accept better tools, and be open to alternative estimates for inputs into value. Thus, the capital asset pricing model is a tool for estimating risk, which we should be willing to adapt, modify or even abandon, if the data so directs us. The beta that we come up with for a firm, in the context of measuring risk, is an estimate and there should be no one approach that dominates. In summary, then, we will remain steadfast in our belief that risk should affect value, open to new

developments when it comes to models for estimating that risk and always be on the lookout for better and more consistent estimates of beta or other risk parameters.

Proposition 2: Pay heed to markets but don't let markets determine your valuations.

Many of the companies that we are called upon to value are traded in financial markets and have a market price. Without making any judgments about markets or their efficiency, we believe that there is valuable information in how the market is pricing assets. In fact, our estimates of riskfree rates, in chapter 6, and equity risk premiums, in chapter 7, came from financial markets – the pricing of the treasury bond for the first and the level of the equity indices in the latter. When valuing a company, it therefore behooves us to pay attention to the market price for three reasons:

1. The information we extract from markets on implied growth rates, risk and cash flows can be used to improve our valuation.
2. Ultimately, we make money, not from our intrinsic values, but from the market price moving to those intrinsic values. Thus, we need to understand why market prices may deviate from value and how they will adjust over time.
3. Warren Buffett is fond of the adage of Mr. Market, where we, as long-term intrinsic value investors, try to take advantage of an unpredictable and unstable market that goes from under pricing to over pricing securities.

At the same time, paying too much attention to markets (and prices) can lead to paralysis, since there is always an alternative set of assumptions under which the market price is justified.

When valuing companies, we face a balancing act, where use market inputs for some numbers (riskfree rates, equity risk premium, market weights for debt and equity in the cost of capital) and argue that the market is wrong on other dimensions. To preserve sanity and derive reasonable values for companies, we have to decide early on in a valuation, which inputs we will accept market inputs on, and which ones we will contest

the market consensus. In chapters 6 and 7, we argued that when valuing individual companies, we should accept the market consensus on riskfree rates and equity risk premiums and focus our attention or estimating the earnings and cash flows of the company. In chapter 14, when valuing an oil company, we made a case for using market consensus prices for oil prices (obtained from either the spot or the forward and futures markets) and limiting our valuation to how much oil the company would produce and its cost structure.

As a general rule, the areas where you choose to accept the market wisdom and the points at which you deviate will depend on two key variables. The first is in the competitive advantages you see yourself having as an analyst and where you think these advantages will generate the biggest payoff. If your edge is in forecasting macroeconomic movements and oil prices, you will replace the market prices with your own and leverage your edge to generate better valuations. If, on the other hand, your biggest strength is in assessing the internal strengths of a company, you should not be expending resources on estimating macro parameters. The second factor is your "job description". If your job is to assess the value of individual companies and not to assess the overall market or commodity prices, you will be doing yourself and your clients a disservice by letting your views on macro economic variables seep into your company valuations.

Proposition 3: Risk matters

It is true that considerable resources have been invested both in academia and practice in coming up with models to assess risk and derive expected returns. It is also true that these models often make assumptions about the real world that do not withstand close scrutiny. In the capital asset pricing model, for instance, we assume no transactions costs and private information to derive the relationship between expected returns and betas. Analysts, looking at these models, find themselves in disagreement with the assumptions and that is perfectly

understandable, but jumping from that disagreement to a conclusion that risk does not matter in value is not.

Risk affects value, no matter which approach you use to derive that value. In discounted cashflow models, the risk is manifested either in the discount rate or in a risk-adjusted cash flow (which is more than just an expected return). In relative valuation, firms that are riskier should trade at lower multiples of earnings, revenues and book value than firms that are safer. If there is one lesson that we should take away from the last few decades of examining risk, it is that all risks are not equal. Some risks affect only a firm or a few firms, whereas other risks have consequences for a wider subset of firms and for the overall market. When assessing risk for valuation, it is therefore imperative that we first identify the marginal investors in a firm and look at risk through their eyes. If the marginal investor is diversified, the only risks that should affect value are the market or macro risks that cannot be diversified away. In contrast, though, if the marginal investor is undiversified or only partially diversified, we should be looking at the much broader set of risks that include firm specific risks. In chapter 9, this was the insight we used to derive total betas and costs of equity for entrepreneurs (who tend not to be diversified) and venture capitalists (who are only partially diversified).

Risk can also be categorized based into continuous risk, i.e. risk that affects cash flows and value continuously over time (interest rate risk and exchange rate risk, for example) and discrete or discontinuous risk, i.e., risk that may lie dormant for long periods but affect value significantly when it occurs (the risk of default and nationalization, for instance). We argued that continuous risk is best captured in discount rates, whereas discrete risk is more easily measured by assessing both the probability and the consequences of the event occurring. We used the latter technique the values of young, growth companies and distressed companies for survival risk and emerging market companies for nationalization risk.

As we build in risk into valuation models, it is also important that we adopt the following practices:

1. Isolate the risk effects: Rather than let risk pervade all of the inputs into a valuation, the effects of risk should be in one or two variables. Thus, the

effect of emerging market risk is captured in the country risk premium but does not affect risk free rates, betas or cash flows. There are two reasons why this is a good practice. The first is that it prevents the double or triple counting of risk. The second is that it allows for more transparency. Thus, those who use the valuation can assess the risk adjustment made and decide whether they agree with it or not; if they disagree, it is relatively simple to make the change and re-estimate value.

2. Be consistent: When valuing firms, especially growth firms or firms in trouble, we have to recognize that the firm's risk profile will change over time and adjust the risk parameters accordingly. With growth firms, for instance, the high discount rates we use in the early years (reflecting the risk in those years) have to come down as growth declines. In almost all of the valuations that we have presented in this book, the betas and discount rates that we use for the terminal value reflect this judgment.

Proposition 4: Growth is not free and is not always value adding.

If there is a theme to our discussions of growth across the chapters, it is that growth is not free. Ultimately, the expected growth in earnings and cash flows in a company have to come either from new investments or improved efficiency. The latter is finite growth – there is a limit to how efficient you can become as a firm – whereas new investment growth is long term. This is the reason why growth in the terminal value computation is estimated purely from new investments in every valuation done in this book.

It is also critical to remember that growth by itself is not always a plus for a company, since the value added by growth is a function of the quality of the investments that generated that growth. That is the reason that we have focused so intensely on excess returns, estimated by comparing the return on capital to the cost of capital or the return on equity to the cost of equity, in our assessments of growth. A firm that grows by investing in new assets that generate a return on capital equal to its cost of capital will become larger over time but will not add value to its investors. A firm that grows by investing in poor investments, i.e., the return on

capital (equity) is less than the cost of capital (equity) will become less valuable over time, even as it grows. In fact, a key test of managers is whether they are able to generate high excess returns, while reinvesting significant amounts back into the business.

In the context of relative valuation, we may not be directly focused on excess returns but we should pay attention to them. Thus, if two companies are expected to deliver the same expected growth in earnings in the future, the company that has the higher excess returns accompanying this growth should be valued more highly.

Proposition 5: All good things come to an end.

In addition to looking at the value added or destroyed by growth, we often have to make estimates of future growth rates for companies. In making this assessment, there are two key variables that we have to pay attention to. The first is the scaling effect. As companies become bigger, it becomes more and more difficult for them to keep delivering the excess returns and growth rates that they used to in the past. Thus, it is almost a certainty that a firm that has grown 100% a year for the last 3 years will grow at a slower rate in the next 3 years. With young growth companies, in chapter 9, we reflected this reality by lowering expected growth rates in revenues and earnings as we move through time. The other is competition. When firms are successful, they attract attention, which in turn leads to imitation and increased competition. While firms may be successful in keeping this competition at bay for extended periods (using legal and business tools), it will inevitably chip away at profitability and growth.

One way we reflected the inevitability of declining growth was by assuming that all firms, no matter how highly regarded they might be, eventually become stable growth firms (at which point we estimate a terminal value). We captured the effects of competition over time by pushing the margins of individual firms to the industry average and the returns on capital (equity) towards the cost of equity (capital).

Proposition 6: Watch out for truncation risk

In conventional discounted cash flow valuation, we value firms as going concerns, with cash flows continuing into perpetuity, often growing each year. While this may not be an unreasonable assumption for some firms, the reality is that most firms do not make it. Many young firms run out of cash, some mature firms become the targets of leveraged acquisitions and most distressed firms end up defaulting on debt (and going out of business). The optimism that underlies discounted cash flow valuation can lead us to over estimate the value of firms, where the risk of not making it (truncation risk) is high.

In the chapters where we looked at firms across the life cycle, we argued that the risk of truncation is greatest for firms at either end of the life cycle – very, young growth companies and declining or distressed companies. With these firms, we adopt a two-step approach to valuation. In the first, we assume that the firms would not only survive but also revert to good financial health over time, and value them on that basis. In the second, we estimate the likelihood that they would not survival and the values that we would assign the firm and equity, in the event of failure. Our final estimate of value is a weighted average of the two numbers.

Proposition 7: Look at the Past, But Think About the Future

One of the conundrums we face in valuation is that although almost all the data we have available to us is about the past, reflecting the company's history (past financial statements, betas), the sector (industry average margins and returns on capital), and macroeconomic variables (interest rates, exchange rates, stock returns), all the forecasts we have to make are for the future. While we cannot manufacture data for the future, we can follow some simple rules to minimize the damage:

<u>Use historical data, but do not be bound by it:</u> One of the key components in valuing companies well is recognizing when to use past data and when to look at alternatives. Thus, for a profitable company with little volatility in year-to-year numbers, there is no harm in building off last year's numbers when valuing the company. For the cyclical and commodity companies we valued in Chapter 13, this

base-year fixation can lead to valuations that are too high (or low) if the most recent year represents the peak (or bottom) of the cycle. Consequently, we normalized earnings for these companies using the information in the company's own history as well as in the commodity price cycle.

Trust in mean reversion, but watch out for structural breaks and changes: Whenever we use historical data to make forecasts, we are assuming that mean reversion exists. In other words, both macroeconomic numbers, such as inflation and interest rates, and company-specific information, such as margins and reinvestment rates, revert to historic norms. While mean reversion has strong empirical support, two dangers are associated with assuming that it will happen. The first is that there can sometimes be significant breaks in history, where the circumstances change so much that the numbers are unlikely to revert for a very long period to past averages. This was the case, for instance, in the U.S. in the 1970s , when inflation increased and pushed up interest rates and commodity prices for almost a decade. The second danger is that there is no consensus on what exactly the historical norm is for most variables. Different analysts can look at the same data and come to very different judgments about what constitutes normalcy.

Use forward estimates as alternatives or checks: With a few variables, we may be able to get forward-looking estimates that can be used either directly in valuation or to check the historical data. With equity risk premiums in Chapter 7, for instance, the implied equity risk premium we estimated, based on the current level of the equity index and the expected future cash flows, offers an alternative to historical risk premiums. With oil prices, futures and forward markets provide estimates that can be used to value companies, rather than past data. We can either replace the historical estimates with these forward-looking numbers or adjust them to reflect differences.

In closing, we should be glad that valuation is not just an exercise in taking last year's numbers and putting them through a model. After all, if this were the case, there would be no need for human intervention, since a computer could very easily accomplish this task. Our role in valuation is to look at past data, assess its usefulness, and then make our best predictions for the future.

Proposition 8: Draw on the Law of Large Numbers

One of the issues we have raised as a manifestation of the dark side is the use of rules of thumb in valuations. These rules can range anywhere from a fixed equity risk premium (obtained from a service) to adding premiums to value for control, synergy, and other "good" factors and subtracting discounts for illiquidity and poor corporate governance. In many cases, the numbers that are used in these adjustments come from looking at past data. The historical equity risk premium, for instance, is estimated by averaging the premium earned by stocks over treasuries over long time periods. The control premium (of 20%) is roughly what acquirers have paid to acquire publicly traded target firms over an extended period. Even if these estimates are constantly updated and are from services that have impeccable reputations, these numbers cannot be treated as facts. They are estimates, based on samples, and they come with substantial standard errors. Thus, a historical equity risk premium of 4%, estimated with 50 years of data in the U.S., has a standard error of about 3% associated with it, just as the acquisition premium of 20% has a standard error of about 5% attached to it.

While making estimates based on data will always be subject to noise, we can do some things to make these estimates more precise:

Use larger samples: When we can, we should expand our data to include more data points. With historical risk premiums, for instance, an equity risk premium estimated over 100 years of data will have a standard error of only 2% associated with it. This push toward larger sample sizes permeates much of what we have done in this book to combat bad practices. From bottom-up betas (where we replace a single regression beta, with a large standard error, with an average of many regression betas, with much smaller standard errors) to the use of industry-average profitability measures in forecasting future earnings for a company, we substitute averages of large samples for individual company data. In relative valuation, where we look at the average multiple at which other firms in a sector trade, we suggested ways of loosening the definition of comparable firms to allow for larger samples. In

effect, we would rather have a larger sample of less comparable firms than a sample of a few firms that look more like the firm we are valuing.

Use statistical tools to improve estimates: In many cases, when analysts use historical data, they settle for the average. While this is a useful number, there may be ways in which the data can be utilized to deliver more precise estimates. Thus, when comparing a firm's PE ratio to the rest of the sector, using a predicted PE from a sector regression should yield a more precise estimate than just using the average PE for the sector.

We should make a final point about the law of large numbers. Since the values we obtain for individual companies are estimates, the likelihood that we will be wrong about any one company in any one period is high. However, if we assess value well, our chances of being right will improve if we extend to multiple periods (longer time horizons) and across multiple stocks (portfolios).

Proposition 9: Accept Uncertainty, and Deal with It

We noted early in this book that one of the reasons we are attracted to the dark side of valuation is because we feel overwhelmed by the uncertainties we face with some groups of companies. In fact, much of this book has been directed at developing healthier responses to uncertainty:

Accept uncertainty: Accept uncertainty as a given, understanding that no matter how carefully we construct models, we cannot make uncertainty go away. Thus, adding more detail to models or making models more complicated, both of which are tactics that analysts use to reduce uncertainty, often does little to alleviate the problem and may actually make it worse.

Ask what-if questions, but for the right reasons: As access to data improves and more sophisticated tools for dealing with uncertainty (Monte Carlo simulations, decision trees) are introduced, we are also seeing the potential downside of allowing analysts to subject their valuations to these tools. The first problem is that these tools often required complex inputs, and the quality of the output is a function of the quality of the inputs. Simulations, for instance, require us to choose probability distributions for our inputs and parameters for the distributions. This is

a difficult task considering the data we can access. The second problem is that some analysts use these as tools for risk adjustment rather than risk assessment. In the process, they double-count risk. Consider a discounted cash flow valuation of a firm, where expected cash flows are discounted back at a risk-adjusted rate to arrive at a value that is higher than the current market prices. Assume that you put this model through a Monte Carlo simulation, derive a distribution of values (which suggests that there is a 40% chance that the stock is overvalued), and decide not to invest in the firm on this basis. You have double-counted risk once by using risk-adjusted rates, and again by drawing on the results of the simulation.

In summary, uncertainty and risk are part of life and investing. When valuing companies, we can demand a premium for taking this risk in the discount rate and arrive at a risk-adjusted value, but it will change as circumstances change.

Proposition 10: Convert Stories to Numbers

One of the critiques of discounted cash flow valuation is that it is so focused on numbers that it misses the qualitative variables. Included in the list of qualitative variables are items such as customer loyalty, brand name, and good management. This can be true in some cases, where few questions are asked about the numbers and the inputs are mostly exogenous.

We hope that we have addressed this criticism directly, as we have valued individual companies. Every number in a discounted cash flow valuation should have an economic rationale a good economic story behind it. Thus, when we set a firm's return on capital at 15%, well above its cost of capital, it behooves us to think about this firm's competitive advantages (most of which are qualitative) and how they translate into the return on capital. On the flip side, every solid economic story should find a place in the numbers. Brand name affects value, but it does so by increasing margins and excess returns, not as a premium at the end of the process.

In relative valuation, the tendency toward storytelling sometimes overwhelms the numbers. One reason for our use of statistical tools (regressions and correlations) in relative valuation is not only to assess whether the underlying story makes sense, but also to quantify the effects in our predictions. Thus, by regressing PE against

expected growth, we can measure how much the PE ratio increases for every 1% increase in the growth rate, at least given how the market is pricing stocks in this sector.

Conclusion

We will end this book how we began it by noting that valuation has only a few first principles, and that the basic structure of valuation has not changed much over time. However, the kinds of firms we are called on to value pose more challenges today than ever before. As firms and investors globalize, companies have become more complex in terms of both their operations and the financing choices they make to fund those operations.

When called on to value companies, we should recognize that rigid rules do not serve us well, since valuing individual companies requires us to be flexible and devise new rules when faced with unusual scenarios. Rather than let estimation issues overwhelm us and force us into bad choices, we should step back and assess commonsense ways, constrained by the data we have available, to deal with difficult issues.